



Vehicle Electric Container (VEC)

prostep ivip Documentation

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Abstract

This is the printable version of the Vehicle Electric Container (VEC) Online Model Description, available at <https://ecad-wiki.prostep.org/specifications/vec/>. Its main purpose is to provide a useful representation for reading and reviewing.

This is the not officially published VEC PSI recommendation.

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1 Model of the VEC data format

1.1 Key Concepts

NA

1.1.1 Parts, Documents and Resources

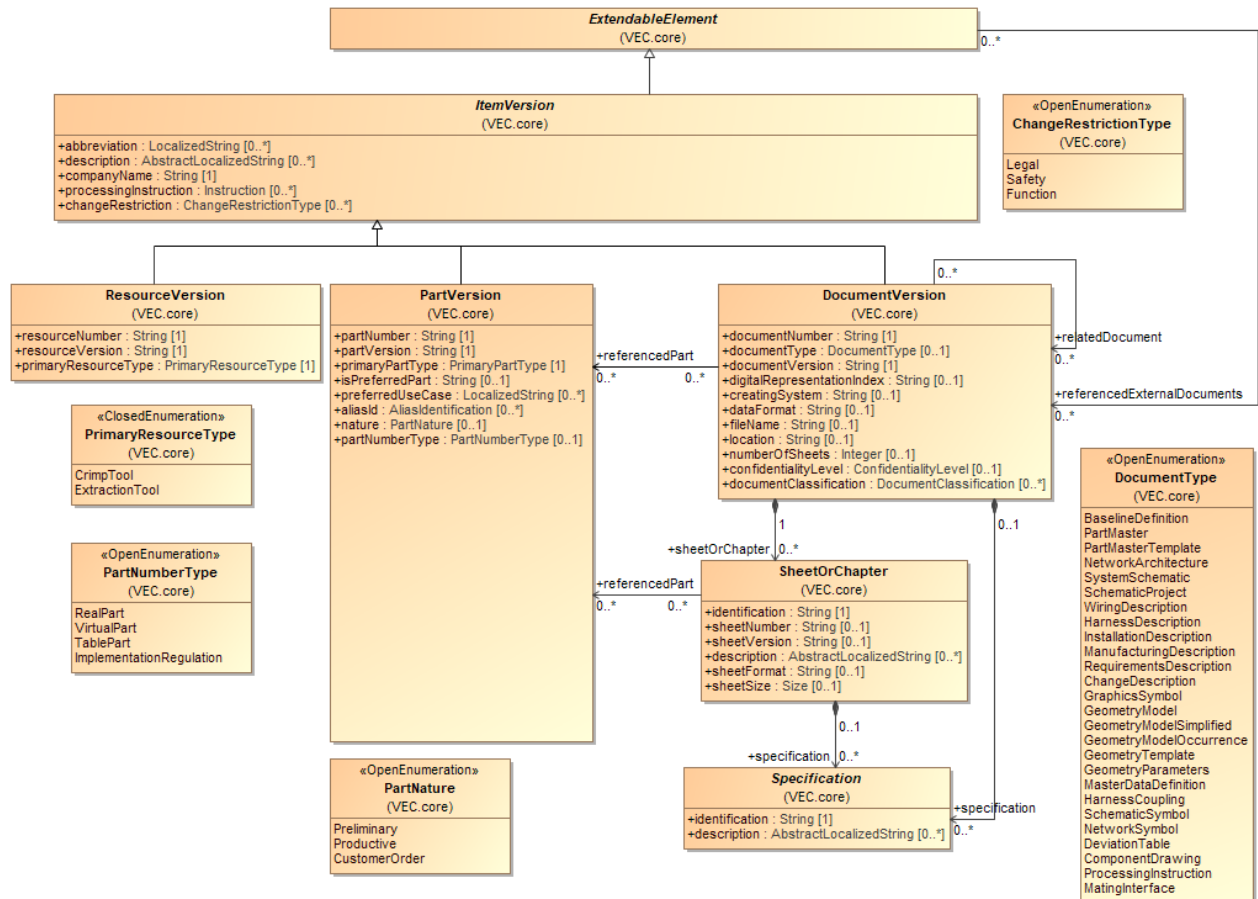


Figure 1: Parts, Documents and Resources

DocumentVersion, *PartVersion* and *ResourceVersion* are representing one of the most fundamental key concepts of the VEC. They are the only classes in the VEC model that are defining version information. In combination with the attributes *companyName* and *partNumber* or respectively *documentNumber* or *resourceNumber* they define an unambiguous context for data exchange. If two instances of the VEC are containing *PartVersions*, *DocumentVersions* or *ResourceVersions* with equal values of the attributes *companyName*, *partNumber* and *partVersion* or *companyName*, *documentNumber* and *documentVersion*, or *companyName*, *resourceNumber* and *resourceVersion*, then the two objects can be trusted to be unchanged.

Unchanged in this context does not mean binary identical content of the resulting XML-Subtree, but unchanged user data. So if the content of a *DocumentVersion* is transferred from one VEC-file to another (e.g. for documents describing part master data), the binary representation can be different (e.g. order of the xml elements, technical ids for xml idrefs), but the information represented by the *DocumentVersion* must not be different (e.g. property values of a *Specification*). However, it is explicitly allowed that the scope of the information contained in different instances of the same *DocumentVersion* may differ. For example, in most processes a component (e.g. a connector) is specified by a technical document, in the VEC represented as a *DocumentVersion*. In the use case of part master data exchange the VEC for the component might contain very detailed information. For the use case of harness description not all of this detailed information is required (e.g. for a connector the information about slots and cavities might sufficient) and therefore it should not be included. However, the component is still specified by the same technical document. Another example of a

scenario where this might occur, is the update of a VEC exporting system. A later export might contain more information, even if the original document has not changed.

These cases are allowed with an unchanged *documentNumber*, *documentVersion*, if the shared part of the represented information in both *DocumentVersion* instances is still consistent.

More specific possibilities to track changes of *DocumentVersions* are supported by the *digitalRepresentationIndex*. In some cases, the *DocumentVersion* represents a version of a real document in the domain. The *Specifications* in the *DocumentVersion* represent the values defined in the original document. Thus, the *DocumentVersion* in the VEC is not a managed document by itself, but only a specific digital representation of another document. In other words, it is only a translation of the original document into the language of the VEC. For example, there is a connector described by a drawing / data sheet, which is available in some arbitrary format (e.g. PDF). This document has a number and a version. When the document is transferred (e.g. with a converter or manual input) into the VEC, the *DocumentVersion* containing the *Specifications* might have the same *documentNumber* and *documentVersion* as the original data sheet, since it is not a managed document in the process, but only a different representation of the same information. In such a scenario, it is possible that the content of a *DocumentVersion* changes without a change of the original document (e.g. there was an input error during the manual transfer of information, there is a new version of the converter / exporter and many more). However, the original document has an unmodified *DocumentNumber* & *DocumentVersion*. To indicate such (potential) changes of the content of a *DocumentVersion* the *DigitalRepresentationIndex* was introduced. *DocumentVersions* that contain a different amount of information (see above) shall also have different *digitalRepresentationIndex* values (if defined).

That means, if no *DigitalRepresentationIndex* is defined, the rules from above apply. If a *DigitalRepresentationIndex* is defined, the content of the *DocumentVersion* can only be trusted to be unchanged, if *companyName*, *documentNumber*, *documentVersion* and *digitalRepresentationIndex* are the same.

Since all tree classes can define PDM information they are derived from the abstract class *ItemVersion*.

The differentiation between *PartVersion* and *DocumentVersion* in the VEC is necessary, because a part can be described by multiple documents (e.g. a technical data sheet, a drawing and a 3D-model) and one document can describe multiple parts (e.g. a drawing for a contact family that displays multiple terminals and their corresponding seals). As a result, a part can be changed (requiring a new *PartVersion*) without the need of changing all of its describing *DocumentVersions*. For example, if a single property of a component changes, a new version of the part itself and its technical data sheet is necessary, but its 3D model might remain unchanged. Another example is, that a document describing multiple parts might be changed (requiring a new *DocumentVersion*), but in fact only one of its represented parts is changed (requiring a new *PartVersion*). This relationship between a *PartVersion* and its describing *DocumentVersions* is expressed in the model by the association *DocumentVersion.referencedPart*.

The relationship *DocumentVersion.relatedDocument* can be used to describe general relationships between *DocumentVersions* (e.g. the relationship between a technical drawing and a given standard the drawing is compliant with). The composition of *SheetOrChapter* considers the fact that *DocumentVersions* may be composed of several sheets or chapters.

The VEC is designed as a format for data exchange. This requires that all content data can be related to a certain unambiguous version context, in order to allow a receiver of the information to easily check if and what has changed since the last data exchange. For this reason, all content data that is not constant is defined with information specific subclasses of the class *Specification*. A *Specification* is always contained in a *DocumentVersion* and thereby related to an unambiguous version context.

1.1.2 Assignment Group

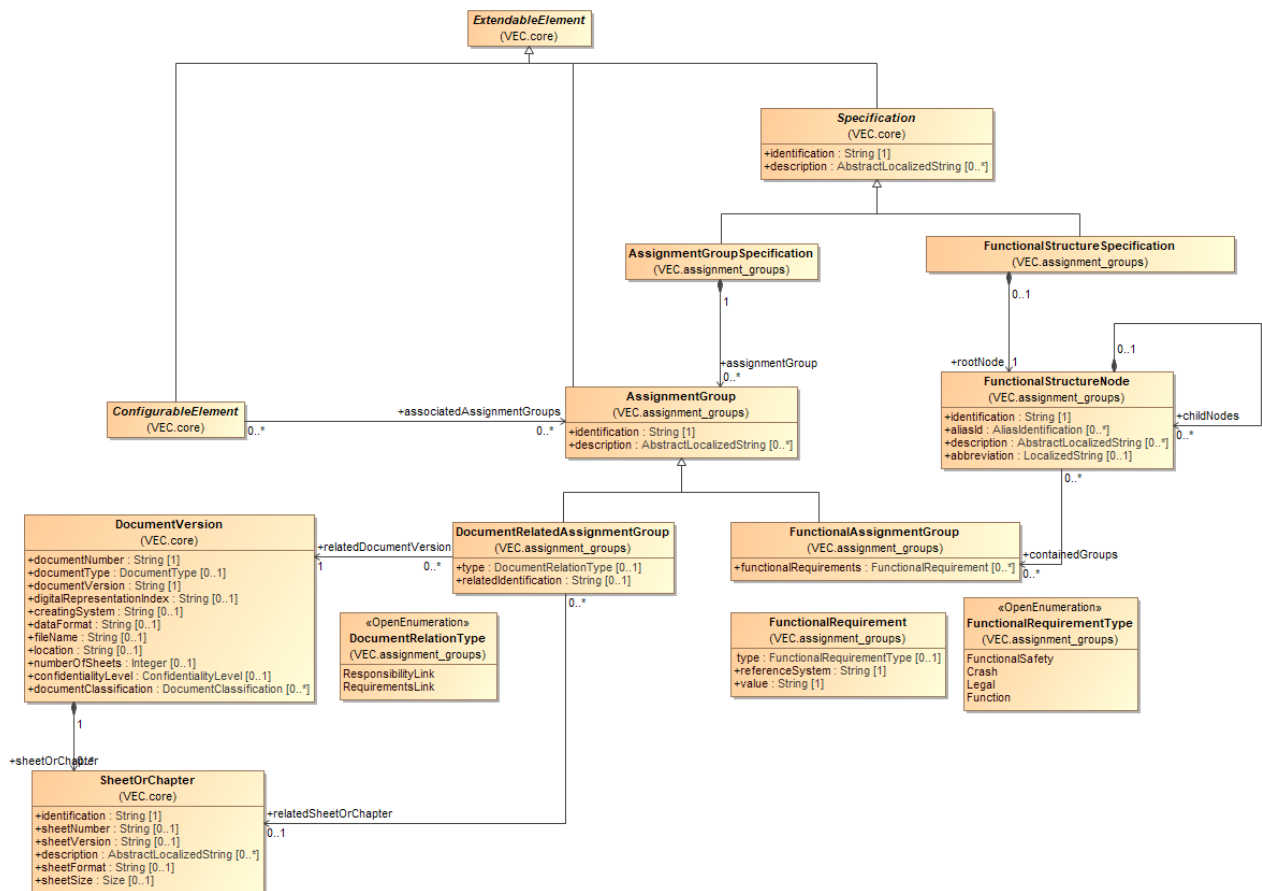


Figure 2: Assignment Group

An *AssignmentGroup* is a concept that allows the clustering of arbitrary elements in ways that are orthogonal to hierarchical and semantic structure of the VEC. Specific use cases are represented as subclasses of the *AssignmentGroup* and enriched with additional information. Proprietary groupings can be created with the *AssignmentGroup* itself and additional information in custom properties.

The *FunctionalAssignmentGroup* can be used to assign elements to a function across the different layers of abstraction.

The *DocumentRelatedAssignmentGroup* can be used to relate elements to a specific document or an element within the document (e.g. a requirement).

The assignment to a group originates from the *ConfigurableElement* itself. This is designed intentionally and has two reasons:

1. In the use cases for this grouping concept, it is intended that the groups exist first (e.g. a function) and the assignment is created together with elements along the process. For example, a connection in a system schematic is assigned to a function. When a wire is derived from the connection, the wire inherits this assignment. Therefore, it wouldn't be practicable to be enforced to modify the *DocumentVersion* containing the *AssignmentGroup* for every added element.
2. In today's processes the assignment of elements to such groups are done in the same documents where the elements themselves are defined.

This concept adds another layer of extensibility to the VEC. *CustomProperties* allow the addition of user-defined properties to a single object. The assignment groups allow the creation of custom grouping of objects. However, it is not permitted to use this concept to express relationships that have well defined concepts in the VEC (e.g. assignment of occurrences to composite parts or usage nodes).

1.1.3 Variants

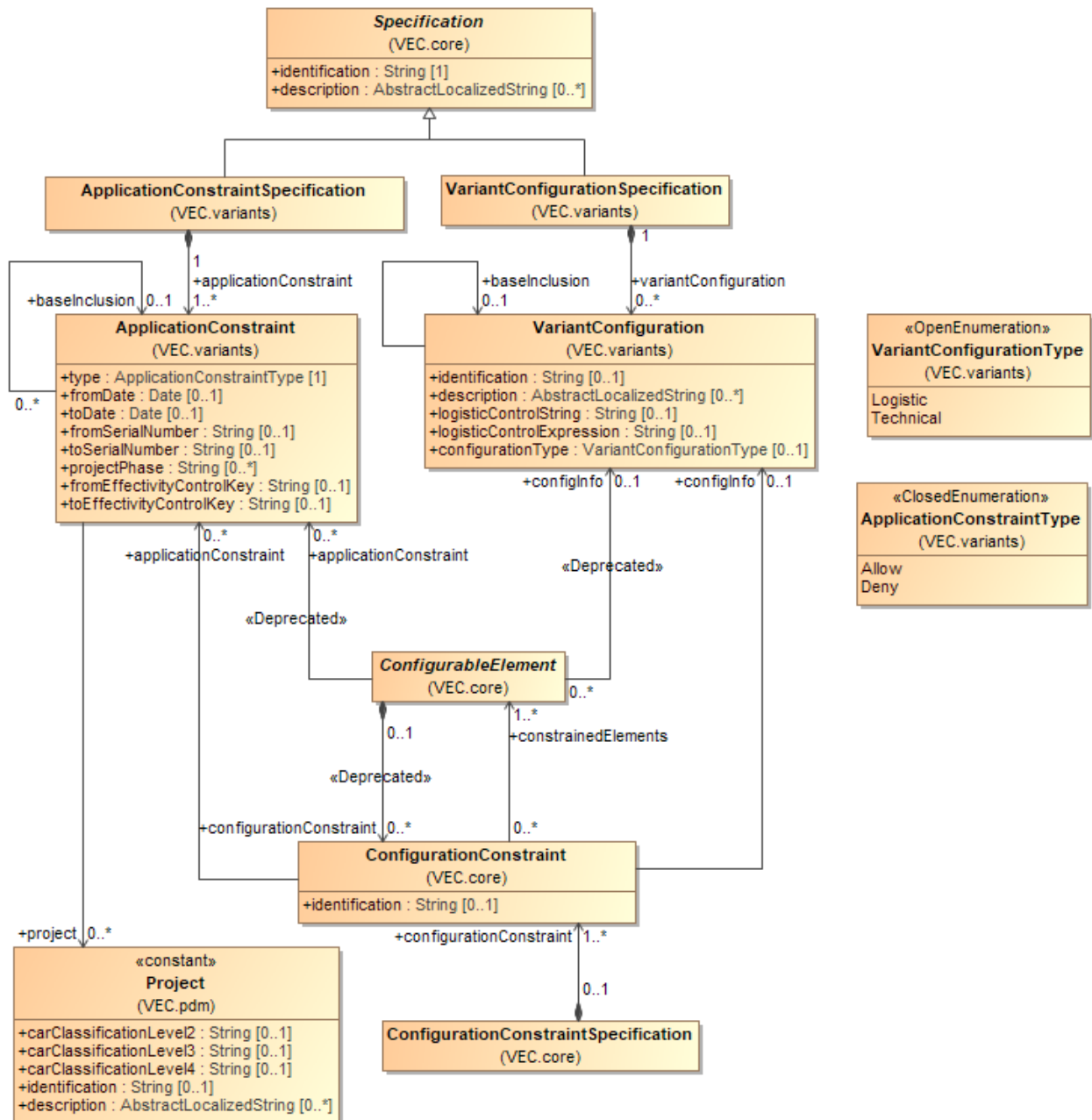


Figure 3: Variants

With variance management features, the VEC supports two dimensions of variance. *VariantConfigurations* and *ApplicationConstraints*. *VariantConfigurations* define the validity of an element based on characteristics of the product itself, such as features, functions, options or technical characteristics. *ApplicationConstraints* define the validity of an element based on conditions "around" the product like manufacturing schedules and change management, association with a specific product derivative or factory dependencies. *VariantConfigurations* are expressed by a *LogisticalControlString* or *LogisticalControlExpression*, typically some kind of Boolean logic in a company specific syntax. *ApplicationConstraints* are expressed by a concise semantic with attributes and relations or by using *EffectivityControlKeys*, which are unique references in the logistics/change/scheduling control systems of an organization.

All elements in the VEC that could be subject to product variance are *ConfigurableElements* (classes that are derived from *ConfigurableElement*). Since most variant management conditions do not exclusively affect only

one element, *ApplicationConstraints* and *VariantConfigurations* are embedded in their own container *Specifications*, the *ApplicationConstraintSpecification* and the *VariantConfigurationSpecification*.

Till VEC 2.0 a *ConfigurableElement* could only reference a single *VariantConfiguration*. This is true, if you consider only one specific context, however, there can be different variant management contexts for a single element. Different contexts might be defined by *ApplicationConstraints*, such as differing variant conditions for different product derivatives or the change of *VariantConfigurations* over the time. Another reason for different contexts might be, that *VariantConfigurations* are based on a defined set of variant codes (a universe of variance). Multiple universes can coexist in an organization and there are a variety of reasons that justify their existence, for example product lines might define their own variant codes and semantics, the variant codes (e.g. feature model) might different for a set of building blocks compared to an actual vehicle, or the variant codes of a generic E/E-component might different from the ones in a specific usage.

Such a context is defined by a combination of *ApplicationConstraints* and a specific *VariantConfiguration.ConfigurationType* and is represented by a *ConfigurationConstraint*. With the *ConfigurationConstraint*, a *ConfigurableElement* can have multiple *VariantConfigurations*. However, an element shall only have one *VariantConfiguration* for a specific context (combination of *ApplicationConstraints* & *VariantConfiguration.ConfigurationType*). If an element has multiple variant conditions within a context, those shall be expressed by the Boolean logic within the *VariantConfiguration.LogisticControlExpression*.

Note: With VEC 2.0.2 the direction of the association between *ConfigurableElement* and *ConfigurationConstraint* has been inverted and the *ConfigurationConstraintSpecification* has been introduced as container for the *ConfigurationConstraints*. The former approach, with the *ConfigurationConstraint* contained in the *ConfigurableElement*, had the effect, that a change of the configuration information resulted in a change of the *ConfigurableElement* itself. This in turn means a change or addition of configuration information results always in change of the original document. However, when reusing engineering artifacts, it is a common approach to configure the elements not until they are used. Configuration information is added in the context of usage and not in the context of definition. By extracting the *ConfigurationConstraint* into an independent specification it is now possible to add configuration information in a later stage of the process.

Note: The direct associations between *ConfigurableElement* and *ApplicationConstraints* and *VariantConfigurations* have been deprecated with VEC 2.0 and have been kept only for backwards compatibility. They will be removed in a future version. It is **not** valid to use these associations in combination with *ConfigurationConstraints*.

Note: Please refer to the detailed class description for information about which elements inherit from *ConfigurableElement*.

Note: *VariantConfiguration.logisticControlExpression* has been introduced as an alternative attribute for *VariantConfiguration.logisticControlString*. For the *logisticControlExpression* the prostep ivip project group "VES-WF" wants to introduce a defined syntax in the future.

1.1.4 Variant Structure

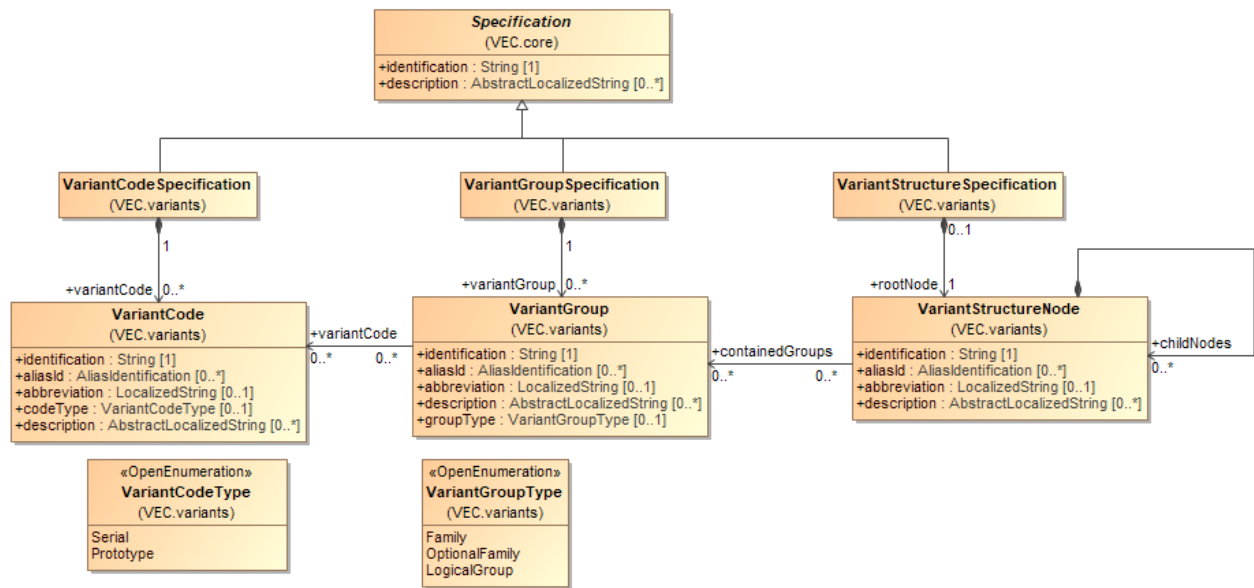


Figure 4: Variant Structure

The classes displayed in this diagram can be used to describe meta information about the elements used to describe variance properties. The description of the variance information of a concrete element (*ConfigurableElement*) is done with a *VariantConfiguration* and the contained *logisticControlString*. These strings are normally created from *VariantCodes* and a proprietary Boolean syntax.

A *VariantCodeSpecification* is a container for various *VariantCodes*. Each *VariantCode* describes an elementary literal which can be used as part of a *VariantConfiguration.logisticControlString* or *VariantConfiguration.logisticControlExpression*.

A *VariantGroupSpecification* is a container for various *VariantGroups*. A *VariantGroup* describes a grouping for *VariantCodes*. There can be a wide variety of reasons to group *VariantCodes*. However, the most common is to mark *VariantCodes* that belong to the same family (mutually exclusive) variant codes.

A *VariantStructureSpecification* with its *VariantStructureSpecificationNode* can be used to define any kind of hierarchical structure upon the *VariantGroups*.

Note: The VEC does not constrain a *VariantCodeSpecification* or a *VariantGroupSpecification* to a precondition for use in *VariantConfigurations*. *VariantCodeSpecification* and *VariantGroupsSpecification* are intended as an optional means to exchange information independently about the “vocabulary” which is possible to express *VariantConfigurations*.

A *UsageConstraintSpecification* is a container for various *UsageConstraints* each representing a single constraint. The sequencing of the *UsageConstraints* specifies the priority: a general denial may be relativized by a following *UsageConstraint* and vice-versa. In other words, the ordering of the *UsageConstraintSpecification.partUsageConstraints* collection has a semantic and is highly relevant. *UsageConstraints* further back in the collection have a higher priority than *UsageConstraints* further ahead. Successive *UsageConstraints* of the same type shall be interpreted as concatenation, whereas successive *UsageConstraints* of different types shall be interpreted as exceptions to the preceding statement.

The relationships *UsageConstraint.usageNode* and *UsageConstraint.project* make it possible to specify at which *UsageNodes* respectively in which *Projects* the usage of the constrained *PartVersions* is allowed/denied.

1.1.7 VEC-Root

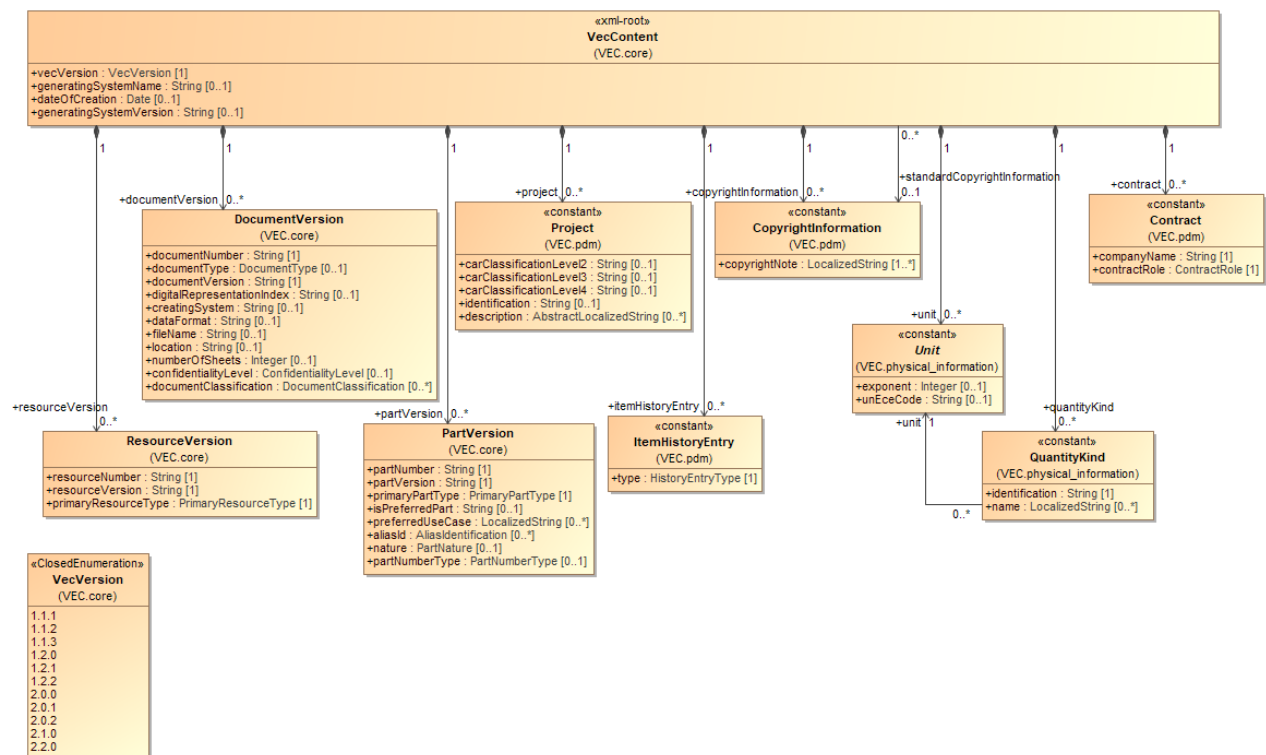


Figure 7: VEC-Root

The diagram above defines the class *VecContent* which is intended to be regarded as the top-level root element for VEC documents. It is the declared entry point to the information that a VEC file contains.

A *VecContent* is a container for various *DocumentVersions*, *PartVersions*, *Projects*, *ItemHistoryEntries*, *CopyrightInformations*, *Units*, and *Contracts*. *DocumentVersion* and *PartVersion* are the only two classes in the VEC model that define and contain version information. Almost all information a VEC document can contain is subordinated to one of those two classes. This means, that every piece of information which is represented as an instance of a class of a certain attribute value is distinctly related to a dedicated *PartVersion* or *DocumentVersion*.

The only exceptions are classes with the stereotype *<<constant>>* as shown in the diagram.

1.1.8 Information Reuse

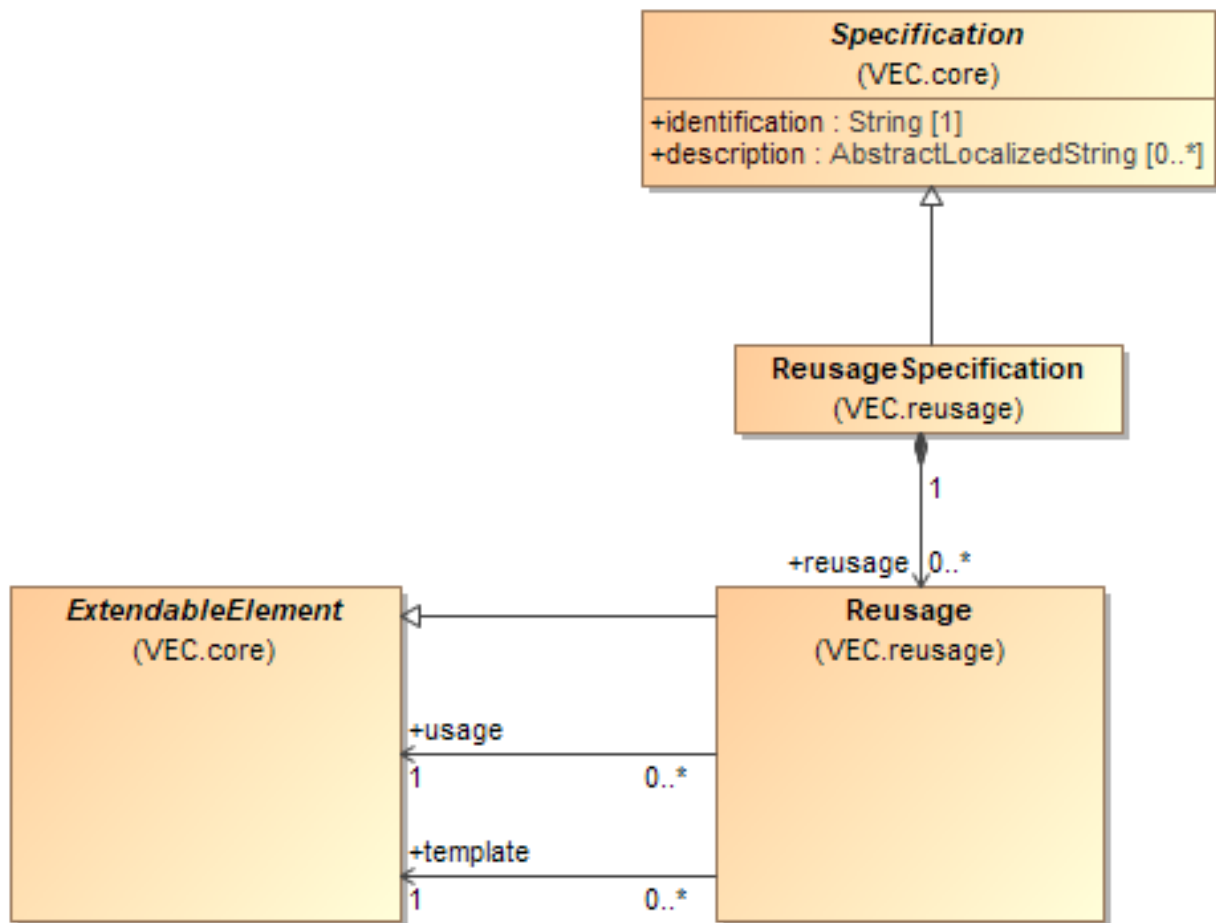


Figure 8: Information Reuse

In engineering processes - particularly in harness development - the reuse or derivation of new information from existing data is a common practice. Specific examples include libraries, building blocks, or modular systems, where certain information, such as a partial system description, is predefined and then adapted to fit the context of a specific vehicle. In the following sections, the source of the information will be referred to as the *template*, while the location where it is reused or the derived information itself will be referred to as the *usage*.

Information reuse differs from a classic instantiation model that typically involves distinct classes for type and instance, such as *PartVersion* and *PartOccurrence*. In an instantiation model, the structure clearly specifies which attributes are defined by the type and which by the instance. For example, the cross-section area of a wire might be defined by the "wire type," while the length is defined by the "wire instance."

In a reuse scenario, however, the same model elements (VEC classes) are employed both in the template and in describing the usage. For instance, a system schematic building block is described using *Connections* and *ComponentNodes*, and its usage is also described using *Connections* and *ComponentNodes*. Essentially, information reuse involves copying the original data, which can then be supplemented with additional information, redefined, or even overridden. The extent to which the usage is allowed to deviate from the template is specific to each case and depends on business rules that apply to the scenario. Typically, the usage is fully self-contained and can function independently.

The traceability provided by the *ReusageSpecification* and *Reusage-Elements* allows the enforcement of business rules between template and usage. Scenarios that exemplify this traceability include:

- **Downstream Change Propagation:** When the template is updated, changes could be propagated to the derived elements.

- **Limitation of Deviation:** Information defined in the template might be restricted or flexible for modification in the usage or only changeable in specific ways. For instance, a template could define standard aspects of a component (such as a standard cavity or conductor). Some template attributes might be mandatory in the usage, while others could be redefined.

An individual *ReusageSpecification* shall be defined when reusing a specific block of information within a particular context. For example, reusing a partial schematic within a vehicle-specific schematic represents one *ReusageSpecification*. Adding a second partial schematic to the vehicle constitutes a second *ReusageSpecification*. Reusing the same partial system schematic in another vehicle-specific schematic would require a third *ReusageSpecification*.

Within a *ReusageSpecification*, a *Reusage-Element* is created for every *ExtendableElement* from the template that is utilized in the usage.

1.2 Basic Datatypes

The VEC supports the following a number of primitive standard data types:

- Boolean
- Date
- Double
- Integer
- String

In addition, the VEC defines various more complex general-purpose data types that are used within the specification. Those are defined in the following sections.

1.2.1 Structured Primitives

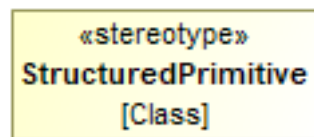


Figure 9: Structured Primitives

The *StructuredPrimitive* stereotype is used to mark model classes that represent structured value containers. These classes group multiple primitive attributes (e.g., strings, numbers, enumerations) into a cohesive structure, typically for the purpose of expressing a concept such as a value with unit, a localized string, or a categorized length.

StructuredPrimitives are semantically like primitive types in that:

- They are used as attribute types in other model elements.
- They are not independently identifiable.
- They shall not be referenced from other parts of the model.
- They shall not be an *ExtendableElement*.

Since they are not independently identifiable and shall not be referenced from other parts of the model, the following is true:

- *StructuredPrimitives* are compared by value in a diff- or change-scenario and not by ID. Meaning, if all attributes are the same, the object is the same.
- They do not require an object identity in an overall scenario (no IRI/URI).
- In RDF representations those objects can be represented as BNodes.

1.2.2 Localization of Strings

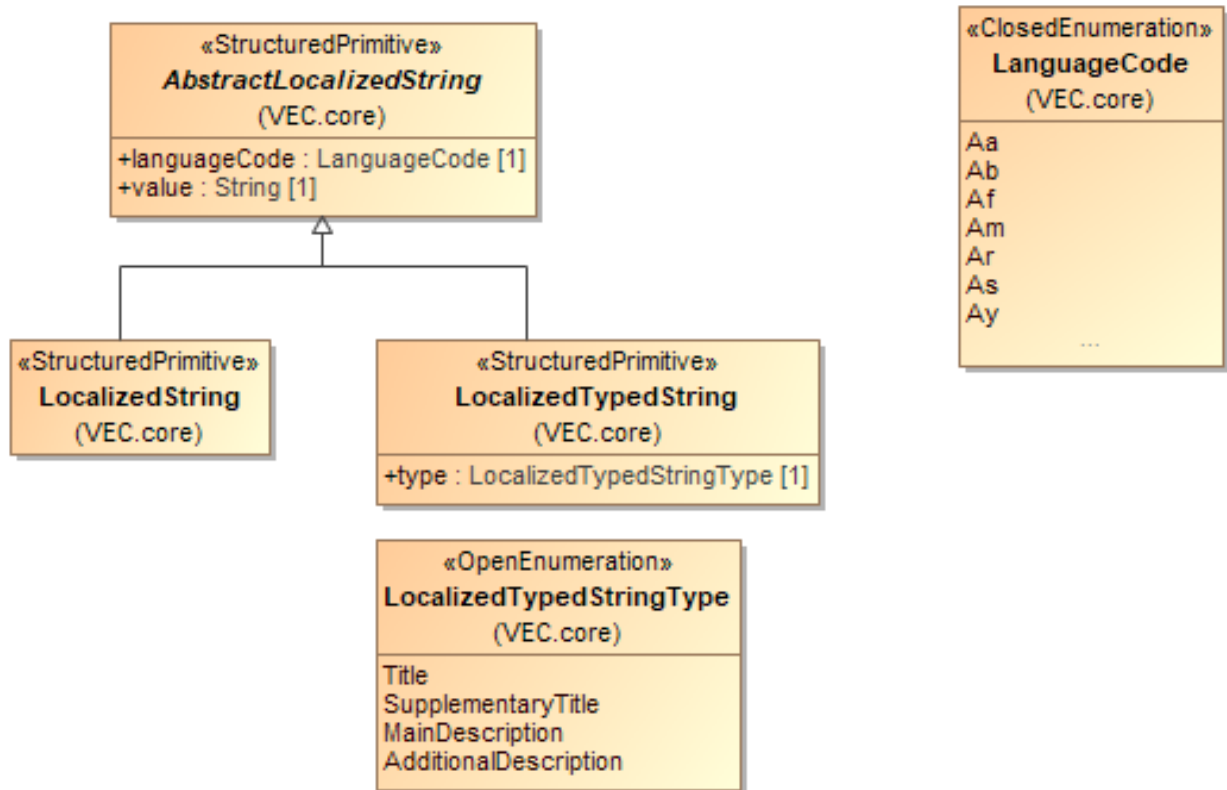


Figure 10: Localization of Strings

The VEC provides a concept for the localization of string values. Every attribute that can be localized has the type of *AbstractLocalizedString* or one of its subclasses.

An *AbstractLocalizedString* consists of an ISO *languageCode* and a *value*. Attributes of this type have a multiplicity of [0..*].

In the case of a *LocalizedString* the attribute represents a human readable text with a specific semantic. All *LocalizedString* instances for a single attribute must represent the same meaning in different languages.

When a *LocalizedTypedString* is used, the semantics of the attribute values can be further detailed by the *type* attribute. For example, an *ItemVersion* can have more than one description with a more specific semantic of each description type. In this case, all *LocalizedTypedString* instances for an attribute must represent the same meaning for their *type* value in different languages.

1.2.3 Numerical Values & Units

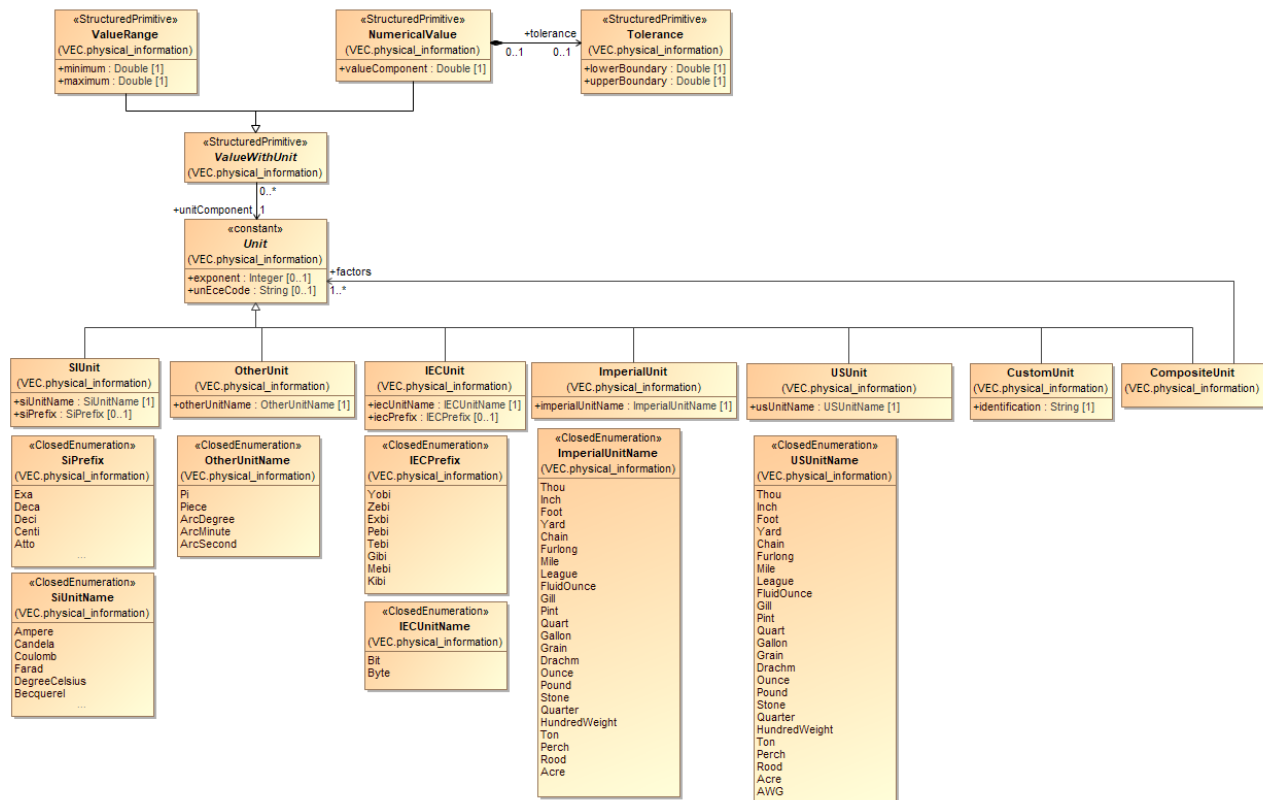


Figure 11: Numerical Values & Units

With Version 1.1.0 a more flexible and meaningful unit system has been introduced. A unit is used to define the dimension of a numerical value. In the VEC a unit is expressed by the abstract class *Unit*. The different existing types of units are represented by its concrete subclasses. Every unit can have an optional exponent. If no exponent is defined, this is interpreted as an exponent of 1.

The class *SIUnit* defines a unit in the terms of the SI-System. A *SIUnit* is defined by a combination of an optional *SiPrefix* (e.g. Milli) and *SIUnitName* (e.g. Metre).

The class *IECUnit* defines a unit in the terms of the IEC-System which is used for measurement of digital data amounts.

The class *ImperialUnit* defines a unit in the terms of the imperial unit system (e.g. Inch).

The class *OtherUnit* is used to define units that do not belong to any standardized unit system, but which are relevant in the context of the VEC. Currently these are the units Pi (as a unit for angular frequency or circular measure) and Piece.

The class *CustomUnit* can be used to define units that are necessary for a specific use case and that are currently not considered in the VEC.

The class *CompositeUnit* is used to define units that are created by the multiplication of other units (Association *CompositeUnit.factors*). For example, the Unit "Ohm per Metre" will be created with two instances of *SIUnit*. One *SIUnit* Ohm (without prefix and *exponent*) and one *SIUnit* Metre (without prefix and an *exponent* of "-1").

Mainly these *Units* are used by the class *ValueWithUnit* which can be either a *ValueRange* or a *NumericalValue*. A *ValueRange* defines a range of a value between *minimum* and *maximum*. A *NumericalValue* defines a single value with an optional *Tolerance*.

Note:

- For the purpose of clarity, not all literals of all enumerations are displayed. If literals are omitted in the graphical representation a "..." is shown.

- Currently the VEC does not support the definition of the same attribute value in different units at the same time (e.g. a single length value in m and ft at same time).

1.2.4 Multidimensional Values

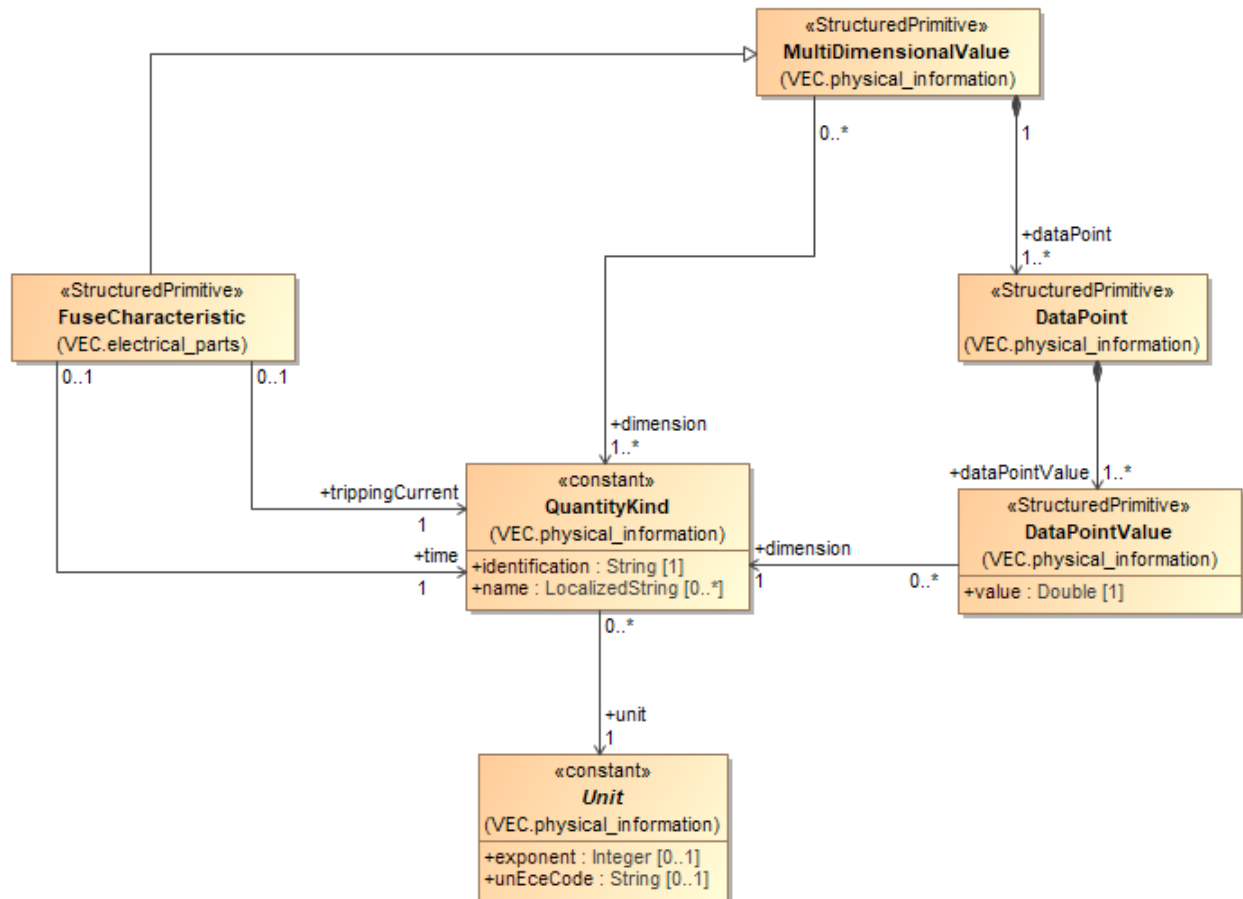


Figure 12: Multidimensional Values

The *MultiDimensionalValue* concept is used to describe properties that are represented by multiple values across different dimensions — for example, a characteristic curve.

The *dimensions*, e.g., tripping time and current for a fuse characteristic, of a *MultiDimensionalValue* are defined by referencing the corresponding *QuantityKinds* in which the dimensions are expressed. The *QuantityKind* also defines a consistent *Unit* of measurement for values in a specific dimension.

To provide a standardized semantic meaning for specific use cases, subclasses of *MultiDimensionalValue* are introduced. These subclasses reference the defined *QuantityKind* in role-specific contexts. However, those references only define a specific semantic for the *dimensions* defined by the *MultiDimensionalValue* itself. That means, those specific *QuantityKinds* shall also be included in the *dimension*-association of the *MultiDimensionalValue*.

For instance, a *FuseCharacteristic* must reference at least two dimensions:

- I** = the tripping current
- T** = the time it takes for the fuse to disconnect

Additional dimensions may be included in certain use cases (e.g., temperature). By explicitly referencing the required dimensions in the *FuseCharacteristic*, these are marked as mandatory and receive a clearly defined semantic meaning.

The value definition consists of a series of *DataPoints*. Each *DataPoint* of a *MultiDimensionalValue* shall define exactly one *DataPointValue* for each *QuantityKind* that is defined as *dimension* of the *MultiDimensionalValue*, and only for those.

1.2.5 Extensibility with Custom Properties

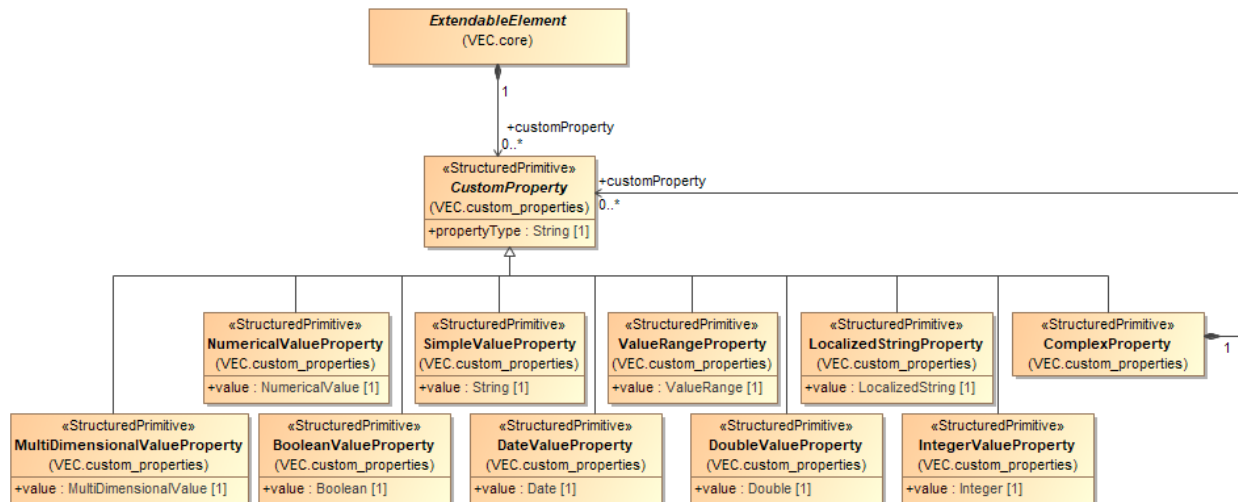


Figure 13: Extensibility with Custom Properties

CustomProperties have been introduced to the VEC as the dedicated extension mechanism. All subclasses of *ExtendableElement* class are extendable by providing the possibility to define *CustomProperties*. *CustomProperties* allow the definition and transport of almost any piece of data in an embedded way, for which the VEC does not define a different means of expression.

If a VEC entity shall be extended with multivalued property this is done by adding multiple *CustomProperties* with the same *propertyType*. In cases where a custom property value consists of a tuple of other values, a *ComplexProperty* can be used. This is especially useful, when the respective property is multivalued. An example for such a property / structure that is already represented in the VEC are *Colors* where each color consists of a referenceSystem and a key in that referenceSystem and each entity can have multiple colors (the same color in different reference systems).

Note: According to this data format specification it is strictly forbidden to store data within *CustomProperties* for which the VEC knows a special predefined way of expression. VEC-Files that do not obey to this rule are not compliant to this data format specification.

Note: Please refer to the detailed class description for information about which elements inherit from *ExtendableElement*.

1.2.6 Foundation Classes for Physical Properties



Figure 14: Foundation Classes for Physical Properties

The diagram displays the common foundation classes necessary for the description of physical properties of a part (e.g. material, mass and size). Most of these classes either define a type of their value (e.g. *TemperatureInformation.temperatureType*) or a reference system (e.g. *Color.referenceSystem*). In these cases attributes of these types normally have the multiplicity of [0..*] in order to define different values for different types (e.g. environment temperature, storage temperature, operating temperature) or to define the same value in different references systems (e.g. the same color in RAL, RGB or a company specific value).

1.2.7 Alias Identifications

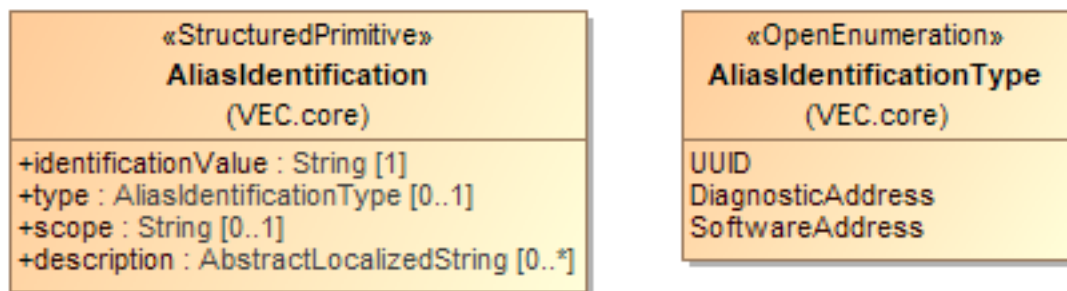


Figure 15: Alias Identifications

There are many applications where an element can have various technical and functional keys at the same time. The *AliasIdentification* provides a possibility to transfer these keys along with meta information about their type and scope.

1.2.8 Open and Closed Enumerations



Figure 16: Open and Closed Enumerations

In the VEC model we distinguish between two kinds of enumerations:

- Closed lists
- Open lists

For a closed list, the enumeration values are fixed. At the time of the UML model generation, it is known which values an attribute can take. There is no possibility to use other values later without changing the XML schema definition as the model defines the acceptable values inherently. An example would be the *PartVersion.primaryPartType* as it closely related to the existing *PartOrUsageRelatedSpecification-class*. Another example would be the *referenceNode* for *Locations* on *TopologySegments*. Since one *TopologySegment* only has one *StartNode* and one *EndNode* this is a typical *ClosedEnumeration*.

For an open list the allowed values for an attribute are not completely known at the point of the UML model generation or might change in future due to new technical developments. There may be a pre-defined set of values. But it must be possible to add further values in a later stage. An example might be the Gender of a Terminal which is currently restricted to *Male* and *Female*. In future new technologies might emerge, where the existing values are not enough. Motivation for this open list concept is, that for known values, standardized literals shall be used, but reading systems must be able to handle additional literals, since they can emerge quite quickly due to technology improvements.

The diagram shows the representation of open and closed enumeration lists as stereotypes.

During the UML to XML transformation, closed enumerations are transformed into XML “*xs:enumeration*”. Open enumerations will be translated into XML “*xs:string*”.

The VEC model defines for open enumerations a pre-defined list of commonly agreed values. The idea behind is, that these pre-defined values are used if applicable, and further values should be used only, if the pre-defined ones do not fit. However, this cannot be enforced by the XML schema. If an open enumeration is translated into “*xs:string*”, it is not possible to check if the predefined values are used.

Therefore, in addition to the standard VEC XML schema, a further XML schema, called the “Strict VEC XML schema” is defined. In this schema, an open enumeration is transformed into XML “*xs:enumeration*”, too. The enumeration values are restricted to the pre-defined ones. With this schema it is possible to check if a VEC used pre-defined values, only.

1.2.9 Open and Closed Pattern Restrictions



Figure 17: Open and Closed Pattern Restrictions

The concept of open enumerations mentioned before is useful in cases where a restriction to a limited number of well-known values is necessary. However, the continuous evolution of the VEC has shown, that this approach is not feasible in all situations.

Therefore, the concept of pattern restrictions has been introduced. A pattern restriction is a regular expression that defines valid attribute values. Those restrictions can be used when a valid pattern is known, but the explicit

enumeration of the values is not possible. For example, it is known that valid values for the coding of a connector are only strings with two digits of alphanumerical characters. To represent this as an enumeration would require 1296 enumeration literals.

In the UML model attributes that shall be restricted with a pattern are defined like regular class attributes with a custom *PrimitiveType* as datatype (e.g. `+coding: CodingName[1]`). The *PrimitiveType* is stereotyped either with *ClosedPatternRestriction* or *OpenPatternRestriction*. The semantic for open and closed is the same as for enumerations. *ClosedPatternRestrictions* are included in all schema versions, *OpenPatternRestrictions* only in the strict schema versions. The actual pattern of the restriction is defined as "class constraint" on the *Primitive Type*.

In the XML Schema the *PrimitiveTypes* are translated as `<xs:simpleType>` based on `xs:string` with a `xs:pattern` restriction.

1.3 PDM Information

The content of this chapter describes the various aspects of PDM information contained in the VEC. PDM information is every information that is no actual content data, but all meta data about *ItemVersions* contained in the VEC.

Since most of the PDM information (product data management relevant information) is in principle equally relevant for both, parts and documents, the VEC defines the abstract superclass *ItemVersion* as anchor for the definition of PDM information.

1.3.1 Lifecycle Information

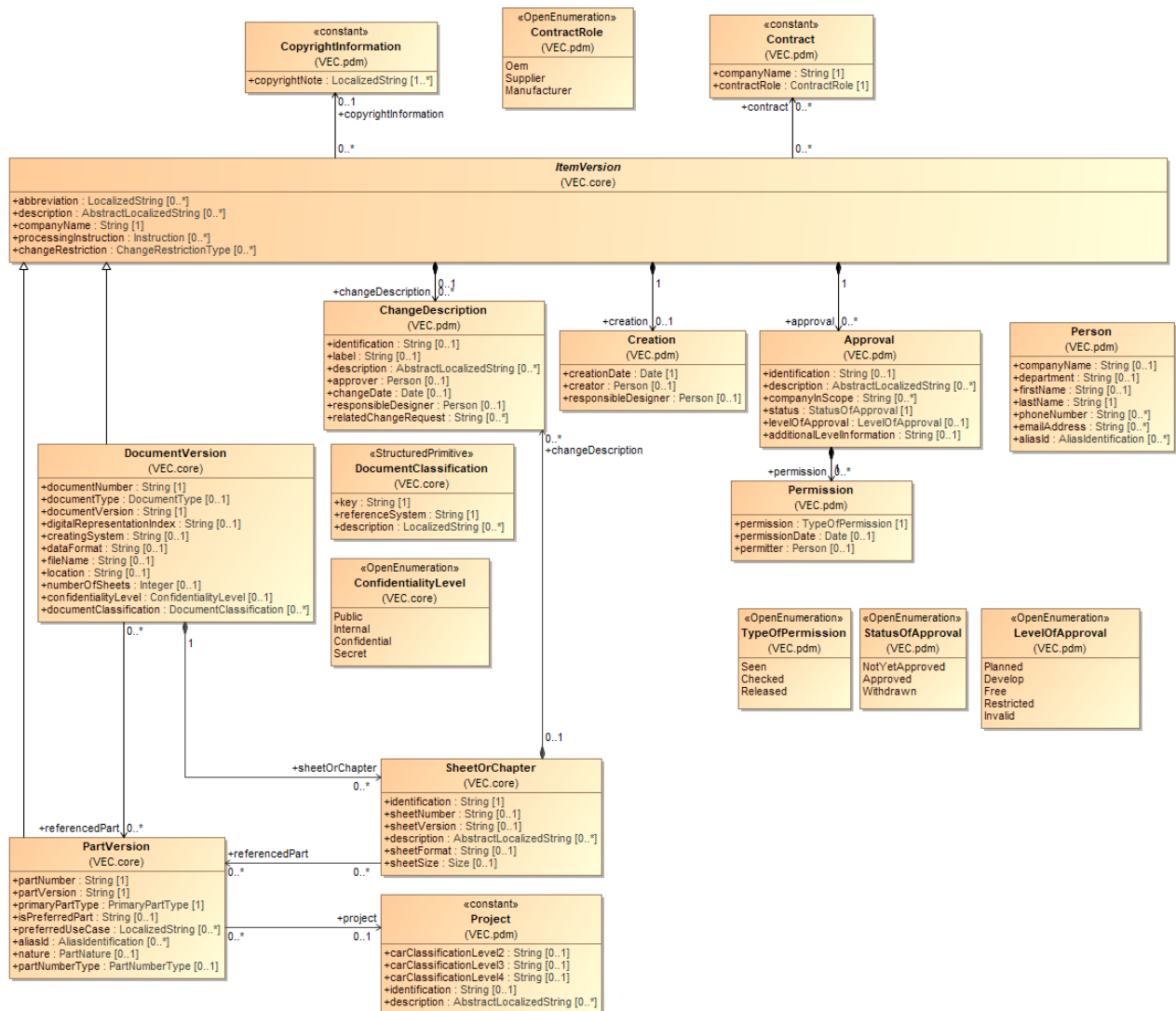


Figure 18: Lifecycle Information

An *ItemVersion* can carry information about the creation of the item. This means the creating timestamp and an optionally information regarding the creator or responsible person.

An *ItemVersion* normally has approval information and can be specified within the VEC by the relationship *ItemVersion.approval*. The class *Permission* allows the documentation of the persons involved in the approval-process.

The class *ChangeDescription* expresses the changes of an *ItemVersion* compared to its direct predecessorVersion(s).

The class *CopyrightInformation* enables the description of copyright information of *ItemVersions*. Instances of the class *CopyrightInformation* are assumed to be constant and so do not require a dedicated belonging to a versioned instance (*PartVersion* or *DocumentVersion*).

An *ItemVersion* can have contract information with company role information. Instances of the class *Contract* are assumed to be constant and so do not require a dedicated belonging to a versioned instance (*PartVersion* or *DocumentVersion*).

A *PartVersion* can reference a *Project* in order to describe for which *Project* it is relevant. Instances of the class *Project* are assumed to be constant and so do not need to have a dedicated belonging to a versioned instance (*PartVersion* or *DocumentVersion*).

1.3.2 Baselines

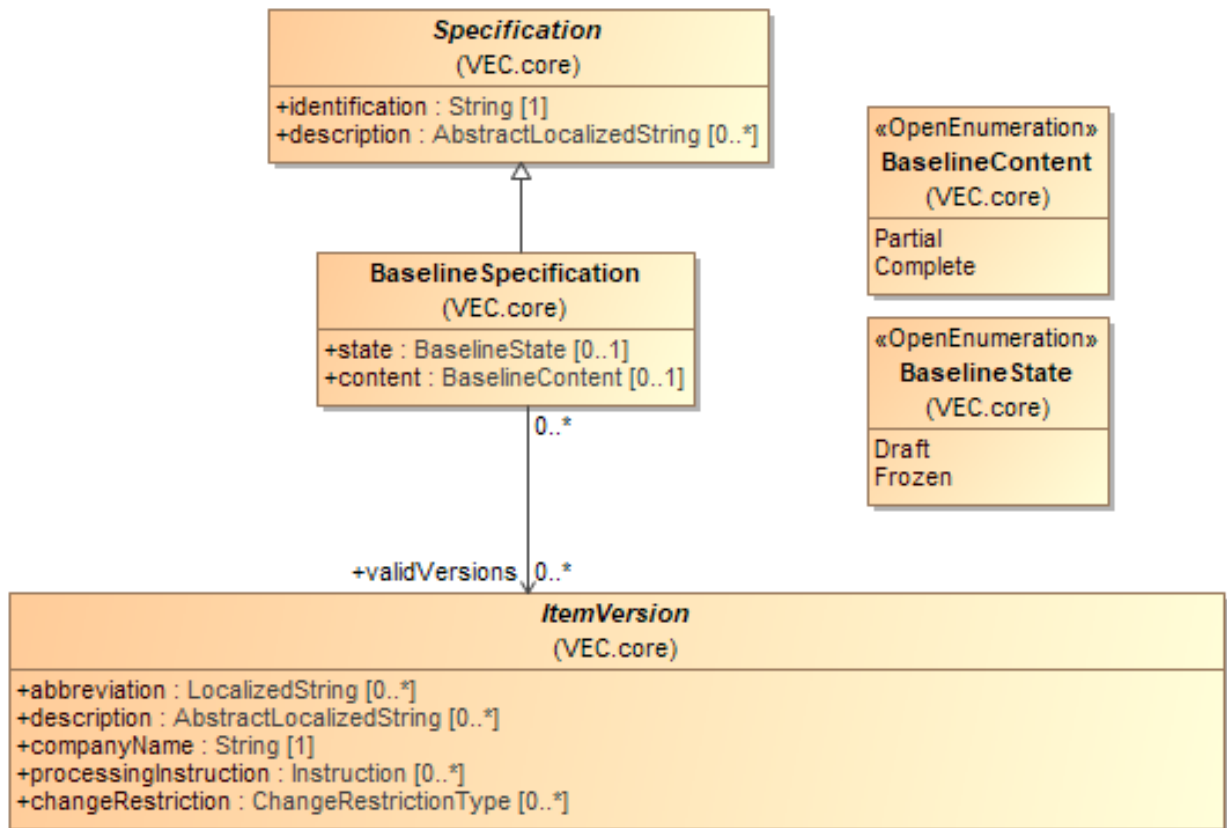


Figure 19: Baselines

A *BaselineSpecification* defines a set of *ItemVersions* (*Document*- and *PartVersions*) that relate to each other in a certain way e.g. all parts and documents in their specific versions that contributed to a specific manufactured result.

Baselines are a standard mechanism to support change, release and configuration management and are often used throughout the whole product life cycle. In the VEC, baselines themselves are *Specifications* and thus content of a *DocumentVersion*, which is equivalent to the fact that they are versioned information themselves.

In a product development for wiring harnesses, the content of a baseline is growing over a period (e.g. system schematics are created before the definition of the physical wiring harness). In this process, items are created and added to the baseline (based on other items from the baseline). Additionally, items in the baseline can be changed, following the rules of the respective development process. Such a baseline has the *BaselineState.Draft*. At a well-defined point in time, the content of baseline is checked and released. After this, the baseline cannot be changed without creating a new version of the baseline itself. Such a baseline has the *BaselineState.Frozen*.

Baselines are normally defined for a certain scope (e.g. a specific vehicle, a specific product). During the product development it is possible, that a baseline is incomplete. A reason for this might be, that not all development artefacts (e.g. wiring harnesses) are defined in this phase of the development process even though all artefacts are required for the final product. Such *BaselineSpecifications* have a *Partial content*.

1.3.3 Item History

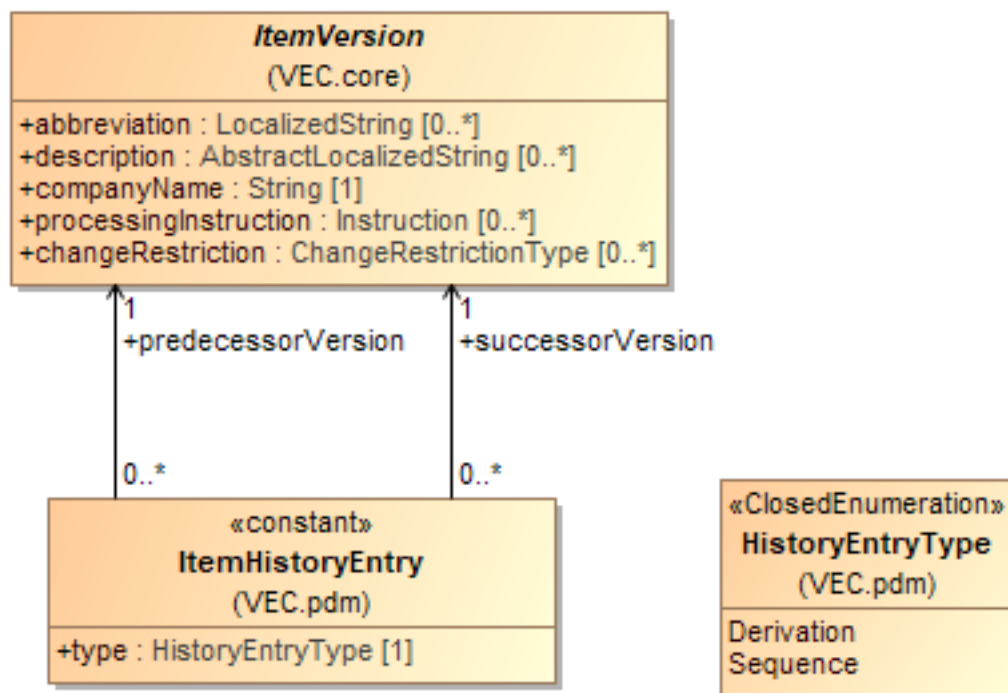


Figure 20: Item History

In a development process items (parts or documents) are often refined incrementally, so one *ItemVersion* originates from another. The class *ItemHistoryEntry* enables the description of predecessor / successor relationships between *ItemVersions*. The description of both linear and branching relationships is possible. The relationship can be classified in order to distinguish between two cases:

1. The successor *ItemVersion* is an instance of the same item (*type* = "Sequence")
2. The successor *ItemVersion* is an instance of a new but somehow similar item (*type* = "derivation").

1.3.4 Item Equivalence

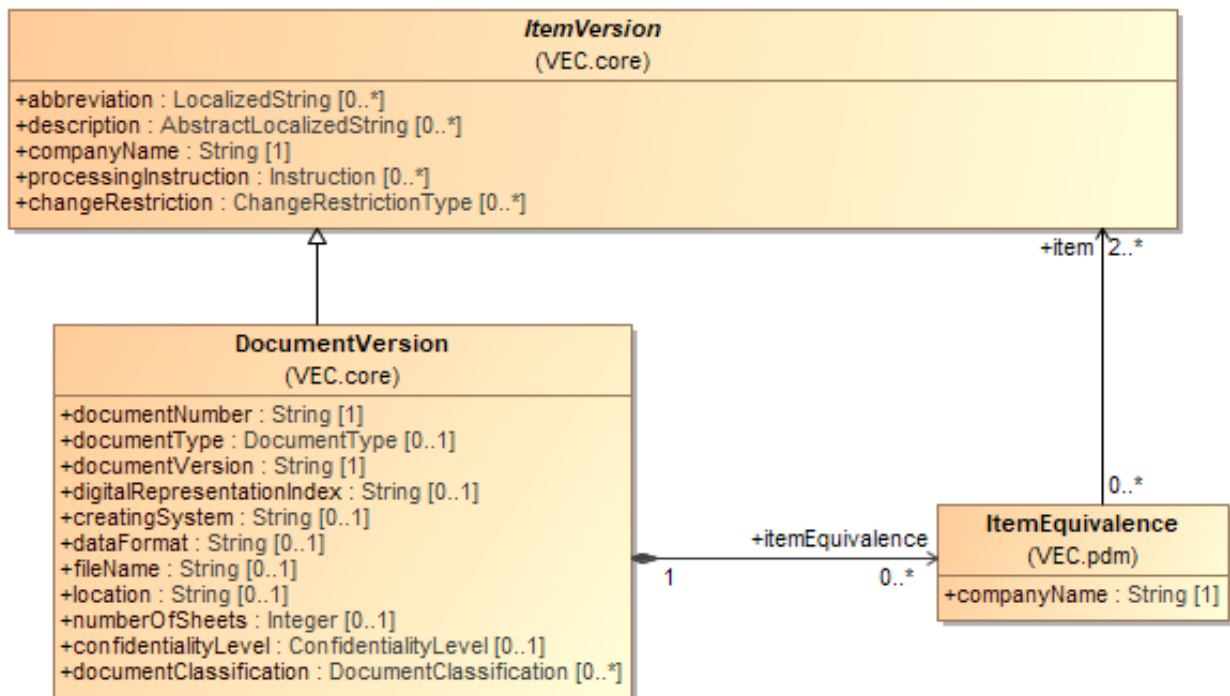


Figure 21: Item Equivalence

ItemVersions (*DocumentVersions* and *PartVersions*) are company specific keys / identifiers for a part (component) or a piece of information. However, this also means that the same part, or piece of information, can have different identifiers depending on the context. The keys refer semantically to the same element or in other words, they are equivalent. The class *ItemEquivalence* enables the description of such equivalence relationships between two or more *ItemVersions*. A standard use case is to relate *PartVersions* defined in the context of different companies for part number translation (e.g. OEM part number -> Tier 1 part number -> component supplier part number).

Note: That *equivalent ItemVersions* refer to the semantically same element does not mean, that the information associated with the *ItemVersions* has to be identical. For example, the same component might have different approved operating conditions depending on the company scope, which results in differing technical attributes.

1.4 General Component Data

The description of components and their properties is quantitatively the largest part of the VEC model. The following sections explain the basic methods for describing components and defining the information that applies to all types of components.

The properties of certain component types are defined in the next chapter.

1.4.1 Description of Parts

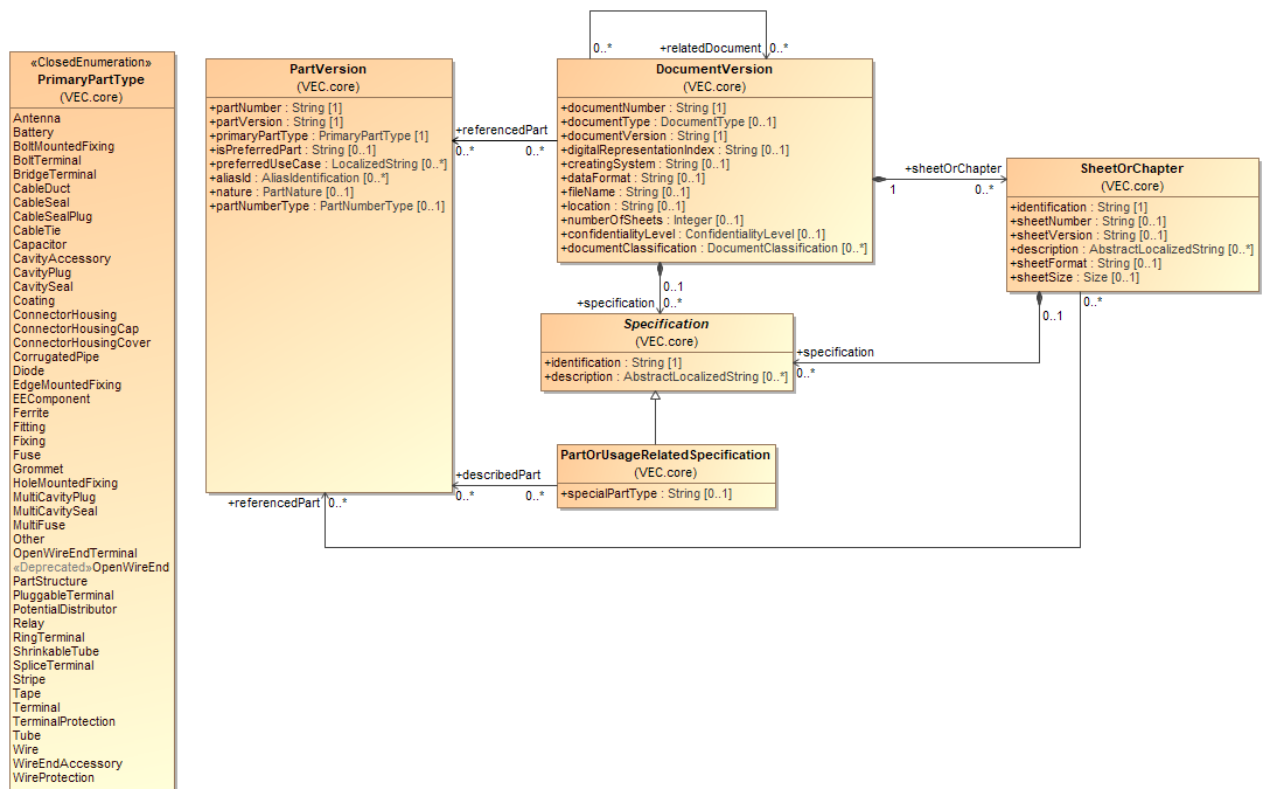


Figure 22: Description of Parts

In traditional data formats the diverse types of parts (e.g. connector housing, terminal, wire) are differentiated with individual subclasses for each type inheriting from the type corresponding to *PartVersion*. These subclasses define all specific information of the component type (e.g. the cavities of a connector). Due to the great diversity of components used in wiring harnesses a different approach is implemented in the VEC.

The general technical / physical properties of a component are described by *PartOrUsageRelatedSpecifications* (and its subclasses). For each aspect of a component an independent subclass of *PartOrUsageRelatedSpecification* is defined (e.g. a *ConnectorHousingSpecification*, a *TerminalSpecification*). This approach allows to describe a *PartVersion* by more than one *PartOrUsageRelatedSpecifications* (Association *PartOrUsageRelatedSpecification.describedPart*). It offers the possibility to extract information, which is shared between component types, into separate specifications so that the structural definition can be reused without having to modify the inheritance tree (e.g. *GeneralTechnicalPartSpecification*). Furthermore, some components cannot be mapped onto such a strict type hierarchy (e.g. a connector housing used as an inliner might have a fixing bolt for vehicle body in order to keep it in place). Representing this by an inheritance tree would result in a huge variety of different classes. With the concept of *PartOrUsageRelatedSpecifications* a component like the mentioned connector housing can be easily described by using a *ConnectorHousingSpecification* for the connector housing aspects and a *FixingSpecification* for the fixing aspects. However, most parts have a primary characteristic. For example, the mentioned connector housing will most probably only be used if a connector is necessary. This primary characteristic is defined by the *PartVersion.primaryPartType*. The *PrimaryPartType* always refers by a naming convention to a corresponding *PartOrUsageRelatedSpecification*. If the *PrimaryPartType* is ABC, the corresponding *Specification* is named *ABCSpecification* (e.g. *ConnectorHousing* and *ConnectorHousingSpecification*). The *PrimaryPartType* serves the purpose of defining the primary characteristic of a part. It is not required to describe every *PartVersion* in a specific VEC file with a corresponding *PartOrUsageRelatedSpecification*, if the information is not required in the context of the VEC file. For example, in the context of geometry data exchange it might be perfectly valid to have *PartVersions* with *PrimaryPartType=ConnectorHousing* that are only associated with a *PlaceableElementSpecification* (and/or a *GeneralTechnicalPartSpecification*) while the *ConnectorHousingSpecification* is not provided.

In some cases, there might be a need to specify attributes of a part that is not related to any of the existing subtypes of *PartOrUsageRelatedSpecification*. (e.g. it might be relevant to describe a screw nut used as an accessory for a connector housing). In this case the class *PartOrUsageRelatedSpecification* is instantiated, and the relevant attributes are added as *CustomProperties*.

Since the *PartOrUsageRelatedSpecification* is a subclass of *Specification*, which is a sub element of a dedicated *DocumentVersion*, all information defined by it can be related to a precise development state.

The relationship *PartOrUsageRelatedSpecification.describedPart* might be seen as redundant to the relationship *DocumentVersion.referencedPart* but considers the fact that there are use cases, where only information on the level of *PartVersions* and *DocumentVersions* is exchanged, but not on the very detailed level of *PartOrUsageRelatedSpecifications*. In addition, it is possible, that several *PartVersions* are described by a single *DocumentVersion* containing a couple of specifications, each of them describing a single *PartVersion* (e.g. a drawing with a major part and its accessory parts).

1.4.2 General Technical Part

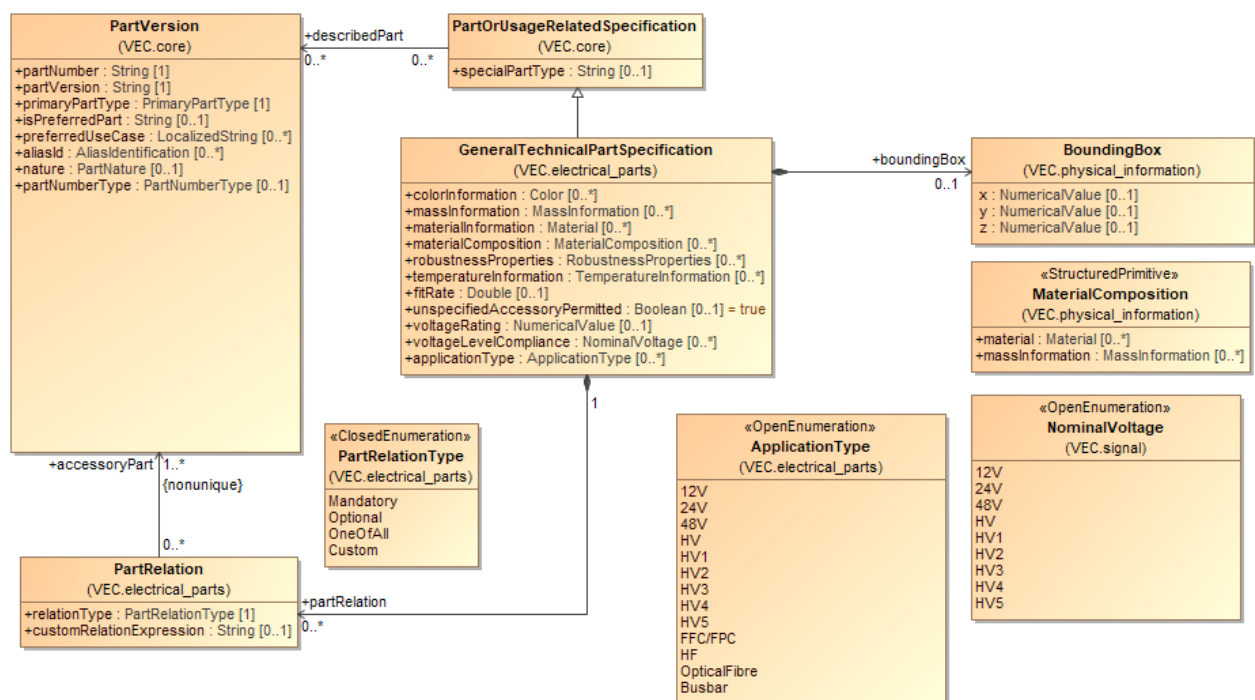


Figure 23: General Technical Part

The class *GeneralTechnicalPartSpecification* is intended to aggregate common technical attributes that are relevant for most kinds of parts (e.g. *massInformation* or information about valid temperature environments).

1.4.3 Supplementary Parts

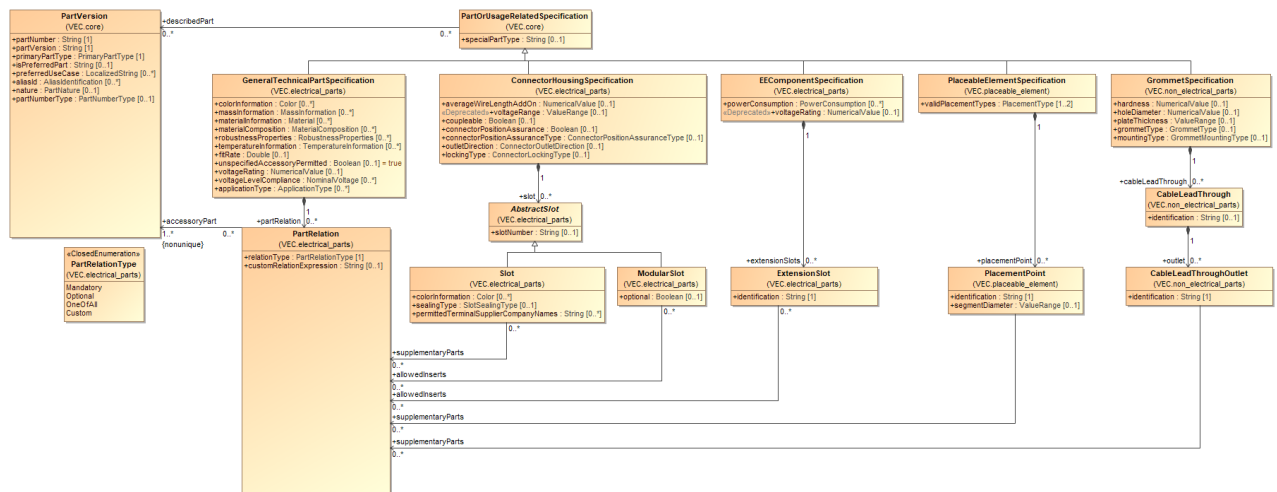


Figure 24: Supplementary Parts

With the *PartRelations* other *PartVersions* (those) can be defined as supplementary parts for the *PartVersions* that are described by the containing *GeneralTechnicalPartSpecification* (this).

The semantic is, that if *this PartVersion* shall be used in a correct way in a product, those referenced *PartVersions* can or shall be added to the product as well (depending on the used *PartRelationType*). That can be for example caps, bars or levers in case of a connector.

The *PartRelation.relationType* defines how the set of referenced *PartVersions* of one *PartRelation* must be interpreted. For details see the description of class *PartRelation*.

The association via the *PartRelation* just defines what other components may be used together with this component. It does not further specify the location of the usage or its function. However, in some cases a more detailed specification of the location is needed, e.g. if the supplementary part is required for a specific slot in a connector housing. In such a case, the *PartRelation* is specified by the *GeneralTechnicalPartSpecification* and may be referenced by its specific usage location (currently from a *Slot*, *ModularSlot*, *ExtensionSlot* or a *PlacementPoint*).

1.4.4 Placeable Elements

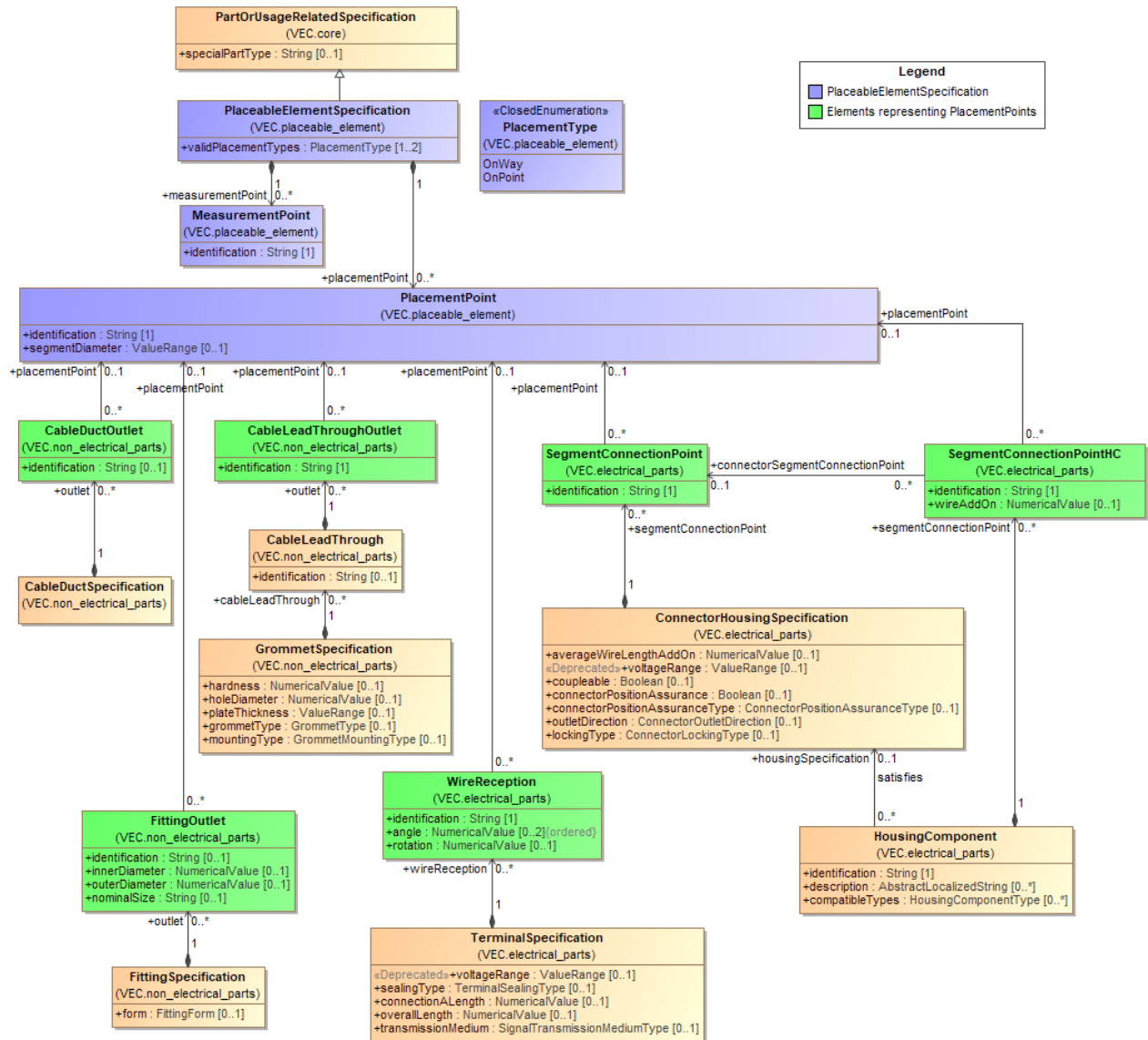


Figure 25: Placeable Elements

In a wiring harness many components can be placed directly onto the topology (e.g. connectors, fixings). To specify the aspect of placeability the VEC defines a *PlaceableElementSpecification*. Every component to be placed directly onto the topology must have a *PlaceableElementSpecification* in addition to its regular type specific *PartOrUsageRelatedSpecification* (e.g. *ConnectorHousingSpecification*).

A *PlaceableElementSpecification* defines at minimum the *validPlacementTypes* for the component (e.g. a connector can be placed on a point, a tape can be placed on a path in the topology).

In addition, the *PlaceableElementSpecification* can define important points of the component, *MeasurementPoints* and *PlacementPoints*.

MeasurementPoints are locations on the component that can be used to define a dimensioning. To make a *MeasurementPoint* locatable by a person more information is needed. This information is normally provided with a drawing of the component. The *identification* can be used to find the *MeasurementPoint* on the drawing (if it is labelled). To make a *MeasurementPoint* locatable by a system, the external mapping mechanism can be used.

PlacementPoints are locations on the component, where it can be attached to the wiring harness. Therefore, it can define a *segmentDiameter* for which it is usable. To locate a *PlacementPoint* on the component, the same approach as for *MeasurementPoints* can be used.

Often these *PlacementPoints* have additional properties depending on the component type. In these cases, the corresponding *PartOrUsageRelatedSpecification* redefines these elements, with a reference to the *PlacementPoint* which represents them (e.g. the *SegmentConnectionPoint*). However, a *PlacementPoint* can be defined without such a corresponding element, if no additional information is necessary.

1.4.5 Coordinate Systems of Components

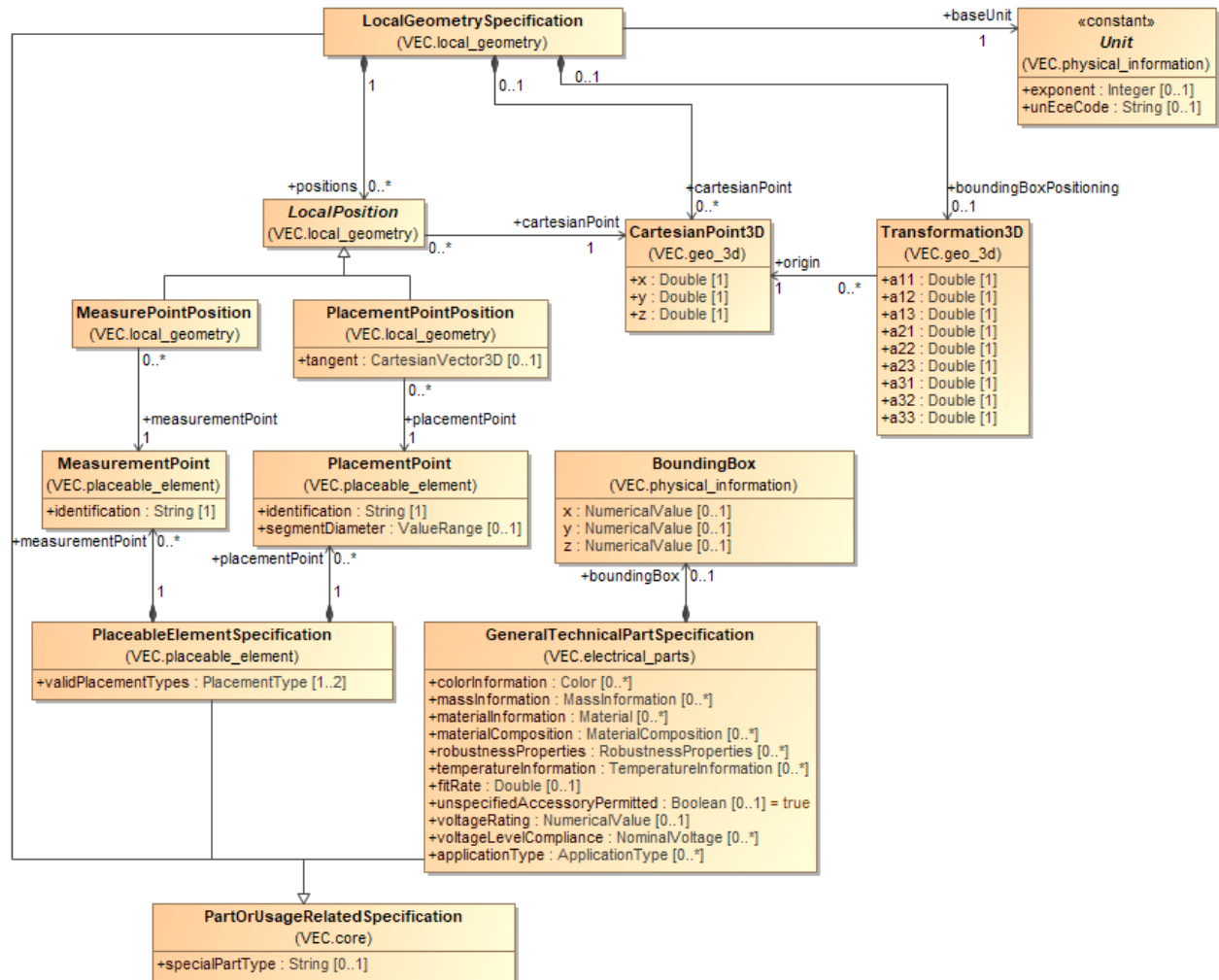


Figure 26: Coordinate Systems of Components

A *LocalGeometrySpecification* defines a mapping between the entities of the VEC and the 3D model of a component. The 'Local' in the name refers to the fact that all definitions within this specification are 'local' to the 3D model of a specific component (a component in a library, not in a specific usage).

The coordinate system of a component is defined by its 3D model, which often follows certain guidelines or conventions (e.g. the coordinate systems of connectors have their origin in the center of cavity 1). The *Transformation3D* of an *OccurrenceOrUsageViewItem3D* (see section 3D-Geometry) defines how the 3D model of the component shall be transformed in order to represent the correct positioning of its usage. This transformation is defined in regard to the origin and the axis of the component's coordinate system.

The *BoundingBox* specifies a box that has the extent to completely enclose the volume of the component. It is defined to have one corner in the origin of its coordinate system and to grow along the respective axes in positive direction. This means, that the origin of bounding box is on a corner of the represented volume, the origin of the 3D model of the component is at some arbitrary point within the represented volume. As a result, the bounding box will most likely not enclose the component's 3D model in "usage space" when placed with the same transformation as the original 3D model. To overcome this problem, the *LocalGeometrySpecification* defines a *Transformation3D* as *boundingBoxPositioning*, which puts the generally defined bounding box into

the coordinate system of the component. The transformed bounding box is then placed into the "usage space" with the same transformation as the 3D model of the component.

In addition, the *LocalGeometrySpecification* also defines *LocalPositions* of important points of the component. The position of these points is defined within the coordinate system of the component. Positions can be defined for:

- *PlacementPoints*: Points where a *TopologySegment* is attached to the component (e.g. a *Bundle / Segment Connection Point* of a *Connector*).
- *MeasurementPoints*: Significant points on the component on to which dimension can be defined (e.g. for QA purposes).

1.4.6 Part Substitutions

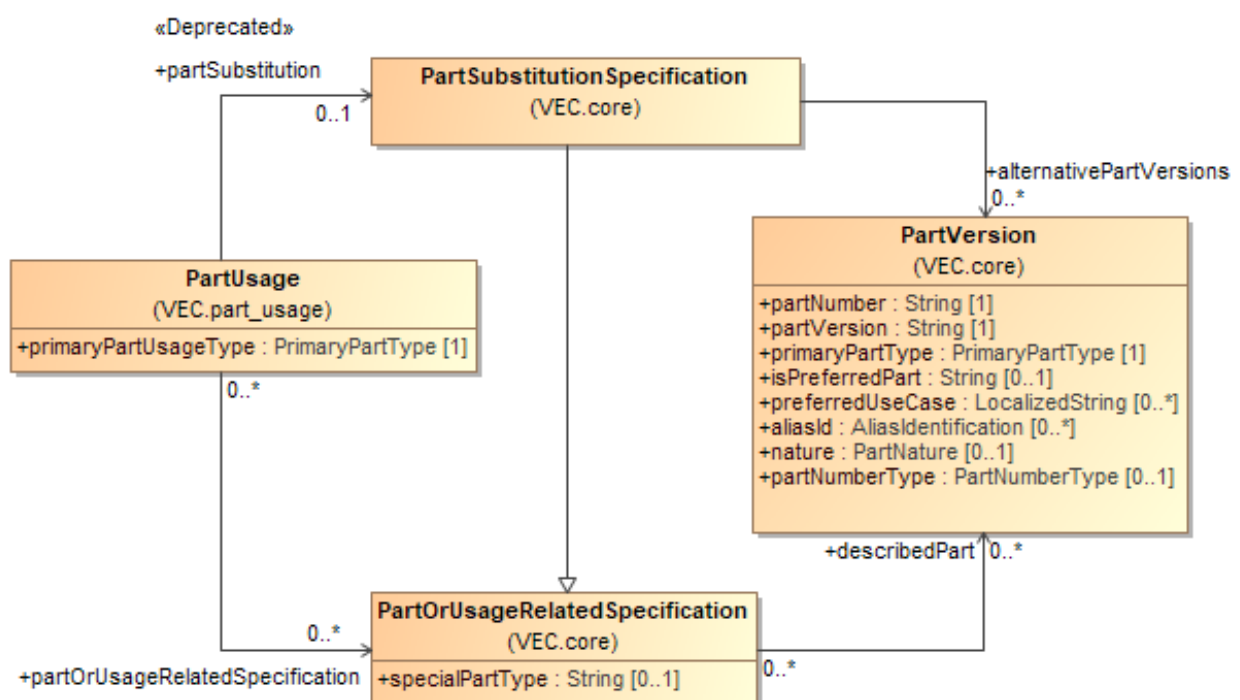


Figure 27: Part Substitutions

A *PartSubstitutionSpecification* defines a set of *PartVersions* that can be used alternatively, due to an incomplete specification for the 150% product description. For a concrete wiring harness only one valid *PartVersion* remains. The selection logic for valid *PartVersions* is not included in the VEC. It is NOT valid to use a *PartSubstitutionSpecification* for *PartVersions* with identical properties that just have different *PartNumbers* in different contexts (e.g. multi supplier topics). For these cases an *ItemEquivalence* shall be used.

A *PartSubstitutionSpecification* can be used e.g. for tubes or ring terminals, where a part of the specification is known at design time, but not yet the complete specification. E.g. for tubes the required inside diameter is not known at design time, since it depends on the bundle diameter of a specific configuration.

In order to represent a component instance that utilizes such a *PartSubstitutionSpecification* a *PartUsage* or a *PartOccurrence* can be used.

The *PartUsage* is the element in the VEC which defines instances of components, where a specific *PartVersion* is not yet known. It references one or more *PartOrUsageRelatedSpecifications* to describe the known properties of the component. If the *PartUsage* references an additional *PartSubstitutionSpecification* the set of valid *PartVersions* can be further constrained.

However, there are business processes that utilize actual *PartVersion* to define such replacement options in a part master data management (compare *PartVersion.partNumberType*).

1.4.7 Conformance to Requirements

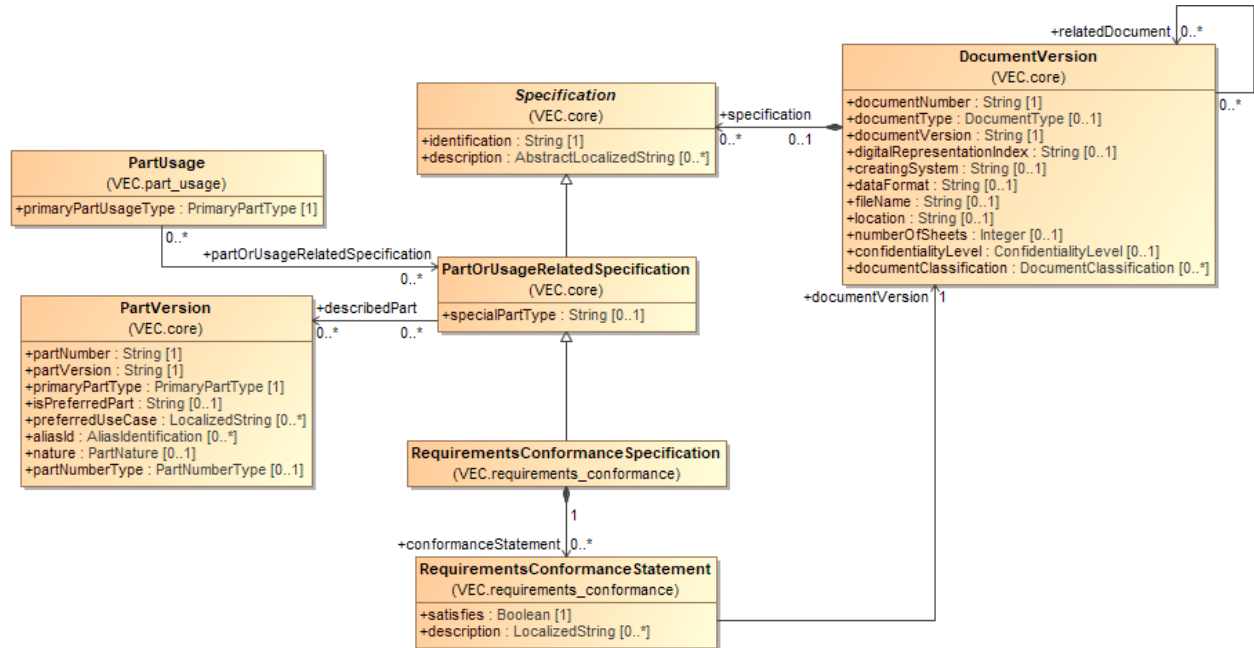


Figure 28: Conformance to Requirements

During the life cycle of a given *PartVersion* manifold situations are existing, where the conformance of that *PartVersion* to a set of requirements must be demonstrated or is required to be known. To represent the result of such a conformance test, the VEC offers the *RequirementsConformanceSpecification* and the *RequirementsConformanceStatement*.

As the *RequirementsConformanceSpecification* is a *PartOrUsageRelatedSpecification*, it can be used to express conformance for a set of *PartVersions* or *PartUsages*. The *DocumentVersion* in which the *RequirementsConformanceSpecification* is contained, normally represents the original document by which the conformance was approved (e.g. a type rating).

The actual conformance is expressed with *RequirementsConformanceStatements*. It can be used to document a successful or explicitly failed conformance test. The requirements for which the state is valid, are referenced as a *DocumentVersion* via the *requirementsSpecification* association.

If a *RequirementsConformanceStatement* is omitted for a *PartVersion*, this does not imply any conformance information at all. The *RequirementsConformanceStatement* can be missing because the component has never been tested according to the specification, or it has been tested, but the information was not transferred within this specific VEC instance, or the information is just not available in a digitally analyzable form.

The VEC does not impose any restrictions on the kind of requirements specifications for which conformance can be expressed. This can be for example:

- A standard (company or public) like an ISO or a DIN.
- A definition of requirements in the sense of a specification sheet.
- Requirements that allow certain handling procedures (e.g. suitability for production automation)

The VEC does not impose any restrictions on how the conformance must be demonstrated. This can be done with well-defined testing procedure, a manual assessment, an audit, a type examination or anything else. Normally the requirements specification itself defines how conformance to it must be demonstrated.

1.4.8 Labeling

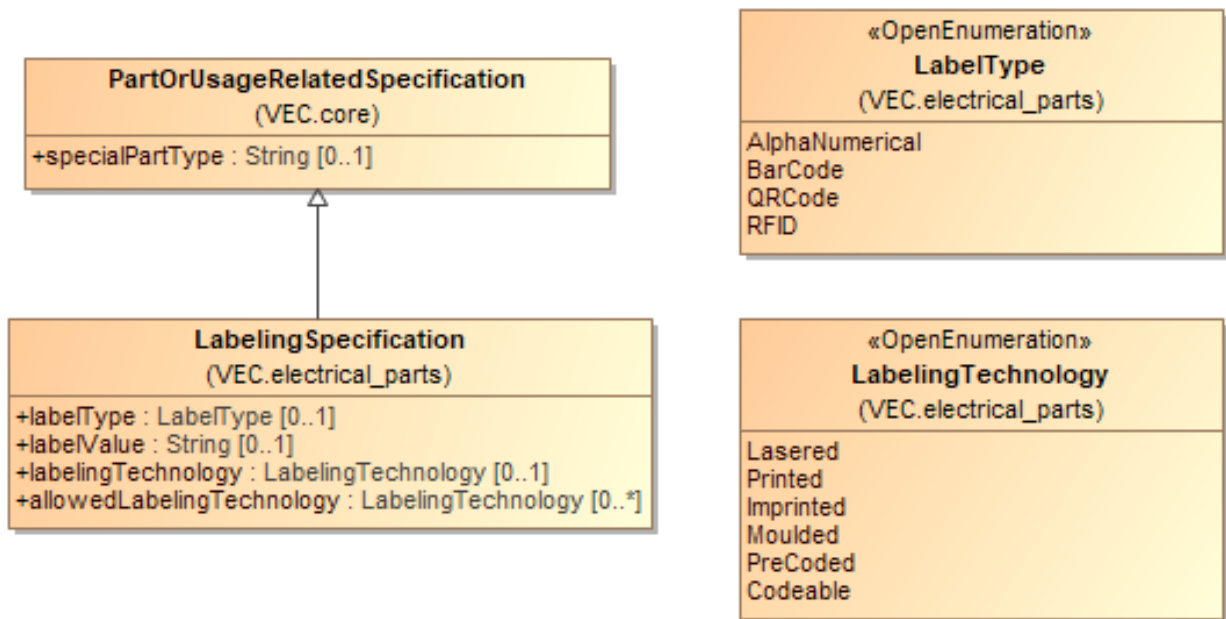


Figure 29: Labeling

A component (of any type) can be labeled in many ways either during the manufacturing process, for example a label that is imprinted on the component during the general manufacturing process (e.g. the part number), or in its concrete use (e.g. with wire number or a usage node id). The *LabelingSpecification* defines general labeling properties of the component. Labeling properties in its concrete use are defined with the corresponding role.

1.5 Component Characteristics

NA


```

classDiagram
    class PartOrUsageRelatedSpecification {
        +specialPartType : String [0..1]
    }
    class WireSpecification {
        <<Deprecated>>
    }
    class WireElement {
        +identification : String [1]
        +index : Integer [0..1]
        +layer : Integer [0..1]
    }
    class WireElementSpecification {
        +type : WireType [0..*]
        +minBendRadiusDynamic : NumericalValue [0..1]
        +minBendRadiusStatic : NumericalValue [0..1]
        +outsideDiameter : NumericalValue [0..1]
        +suitedForDynamicUse : Boolean [0..1]
        +impedance : NumericalValue [0..1]
        +size : Size [0..1]
        +validWireReceptionTypes : WireReceptionType [0..*]
        +gridSpacing : NumericalValue [0..1]
        +shape : WireElementShape [0..1]
        +transmissionMediumType : TransmissionMediumType [0..1]
    }
    class InsulationSpecification {
        +wireType : WireType [0..*]
        +baseColor : Color [0..1]
        +firstIdentificationColor : Color [0..1]
        +secondIdentificationColor : Color [0..1]
        +labelIdentificationColor : Color [0..1]
        +labelIdentificationType : LabelType [0..1]
        +labelIdentificationValue : String [0..1]
        +material : Material [0..1]
        +thickness : NumericalValue [0..1]
        +labelingTechnology : LabelingTechnology [0..1]
        +allowedLabelingTechnology : LabelingTechnology [0..*]
    }
    class ConductorSpecification {
        +wireType : WireType [0..1]
        +crossSectionArea : NumericalValue [0..1]
        +massInformation : MassInformation [0..1]
        +material : Material [0..1]
        +resistance : NumericalValue [0..1]
        +structure : ConductorStructure [0..1]
        +type : ConductorType [0..1]
        +numberOfStrands : NumericalValue [0..1]
        +platingMaterial : Material [0..1]
        +strandDiameter : NumericalValue [0..1]
        +voltageRange : NumericalValue [0..1]
        +transmissionMedium : SignalTransmissionMediumType [0..1]
    }
    class WireGroupSpecification {
        +groupType : WireGroupType [1]
        +lengthOfTwist : NumericalValue [0..1]
    }
    class FillerSpecification {
        +material : Material [0..1]
        +diameter : NumericalValue [0..1]
    }
    class CoreSpecification {
        +outsideDiameter : NumericalValue [0..1]
    }
    class FlatCoreSpecification {
        +size : Size [0..1]
    }
    class ShieldSpecification {
        <<Deprecated>>
        +opticalTissueDensity : NumericalValue [0..1]
        +opticalCoverage : Double [0..1]
        +windingType : FoilShieldWindingType [0..1]
    }
    class ConductorCurrentInformation {
        +maxCurrent : NumericalValue [1]
        +environmentTemperature : NumericalValue [1]
        +voltage : NumericalValue [1]
    }

    PartOrUsageRelatedSpecification --|> WireSpecification
    WireSpecification --|> WireElement
    WireElement -- "0..*" -- "0..1" WireElementSpecification : +wireElementSpecification
    WireElement -- "0..1" -- "0..1" WireElementSpecification : +wireElementSpecification
    WireElement -- "0..*" -- "0..1" WireElementSpecification : +subWireElement
    WireElementSpecification -- "0..*" -- "0..1" WireElementSpecification : +insulationSpecification
    WireElementSpecification -- "0..*" -- "0..1" WireElementSpecification : +conductorSpecification
    WireElementSpecification -- "0..*" -- "0..1" WireElementSpecification : +wireGroupSpecification
    WireElementSpecification -- "0..*" -- "0..1" WireElementSpecification : +fillerSpecification
    WireElementSpecification -- "0..*" -- "0..1" WireElementSpecification : +currentInformation
    WireElementSpecification -- "0..*" -- "0..1" WireElementSpecification : +coreSpecification
    WireElementSpecification -- "0..*" -- "0..1" WireElementSpecification : +flatCoreSpecification
    WireElementSpecification -- "0..*" -- "0..1" WireElementSpecification : +shieldSpecification
    WireElementSpecification -- "0..*" -- "0..1" WireElementSpecification : +conductorCurrentInformation
    
```

The diagram illustrates the UML class structure for electrical parts specifications. It starts with a base class **PartOrUsageRelatedSpecification** (VEC core), which has a **+specialPartType : String [0..1]** attribute. This class is specialized by **WireSpecification** (VEC electrical_parts), which is marked as **<<Deprecated>>**. **WireSpecification** further specializes into **WireElement** (VEC electrical_parts). **WireElement** has attributes **+identification : String [1]**, **+index : Integer [0..1]**, and **+layer : Integer [0..1]**. It also has associations with **WireElementSpecification**: **+wireElement** (multiplicity 1 at WireElement, 0..1 at WireElementSpecification), **+subWireElement** (multiplicity 0..* at WireElement, 0..1 at WireElementSpecification), and **+wireElementSpecification** (multiplicity 0..* at WireElement, 0..1 at WireElementSpecification). **WireElementSpecification** (VEC electrical_parts) contains numerous attributes such as **+type : WireType [0..*]**, **+minBendRadiusDynamic : NumericalValue [0..1]**, **+minBendRadiusStatic : NumericalValue [0..1]**, **+outsideDiameter : NumericalValue [0..1]**, **+suitedForDynamicUse : Boolean [0..1]**, **+impedance : NumericalValue [0..1]**, **+size : Size [0..1]**, **+validWireReceptionTypes : WireReceptionType [0..*]**, **+gridSpacing : NumericalValue [0..1]**, **+shape : WireElementShape [0..1]**, and **+transmissionMediumType : TransmissionMediumType [0..1]**. It also has associations with other classes: **+insulationSpecification** (multiplicity 0..* at WireElementSpecification, 0..1 at InsulationSpecification), **+conductorSpecification** (multiplicity 0..* at WireElementSpecification, 0..1 at ConductorSpecification), **+wireGroupSpecification** (multiplicity 0..* at WireElementSpecification, 0..1 at WireGroupSpecification), **+fillerSpecification** (multiplicity 0..* at WireElementSpecification, 0..1 at FillerSpecification), **+currentInformation** (multiplicity 0..* at WireElementSpecification, 1 at ConductorCurrentInformation), **+coreSpecification** (multiplicity 0..* at WireElementSpecification, 0..1 at CoreSpecification), **+flatCoreSpecification** (multiplicity 0..* at WireElementSpecification, 0..1 at FlatCoreSpecification), **+shieldSpecification** (multiplicity 0..* at WireElementSpecification, 0..1 at ShieldSpecification), and **+conductorCurrentInformation** (multiplicity 0..* at WireElementSpecification, 1 at ConductorCurrentInformation). The diagram also includes several **OpenEnumeration** classes: **ConductorType** (Rigid, Stranded, Foil, Braided), **ConductorStructure** (Symmetric, Asymmetric, MultiStrand), **FoilShieldWindingType** (Overlap, Longitudinal), **LabelType** (AlphaNumerical, BarCode, QRCode, RFID), **LabelingTechnology** (Lasered, Printed, Imprinted, Moulded, PreCoded, Codeable), **TransmissionMediumType** (Electrical, Optical, Hydraulic, Pneumatic), and **WireElementShape** (Circular, Flat, Elliptical).

In the VEC wires are defined through a *WireSpecification*, regardless of their type. This means, for all types of wires (e.g. normal single core, multi core or coax wires) the same structure is used to describe them. Since a wire can be a hierarchical structure the actual definition of the structure is delegated to a *WireElementSpecification*. A *WireElementSpecification* can define a certain wire element and more complex structures by referencing the appropriate *subWireElementSpecifications*.

In order to allow an unambiguous identification of a particular *WireElementSpecification* within the context of a *WireSpecification*, the *WireSpecification* defines a list of *WireElements* for each *WireElementSpecification* that belongs to the hierarchy of the wire. The *WireElement* defines the *identification* of a *WireElementSpecification* within the context of a wire.

For complex wires, it may be necessary to define the order and layering of *WireElement* objects that are *subWireElements* of the same *WireElement*. This is relevant, for example, in multi-core wires where multiple *WireElements* are arranged circularly around an inner *WireElement*. Similarly, the order is also important for flat ribbon cables. This is defined with the *index*- and *layer*-attributes in the *WireElement*. As a consequence, order and layering cannot be reused between *WireSpecifications*, offering the advantage of a significantly simpler model structure.

A *WireElementSpecification* can reference an *InsulationSpecification*, a *CoreSpecification*, a *ShieldSpecification*, a *FillerSpecification* and/or a *WireGroupSpecification* in order to describe its technical details. These aspects are separated into individual *Specifications* in order to allow the reuse of them. For example, the *CoreSpecification* of a FLRY-0.75 is the same for a group of wires, regardless of their insulation color. In turn the *InsulationSpecification* of a blue & green FLRY wire might be the same for a group of wires, regardless of their cross-section area.

When creating the hierarchy of *WireElementSpecifications* for a wire the representation with the minimal amount of *WireElementSpecifications* shall be used. This means that a single core shall be represented by a single *WireElementSpecification* with an *InsulationSpecification* and a *ConductorSpecification* and not with individual hierarchical *WireElementSpecifications* for the insulation and the conductor.

In most cases, a *WireElement* has a physical representation within the wire. However, there are cases where the *WireElementSpecification* is just a group of *WireElementSpecifications* with no real physical manifestation. For example, a twisted pair wire, consists of two single core *WireElementSpecifications* and a parent *WireElementSpecification* that just defines the type of twist.

Note: All Specification in this diagram, where no superclass is displayed inherit directly from Specification.

1.5.2 Wire Tuple

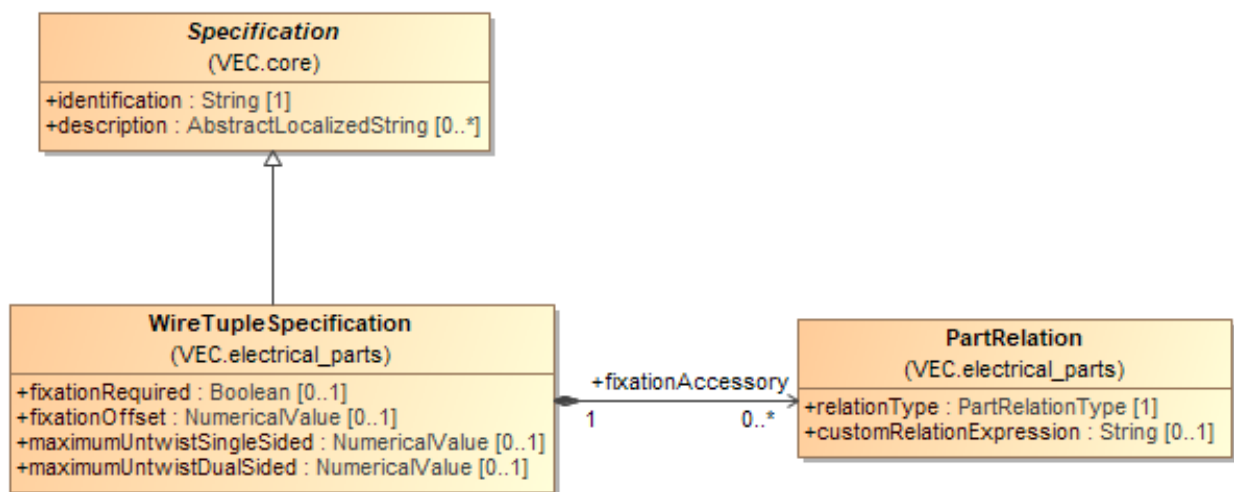


Figure 31: Wire Tuple

A *Specification* to define requirements for the handling of wire / wire element tuples (pairs, triple...) with the focus on twisting. More precisely, the allowed untwisting at the wire ends and required measures against it.

This definition is made independent from the *WireGroup*, as the *WireGroup* might be defined for the part master data (the wire on the roller). The specifications for untwisting are always based on a concrete usage.

1.5.4 Terminal Dimensions

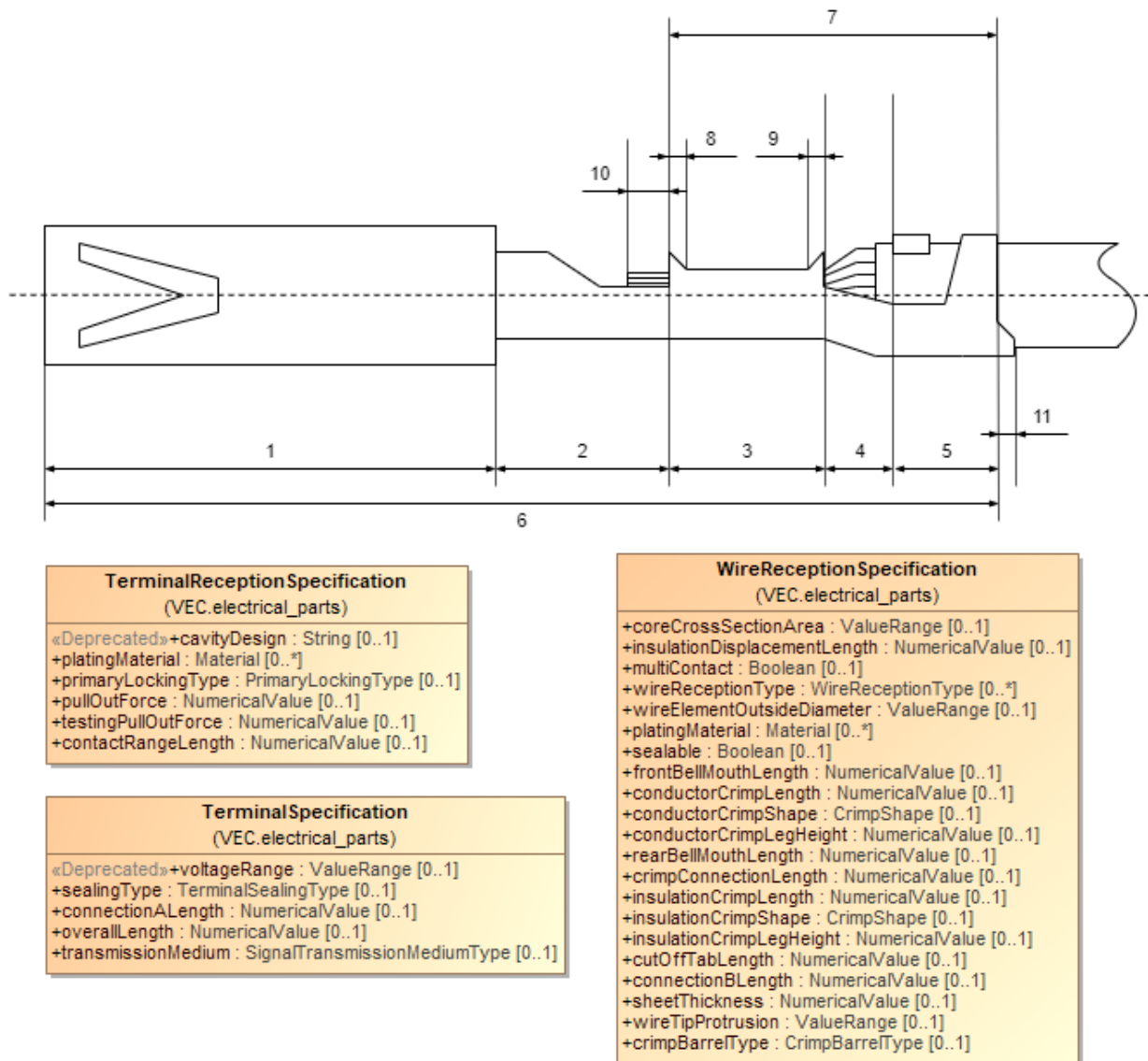


Figure 33: Terminal Dimensions

This diagram shows common terminal dimensions. The corresponding attributes in the VEC are:

1. *TerminalReceptionSpecification.ContactRangeLength*
2. *TerminalSpecification.ConnectionALength*
3. *WireReceptionSpecification.ConductorCrimpLength*
4. *WireReceptionSpecification.ConnectionBLength*
5. *WireReceptionSpecification.InsulationCrimpLength*
6. *TerminalSpecification.OverallLength*
7. *WireReceptionSpecification.CrimpConnectionLength*
8. *WireReceptionSpecification.FrontBellMouthLength*
9. *WireReceptionSpecification.RearBellMouthLength*
10. *WireReceptionSpecification.WireTipProtrusion*
11. *WireReceptionSpecification.CutOffTabLength*

1.5.5 Terminal Coordinate System

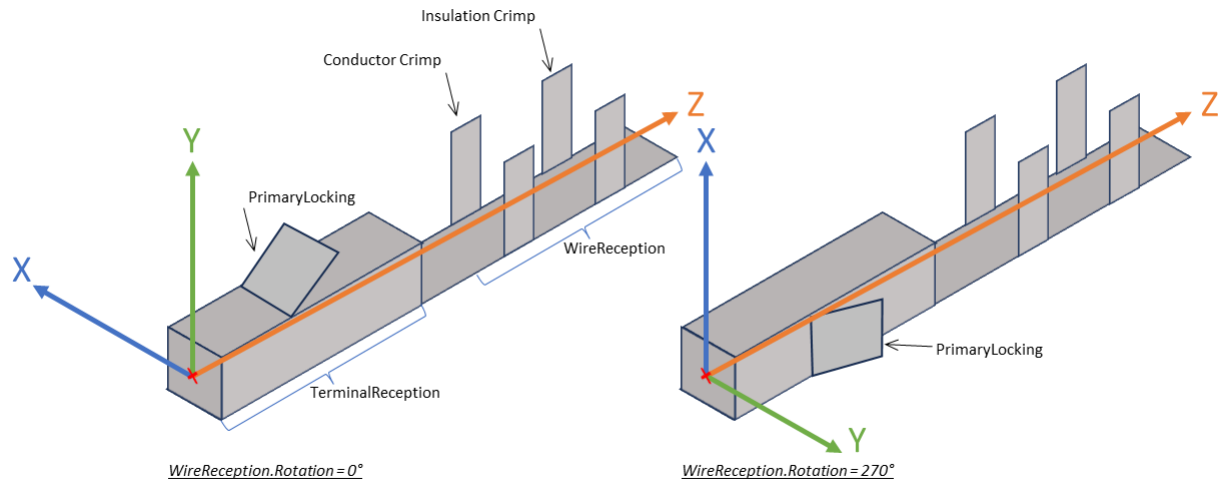


Figure 34: Terminal Coordinate System

The coordinate system of a terminal defines a reference system for the definition of other values (e.g. the *WireReception.Rotation*). The definition is the following (see illustration).

1. The origin (0,0,0) is in the intersection point the centerline of terminal (z-axis) and the perpendicular plane through the most forward end position of the terminal in its cavity.
2. The y-axis points in the direction of the primary locking.
3. The x-axis is the cross product of y and z, $x = y \times z$.

The rotation of the *WireReception* is the angle between the normal vector on base plane of the crimp area and the y-axis, which is a rotation around the z-axis (right hand rule, compare illustration).

1.5.6 Crimp Details

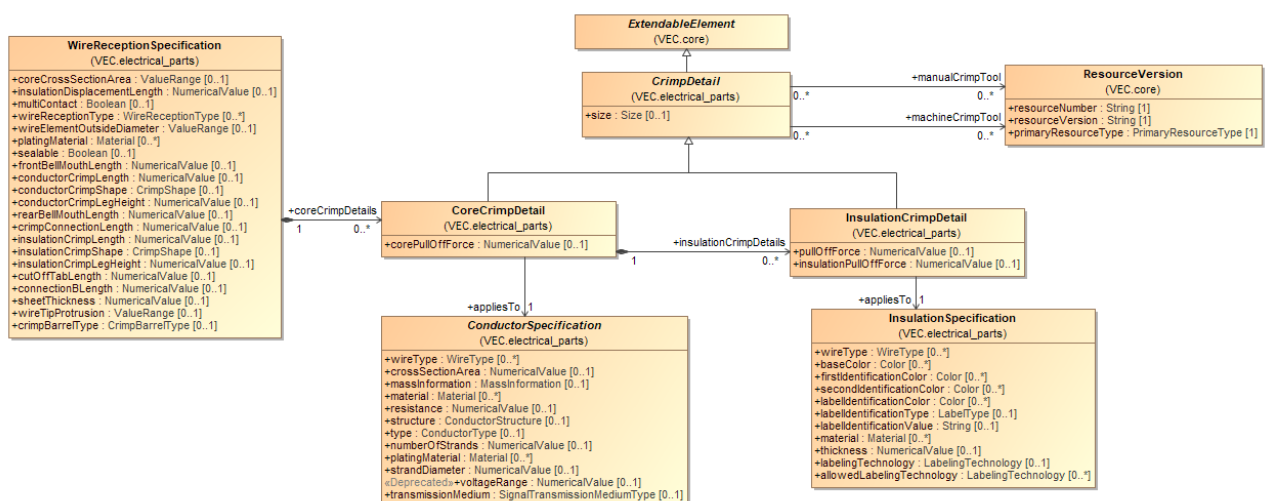


Figure 35: Crimp Details

Height & Width of the crimp are important quality criteria for the crimping process defined in the part master data of the terminal.

The core crimp and the insulation crimp have different heights & width depending on multiple parameter cross-section, material, insulation thickness etc. Each *CrimpDetail* represents a pair of *Height* & *Width* (with optional *Tolerance*) for specific parameter combination.

Those parameters are not defined in the *CrimpDetail* itself, but in the referenced *Conductor-* & *InsulationSpecification*. There are two possibilities here:

1. The *Conductor-* & *InsulationSpecification* can belong to the description of a specific wire. In this case, the specifications will most likely contain more information than the information required to determine the crimp parameters.
2. The *Conductor-* & *InsulationSpecification* can belong to the description of the terminal itself. In this case, the specifications represent a pattern for matching wires, by defining only the relevant attributes (e.g. *CrossSectionArea* & *Material* in the *ConductorSpecification* and *InsulationThickness* & *Material* in the *InsulationSpecification*).

1.5.7 Wire End Accessory and Terminal Protections

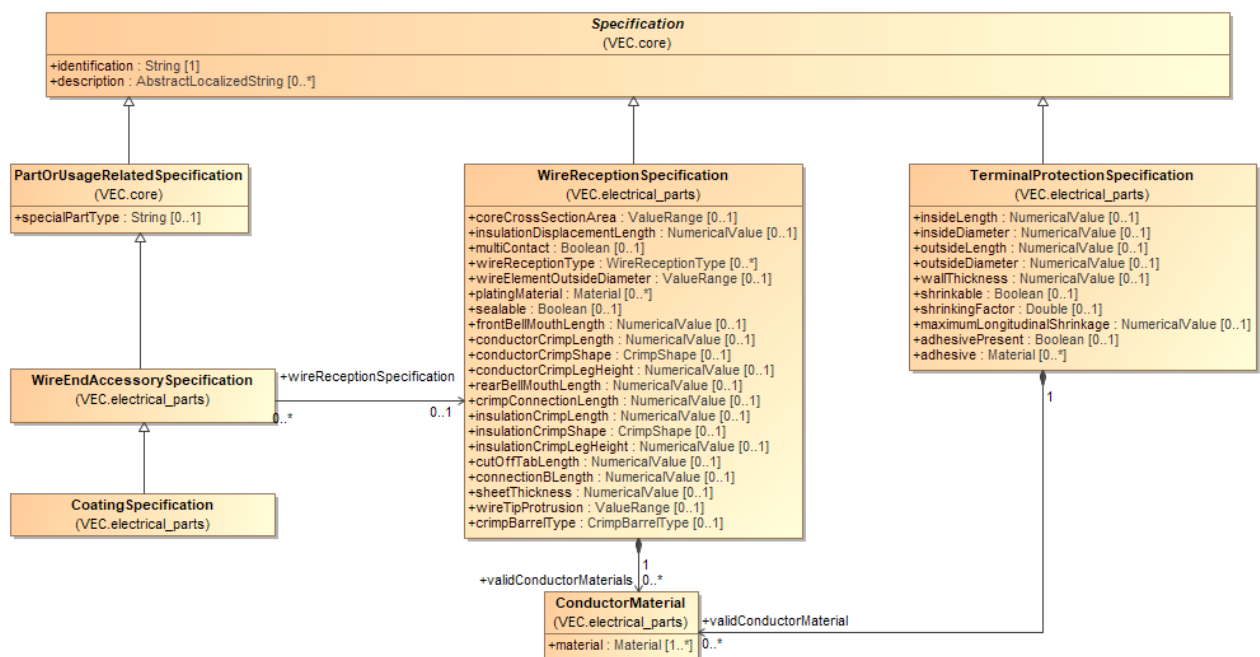


Figure 36: Wire End Accessory and Terminal Protections

WireEndAccessorySpecifications are describing parts that are used for a treatment of a wire end before the actual terminal is attached to the *WireEnd* (e.g. a ferrule). In contrast to this a *TerminalProtection* is something that is applied after the joining process.

1.5.8 Terminal Pairing

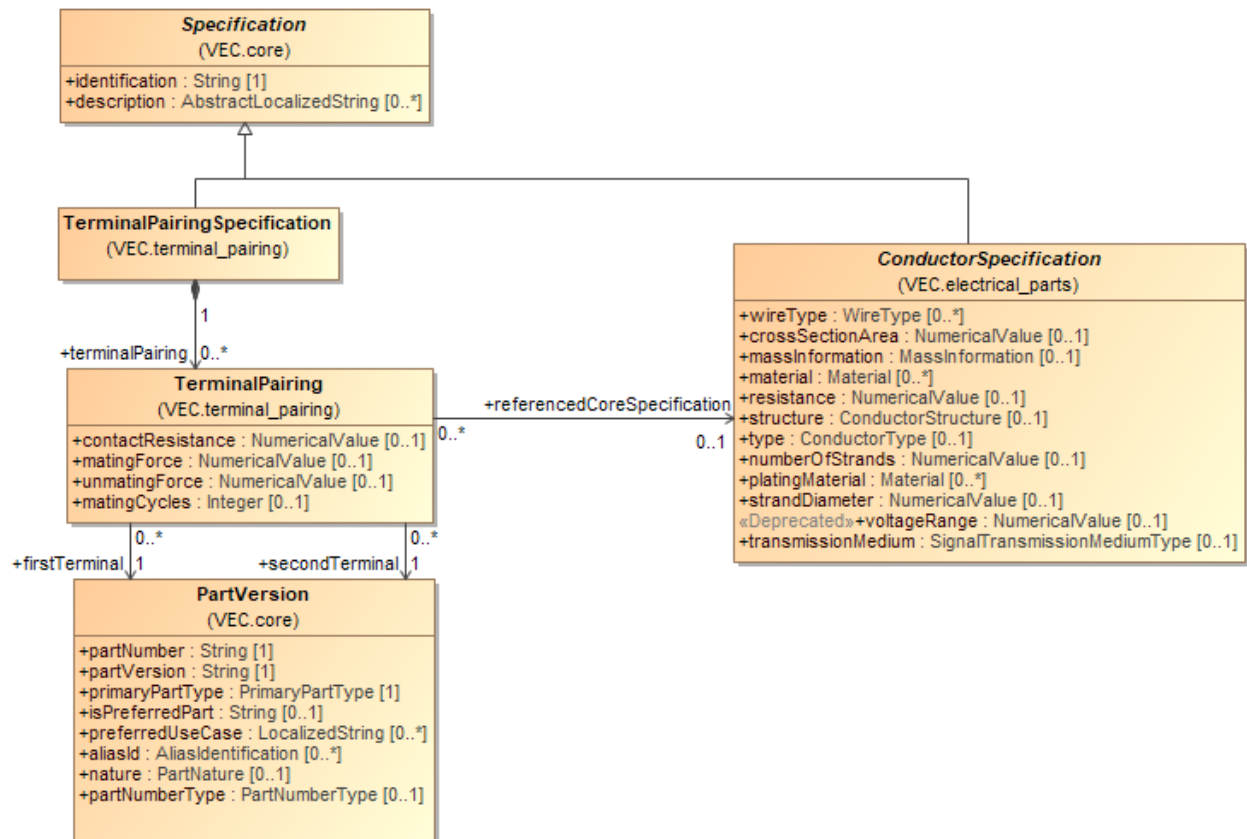


Figure 37: Terminal Pairing

A *TerminalPairingSpecification* is a container for various *TerminalPairing*. A *TerminalPairing* is an aspect of part master definition of terminals and represents a specific pair of terminals pluggable to each other. It specifies properties of that terminal pair in combination with a specific *ConductorSpecification*.

1.5.9 Connector Housings

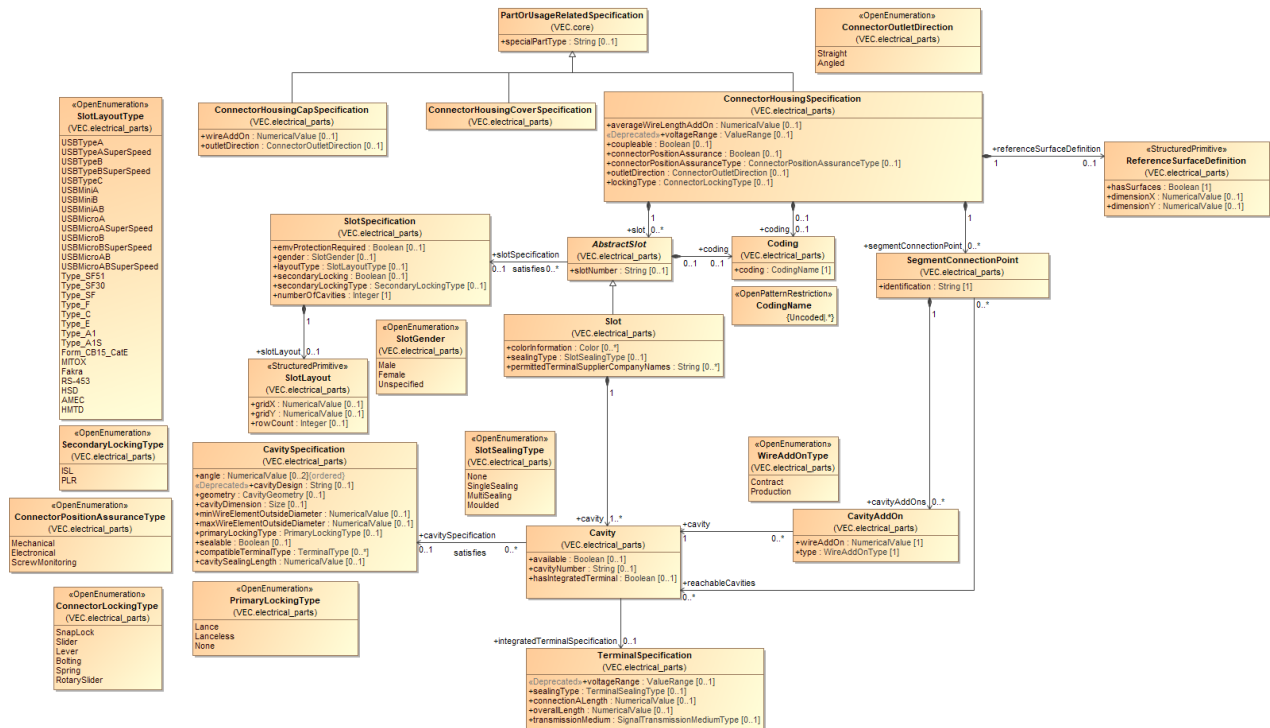


Figure 38: Connector Housings

A *ConnectorHousingSpecification* defines the technical properties of a connector housing or similar items (e.g. modular connectors, contact modules, connector shells). A connector is normally plugged into a mating connector in order to create connectivity (this is in most cases electrical connectivity but not limited to it). The actual electrical connection is normally established by terminals, mated to each other, which are placed in cavities. The VEC supports two primary types of connectors: normal (conventional) connectors and modular connectors.

A conventional connector defines a fixed number of *Cavities*, which are grouped into a fixed number of *Slots*. A modular connector defines a geometrical shape into which another component (a contact module) can be placed (see Diagram "Modular Connector Housings"). The cavities are then defined by the used contact module. The combination of both is possible as well.

Therefore, the *ConnectorHousingSpecification* consists of *AbstractSlots*, which either can be *Slots* or *ModularSlots* in an arbitrary combination.

A *Slot* is used if the cavities are part of the connector housing. Therefore, a *Slot* consists of *Cavities*. A *Cavity* is defined in detail with a *CavitySpecification*.

A *SegmentConnectionPoint* can be used if a connector housing has more than one entry point for wires (segments). The *SegmentConnectionPoint* is used to define which *Cavity* is reachable through which entry point. In the latter this is a necessary information to support auto routing for connectors with more than one *SegmentConnectionPoint*.

1.5.10 Modular Connector Housings

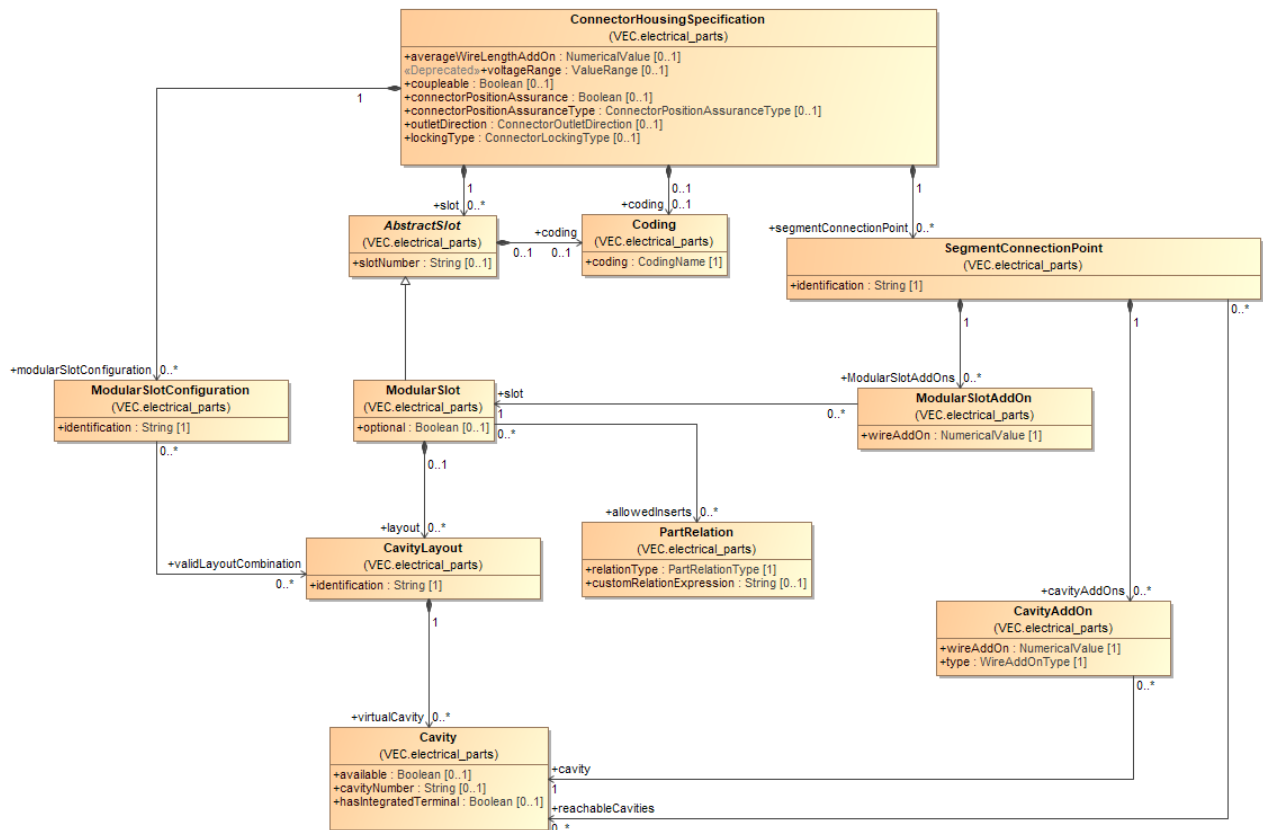


Figure 39: Modular Connector Housings

A modular connector defines a geometrical feature, the *ModularSlot*, into which another component—a contact module—can be inserted. The actual cavities are defined by the contact module used. It is also possible to combine regular *Slots* and *ModularSlots* within the same connector. The inserts themselves are defined by a *ConnectorHousingSpecification*, allowing the creation of recursive structures. A *ModularSlot* can reference multiple inserts, since different insert variants often fit into the same *ModularSlot*. The allowed inserts are referenced via a *PartRelation*, enabling part master data to be distributed across multiple VEC files.

In the perspective of many use cases, an assembled modular connector is treated in the same way as a regular one-piece connector. For example, the system schematic will typically define a single *ComponentConnector* with continuous pin numbering. The same applies in many cases to the mating connector, such as the interface on an E/E component. In both scenarios, the decision whether to use a one-piece or a modular connector is often made later in the process.

The *ConnectorHousingSpecification* of the contact module, on the other hand, is independent of its usage context. This means that both the cavity numbering and any defined *MappingSpecification* apply to the contact module itself, regardless of the collector housing. For example, cavity 1 in a contact module may correspond to cavity 1 in one collector housing, and to cavity 11 in another. Similarly, the same contact module may fit into different *ModularSlots* within a housing, causing its logical cavity positions to vary depending on the context.

Therefore, even in the case of modular connectors, it is necessary to define a consistent cavity numbering across its different assembly variants. This is essential to enable the mapping to potential mating connectors and to ensure continuity with upstream data, such as the system schematic.

The *CavityLayout* serves as a template for a valid configuration of a *ModularSlot*. A *ModularSlot* can support various configuration variants (e.g., single-pole, five-pole, ten-pole, etc.). The cavities defined within the *CavityLayout* are referred to as *virtualCavities*, as they merely act as placeholders for the actual cavities defined by the contact module. The technical properties of the cavities are determined by the contact module itself.

The virtual cavities serve multiple purposes. Using a *MappingSpecification*, it is possible to define a mapping between the cavities of a modular connector and those of a mating connector. The *MappingSpecification* can also be used to map virtual cavities to the actual cavities of a valid contact module (see Diagram "Cavity Mapping").

In addition, virtual cavities allow the assignment of individual *CavityAddOns* to specific cavities within modular assemblies.

Depending on the specific methodology used for such modular connectors, certain *ModularSlotLayouts* may only be valid in relation to the configuration of other *ModularSlots*. For example, if continuous cavity numbering is required, the same insert may have different cavity numbers depending on how many cavities a preceding slot is populated with. The *ModularSlotConfiguration* allows the definition of such valid combinations.

However, it should be noted that applying such a methodology can lead to a rapid, potentially exponential increase in the number of valid configuration variants, depending on the degree of flexibility in the connector's population.

1.5.11 Geometric Properties of Connector Housings (Definitions)

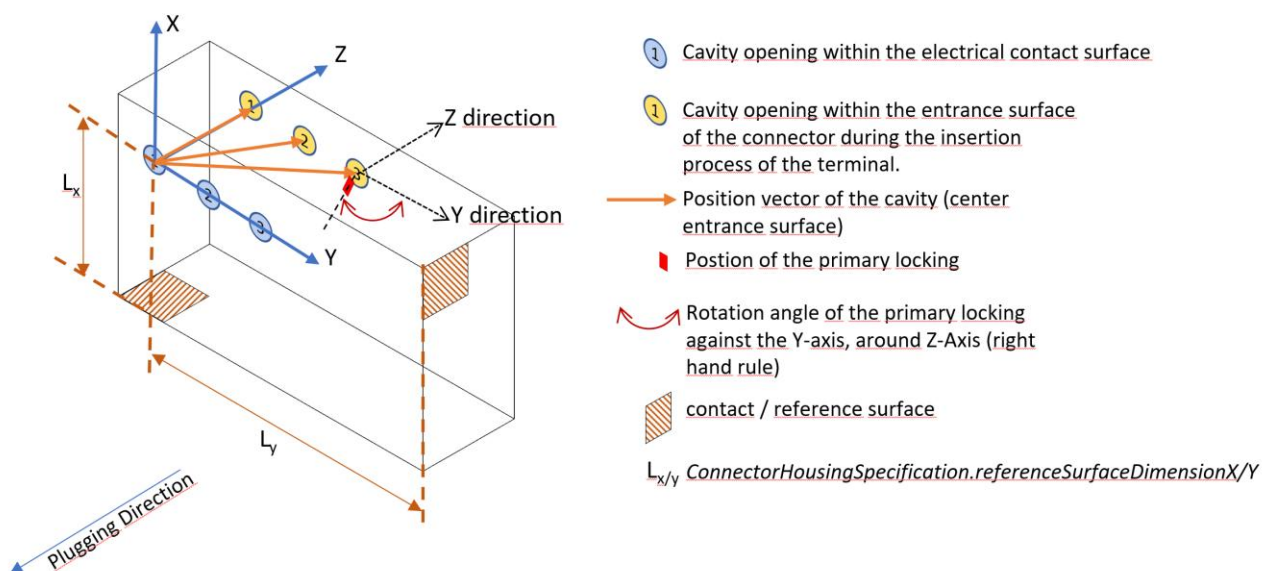


Figure 40: Geometric Properties of Connector Housings (Definitions)

The following section contains all relevant definitions that are required for the corresponding model elements in the VEC.

The geometric properties of connector housings are closely related to the *LocalGeometrySpecification*, described in the section "Coordinate Systems of Components". All coordinates and values are defined within the same coordinate system. All geometric properties that do not apply exclusively to connector housings are defined there (e.g. *BoundingBox*, *PlacementPoints* / *SegmentConnectionPoints*).

The 3D model and coordinate system of a connector housing is defined as follows:

- The origin of the plug is in the plane of the electrical contact surface in the center of the hole of the cavity/pin with the smallest alphanumeric identifier (e.g. cavity/pin 1 or A1).
- The Y-axis is in the plane of the electrical contact surface and points towards the center of the cavity/pin with the next higher alphanumeric identifier (e.g. cavity/pin 2 or A2). For single-pole connector housings, in the direction of the coding, if present.
- The Z-axis is normal to the plane of the electrical contact surface and points into the housing for female connectors and to the outside for male connectors (the connector in the illustration above is a female connector). So, for female connectors the Z-axis points in the opposite direction of the plugging direction. For male connectors it points in direction of the plugging direction.

- The X-axis is the vector product of Y and Z.

For cavities there are two relevant positions in the connector:

- The position in the electrical contact surface (highlighted in blue), and
- The position in the entrance surface during the insertion process (highlighted in yellow).

The rotation of the cavity within the connector will be defined as the angle between the Y-axis and the primary locking in the cavity.

1.5.12 Geometric Properties of Connector Housings (Model)

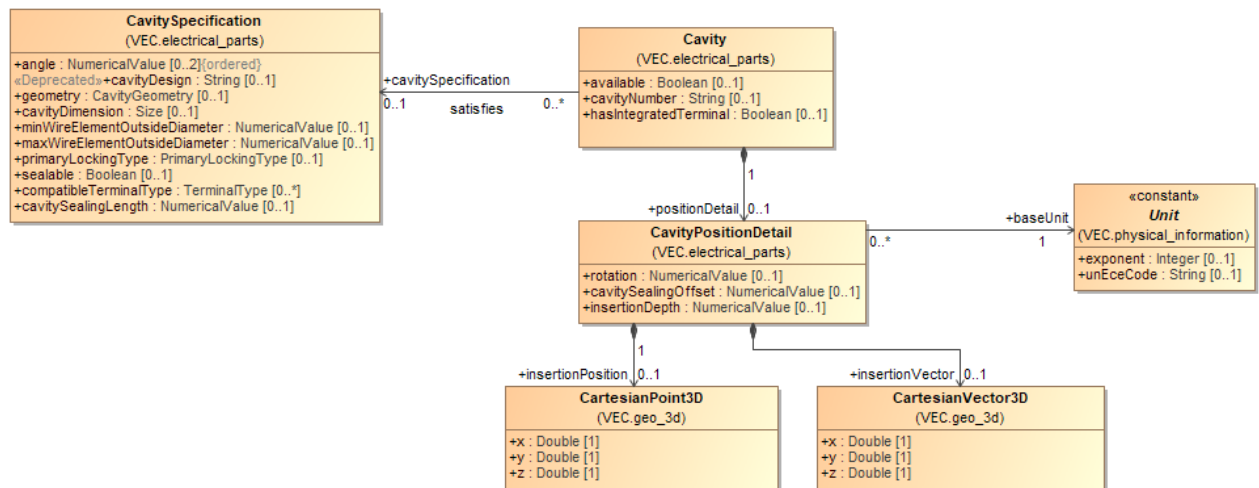


Figure 41: Geometric Properties of Connector Housings (Model)

The geometric properties of connector housings are closely related to the *LocalGeometrySpecification*, described in the section "Coordinate Systems of Components". Positions of common elements (e.g. *BoundingBox*, *PlacementPoint* / *SegmentConnectionPoint*) are defined in a *LocalGeometrySpecification*. Within the *ConnectorHousingSpecification* only those aspects that apply exclusively to connectors are defined. All definitions & conventions that are required to define these geometric properties are described in the previous section.

All geometric properties of a cavity within the context of a connector are aggregated in the optional *CavityPositionDetail*.

1.5.13 Cavity Mapping

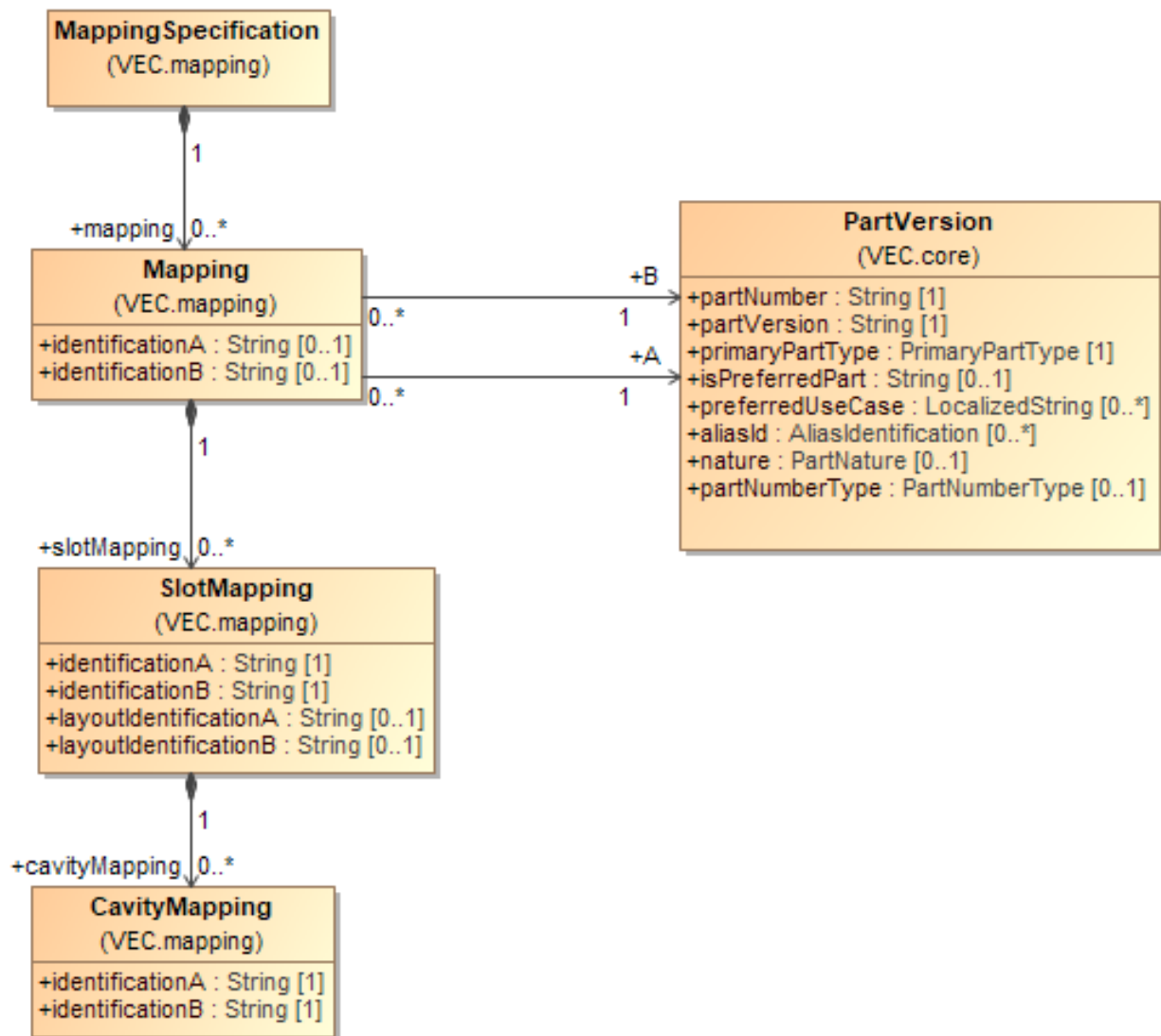


Figure 42: Cavity Mapping

A *MappingSpecification* defines the mapping of cavities in one connector housing onto the cavities in another connector housing. In order to create a compact data structure, the mapping is not done with direct references to *Cavities*, but with *Mapping* elements referencing two *PartVersions* for which the mapping is valid.

The *Mapping* element contains *SlotMappings* and *CavityMappings* to define which *Cavity* of one side gets plugged into which *Cavity* of the other side. The mappings are based on functional keys (e.g. cavity number), not technical keys like *ID/IDREF*, to ensure flexibility when partitioning the data into different VEC files.

For a valid *Mapping*, the referenced *PartVersions* shall represent parts that have "related" *ConnectorHousingSpecifications* (e.g. connectors or E/E-components, inliner). The *identification*-Attributes in the *Mapping* class are used in cases where the "related" *ConnectorHousingSpecification* cannot be resolved unambiguously. Most regular connectors are described directly by a single *ConnectorHousingSpecification*. In these cases, the *identification*-Attributes in the *Mapping* class can be omitted. In the case of E/E-components, it is not clear from the *PartVersion* alone against which *ConnectorHousingSpecification* the mapping is defined, as each *HousingComponent* can reference its own. In these cases, the *identifications* in the *Mapping* are used to reference the relevant *ConnectorHousingSpecification*. If the *Slots* of all relevant *ConnectorHousingSpecifications* have unique identifiers within the E/E-component, the *identifications* in the *Mapping* can be omitted.

the component is only described indirectly by the *ConnectorHousingSpecifications* for each *HousingComponent*.

The identificationA & B of the *SlotMapping* / *CavityMapping* shall have correspondent *Slot.identification* & *Cavity.identification* in the respective part master definition of the referenced *PartVersions*.

1.5.14 Cavity Seals and Cavity Plugs

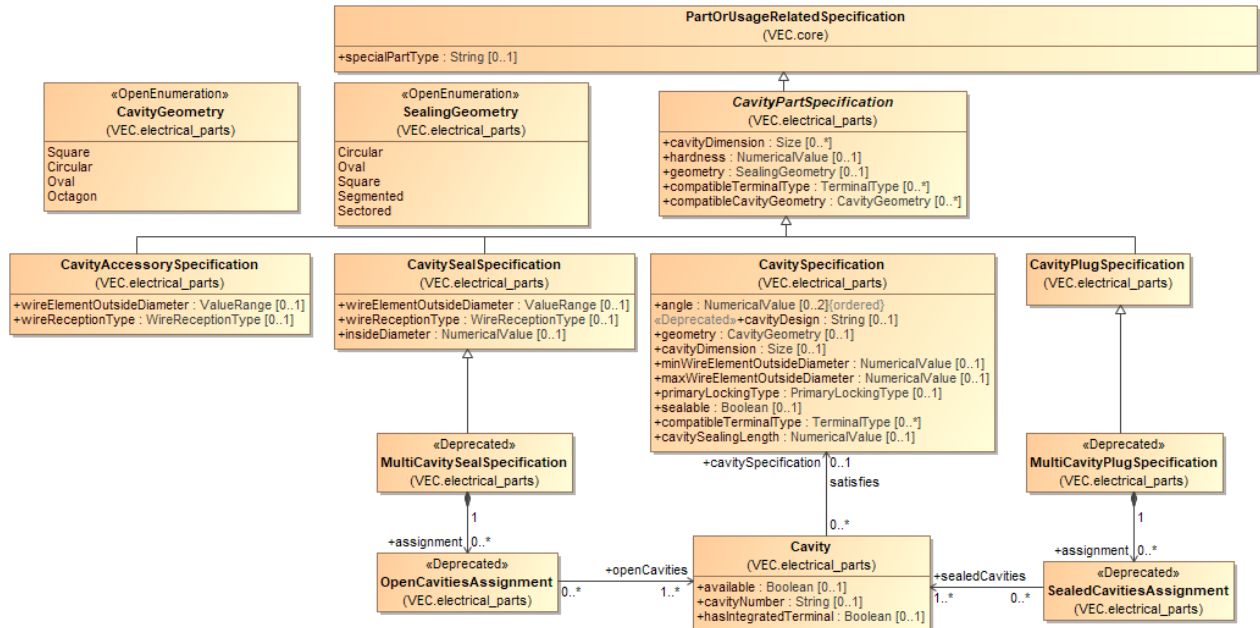


Figure 43: Cavity Seals and Cavity Plugs

The diagram displays the *PartOrUsageRelatedSpecifications* for Cavity seals and plugs. Both are used to seal *Cavities*. A cavity plug is used for a single cavity that has no wire in it, a cavity seal is used to seal a single cavity together with a terminal and a wire (single wire seal).

The *MultiCavitySeal* and the *MultiCavityPlug* represent *parts* which are sealing more than one cavity at the same time (multi wire seal). For more details see the description of the corresponding classes.

1.5.15 Cable Seal & Plugs

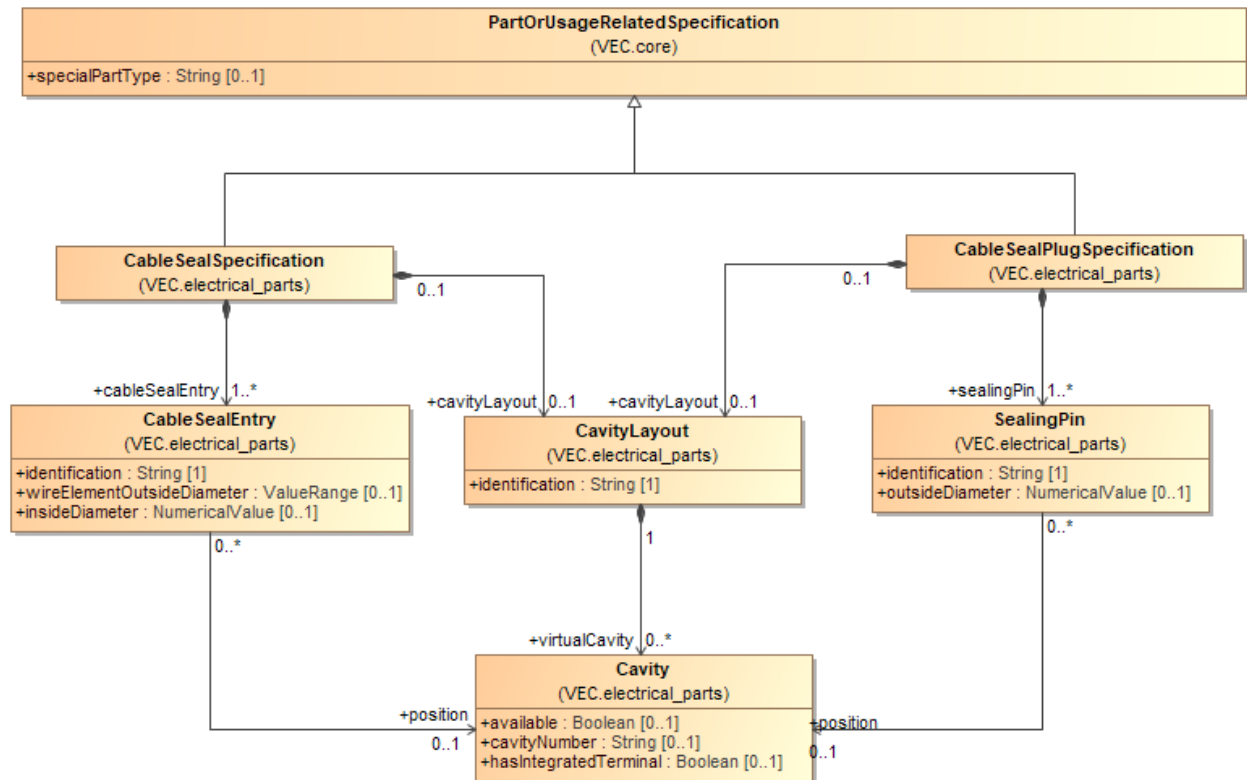


Figure 44: Cable Seal & Plugs

CableSeals are seals that are used on a wire/cable, outside of the actual crimp, but often within a complex connector housing. A *CableSealSpecification* defines one or more *CableSealEntries*. A *CableSealPlug* is a plug that can seal one or more *CableSealEntry* that are not occupied by a wire element. The *CableSealPlugSpecification* defines *SealingPins* for each *CableSealEntry* that it might seal.

Both *Specifications* can define a *CavityLayout* with *virtualCavities*. Those *Cavities* serve multiple purposes:

1. By utilizing a *CavityMapping* it is possible to create a mapping between a *CableSealSpecification* and its compatible *CableSealPlugSpecification* and to define which *CableSealEntry* is plugged by which *SealingPin*.
2. Also, by utilizing a *CavityMapping* it is possible to create a mapping between a *CableSealSpecification/CableSealPlugSpecification* and a *ConnectorHousingSpecification* specifying which *CableSealEntry* is associated with which *Cavity* of the connector housing. This information is required for example to determine which *CableSealEntry* has to be used for specific wire, when it is contacted in a specific cavity of the connector housing.

1.5.16 Wire Protections

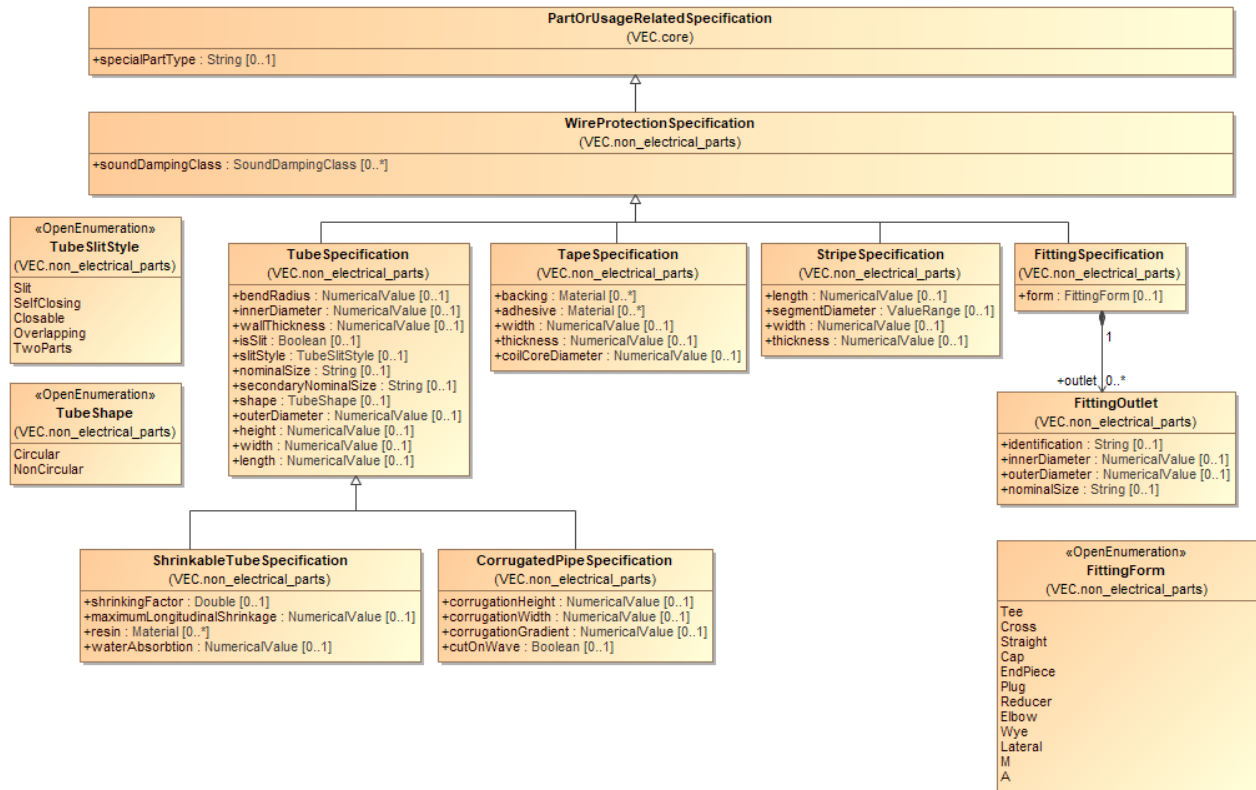


Figure 45: Wire Protections

The diagram displays the type hierarchy for *WireProtectionSpecifications*. These *PartOrUsageRelatedSpecifications* are used to describe the technical aspects of wire protections like tapes, tubes, fittings and stripes.

1.5.17 Fixings, Cable Ducts, Cable Ties and Similar

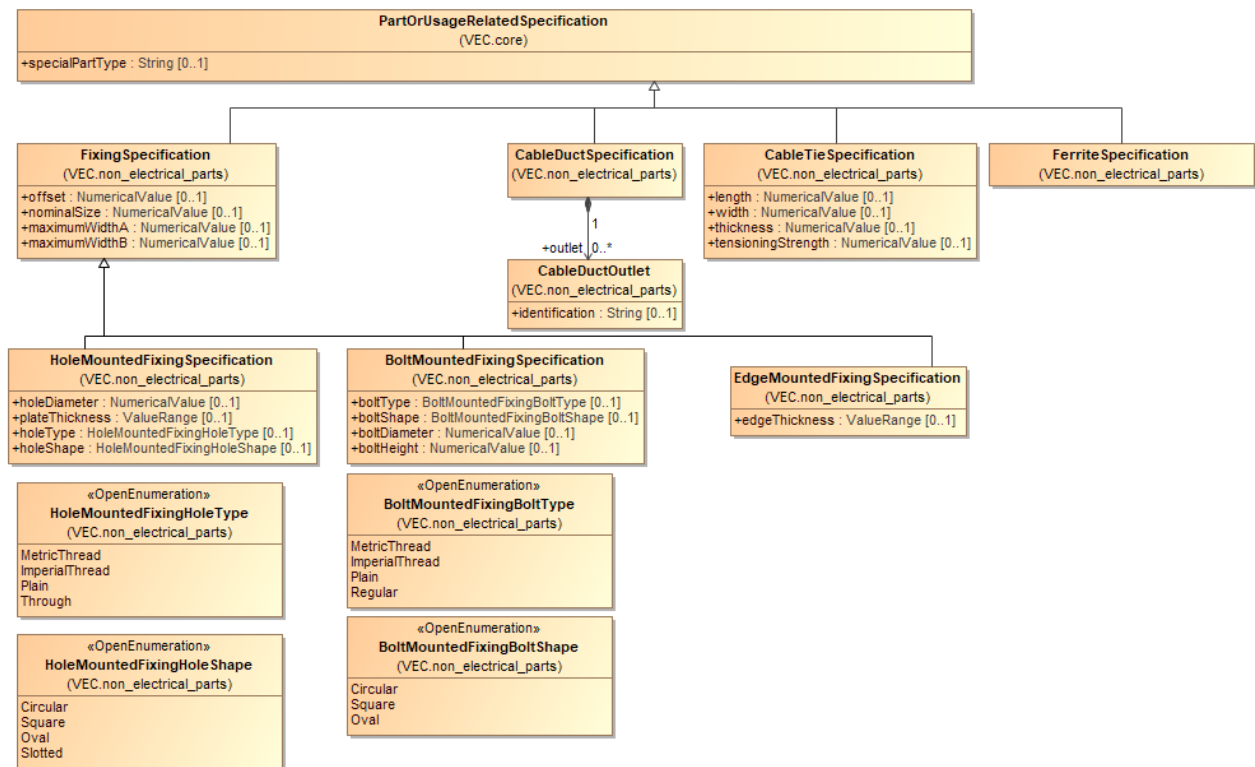


Figure 46: Fixings, Cable Ducts, Cable Ties and Similar

The diagram displays the type hierarchy for parts that are usually mounted onto the topology of a wiring harness. These *PartOrUsageRelatedSpecifications* are used to describe the technical aspects of components like fixings and cable ducts.

1.5.18 Fixings Coordinates and Dimensions

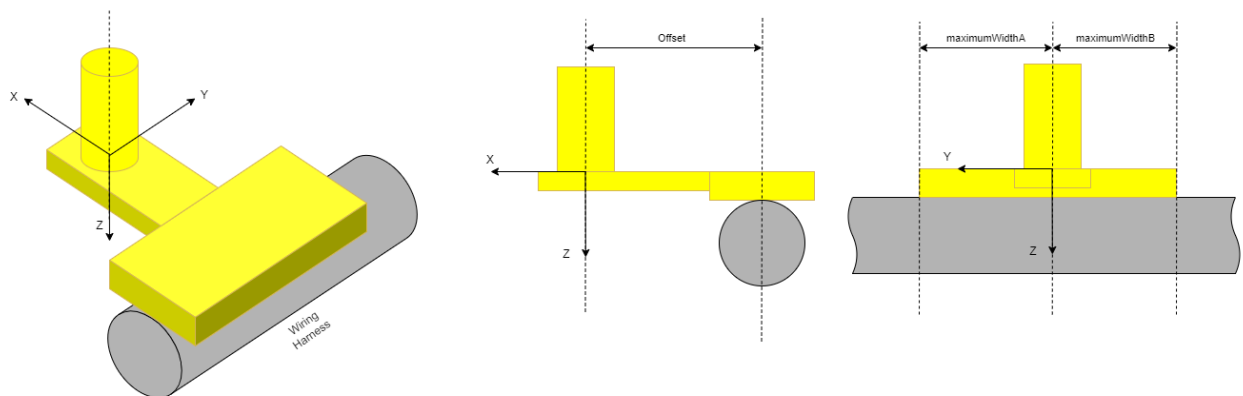


Figure 47: Fixings Coordinates and Dimensions

The following section contains specifications for the definition of coordinate systems of fixings, clips, cable ties and similar (and their 3D models). These coordinate definitions form the basis for various attribute definitions and the placement of components on the wire harness (see *PlacementSpecification*) and in the 3D model (see *BuildingBlockSpecification3D*). For components whose geometry changes upon installation, the installed state must be used.

The orientation of the coordinate system of cable ties shall be as follows:

- The origin of the cable tie lies centrally at the edge of the lock on the side of the cable tie.
- The Y-axis is positioned in the plane of the wire harness support surface and is oriented parallel to the direction in which the wire harness passes through the cable tie.
- The XZ-plane intersects the wire harness.

For fixings with cable tie (including those with attached cable ties), only the fastening is represented. The cable tie itself (which varies in diameter) and its closure are omitted.

The orientation of the coordinate system of fixings shall be as follows:

- The origin of the component lies at the intersection of the support surface of the fixing (in the installed state) with the central axis of the locking pin, bolt holder, or hole/slot (depending on the fastening method).
- The Y-axis is in the plane of the support surface and oriented parallel to the direction in which the cable harness passes through the fixing.
- The Z-axis is perpendicular to the plane of the support surface and points in the opposite direction to the clip's mounting direction.

For fastening elements attached to an edge, the origin of the component is located at the intersection of the support surface of the component (in the installed state) with the center of the side contact surface of the component on the sheet metal.

If the conditions described above cannot be met, the following priorities should be applied:

1. The Y-axis is oriented parallel to the direction in which the cable harness passes through the fixing and lies in the plane of a support or contact surface.
2. The XZ plane must intersect the cable at the point where the position of the component on the segment is dimensioned (e.g., in the middle of the cable tie for cable tie holders).
3. The YZ plane lies in the middle of the component.

Definitions were adopted from the CES guidelines.

1.5.19 Grommets

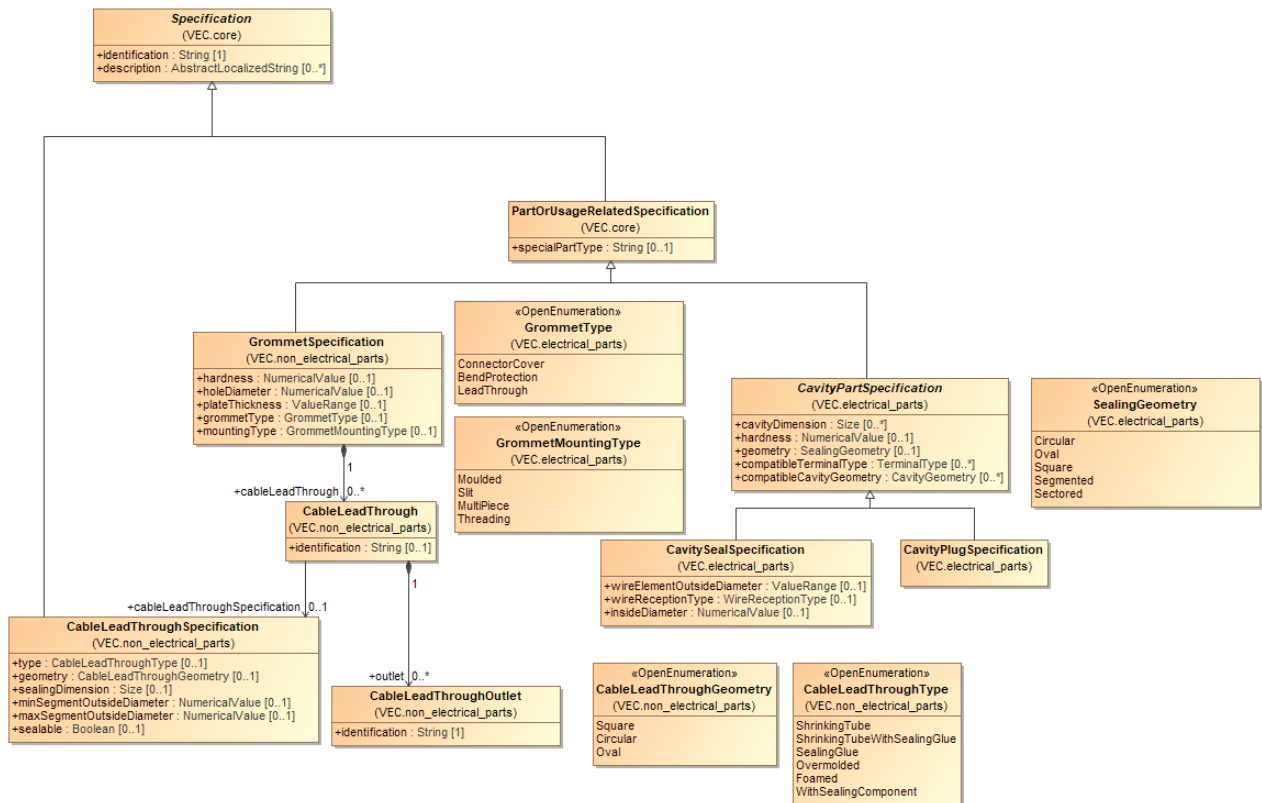


Figure 48: Grommets

This diagram displays the relevant classes for grommets. Basically, grommets belong to the same category of parts like fixings and cable ducts. Grommets are mounted onto a harness topology and are used to guide cables safely, and in some cases sealed, through openings, for example in the car body.

There are cases, where additional sealing of a cable lead through is not done with some kind of manufacturing steps (e.g. Foamed), but with additional sealing components. These are common case e.g. in high voltage applications where one lead through per wire is used. Those are often sealed with an additional single wire seal. In the VEC those seals are represented by a *CavitySeal-* or *CavityPlugSpecification*. Even if the name "Cavity..." seems awkward in the combination with grommets & cable lead throughs, the properties remain the same.

The *CableLeadThroughSpecification* includes all attributes that are necessary to calculate appropriate seals.

1.6 EE-Components

NA

1.6.1 EE-Components

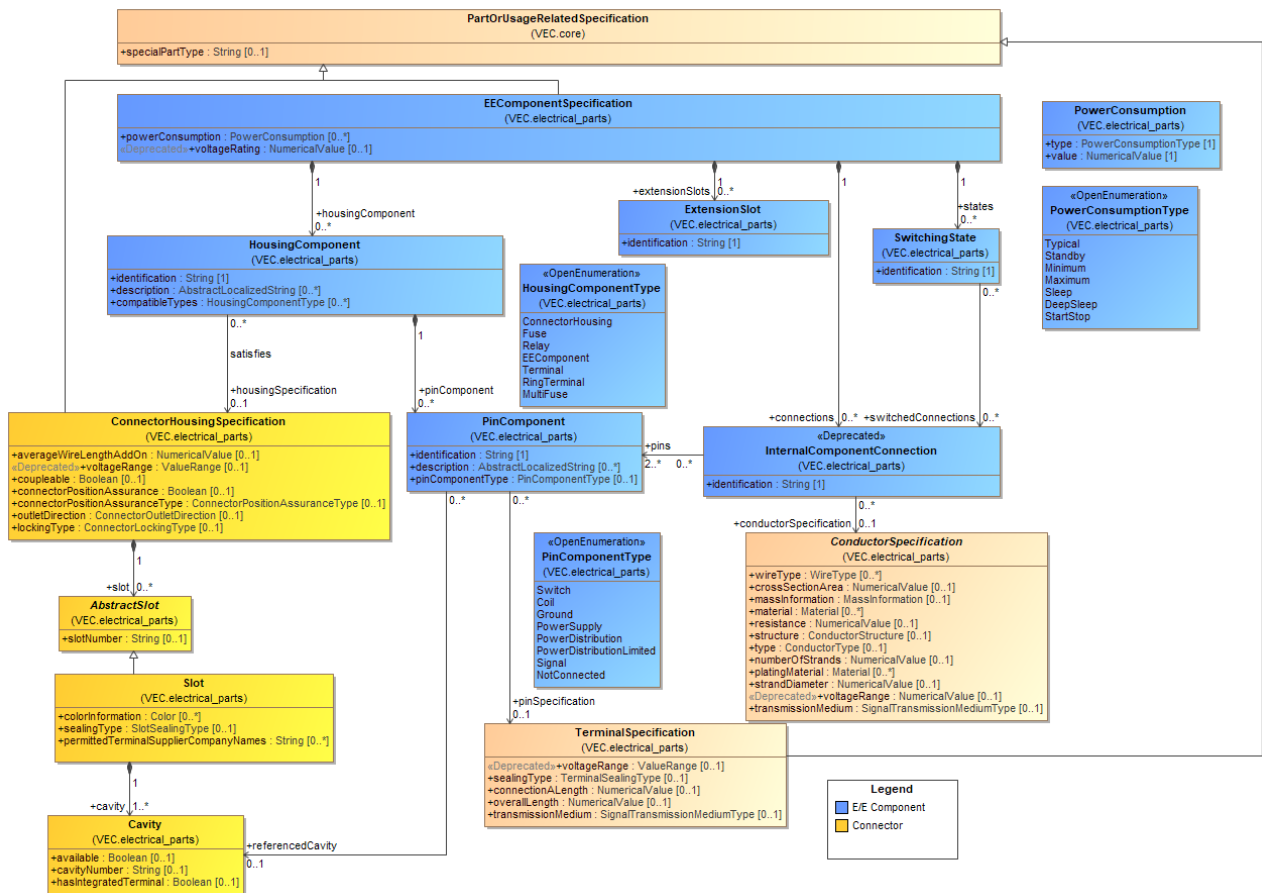


Figure 49: EE-Components

An *EEComponentSpecification* is used to describe the technical aspects of an EE-component. An EE-component is an electrical component to which the wiring harness is attached to (e.g. an ECU, a sensor, a relay, an antenna, a battery, a fuse). All EE-components have in common that they have a certain electrical interface (e.g. to which the wiring harness is attached). To describe this interface, the *EEComponentSpecification* is used. The content of an *EEComponentSpecification* is highlighted in blue.

An *EEComponentSpecification* can define a couple of *HousingComponents*. A *HousingComponent* is one interface to which other components can be attached (e.g. the wiring harness). To describe a *HousingComponent* satisfyingly two aspects are necessary:

- its geometrical shape and,
- its electrical behavior.

In order to describe the geometrical shape of a *HousingComponent* it can reference a *ConnectorHousingSpecification*, so the same concept is used in the wiring harness and in an EE-component (highlighted in yellow). The electrical behavior of a *HousingComponent* is defined by several *PinComponents*. A *PinComponent* references a *Cavity* to define its location in the geometrical shape of the *HousingComponent*. To define its physical properties a *PinComponent* references a *TerminalSpecification*. These properties are defined inherently by the material and the type of the pin. It is a common case that some of the *PinComponents* (sometimes all) of a single *HousingComponent* are referencing the same *TerminalSpecification*. The definition of properties regarding the behavior of a *PinComponent* (e.g. defined by the software deployed on an ECU) is explained in the diagram "Pinning Information & Pinning Variance".

1.6.2 Segment Connection Points

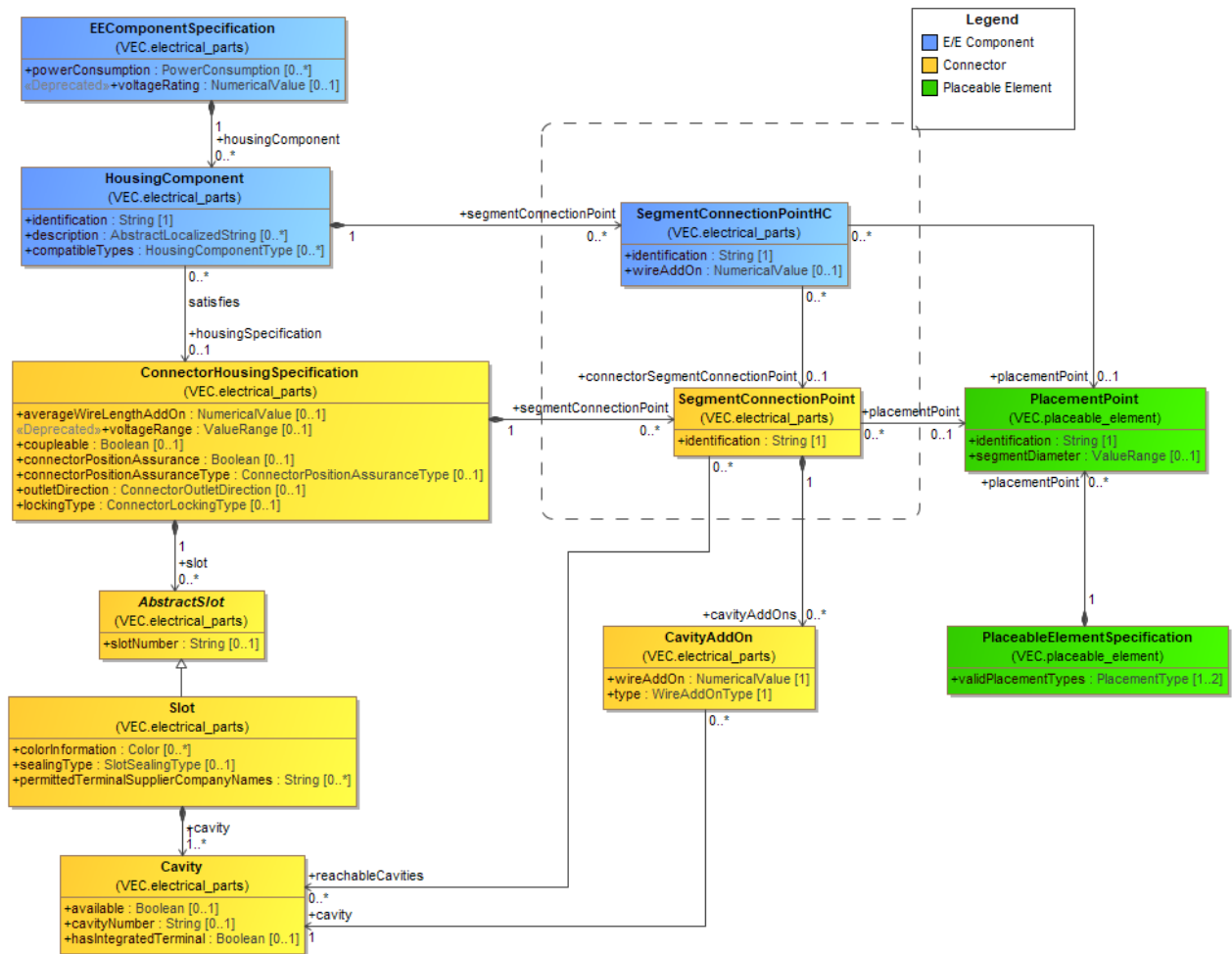


Figure 50: Segment Connection Points

The *HousingComponent* of an E/E-Component has two major use cases:

1. Other E/E-Components can be plugged into it (e.g. Fuses, Relays, Harnesses).
2. Wire can be contacted directly into the *HousingComponent* (without an additional connector)

In the first case, no connection to the harness topology is required. Either because there is no harness topology (e.g. a Fuse), or because the end point of the topology is the harness connector itself (which is plugged into the housing component). However, for the latter case, *HousingComponent* itself is the endpoint of the wires and thus of the harness topology. Like a connector, an E/E-component and its *HousingComponents* can have multiple different points at which *TopologyNode* can be connected and not all *HousingComponents* and *Cavities* are reachable from a connecting point.

Reusing the *SegmentConnectionPoint* of the *ConnectorHousingSpecification* is not sufficient, because a single *ConnectorHousingSpecification* can be used by different *HousingComponents* and the definition would be ambiguous.

Therefore, a *HousingComponent* can define multiple *SegmentConnectionPointHC* representing the valid options for this specific *HousingComponent* to attach to a *TopologyNode*. The definition of cavity reachability and wire addons is specified with the *SegmentConnectionPoint* of the *ConnectorHousingSpecification*. *SegmentConnectionPointHC* can reference the *SegmentConnectionPoint* which is used to "enter" the connector housing.

1.6.3 Internal Connectivity

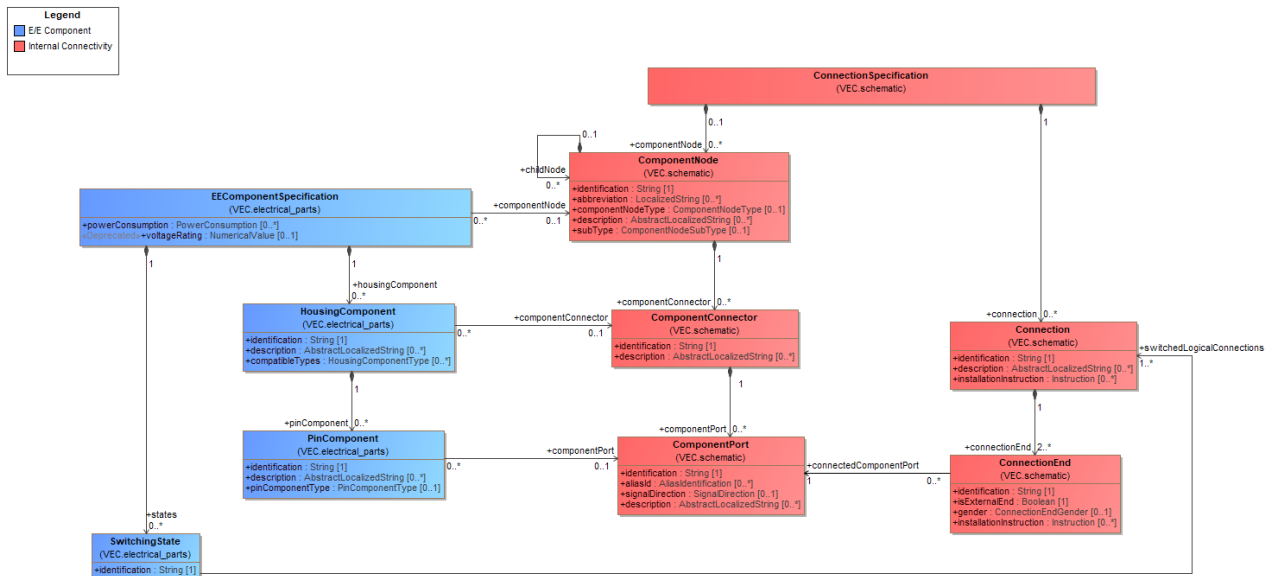


Figure 51: Internal Connectivity

The model elements above enable the explicit modelling of the internal electrical connectivity of E/E-Components in the VEC. The *ConnectionSpecification*, a core concept in the VEC is used to represent the logical internal connectivity between *PinComponents* within an *EEComponentSpecification*. For more details of the *ConnectionSpecification* see "Connectivity / Connection Specification".

The *ConnectionSpecification* should be co-located with the *EEComponentSpecification* in the *DocumentVersion* that defines the E/E-component. It contains a single *ComponentNode* representing the E/E-component itself. The *HousingComponents* and *PinComponents* are represented by *ComponentConnectors* and *ComponentPorts* on this *ComponentNode*. The internal connectivity is represented by *Connections*. If the E/E-component contains integrated subcomponents, those are represented as *childNodes* of the *ComponentNode* of the E/E-component.

Traceability Links

To ensure semantic consistency and enable model traceability, dedicated references are introduced between the following elements:

1. *EEComponentSpecification* → *ComponentNode*
2. *HousingComponent* → *ComponentConnector*
3. *PinComponent* → *ComponentPort*

These links allow navigation between the specification of a component and its internal connectivity representation.

Representation of Physical Properties

The *ConnectionSpecification* is the layer to describe logical connectivity, and it does not contain any physical properties (such as conducting properties of a *Connection*). However, it is possible to define physical properties with *PartOrUsageRelatedSpecifications* and *Roles* and link this information with the logical connectivity (see Wiring Specification).

1. If the physical entity is a component (e.g. a fuse) it is represented as a *ChildNode* of the *ComponentNode* representing the E/E-component.

2. If it is a conductor of any kind, it is represented by a *Connection*.
3. A *PartUsageSpecification* and a *PartStructureSpecification* is added to the definition of the E/E-component. If the subcomponents should be represented as actual *PartOccurrences*, instead of *PartUsages*, a *CompositionSpecification* is used instead of the *PartUsageSpecification*.
4. For each element with physical properties a *PartUsage* (with corresponding *PartOrUsageRelatedSpecification* and *Role*) is created. This includes traceability links to the corresponding *ConnectionSpecification* elements (e.g. *WireElementReference* -> *Connection*, *FuseRole* -> *ComponentNode* (a *childNode*)).

This approach ensures a separation of concerns between logical connectivity and physical embodiment while maintaining full traceability.

1.6.4 EE-Component subclasses

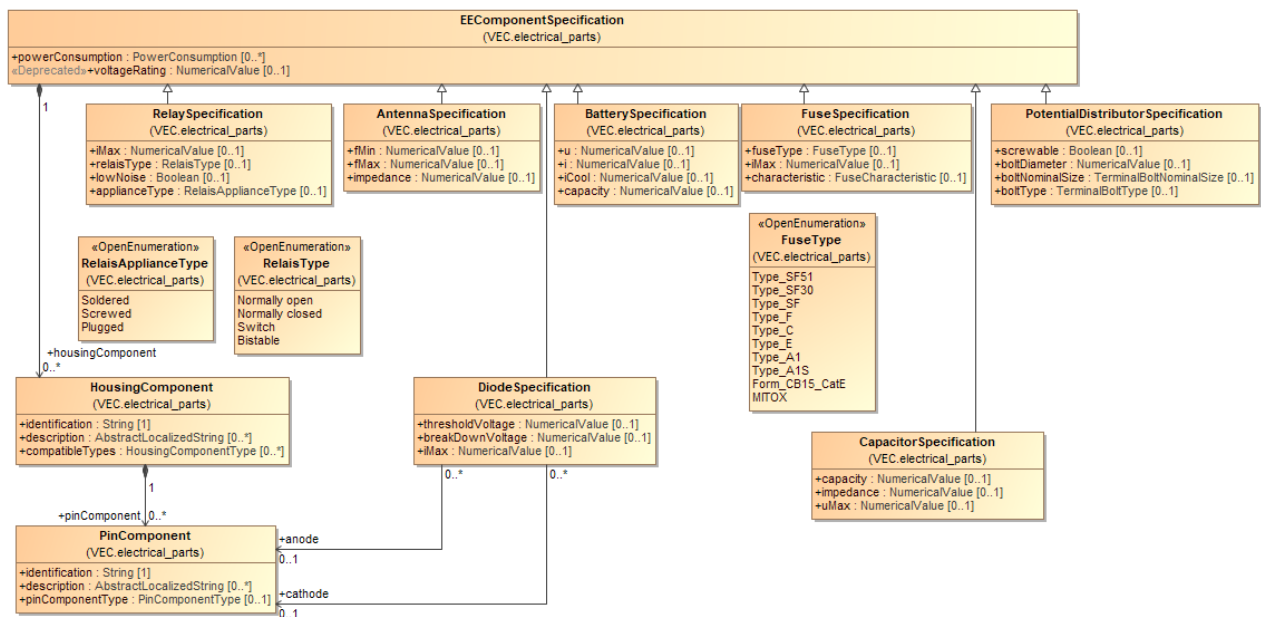


Figure 52: EE-Component subclasses

In order to define additional aspects of an EE-component there are some subclasses of the *EComponentSpecification*. This diagram displays these subclasses.

1.6.5 Multi Fuse

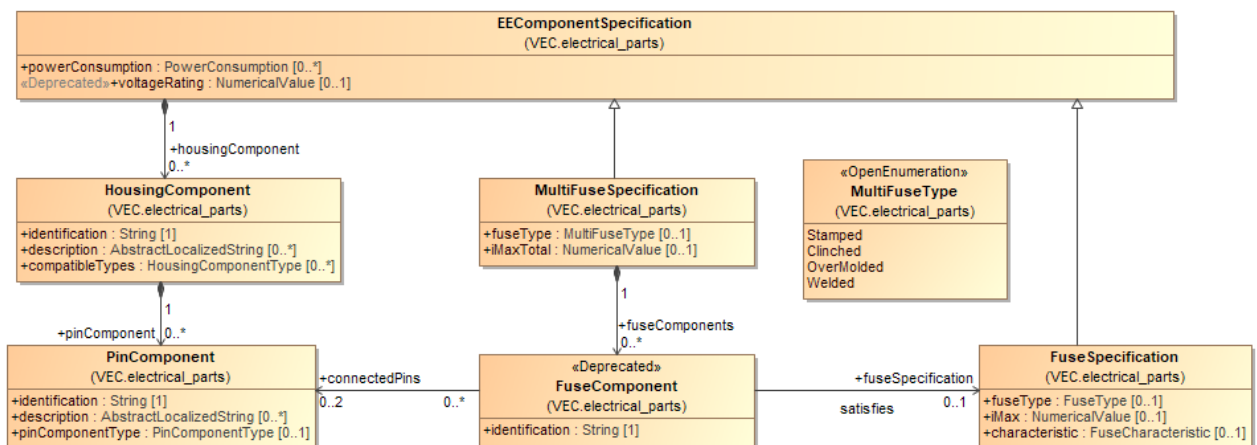


Figure 53: Multi Fuse

A multi-fuse is a metal structure that combines several fuses with a common power feed in a fixed layout. It is used to distribute a power supply to different fuse paths.

1.6.6 Pinning Information & Pinning Variance

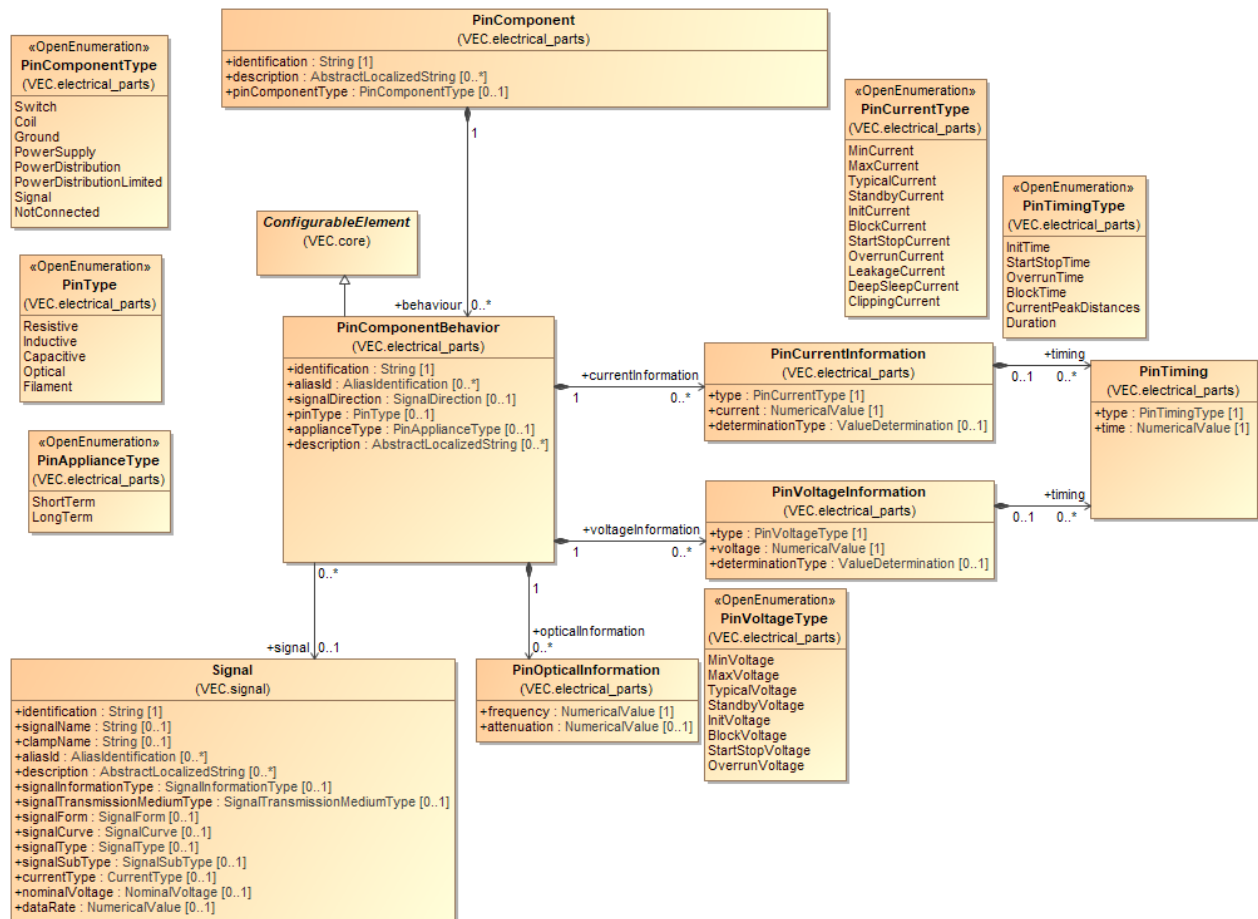


Figure 54: Pinning Information & Pinning Variance

The electrical behavior of a *PinComponent* can be fixed for a certain EE-component but does not have to be fixed. For example, in the case of an ECU, the behavior depends on the deployed software.

Therefore, the electrical behavior of a *PinComponent* is not defined directly and fixed in the *PinComponent*. In fact, a *PinComponent* can define *PinComponentBehaviors*, which are *ConfigurableElements*. This means the actual valid *PinComponentBehavior* for a *PinComponent* can be determined when the necessary variant information is available. When no variant information is necessary (the *PinComponent* has a fixed behavior) then the *PinComponent* just defines a single *PinComponentBehavior*.

The *PinCurrentInformation* and *PinVoltageInformation* can be used to describe the characteristic curves for different types and times of currents and voltages.

1.7 Instances of Components

NA

1.7.1 Instantiation of Components

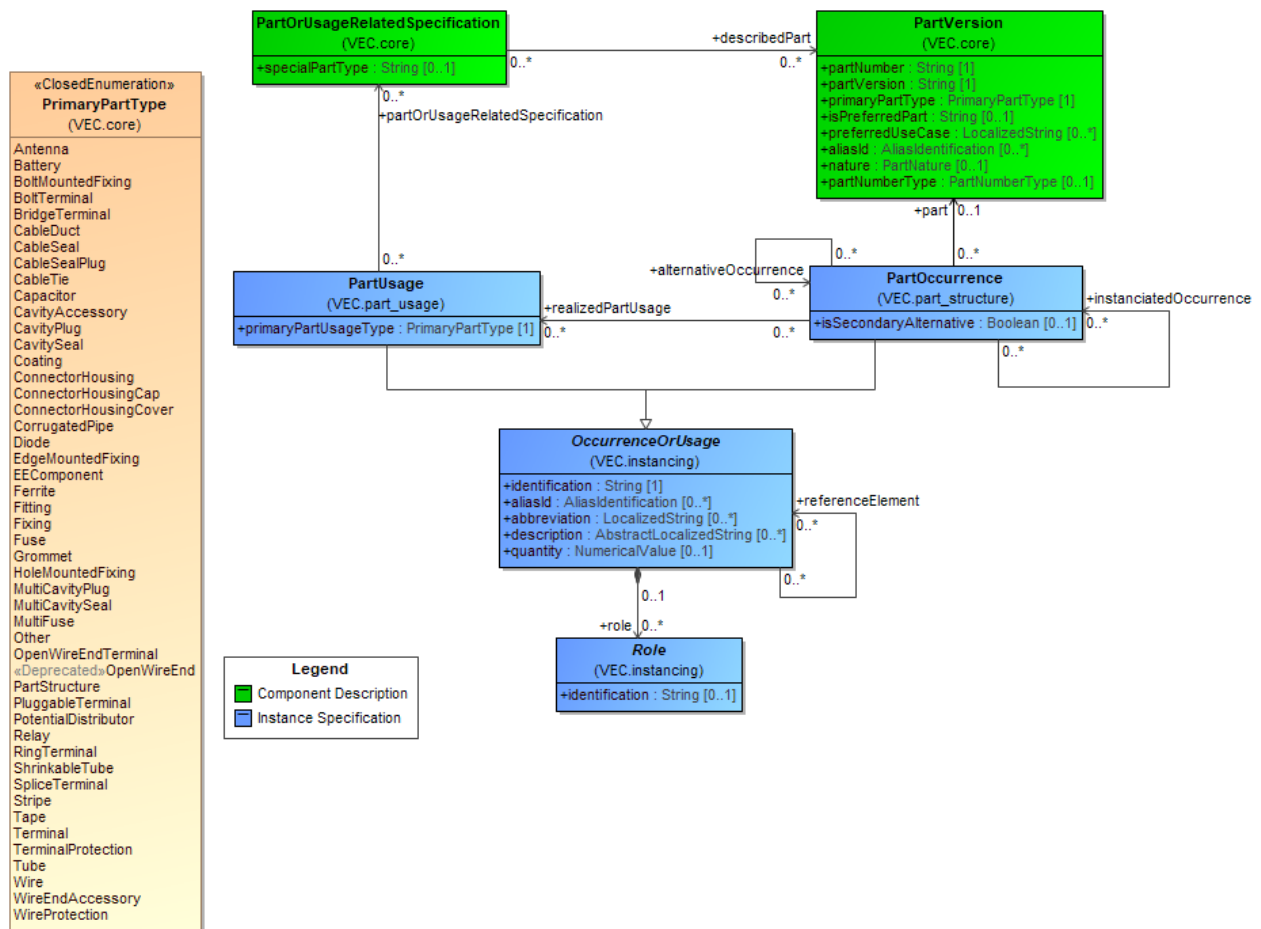


Figure 55: Instantiation of Components

It is a quite common pattern for complex product descriptions, to differentiate between types and their instances. A type is a general description that applies to all instances of this type. In the VEC the types of components are described with *PartVersions* and *PartOrUsageRelatedSpecifications*. They are describing the general properties of a certain component type (e.g. a connector housing with a certain *partNumber*). In order to define a more complex product with these elements, instances of the components are necessary (e.g. the connector housing for the left front light). To create such instances of components, the VEC provides two concepts: *PartOccurrence* and *PartUsage*.

The two concepts are necessary to fulfil the requirement for the VEC, that equal design elements are represented equally in the model, regardless of the level abstraction. This means for example, that the specification of a contacting must be represented always in the same way, regardless of the context either in an electrological specification or in a finalized product specification. The difference is that in an electrological specification in many cases defines only requirements for a component (e.g. the wire color, cross section area or number of cavities). In a finalized product specification, all components normally have assigned part numbers.

A *PartOccurrence* is a concrete occurrence of a certain *PartVersion* in a real product (Metaphorically spoken: if the product is broken into pieces, there will be a good chance that one of the pieces is the *PartOccurrence* one is searching for). Therefore, the *PartOccurrence* references directly a *PartVersion*.

PartUsages shall be used for the specification of the elements on a more abstract level. This is the case either when not all technical properties required to select a correct *PartVersion* are known (e.g. only the cross-section area of a wire is known, but not the color) or when not all usage properties are known that are required to create a *PartOccurrence*. This is true for example for a connectivity description and for a pure geometry description. The final occurrences come first into being when a connectivity description and a geometry description is combined. This is even true in the case of connectivity description where the part numbers for

the wiring connections are already defined. The reason is that the wire lengths are still undefined, because the length information is dependent on a certain geometry. Additionally, it is possible that the same wiring definition is instantiated within different geometries, which might result in different *PartOccurrences*. For reasons of traceability, each *PartOccurrence* can reference back to the *PartUsage* it realizes.

In order to describe technical properties for a *PartUsage* without requiring a concrete *PartVersion* or a pseudo *PartVersion* the *PartUsage* can reference several *PartOrUsageRelatedSpecifications* like the *PartVersion*. As described in the chapter "part master data" the VEC allows the description of components through the combination of different characteristics (the *PartOrUsageRelatedSpecifications*). Therefore, a *PartUsage* can reference multiple *PartOrUsageRelatedSpecification*. Its primary characteristic is defined with the *PrimaryPartType*.

However different characteristics of component types may require different properties for instances of them. Therefore, a symmetrical concept to the *PartOrUsageRelatedSpecification* for instances was created in the VEC: The *Role*. Quite simplified: A *Role* is an instance of a *PartOrUsageRelatedSpecification* in the context of an *OccurrenceOrUsage* which is an instance of an abstract or concrete part (difference between *PartUsage* and *PartOccurrence*).

Roles, their subclasses and the sub elements of them are used whenever type specific usage information is required (e.g. a reference to a cavity in a certain use of a connector). Every other concept in VEC that requires references to instances uses these elements (e.g. the contacting). Since a *Role* can be used without knowing if it belongs to a *PartUsage* or a *PartOccurrence* concepts like the contacting can be reused regardless of the level of abstract (e.g. in a system wiring without concrete part numbers or in the product specification of wiring harness).

1.7.2 Instances of Wires

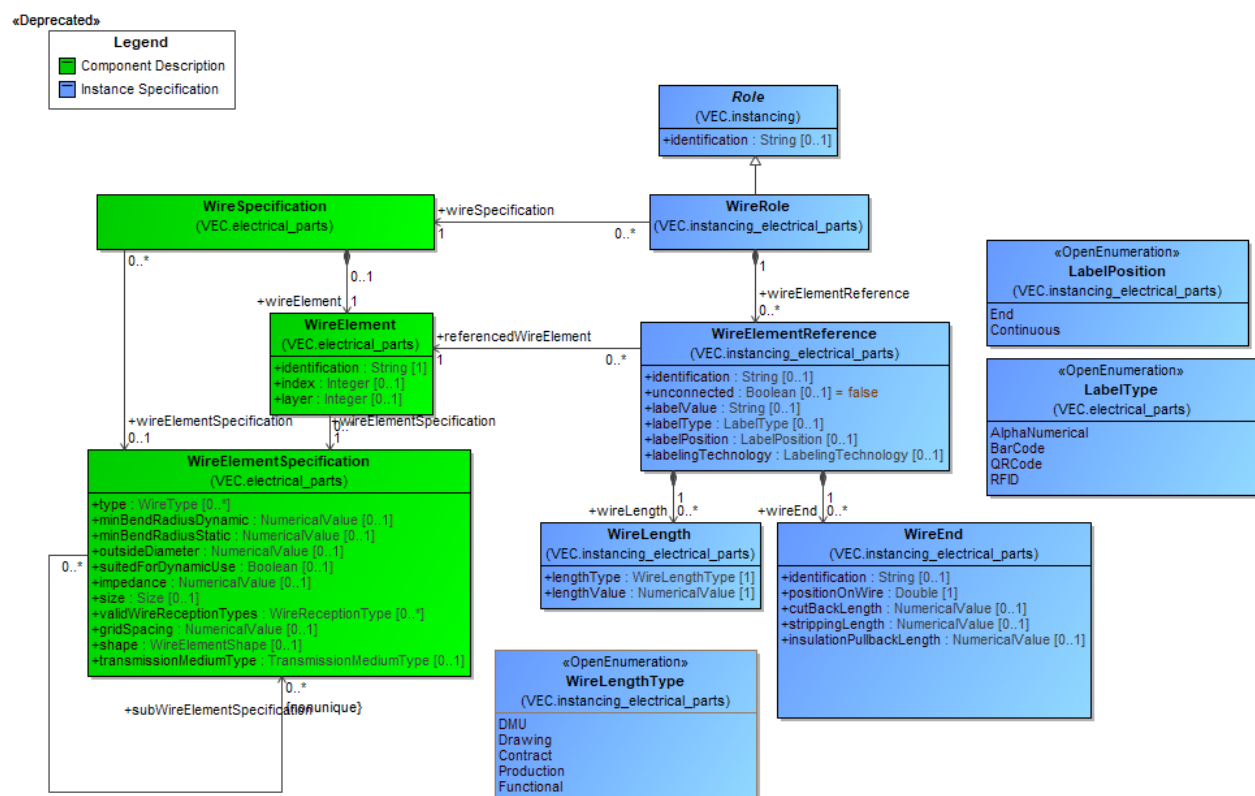


Figure 56: Instances of Wires

The diagram shows the mapping between the part master data of a wire (partly displayed on the left side) and the instance specific information (displayed on the right side).

1.7.3 Instances of Terminals

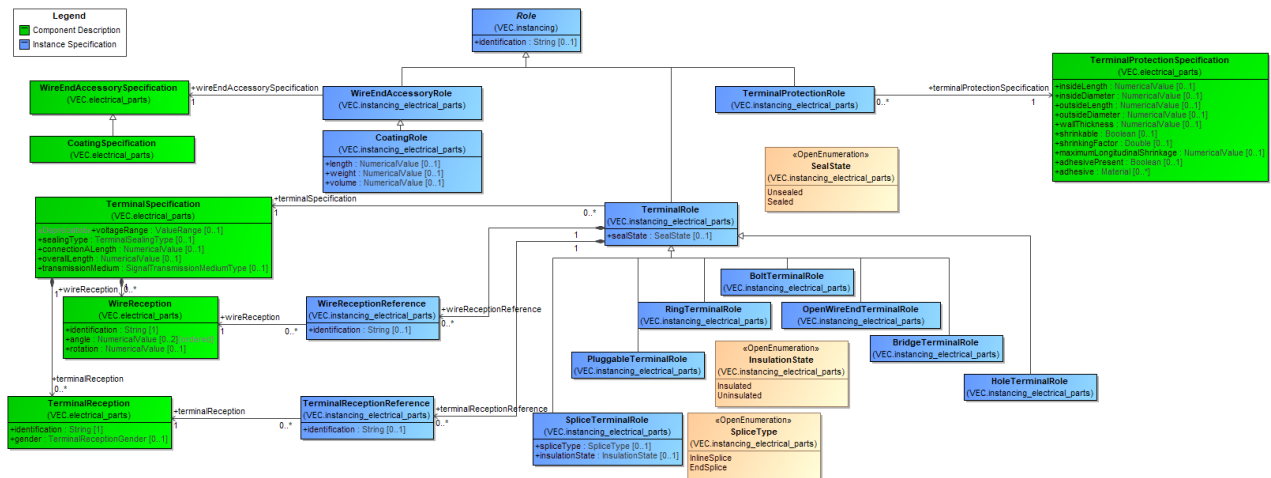


Figure 57: Instances of Terminals

The diagram shows the mapping between the part master data of a terminal (partly displayed on the left side) and the instance specific information (displayed on the right side).

1.7.4 Instances of Connector Housings

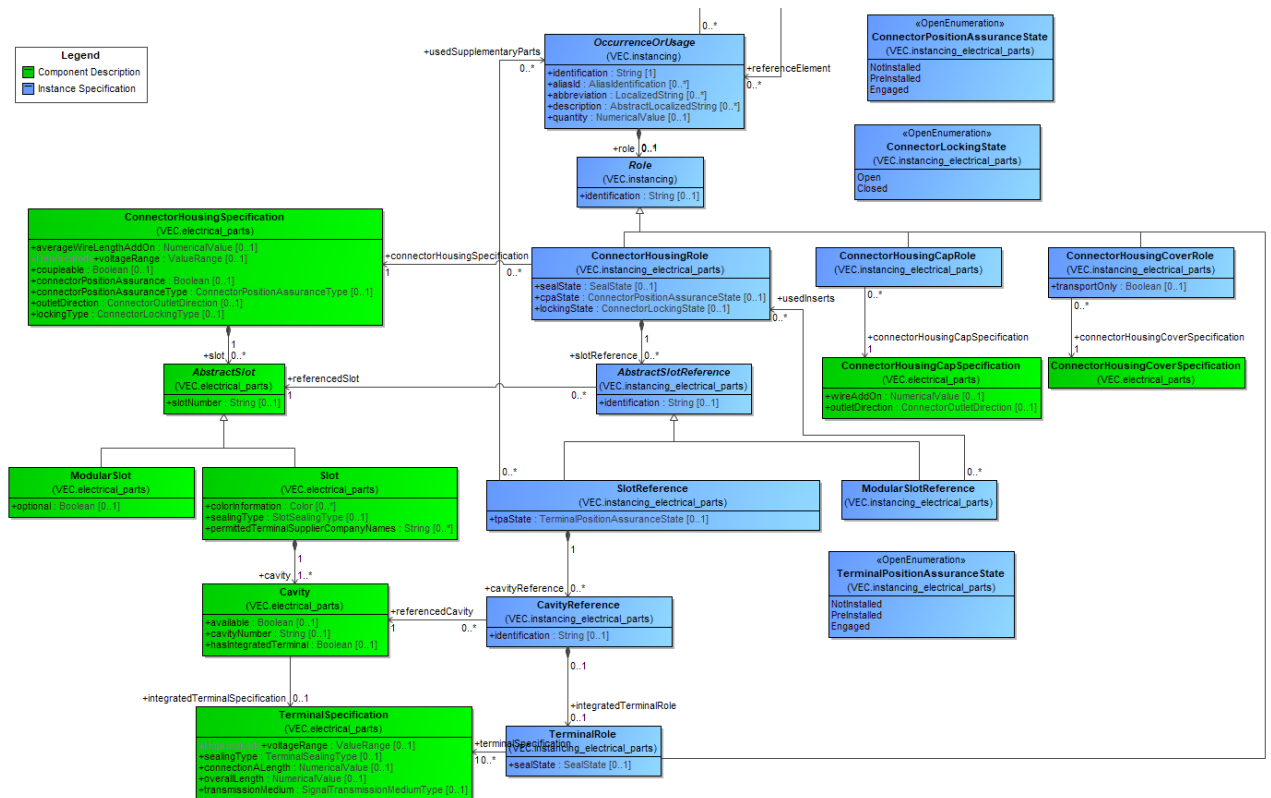


Figure 58: Instances of Connector Housings

The diagram shows the mapping between the part master data of a connector housing (partly displayed on the left side) and the instance specific information (displayed on the right side).

1.7.5 Instances of Cavity Seals and Cavity Plugs

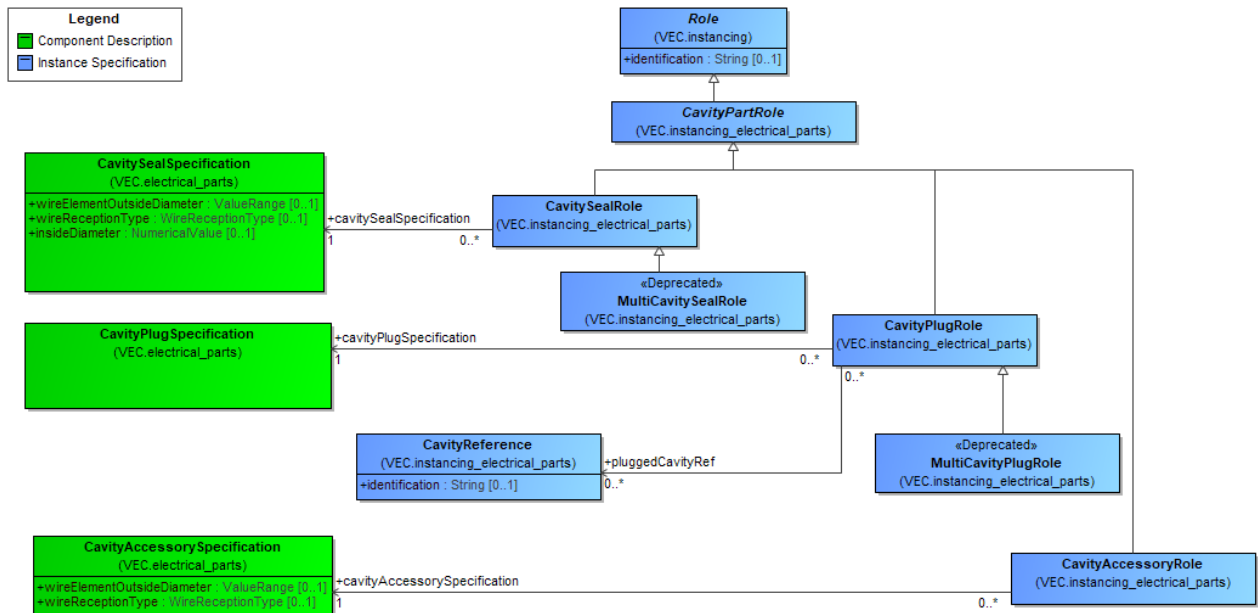


Figure 59: Instances of Cavity Seals and Cavity Plugs

The diagram shows the mapping between the part master data of cavity seals and cavity plugs (partly displayed on the left side) and the instance specific information (displayed on the right side).

1.7.6 Instances of Cable Seals and Plugs

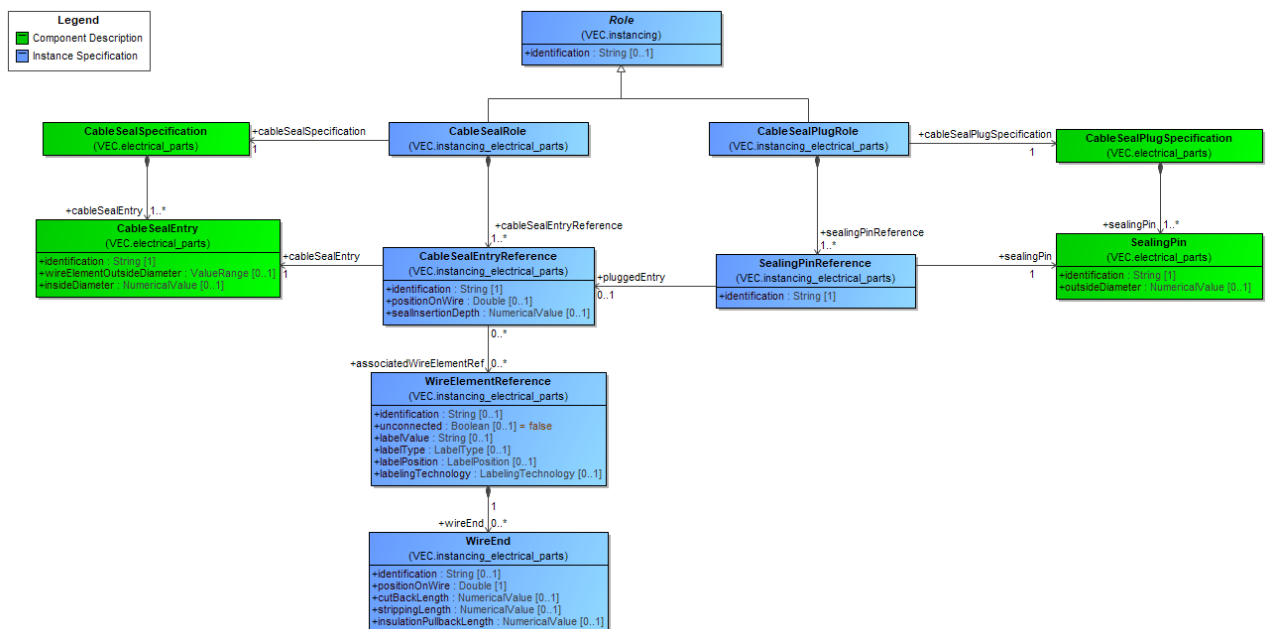


Figure 60: Instances of Cable Seals and Plugs

The diagram shows the mapping between the part master data of cable seals and cable plugs (partly displayed on the left side) and the instance specific information (displayed on the right side).

1.7.7 Instances of EE-Components

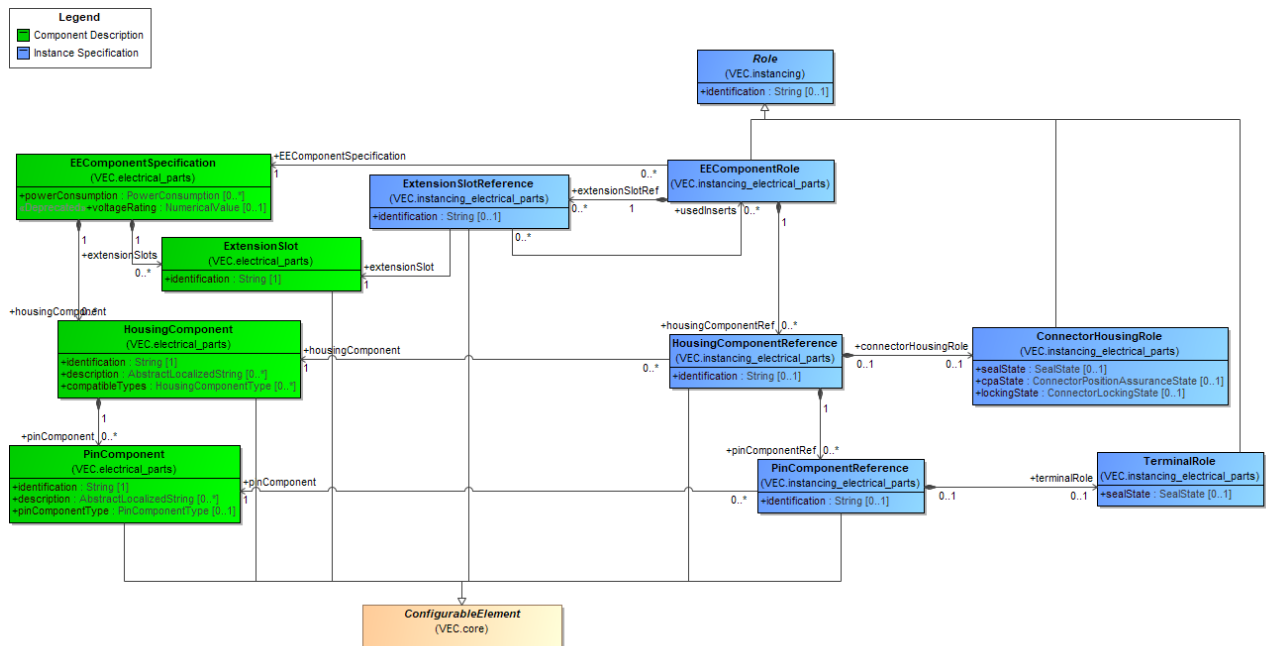


Figure 61: Instances of EE-Components

The diagram shows the mapping between the part master data of an EE-component (partly displayed on the left side) and the instance specific information (displayed on the right side). Additionally, an *EEComponentRole* can reference one or more *ConnectorHousingRoles*, *AbstractSlotRoles* or *CavityReferences* onto which it is mounted.

1.7.8 Instances of EE-Components subclasses

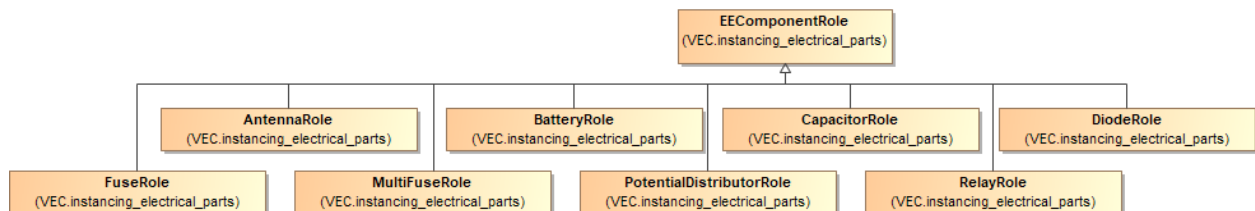


Figure 62: Instances of EE-Components subclasses

In order to define additional aspects of an EE-component instance there are some subclasses of the *EEComponentRole* symmetrical to *EEComponentSpecification* inheritance hierarchy. This diagram displays these subclasses.

1.7.9 Instances of Wire Protections

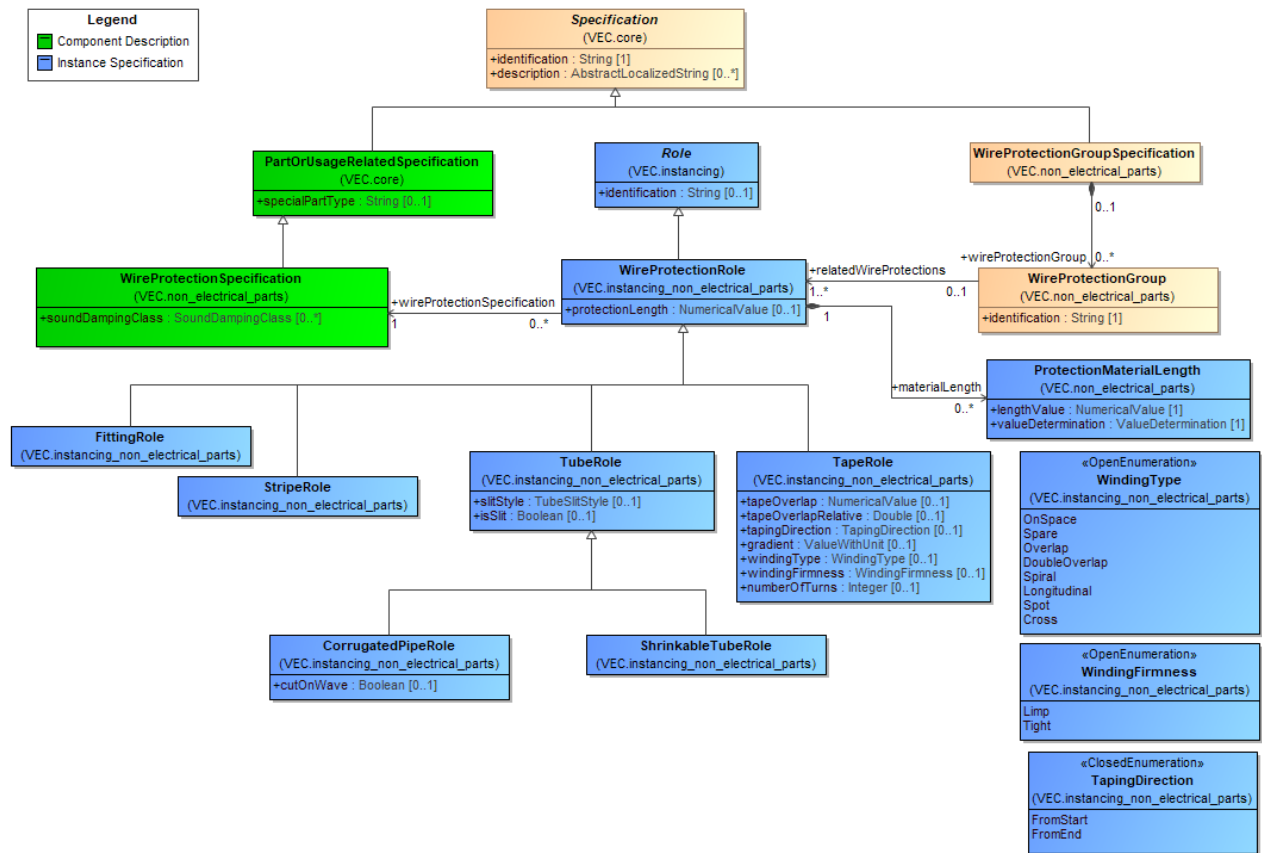


Figure 63: Instances of Wire Protections

The diagram shows the mapping between the part master data of a wire protection (partly displayed on the left side) and the instance specific information (displayed on the right side).

1.7.10 Instances of Fixings, Cable Ducts, Cable Ties and Similar

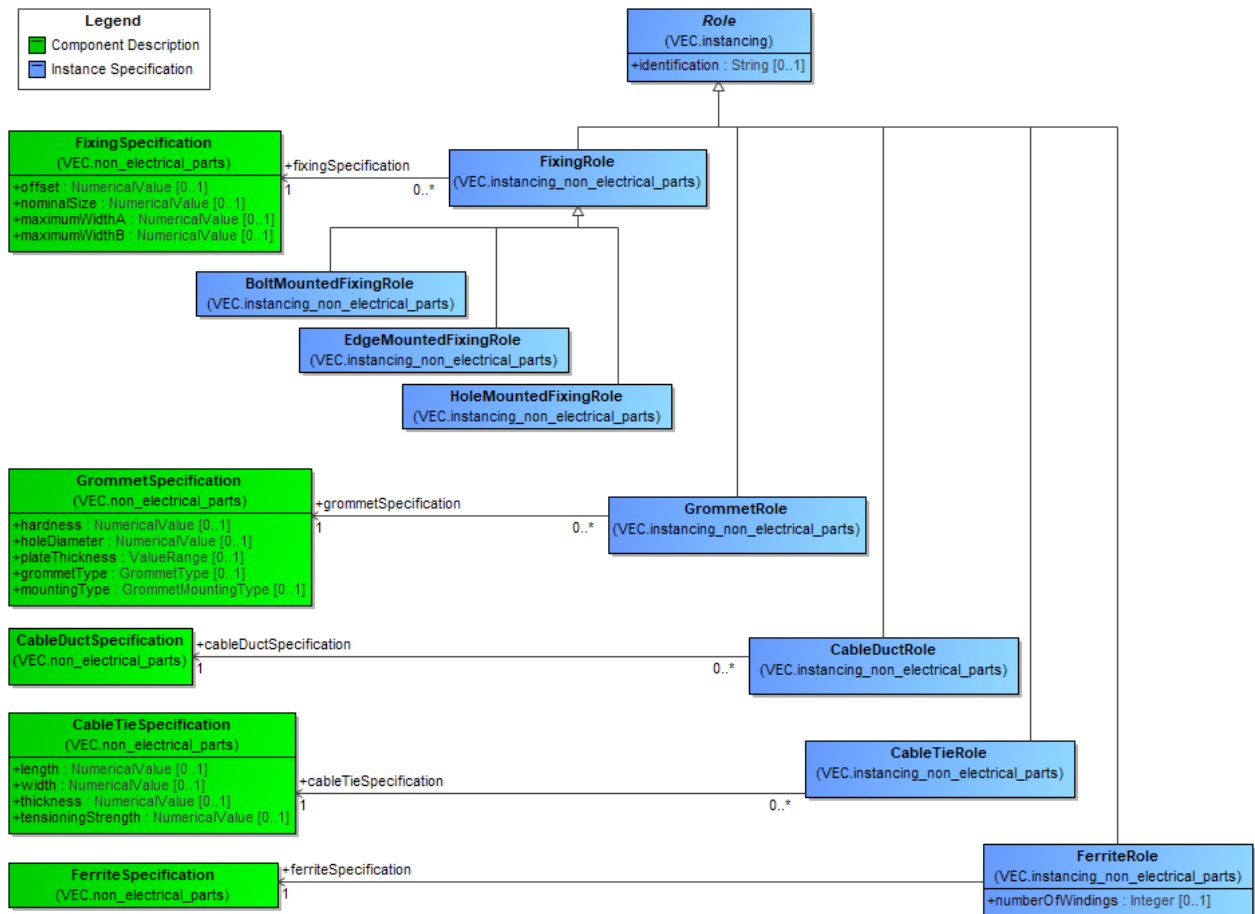


Figure 64: Instances of Fixings, Cable Ducts, Cable Ties and Similar

The diagram shows the mapping between the part master data of fixings and cable ducts (partly displayed on the left side) and the instance specific information (displayed on the right side).

1.7.11 Instances of Grommets

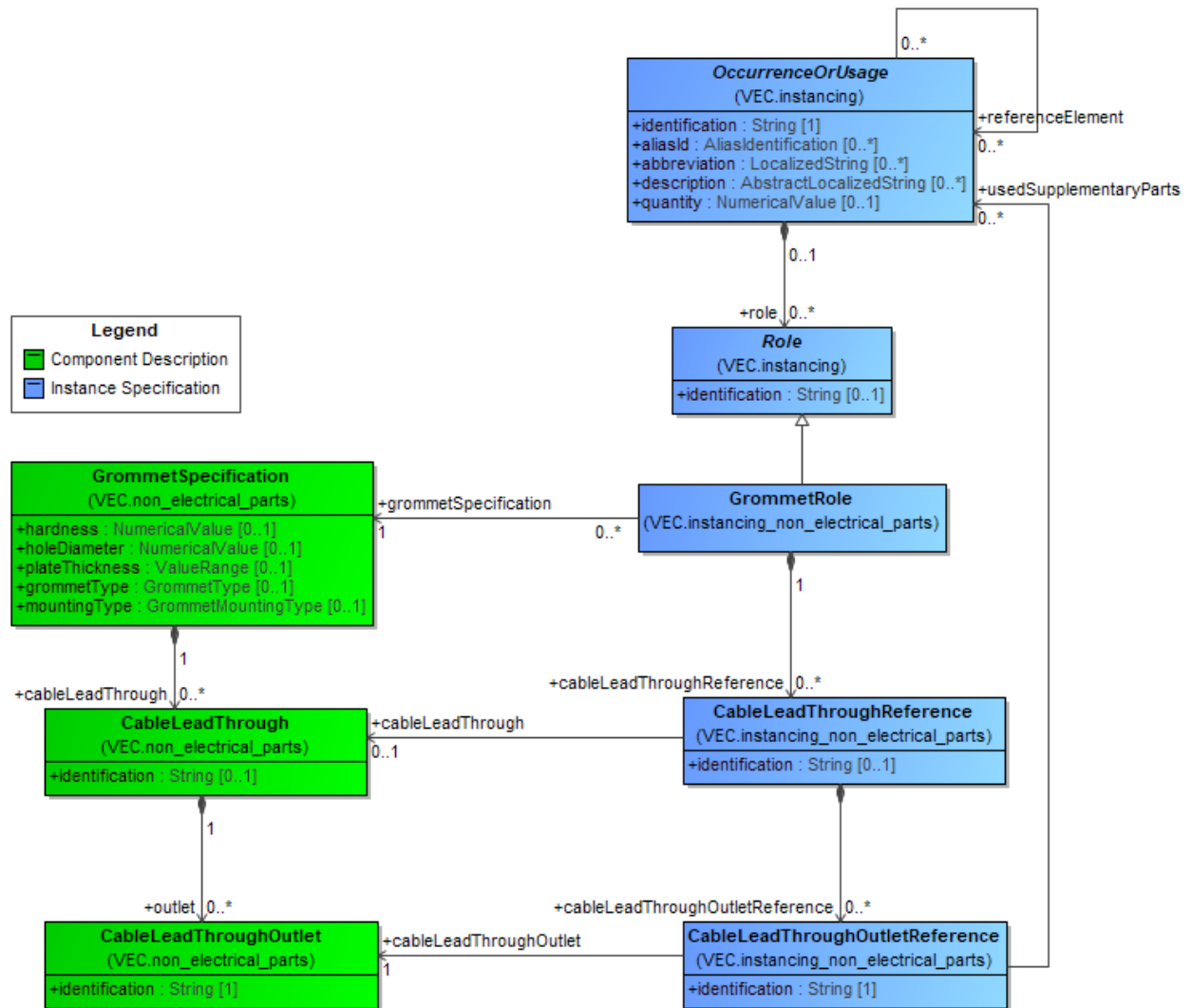


Figure 65: Instances of Grommets

1.7.12 Instances of undefined Components

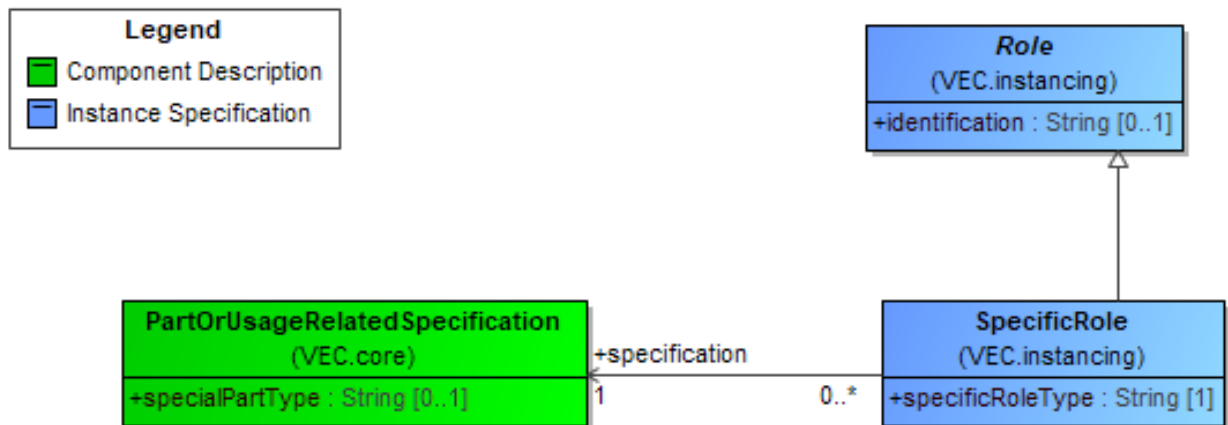


Figure 66: Instances of undefined Components

In some cases, it is necessary to specify and to use a component which is not defined in detail by the VEC. The recommended approach in such a case is to use a *PartOrUsageRelatedSpecification* with *CustomProperties* to specify such a component. When this component must be instantiated, a *SpecificRole* shall be used.

1.7.13 Instances of Placeable Components

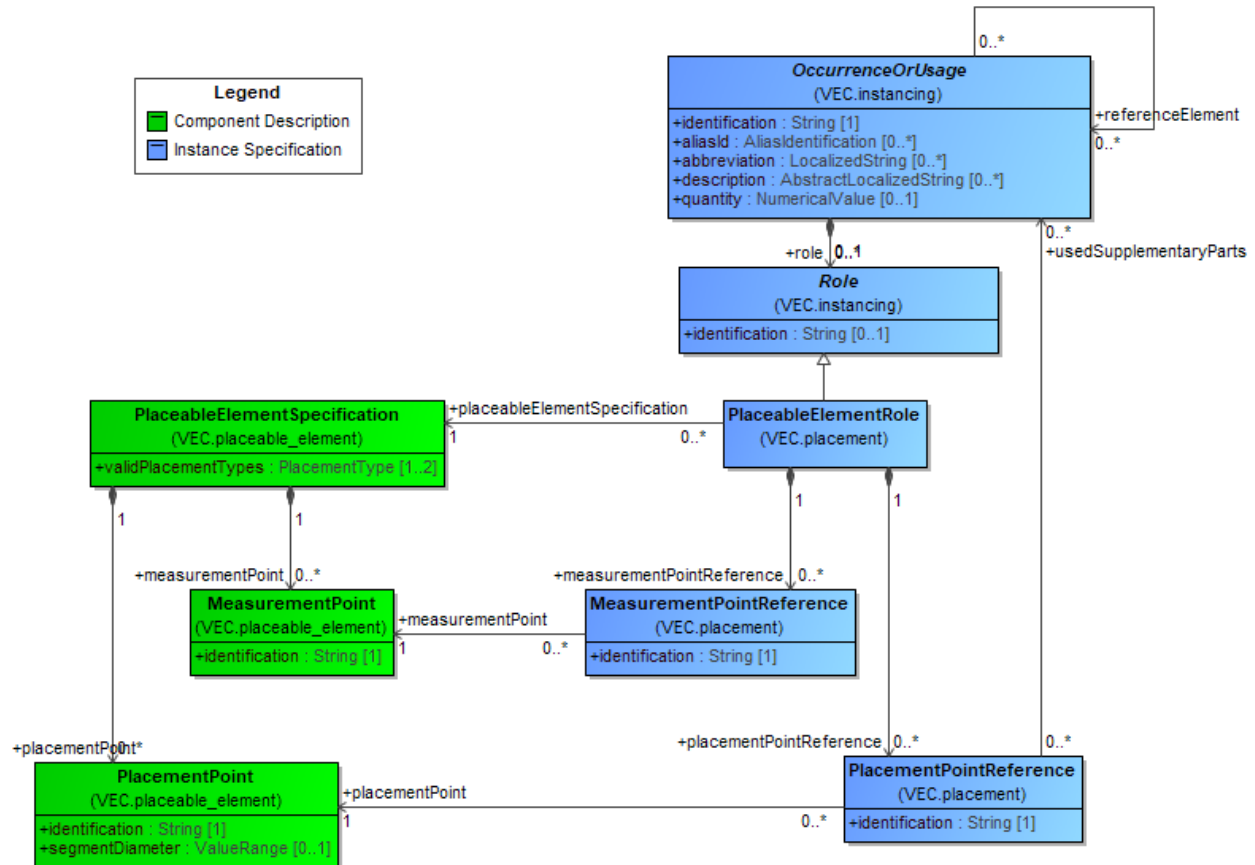


Figure 67: Instances of Placeable Components

The diagram shows the mapping between the part master data of a *PlaceableElement* (partly displayed on the left side) and the corresponding instance specific information (displayed on the right side).

1.7.14 Installation Instructions

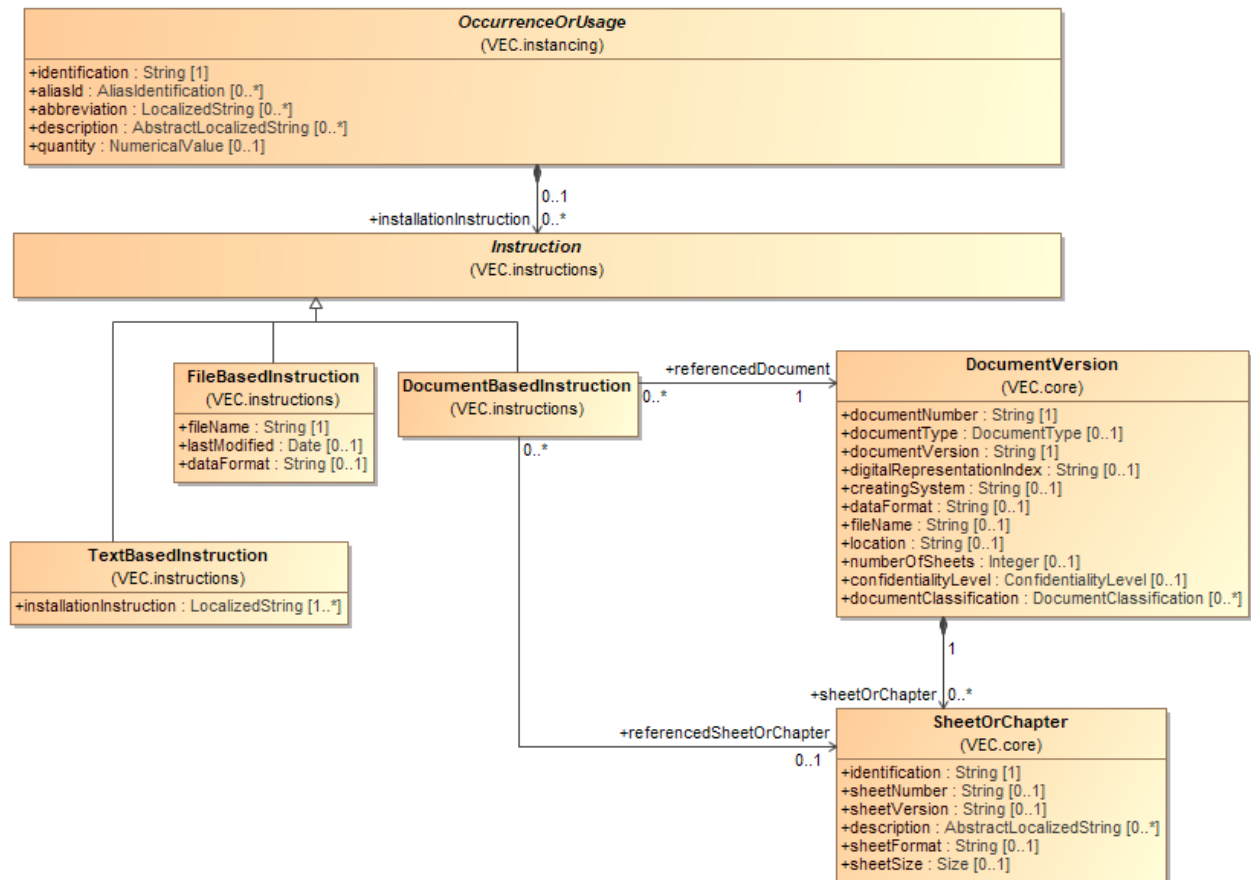


Figure 68: Installation Instructions

Basis for the definition of installation instructions is the abstract class *Instruction* which allows instructions to be specified as *TextBasedInstructions*, *FileBasedInstructions* or *DocumentBasedInstructions*.

Note: This diagram only shows the *Instruction* concept in the context of *OccurrenceOrUsage*. However, the *Instruction* concept is used in the context of *PartVersions* and several elements of a *ConnectionSpecification* as well.

Each *OccurrenceOrUsage* can define various *TextBasedInstructions* in order to exchange (normally) human readable installation instruction information.

Note: Unlike the KBL a *TextBasedInstruction* does not contain a *type* attribute. This is because a *TextBasedInstruction* is explicitly not wanted to be used as an extension mechanism. The declared extension concept of the VEC is the *CustomProperty* concept.

In addition, each *OccurrenceOrUsage* can define various *FileBasedInstructions*. This may be useful to reference for example files that contain graphical information e.g. in form of SVG.

In addition, each *OccurrenceOrUsage* can define various *DocumentBasedInstructions*. This allows to reference a *DocumentVersion* or alternatively a certain *SheetOrChapter* in a *DocumentVersion* that contains the instruction.

1.7.15 Instances of Labels

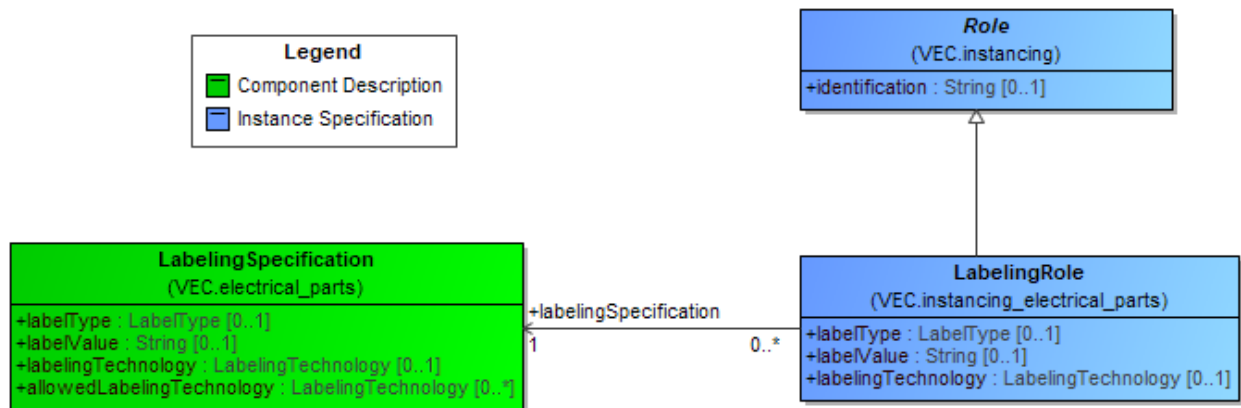


Figure 69: Instances of Labels

If a component is labelled in its concrete usage, label information is defined with the *LabelingRole*.

1.8 Composite Part Descriptions

The KBL defined a fixed hierarchy for composite parts, where different concepts covered specific aspects. These concepts were all represented in the model individually: *Harness*, *Module*, *HarnessConfiguration*, *Assembly*. From a structural point of view, all of them were quite the same, a composition of components (parts), the only real difference is the handling in process. However, the point of view might change the handling in the process.

For example, a module is assembled of components like connector housings, wires and terminals or pre-assembled parts (assembly) that are again assembled of connector housings, wires and terminals. From an abstract point of view a part regarded as a module by one process partner could be regarded as an assembly by another process partner and vice versa.

The VEC harmonizes these different concepts into one general concept for all these aspects. The following sections are describing this concept.

1.8.1 Multi-level Part Structure

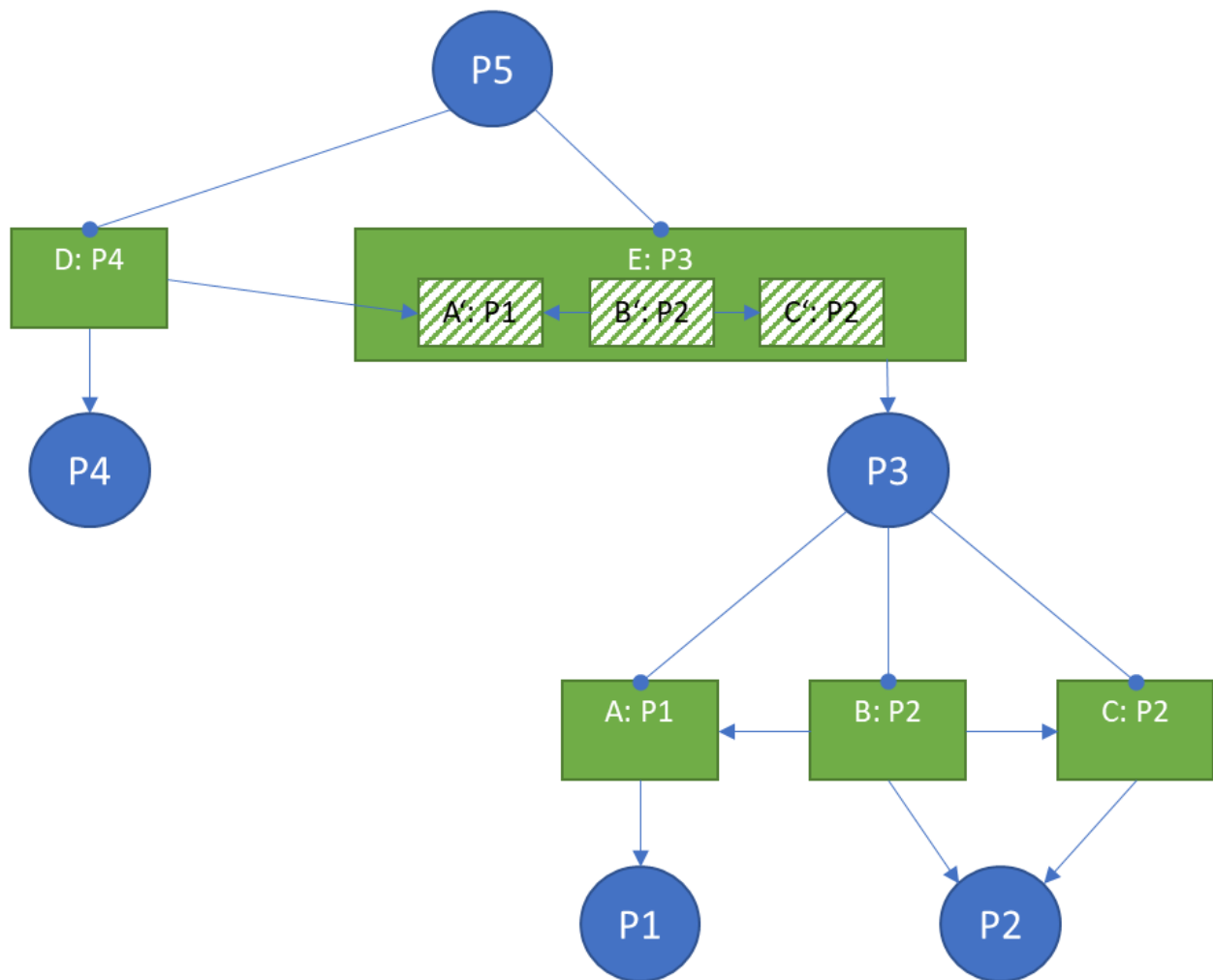


Figure 70: Multi-level Part Structure

The basic idea is the same as for *multi-level bill of materials (BOM)*, where a product is defined in a parent-child, top-down method. It lists all raw materials, semi-finished goods and sub-assemblies that are required to build the product in a hierarchical way. A component in such bill of material can consist of child components, which in-turn can have their own child components, and so on.

However, a regular bill of material just defines quantities of parts required (e.g. 4x P1, 1x P2, ...). Since the VEC not only describes the quantities of materials required to build a product, but also the way how the materials should interact (e.g. a specific terminal on a wire end in a defined cavity), the VEC requires an extended approach.

The diagram above illustrates an example for this and will be explained from bottom to top. P1 and P2 are basic components (parts with a part number). P1 could be a wire, P2 a terminal. To define some sort of composite component, the occurrences A, B and C are necessary (VEC class *OccurrenceOrUsage*). Those are required to specify the details of the composition with relationships among the occurrences and usage specific properties (e.g. wire length and contacting relationships). A composite component P3 is defined as consisting of occurrences A, B, and C.

The composite component P3 is used together with P4 to build P5. Again, occurrences are required. However, a single occurrence E for the composite part P3 is not sufficient, to specify the detailed composition of P5. Individual occurrences A', B' and C' of the sub-components of E are required for this. For example, to correctly specify the usage of a pre-assembled cable in a harness (a wire with connectors already attached to it), references to the sub-components are necessary (e.g. routing of the wire, position of connectors).

The *PartStructureSpecification* is used to define composite components based on occurrences (e.g. P3), the *PartWithSubComponentsRole* is used for the instantiation of such composite parts (e.g. E: P3). For details see the next diagram.

1.8.2 Assemblies, Modules and Harness (Configurations)

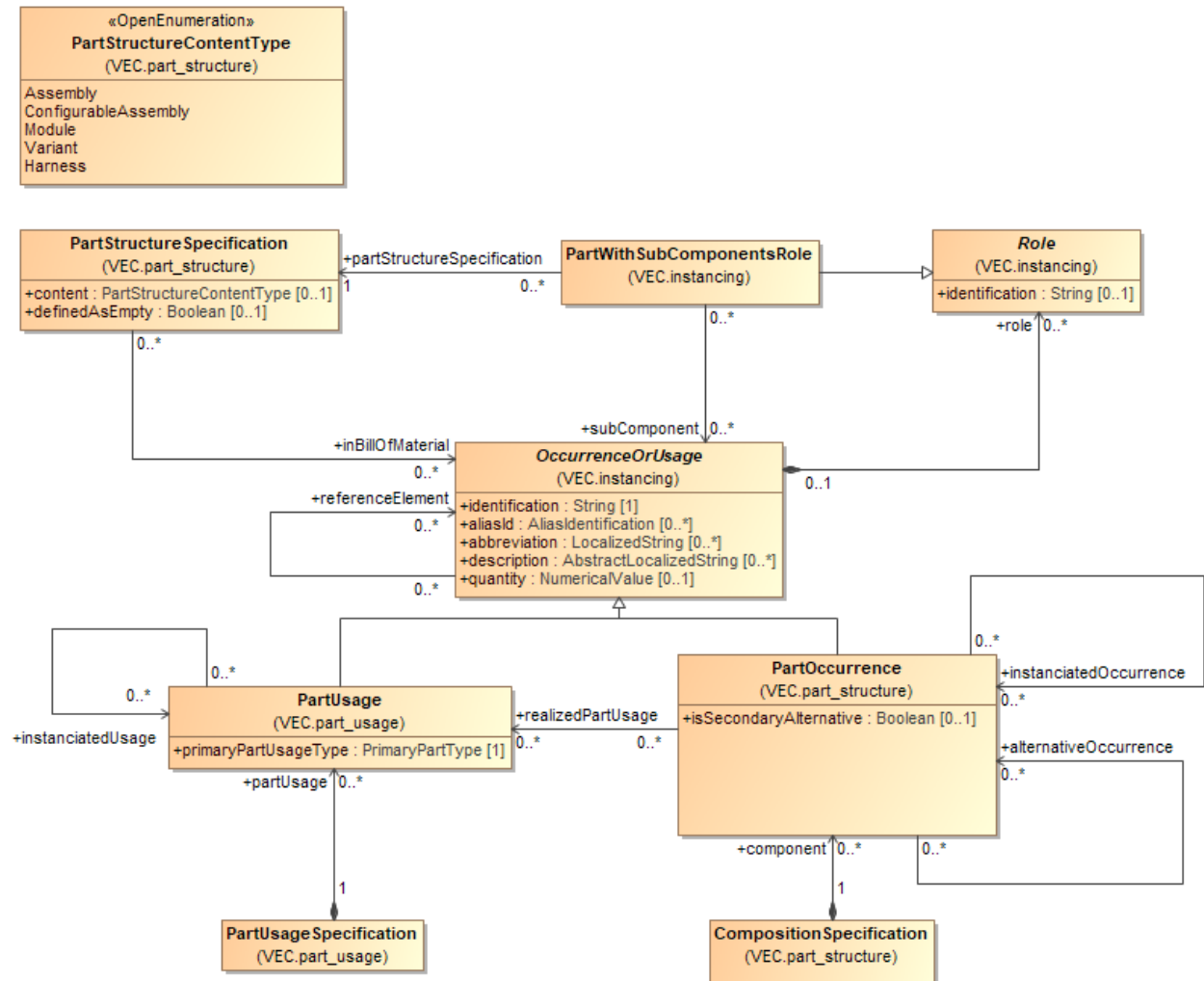


Figure 71: Assemblies, Modules and Harness (Configurations)

The VEC composite part concept has four key classes: the *PartStructureSpecification*, the *PartWithSubComponentsRole*, the *CompositionSpecification* and the *PartUsageSpecification*.

As mentioned in the example before, the *PartStructureSpecification* is used to define the bill of material for a composite part based on occurrences. Specific occurrences are used in favor of just referencing parts, to define not only quantities of material, but also the details of the composition. The *PartWithSubComponentsRole* is the corresponding instance of such a composite part.

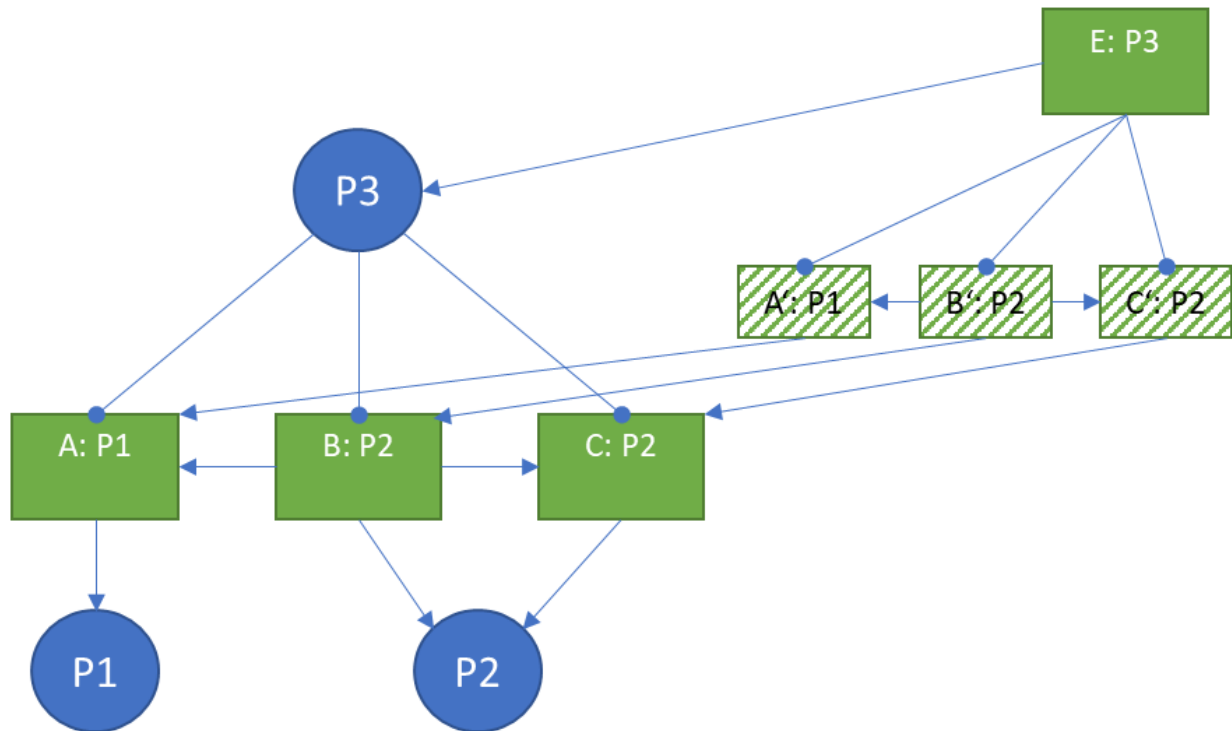
Composite parts consist of *OccurrenceOrUsages*, so they can contain *PartOccurrences* (instances of a specific *PartVersion*) as well as *PartUsages* (instances of components where a specific *PartVersion* is not yet known). From a pure bill of material point of view this seems counter intuitive, since all materials in a bill of material should be well defined. However, there are scenarios where this is necessary. For instance, if a pre-assembled part (e.g. a cable) shall be used in a certain context, it is necessary, as mentioned before, to be able to reference the subcomponents. But if the subcomponents are not known in detail (no specific part number) *PartOccurrences* could not be used. Therefore, it is necessary to allow *PartUsages* in a composite part.

The *OccurrenceOrUsages* used for a composite part are just referenced and not owned (contained) by the *PartStructureSpecification* and *PartWithSubComponentsRole*. As a result, they can be shared among them. This is especially useful in scenarios where a 150% product specification is done. Different variants of the product can reuse the same *OccurrenceOrUsages* without the need to copy them. The available *OccurrenceOrUsages* are defined by a *CompositionSpecification* (in case of *PartOccurrences*) or by a *PartUsageSpecification* (in case of *PartUsages*).

Note: A *PartVersion* without a separate *PartStructureSpecification* shall be regarded as one atomic part out of a bill of material perspective even if it is referenced by a *DocumentVersion* containing a *CompositionSpecifiation* with several occurrences.

1.8.3 Instantiation Approaches

Library Approach



"Inplace" Approach

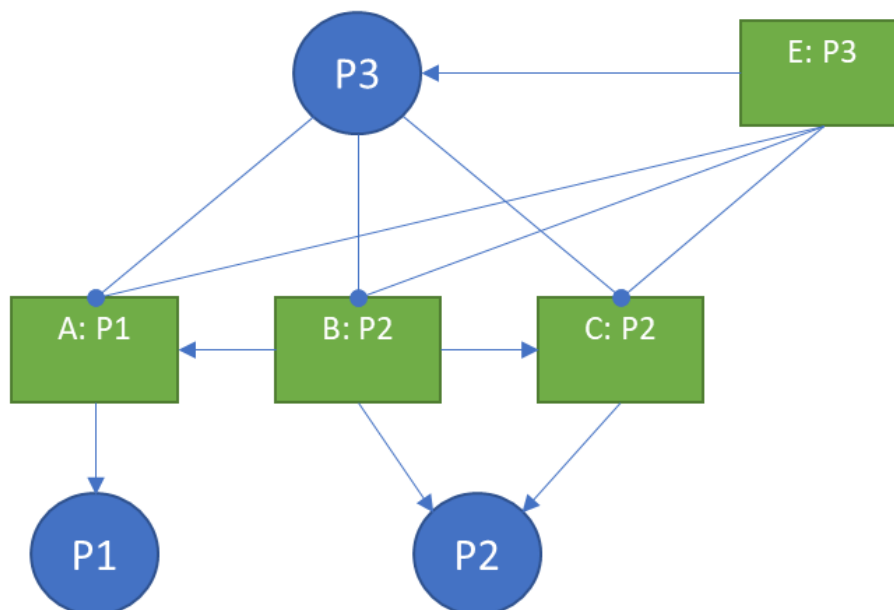


Figure 72: Instantiation Approaches

When a composite part is instantiated, an *OccurrenceOrUsage* with a *PartWithSubComponentsRole* is created and references the instantiated *OccurrenceOrUsages* that belong to the composite part instance. The instantiated occurrences are owned by a *PartUsageSpecification* / *CompositionSpecification* that is related to the context where the composite part is instantiated.

There are two approaches of how the definition and instantiation of composite parts is achieved: the library approach and the "in place" definition.

In the library approach, a composite part (e.g. a predefined USB cable) is defined in the component library and later used in different products. In this case, the part master definition of the composite part contains the *OccurrenceOrUsages* ("A: P1", "B: P2", "C: P2") that are required by the composite part and those are referenced by the *PartStructureSpecification* (P3). When the composite part is used (instantiated), the *OccurrenceOrUsages* from the part master definition are instantiated (copied) into the using context ("A: P1", "B: P2", "C: P2"). Those are then referenced by the *PartWithSubComponentsRole* (E: P3) and additional information can be attached to them. For traceability reasons, the *OccurrenceOrUsages* shall reference their origin in the part master definition via the *instantiatedUsage/instantiatedOccurrence* association.

The "in place" definition is used for example for modules of a 150% harness. In this case, the composite part is not defined in a library, but in its only and single usage. In other words, the part master definition coincides with an instantiation of part. Therefore, the *PartStructureSpecification* (P3) and the *PartWithSubComponentsRole* (E: P3) reference the same set of *OccurrenceOrUsages* ("A: P1", "B: P2", "C: P2"). In this case both relationships of the *PartWithSubComponentsRole* and a *PartStructureSpecification* to the *OccurrenceOrUsage* (inBillOfMaterial & subComponents) shall be maintained consistently.

1.8.4 Module Lists and -Families

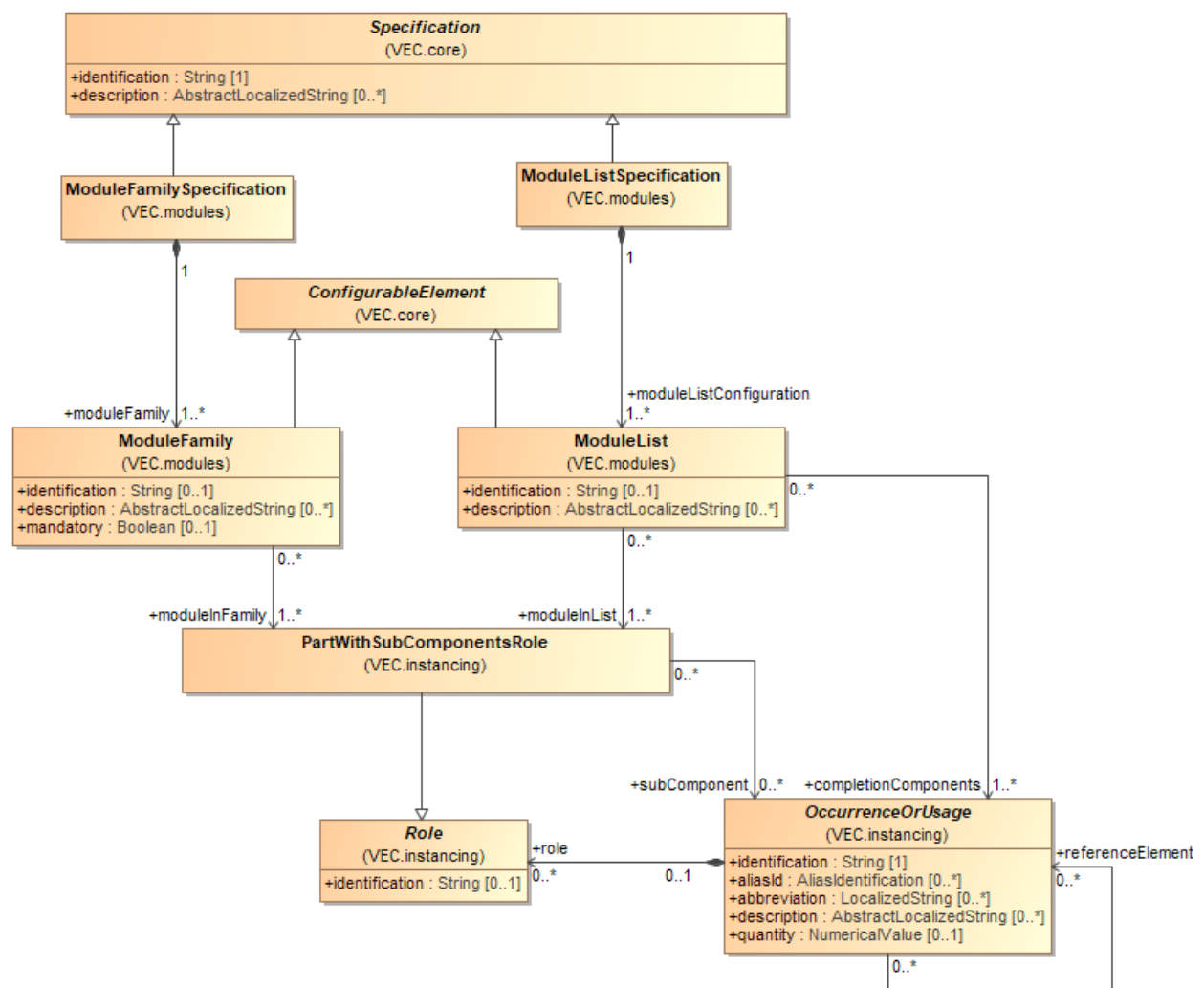


Figure 73: Module Lists and -Families

In addition to the *CompositionSpecification* and the *PartStructureSpecification*, the VEC defines two other concepts to control occurrences, especially composite parts (e.g. modules): *ModuleFamily* and *ModuleList*.

PartWithSubComponentsRoles referenced by a *ModuleFamily* respectively the *OccurrenceOrUsage* these *PartWithSubComponentsRoles* belong to are mutually exclusive.

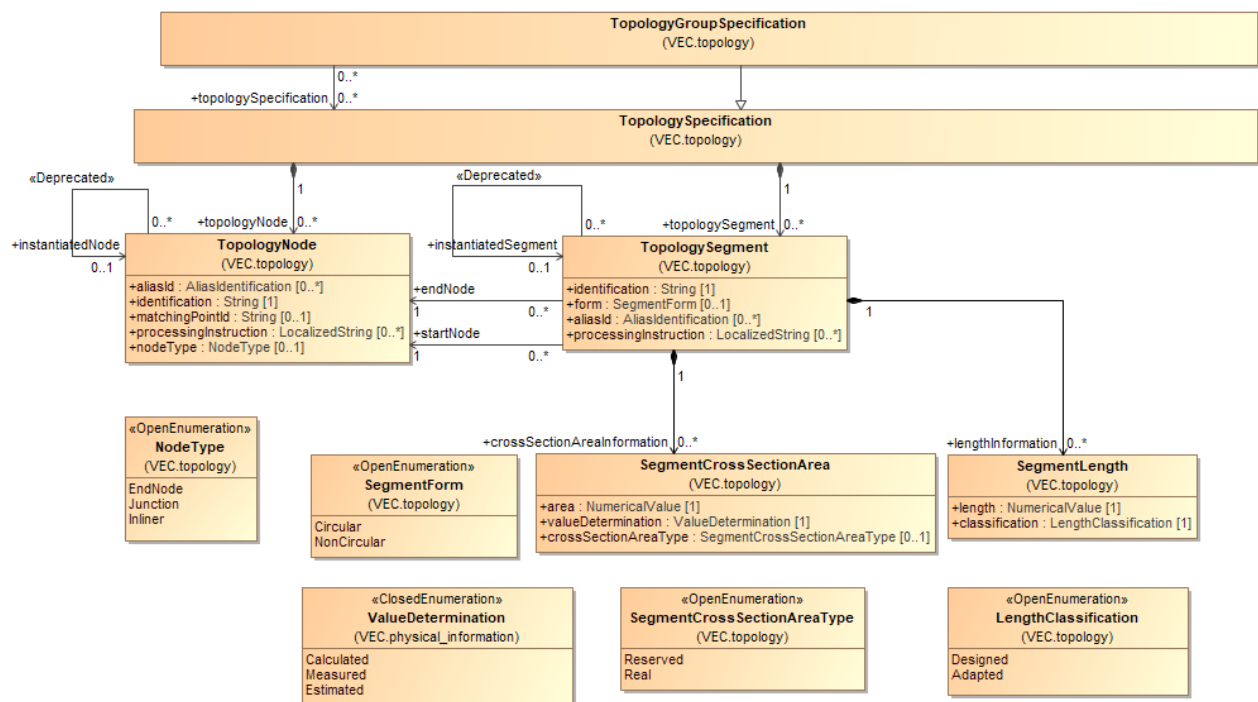
A *ModuleList* specifies a set of *PartWithSubComponentsRoles*. The definition requires that if one or more of this *PartWithSubComponentsRoles* respectively the *OccurrenceOrUsage* these *PartWithSubComponentsRoles* belong to are part of an assembly the referenced *completionComponents* have to be added.

1.9 Topology and Geometry

This section covers the general definition of a topology, the definition geometric views of that topology, placement of *OccurrenceOrUsages* on a topology and other geometric aspects like dimensioning.

All geometric coordinate systems used in the VEC are right-handed (Right-hand rule). This also means, the direction of the z-Axis is in the direction of the cross product (X-Axis x Y-Axis).

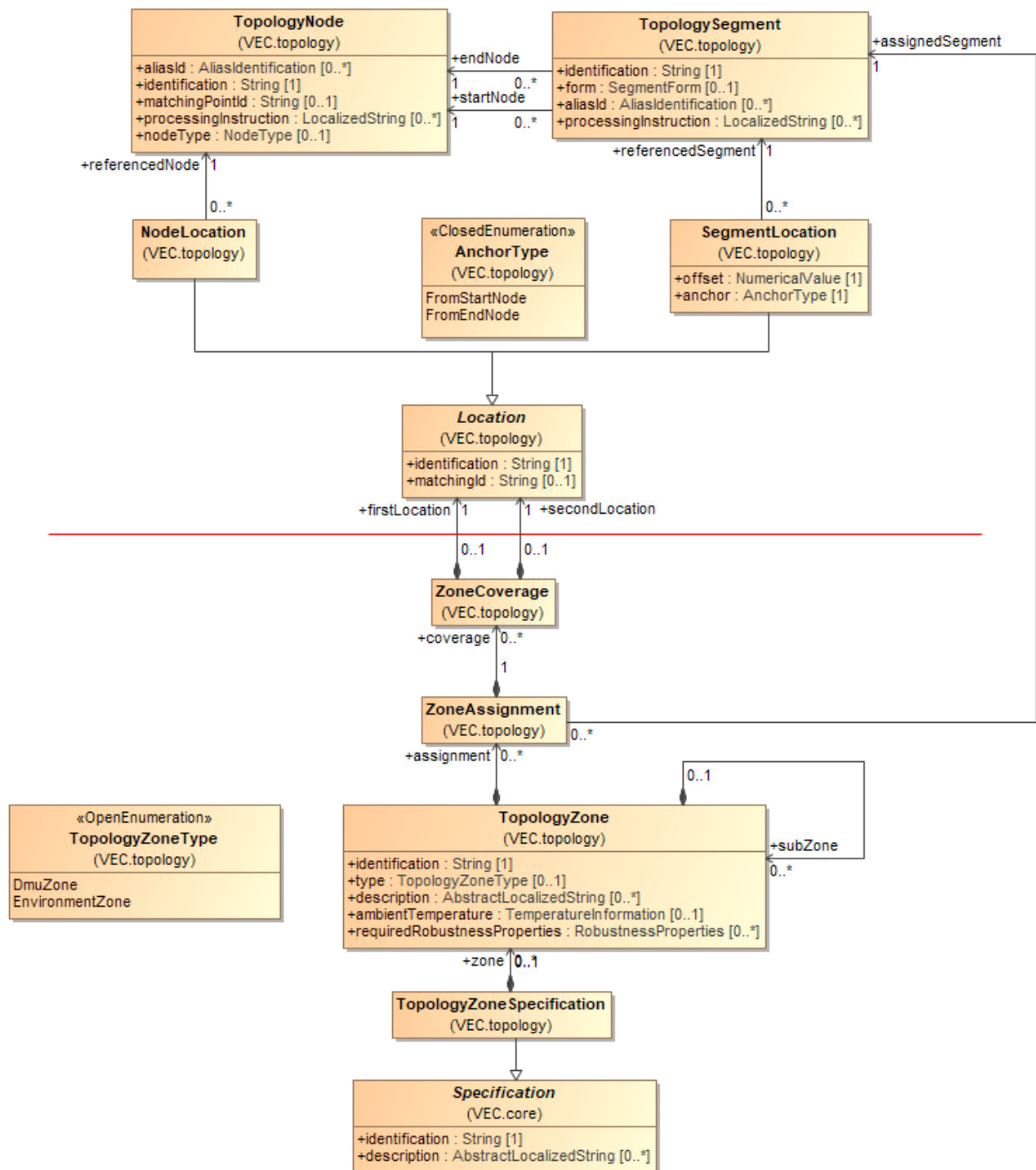
1.9.1 Topology- and Topology Group Specification

**Figure 74: Topology- and Topology Group Specification**

A *TopologySpecification* is a container for various *TopologyNodes*, *TopologySegments* and *Zones*. A *TopologyNode* represents an abstract place (meaning without concrete coordinates) in the wiring harness that can be typed either as *EndNode*, *JunctionPoint* or *Inliner*.

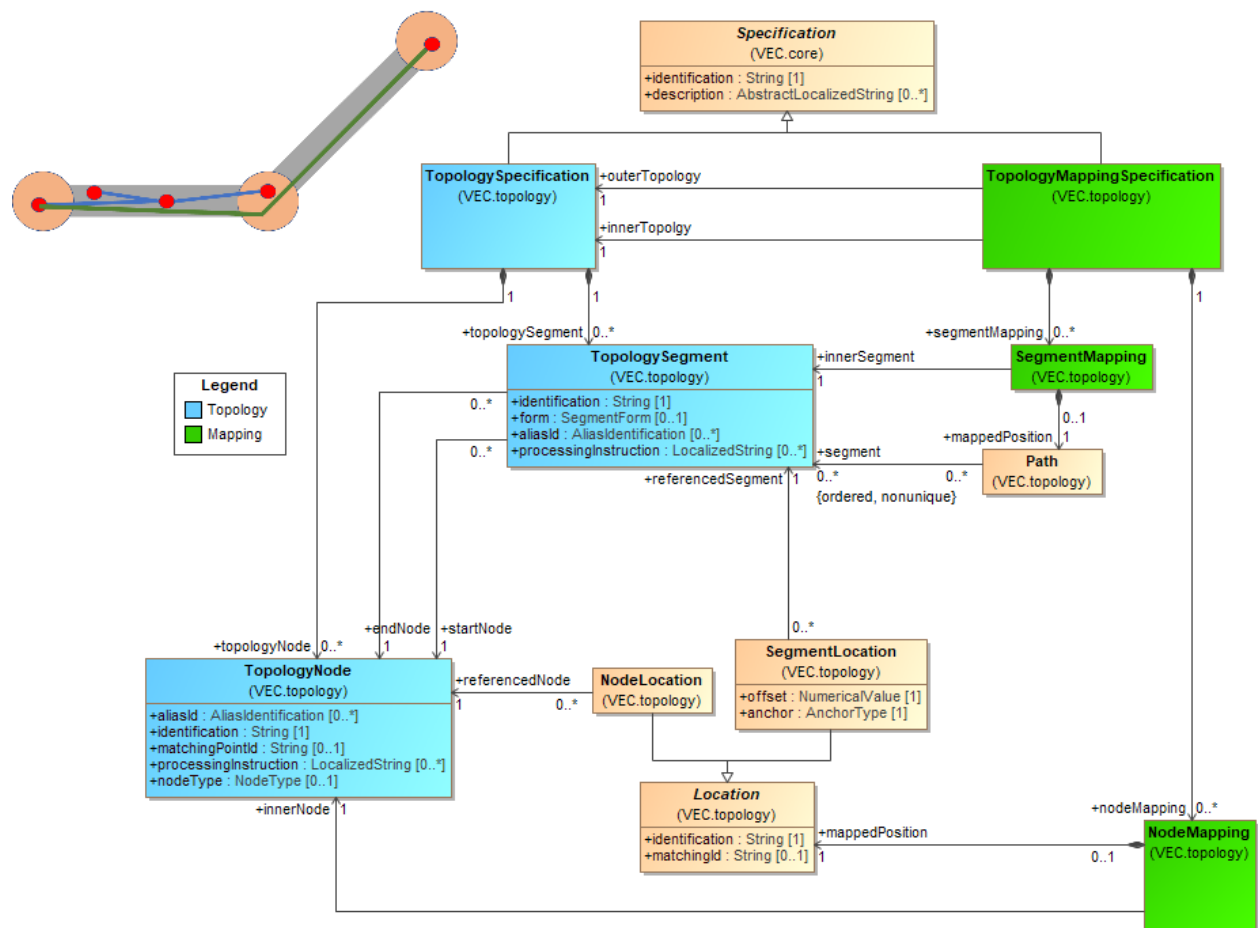
A *TopologySegment* represents a section of a wiring harness where no intermediate electrical contacts appear. In consequence, at the beginning and at the end of a *TopologySegment* the same wires go in and out. The beginning and the end are each defined by a dedicated reference to a *TopologyNode*. A *TopologySegment* can specify various *SegmentCrossSectionAreas*. These *SegmentCrossSectionAreas* are not supposed to express configuration-dependent values but different views (*Calculated*, *Measured* or *Estimated*) on the same type-dependent value whereupon the default-type is “reserved design space”. The same applies to *SegmentLength*.

In some cases, a topology can be an instance of another topology. If a more complex assembly or small harness (e.g. like a door) is defined in some place (e.g. in a component library) and then reused elsewhere it might be necessary to instantiate the topology in the target. To keep traceability in these cases, the *TopologySegment* and the *TopologyNode* have an *instantiated...* association to their source element.



To define the *TopologyZone* unambiguously an assignment of the topology elements to their zones is necessary. *TopologyNodes* are only connecting points for *TopologySegments* or for the placement of components and do not have any own extent in space. Therefore, *ZoneAssignments* are only possible for *TopologySegments*.

1.9.3 Hierarchical Topologies



Hierarchical topologies are a versatile concept to create harness topologies that have an advanced structural design. For historical reasons, the semantics defined by the VEC for topologies are quite strict and simple, yet sufficient for most cases of traditional wiring harnesses. However, with the advent of modern wiring harness requirements, more advanced concepts are needed for certain applications.

With the integration of new technologies (e.g. high voltage, high data rate bus systems) and the improvements of design processes (e.g. integrated 3D design process with traceability and round-trip capabilities) more complex designs must be expressed in a concise, digital evaluable way. Examples for such use cases are:

- Structured harness bundles (e.g. specific wires in a bundle taped together).
- Segments of splices folded into a main bundle in a specific way.
- Reuse of shared parts (e.g. assemblies, right / left back door)

- Refinement of a topology in the design process (e.g. splice positioning after the "geometry design").

To address these requirements, the VEC introduced the concept of hierarchical topologies. In the top left corner of the diagram a simplified illustration shall serve as small example to make the concept more understandable.

A *TopologyMappingSpecification* is used to associate an "inner" topology with an outer topology. In the illustration the "outer" topology is displayed with grey segments and orange nodes. The two "inner" topologies are displayed with blue & green segments and red nodes.

A *NodeMapping* is used to define how the nodes of the inner topology relate to the outer topology. An inner node can be either placed exactly on / in an outer node or somewhere within a segment. For the definition of the position of the inner node, the concept of locations is reused.

A *SegmentMapping* is used to define how the segments of the inner topology relate to the outer topology. An inner segment can run through multiple segments or to stay in a single one. In any case, all outer *TopologySegments* to which it relates are referenced by a *Path*.

Some additional restrictions apply on the mapping:

- The representation of the inner / outer relationship with a *TopologyMappingSpecification* allows the creation of n:m relationships between topologies. However, it is forbidden to create "circular" mappings. If n:m mappings are created, it must be ensured that those mappings shall not exist at same time, or in other words they must be the result of product variance or reuse in different variants.
- "Cross Topology assignment of Components" is forbidden. That means wires are routed strictly in the elements of a single *TopologySpecification* and other components are placed strictly on a single *TopologySpecification*.
- If a *TopologySpecification* is mapped as an inner topology, all its elements shall be mapped.
- A creating system is responsible that the constraints of the topologies are maintained (e.g. that length constraints between inner and outer topologies are satisfied).

Note: This concept was introduced newly with VEC Version 1.2.0. As there are currently no known systems that implement such a concept further detailing of its usage will be done in the Implementation Guidelines accompanying implementations as they occur.

1.9.4 Bending Restrictions

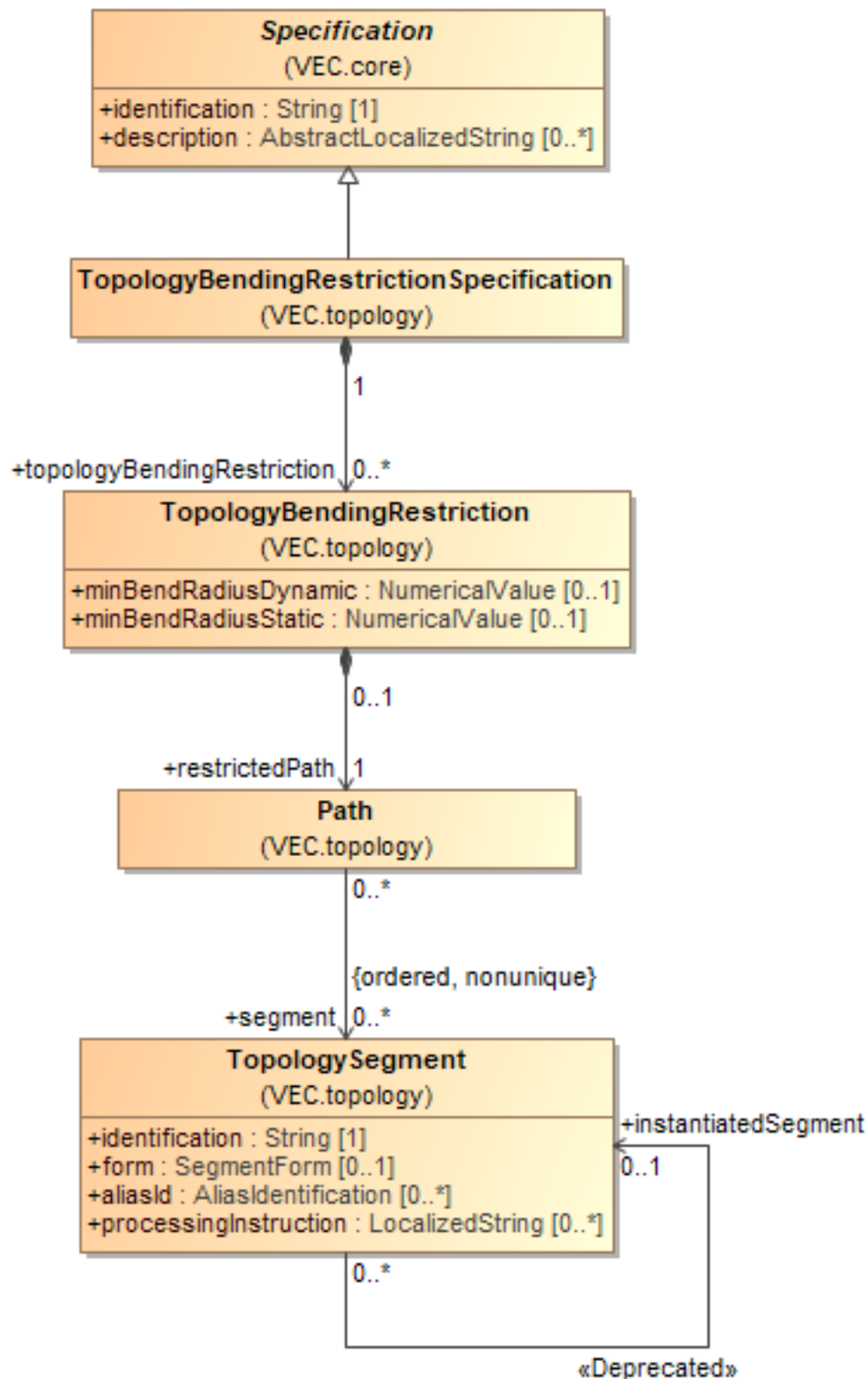


Figure 77: Bending Restrictions

A bending restriction defines restrictions on the bendability of a path in the topology. See *TopologyBendingRestriction* for a detailed description.

Each *GeometrySegment2D* can specify an ordered set of *PathSegments* each defining a certain section of its graphical representation. Moreover, it can specify which *TopologySegment* it represents.

Note: The data model does not constrain a *GeometryNode2D* respectively a *GeometrySegment2D* to reference a *TopologyNode* respectively a *TopologySegment*. This allows a VEC e.g. to transport graphical 2D-information for its own. However, for the description of lengths and dimensions as well as segment cross-section-areas and placements the VEC requires adequate references to the relevant elements in the topology.

Each *OccurrenceOrUsageViewItem2D* can reference which *OccurrenceOrUsages* it represents. Furthermore, it can specify a *Transformation2D* which determines its graphical position and orientation within the *BuildingBlockSpecification2D*.

1.9.6 3D-Geometry

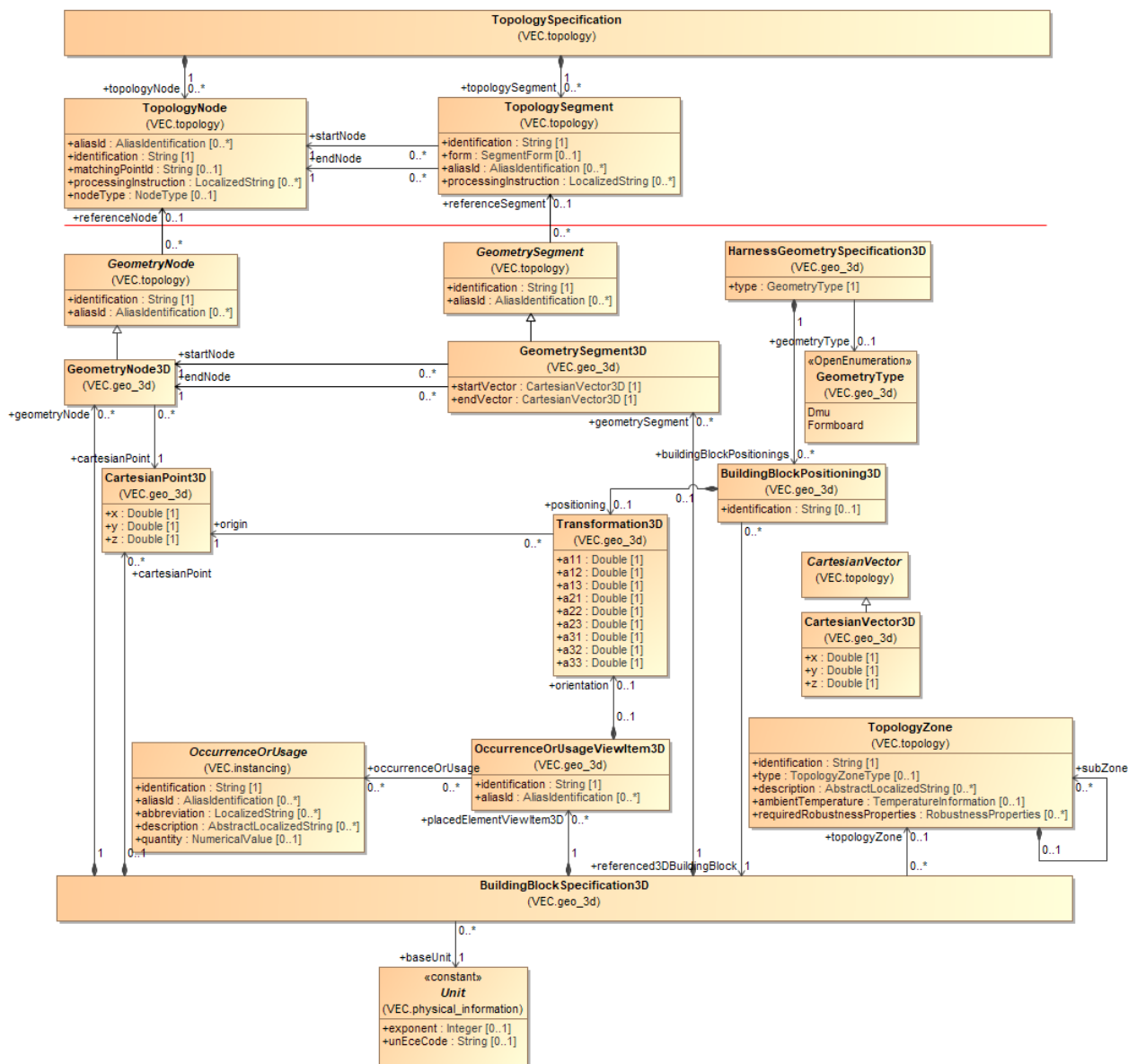


Figure 79: 3D-Geometry

A *HarnessGeometrySpecification3D* represents either a 3D-model in the car coordinate system or a 3D-model in the formboard system. Each *HarnessGeometrySpecification3D* is a container for various instances of the class *BuildingBlockPositioning3D*. Each *BuildingBlockPositioning3D* determines the positioning of a specific *BuildingBlockSpecification3D* in the context of the *HarnessGeometrySpecification3D* by referencing a

BuildingBlockSpecification3D and optionally specifying a *Transformation3D* for the case that a coordinate transformation is required.

A *BuildingBlockSpecification3D* represents a specific geometrical subset normally belonging to a certain *Zone*. Each *BuildingBlockSpecification3D* is a container for various instances of the classes *GeometryNode3D*, *GeometrySegment3D*, *CartesianPoint3D* and *OccurrenceOrUsageViewItem3D*.

Each *GeometryNode3D* references a *CartesianPoint3D* which determines its position within the *BuildingBlockSpecification3D*. Moreover, it can specify which *TopologyNode* it represents.

Each *GeometrySegment3D* can specify an ordered set of *Curves* in order to approximate its geometry (explained in detail on the next diagram). Moreover, it can specify which *TopologySegment* it represents.

Note: The data model does not constrain a *GeometryNode3D* respectively a *GeometrySegment3D* to reference a *TopologyNode* respectively a *TopologySegment*. This allows a VEC e.g. to transport 3D-information independently from a *TopologySpecification*. However, for the description of dimensions including tolerances as well as segment cross-section-areas and placements especially *OnWayPlacements* the VEC requires adequate references to the relevant elements in the topology.

Each *OccurrenceOrUsageViewItem3D* can reference which *OccurrenceOrUsages* it represents. Furthermore, it can specify a *Transformation3D* which determines its position and orientation of the component's 3D model within the space of the *BuildingBlockSpecification3D*. If no *Transformation3D* is defined, the *OccurrenceOrUsageViewItem3D* is meant to be placed implicitly by its position in the topology (e.g. wire protections with an *OnWayPlacement*).

1.9.7 3D Curves

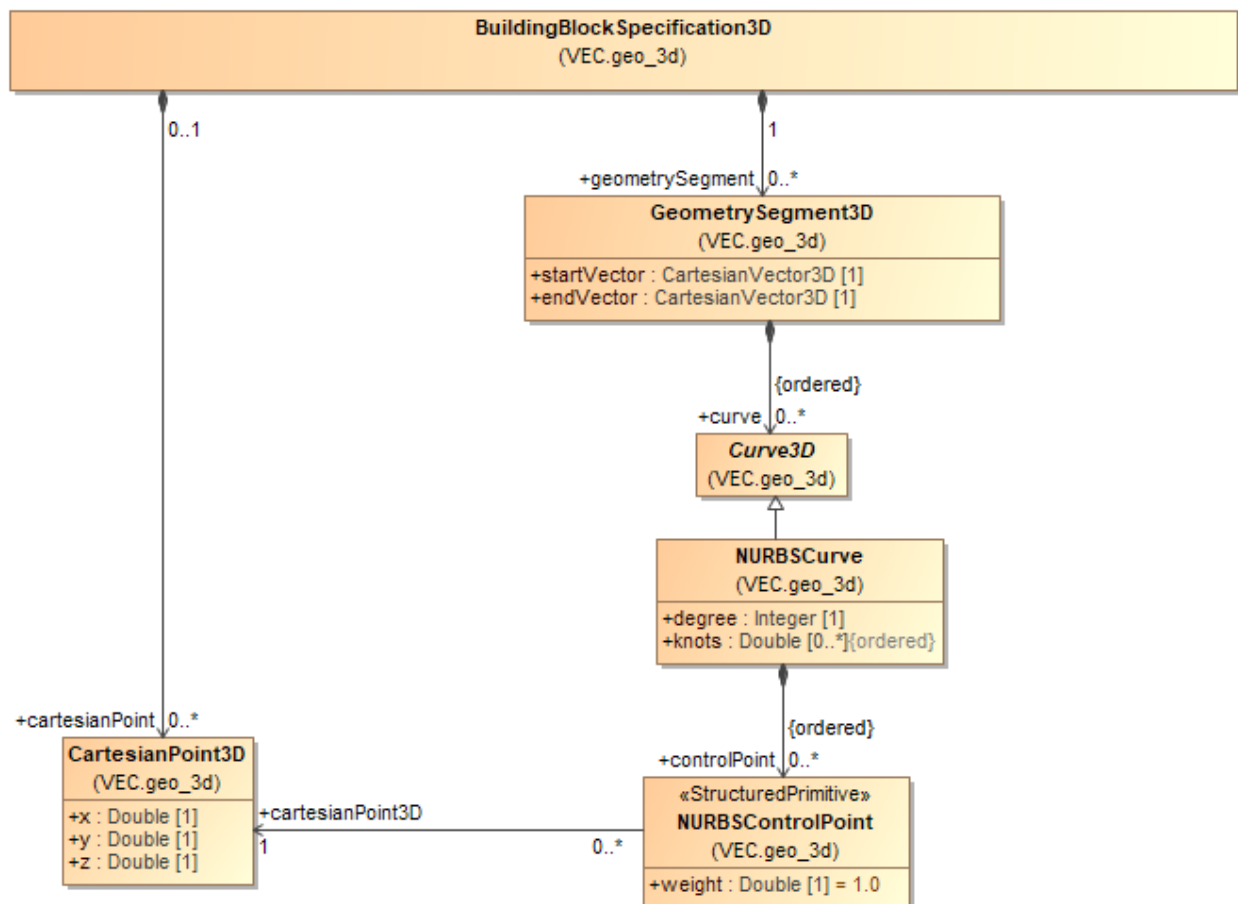


Figure 80: 3D Curves

The center line of a *GeometrySegment3D* is defined by one or more *Curve3D*. This abstract class is designed to be an extension point for different curve implementations. At the moment the only curve implementation

that is supported by the VEC are *NURBS*Curves. The details are described in the documentation of the individual classes.

1.9.8 Locations

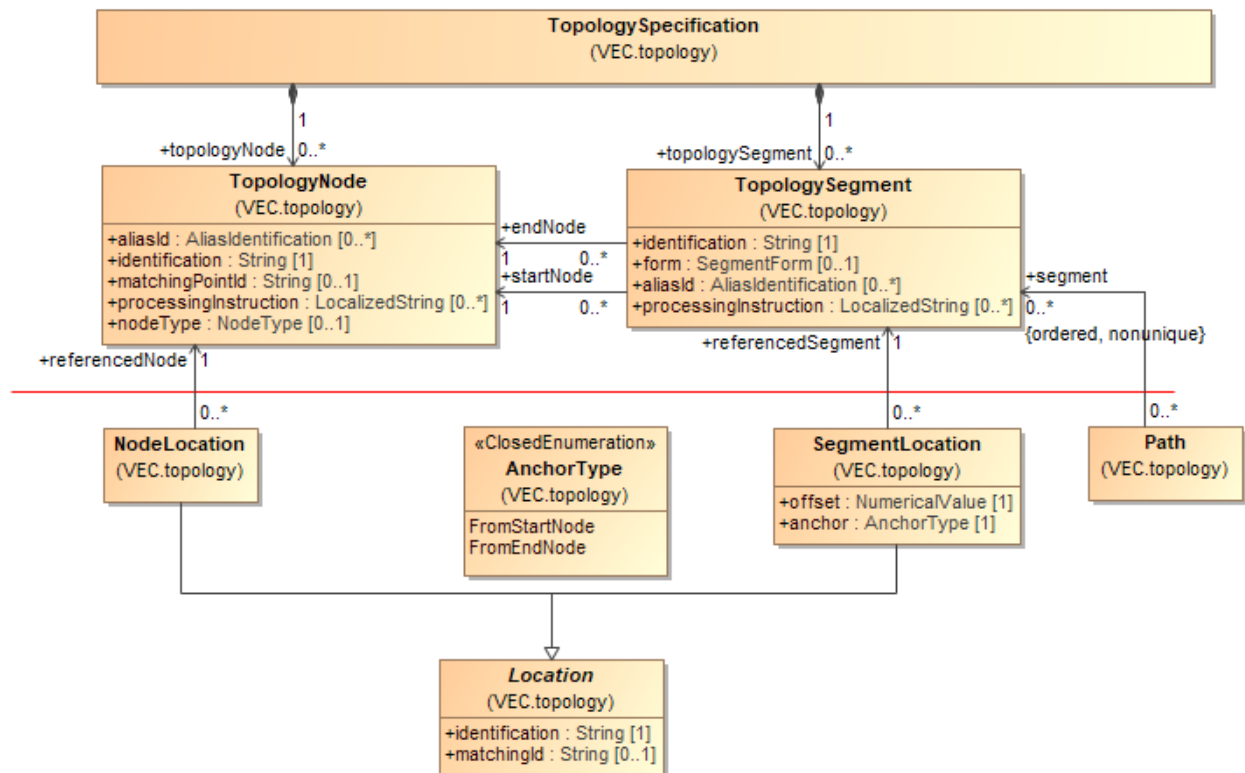


Figure 81: Locations

In order to describe a wiring harness, the abstract information of a topology must be enhanced with information about the concrete positions of components in the topology. For purposes of clarity the explanation of this concept is split into two diagrams. This diagram explains how *Locations* are defined onto the *Topology*. The next diagram shows how these locations can be used to define *Placements* of components and *Dimensions*.

The VEC defines two types of *Locations*: *NodeLocations* and *SegmentLocations*. A *NodeLocation* references one *TopologyNode*. A *SegmentLocation* defines a single point on a *TopologySegment* located between its *startNode* and *endNode*. To achieve this, a *SegmentLocation* references the relevant *TopologySegment* and specifies as anchor the start- or end node as well as the offset from that anchor.

Note: In contrast to KBL 2.3 and GEO the VEC locations on segments are expressed by anchor and offset and not through relative measurements. This enables ease of handling in case of segment-length changes.

1.9.9 Placement and Dimensions

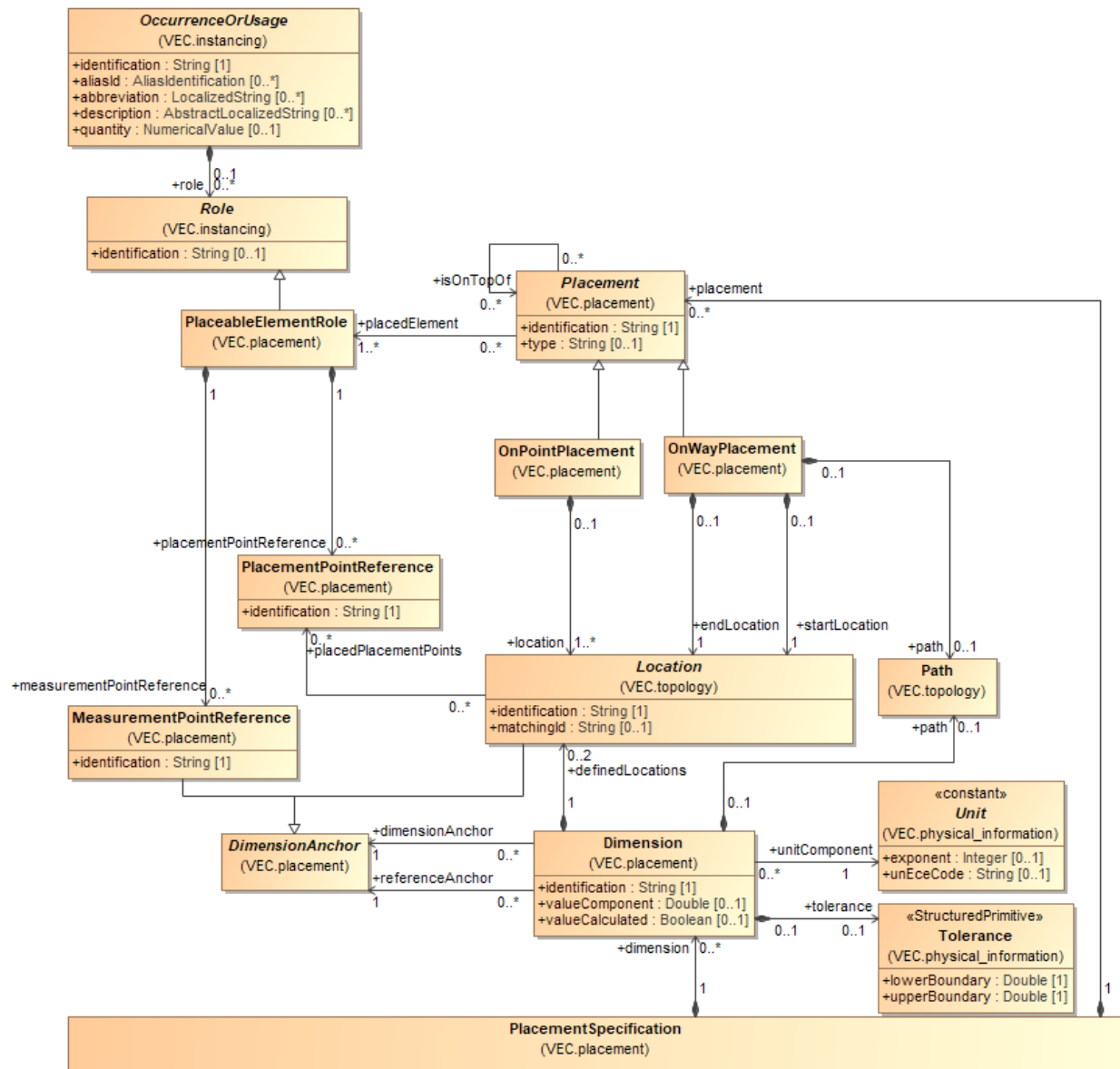


Figure 82: Placement and Dimensions

A *PlacementSpecification* is used to define the *Placement* of *OccurrenceOrUsages* in a topology as physical elements that build this topology and to define *Dimensions* between these elements. Each *PlacementSpecification* is a container for various *Placements* and *Dimensions*.

Each *OccurrenceOrUsage* that is placeable in the topology defines a *PlaceableElementRole*. A *PlaceableElementRole* can define various *PlacementPointReferences* and *MeasurementPointReferences*, which are instances of the corresponding elements *PlacementPoint* and *MeasurementPoint* defined in the part master data of the component. The position in the topology of a *PlaceableElementRole* is defined with a *Placement*. The multiplicity of this association is caused by the support of variance in the VEC. For a concrete configuration, there should be only one *Placement* for one *PlaceableElementRole*.

The VEC specifies two individual types of *Placements*: *OnPointPlacements* and *OnWayPlacements*.

Each *OnPointPlacement* defines one or more *Locations* in order to determine the positioning of the *placedElement*. More than one *Location* is necessary to specify the exact *Placement* of *placedElements* with more than one *PlacementPointReferences*, e.g. the different entries in case of a cable duct. Each *Location* can therefore reference a *PlacementPointReference*.

An *OnWayPlacement* references one *Location* as *startLocation* and another *Location* as *endLocation* in order to determine the positioning of *placedElements* like tapes that cover a way in the topology. The positioning can be described even more precise by using the *OnWayPlacement* with a *Path* through the topology.

Both *OnPointPlacements* and *OnWayPlacements* reference one or more *PlaceableElementRoles* as *placedElements*. The possibility to reference more than just one *PlaceableElementRole* permits to express that configuration dependent the one or other *PlaceableElementRole* is placed. Finally, both *OnPointPlacements* and *OnWayPlacements* can reference other *Placements* on which they are placed on top of in order to determine the arrangement of layering (e.g. a tape around a tube).

Note: When using the *isOnTopOf*-relationship all other underlying *Placements* shall be referenced on which the referencing *Placement* is on top of and not only the direct neighbor. The reason for this is that *Placements* are *ConfigurableElements* and so are potentially not present in all configurations.

Since the VEC can be used to exchange a product definition, it must be possible to define *Dimensions* with tolerances between the elements in addition to its exact positions in the topology.

Each *Dimension* references two *DimensionAnchor*, one as *referenceAnchor* and one as *dimensionAnchor* and defines the distance in between as *dimensionValue*, optionally together with a tolerance (included in the definition of *NumericalValue*). In case of ambiguity regarding the path between the two *Locations*, the *Dimension* shall specify the relevant *Path*.

A *DimensionAnchor* can be either a *MeasurementPointReference* or a *Location*. A *MeasurementPointReference* is used when the dimension refers to a specific point on the component that is not part of the topology, such as a bolt on a cable duct or a seam on a grommet. If a *Location* is used, it can either be a *Location* defined by a *Placement* (when referencing the position of a placed component, such as the entry point of a connector or the start of a wire protection) or a *Location* defined directly by the *Dimension* itself. The latter is used when the dimension references a point in the topology where no physical component is placed, for example to specify a tolerance over a section of the topology, such as across multiple segments.

1.9.10 Default Dimensions

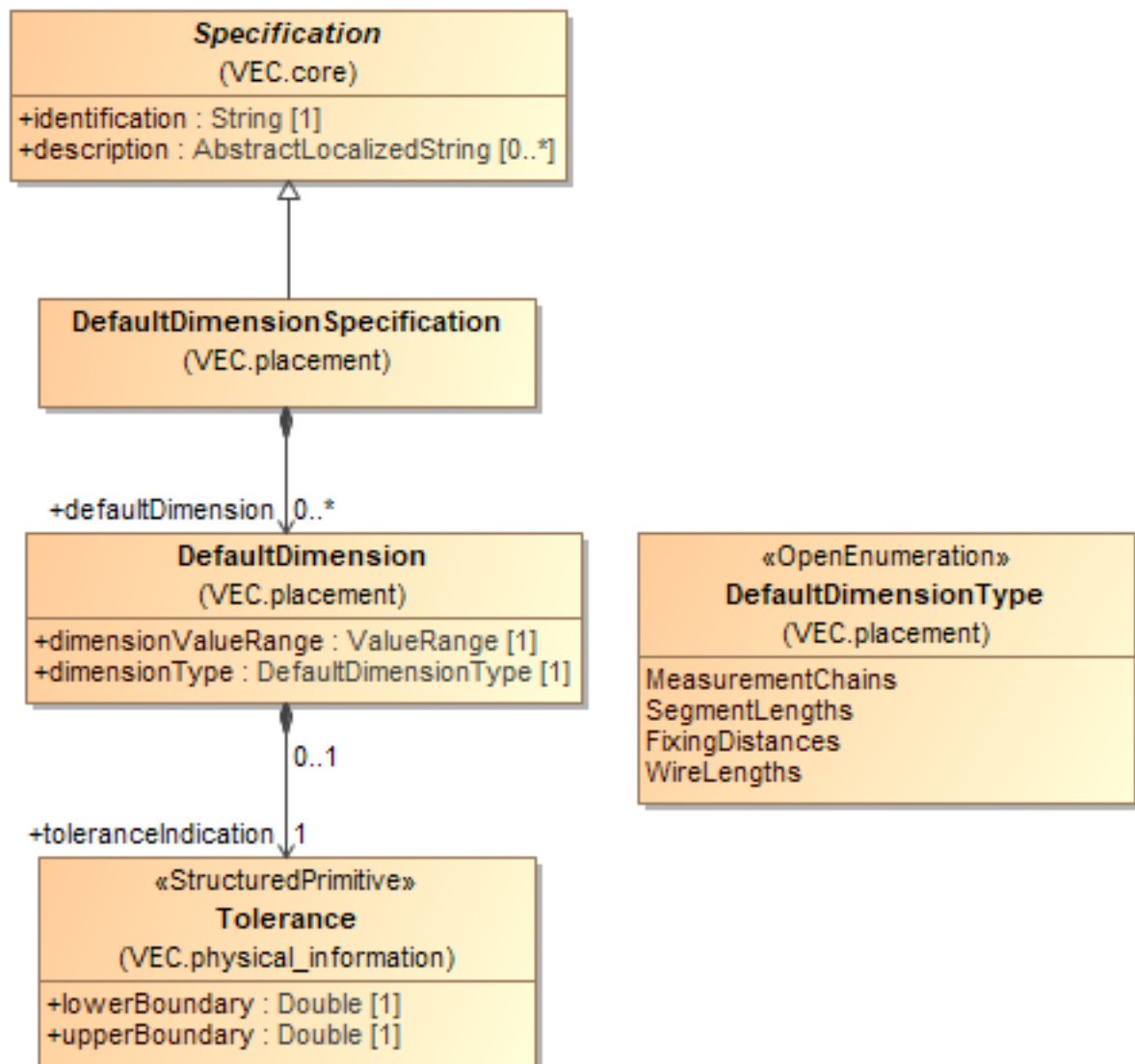


Figure 83: Default Dimensions

DefaultDimension can be used to define standard tolerances that apply to certain elements. Typically, they are represented in a document or a drawing as a table (e.g. for wires with length < x; tolerance a, x < length < y; tolerance b and so on).

1.9.11 Routing

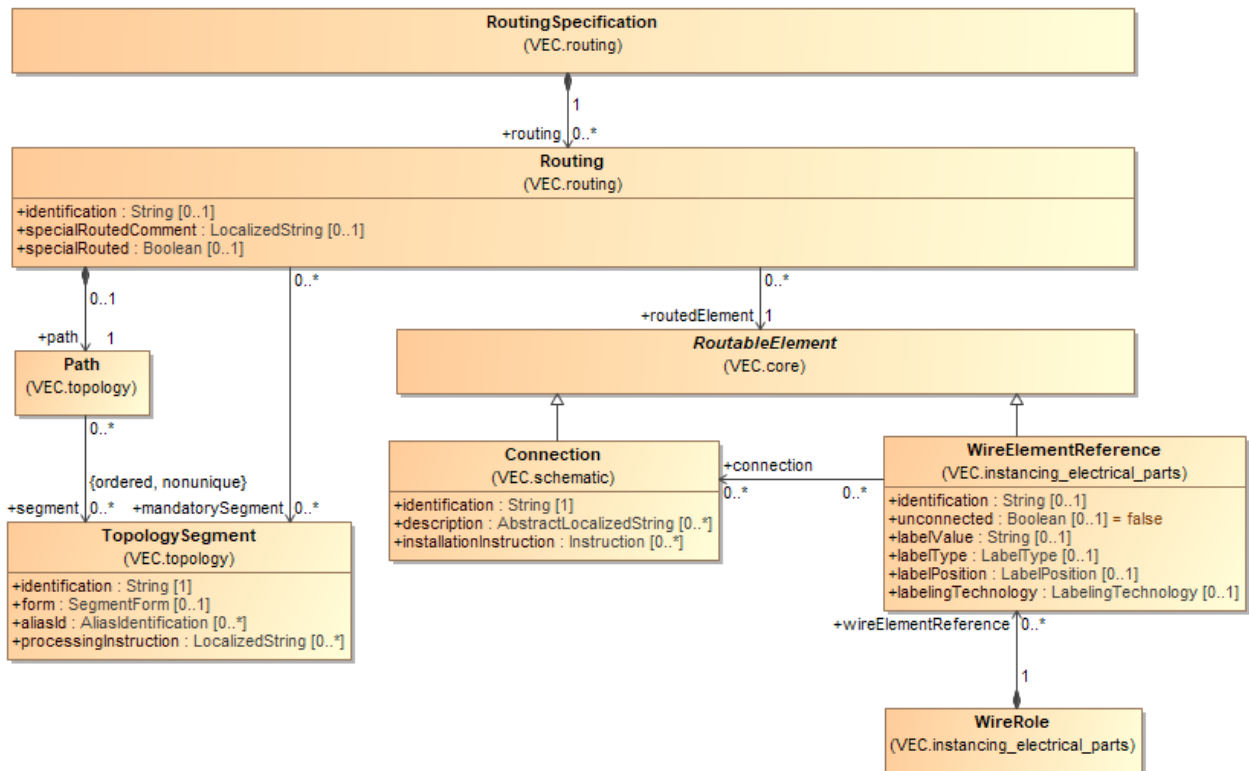


Figure 84: Routing

A *RoutingSpecification* is a container for various *Routings*. Each *Routing* specifies a dedicated *Path* as an ordered list of *TopologySegments* through which the referenced *RoutableElement* is routed. A *RoutableElement* can either be a *Connection* or a *WireElement* (which can alternatively belong to a *PartUsage* or a *PartOccurrence*).

In addition, a *Routing* can reference some segments on the *Path* as *mandatorySegments*. This can be helpful in cases where otherwise a simple redefinition respectively recalculation would not consider these segments.

1.10 Connectivity

NA

1.10.1 Signal Specification

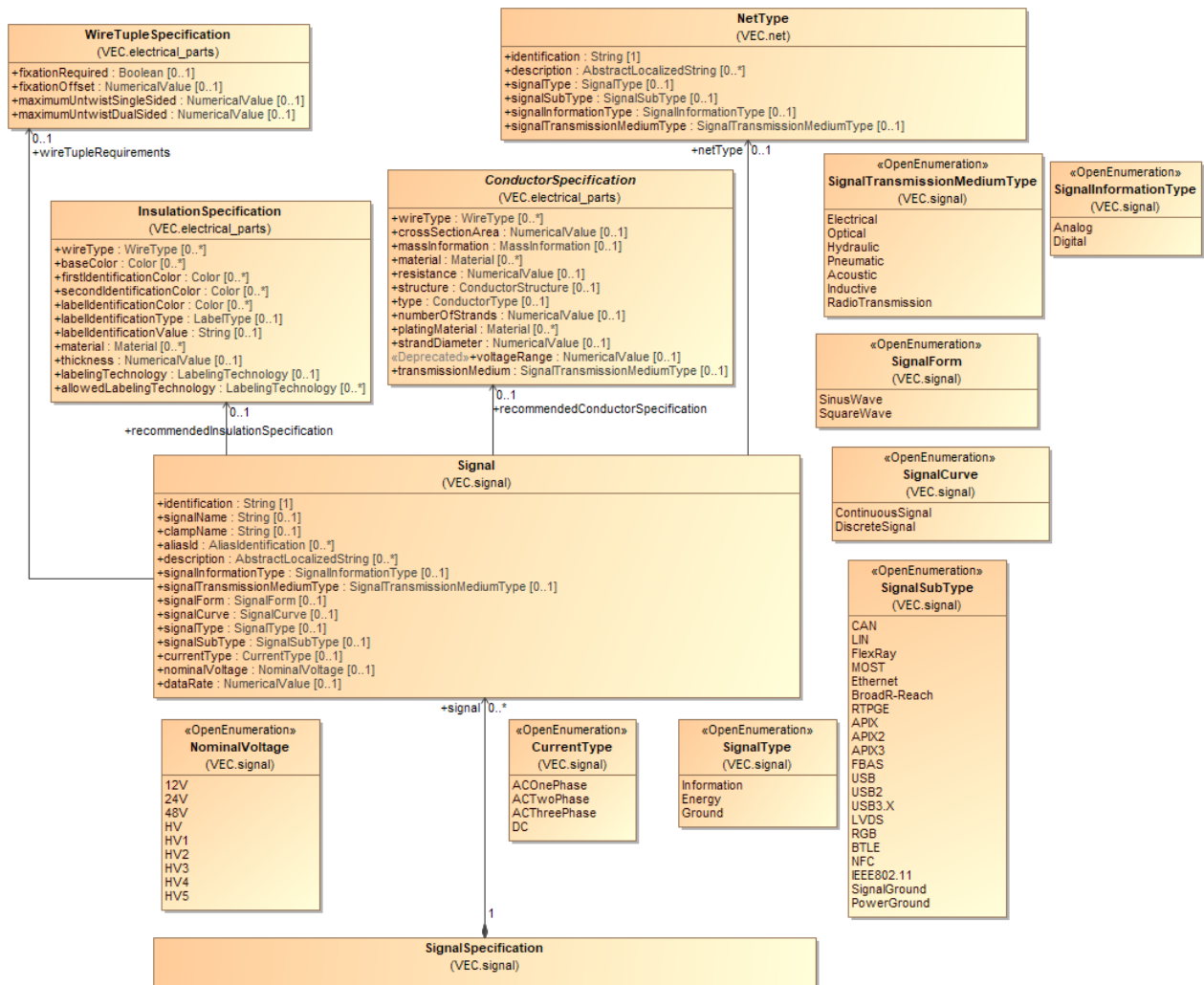


Figure 85: Signal Specification

A *SignalSpecification* defines a set of *Signals* used in the vehicle electrical system. A *Signal* is a more general concept than *Nets*, *Connections* or *Wires*, as they are defined in the scope of a certain vehicle electrical system or a wiring harness. A *Signal* in the VEC represents a physical signal in contrast to abstract messages that are often used to define ECUs and their interfaces (e.g. in AUTOSAR). Typically, the same *Signals* are used in different vehicles or different sections of a wiring harness. A *Signal* can define various properties e.g. the *SignalType* or the *SignalForm*. Furthermore, a *Signal* can define a recommended *ConductorSpecification* and a recommended *InsulationSpecification*. This mechanism can be recommended for example to provide information about the recommended color coding of a certain *Signal*, regardless of the vehicle electrical system or wiring harness section where it is implemented.

1.10.2 Net Specification

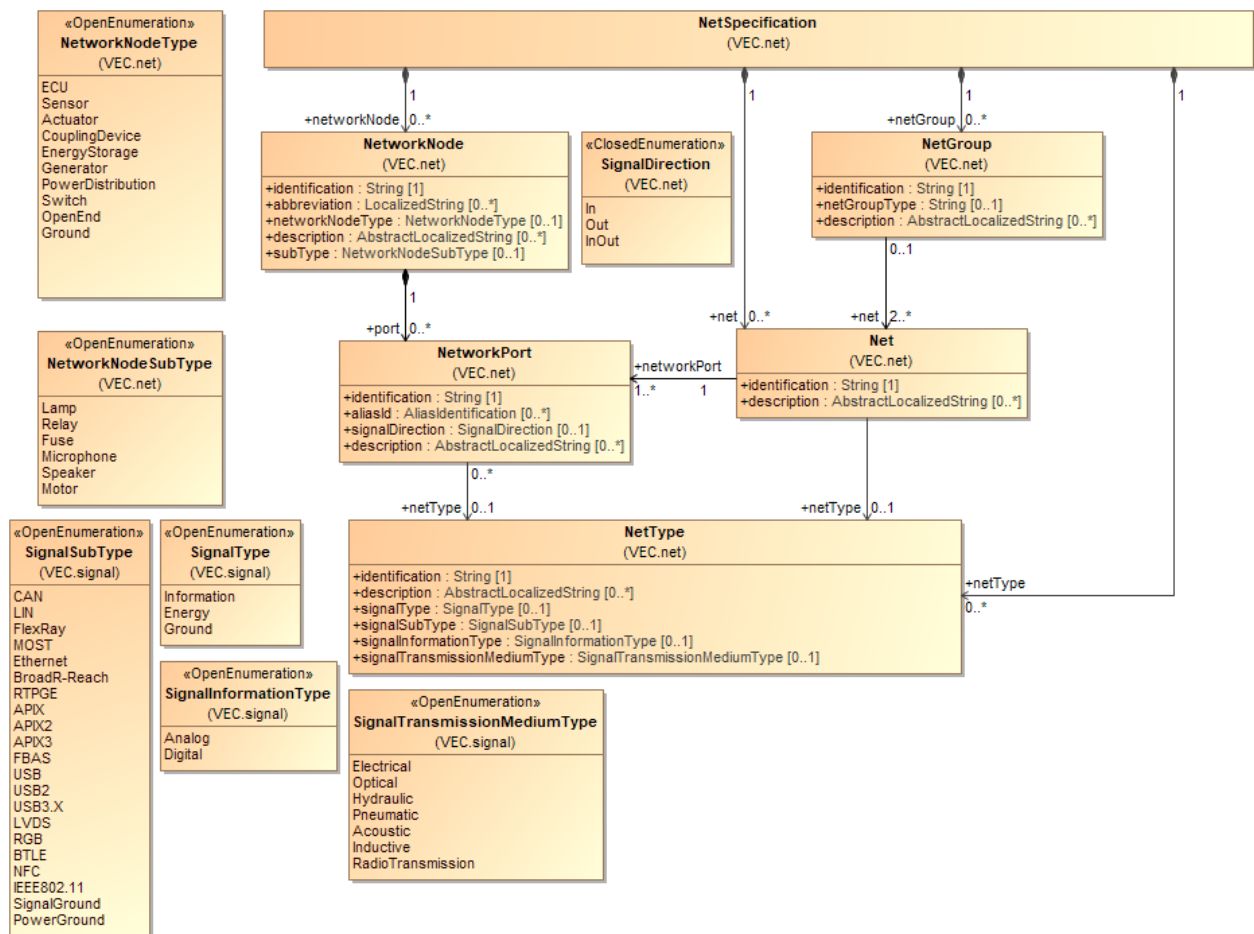


Figure 86: Net Specification

A *NetSpecification* is the most abstract way to describe a vehicle electrical system. It is a container for various *NetworkNodes*, *Nets* and *NetGroups*. A *NetSpecification* is used if the physical links between electrical components are specified without specifying a concrete network topology and a physical realization. In many processes and applications this is defined as "Architectural Layer".

A *NetworkNode* is a representative for an actor in the electric system, e.g. an actuator, a sensor, an ECU. It can define various *NetworkPorts*, which can be classified as a signal source (in the case that the attribute *signalDirection* has got the value *Out*) or as a signal sink (in the case that the attribute *signalDirection* has got the value *In*) or with changing behavior (in the case that the attribute *signalDirection* has got the value *InOut*).

Note: According to the definition above, inliners (normally) and splices are no *NetworkNodes* and so are not represented within a *NetSpecification*. In some cases, there are architectural relevant inliners (*NetworkNodeType* = *CouplingDevice*) that represented in the architectural layer.

A *Net* is a representative of an abstract link between the referenced *NetworkPorts* and can be related to a *NetType*.

Note: A *Net* itself doesn't define:

- How the topology of the conducting connection must be realized (e.g. if three *NetworkPorts* are interconnected, the *Net* makes no definition if this physical three-point connection is realized with a splice, an insulation displacement connector or a double contact.).
- How the physical connections must be realized (e.g. if a certain BUS Connection is realized by a pair of wires or four wires).

Note: Normally, a *Net* references at least two *NetworkPorts*. However, in the very early stages of product development it might be the case that some *NetworkNodes* with *NetworkPorts* which are source of a dedicated

Net are already defined – but the counterparts are not. This is an example where it can be useful to define *Nets* referencing only one *NetworkPort*.

1.10.3 Net View Specification

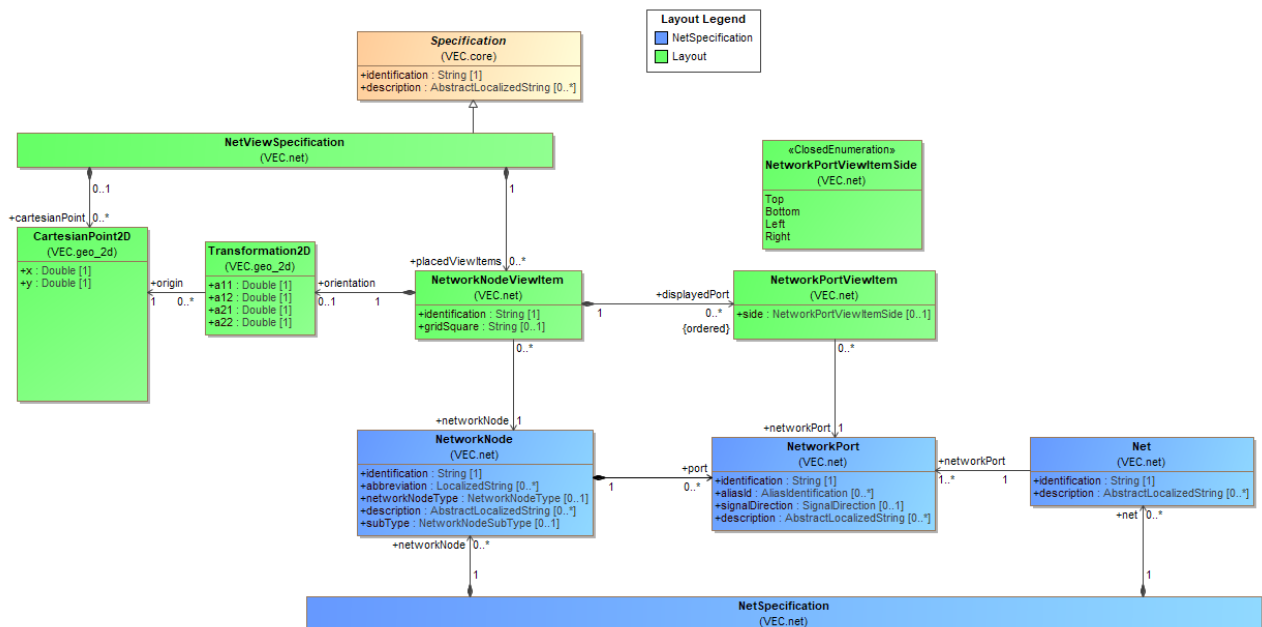


Figure 87: Net View Specification

The *NetViewSpecification* supports the exchange of a basic layout for a *NetSpecification*. The objective is not a pixel perfect representation of a network / architecture diagram in the VEC (with label positions, font sizes etc.). Such use cases are supported via the external mapping with other standards like SVG. However, a reasonable arrangement of the *NetworkNodes* on a diagram sheet greatly improves the understanding and cannot (easily) be found by an automatic algorithm. Therefore, a basic layout has value in the context of data exchange and is in scope of the VEC.

A *NetViewSpecification* represents the layout of a network diagram / sheet. The layout is defined with *NetworkNodeViewItems*, each representing a *NetworkNode*. A *NetViewSpecification* does not have to represent the complete content of a *NetSpecification*. It is perfectly valid to split the layout of a *NetSpecification* into multiple *NetViewSpecifications*, one for each view (e.g. sheet) and each laying out a subset of a *NetSpecification*. It is also valid to have multiple *NetworkNodeViewItems* for a *NetworkNode*.

The *NetworkNodeViewItem* contains *NetworkPortViewItems* for each *NetworkPort* displayed on this *NetworkNodeViewItem*. This can be a subset of all *NetworkPorts* of the corresponding *NetworkNode*.

For *Nets* there are no explicit view items defined. The representation and routing path of a *Net* is not considered as basic information. All *Nets* that are connected to a displayed *NetworkPort* are considered to be display on the diagram represented by *NetViewSpecification*.

1.10.4 Connection Specification

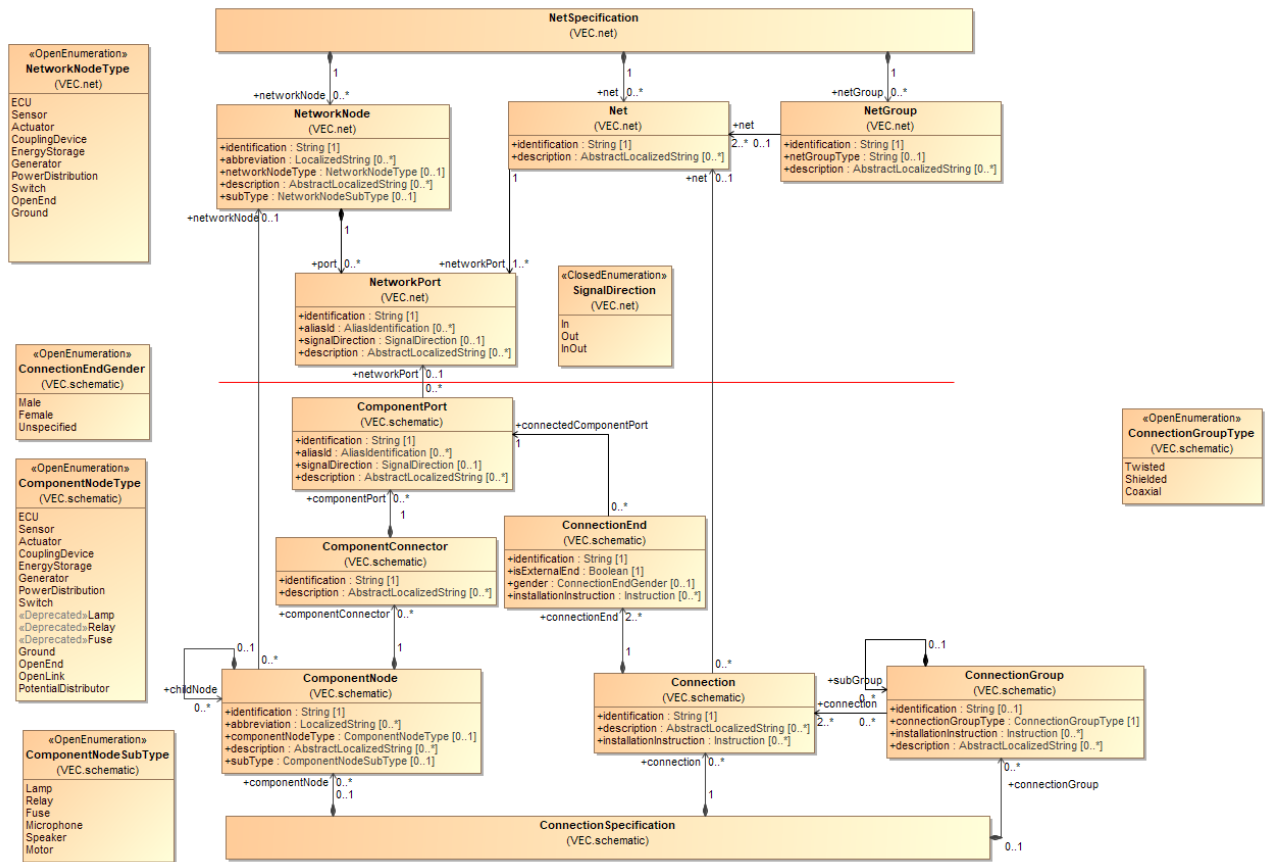


Figure 88: Connection Specification

A *ConnectionSpecification* is intended to contain the logical information that is usually included on an electrical system connection plan (system schematic). It can either stand alone or detail the information of a *NetSpecification*. A *ConnectionSpecification* is a container for various *ComponentNodes*, *Connections* and *ConnectionGroups*. A *ConnectionSpecification* is used, if requirements for realization of the network should be specified (e.g. should a *Net* be realized with a single or multiple connection (BUS-Systems)).

Note: Logical information means a *ConnectionSpecification* doesn't contain layout information. Layout information can be exchanged by other formats like SVG, PDF or DXF.

A *ComponentNode* is a representative for an element in the electric system, e.g. an actuator, a sensor, an ECUs. In this way it is quite similar to a *NetworkNode* and may even reference the corresponding *NetworkNode* in this case. Moreover, a *ComponentNode* can define *childNodes* in order to describe its internal structure.

There is no general rule if inliners shall appear in the system schematic or not. In many cases there is a detail of the physical design of the harness (like splices). Their creation is done implicitly when an electrological design is routed into a specific topology and the routing crosses an inliner definition of the topology. In these cases, the inliner definition should not appear in the system schematic. However, there are other scenarios where the inliner definition is done on purpose and as part of the electrological design (e.g. for providing a pluggable modularity in the system). In these cases, an inliner can appear in the *ConnectionSpecification*.

A *ComponentNode* can specify various *ComponentPorts*. Each *ComponentPort* can reference its corresponding *NetworkPort* (if it has one).

Connections represent the physical realization of a *Net* (without a topology). A *Net* (e.g. the BODY-CAN-BUS) can be realized by one or more physical connections (e.g. a CAN-BUS is normally realized by two physical connections (HI & LOW)). The *Connections* do not define a topology of the realization. This means that a CAN *Connection* with three *ConnectionEnds* can be realized in different ways (e.g. three wires with splice, two wires with a double contact or an IDC / IDS contact).

1.10.5 Connection View Specification



A *ConnectionViewSpecification* represents the layout of a schematic diagram / sheet. The layout is defined with *ComponentNodeViewItems*, each representing a *ComponentNode*. A *ConnectionViewSpecification* does not have to represent the complete content of a *ConnectionSpecification*. It is perfectly valid to split the layout of a *ConnectionSpecification* into multiple *ConnectionViewSpecifications*, one for each view (e.g. sheet) and each laying out a subset of a *ConnectionSpecification*. It is also valid to have multiple *ComponentNodeViewItems* for a *ComponentNode*.

For *Connections* there are no explicit view items defined. The representation and routing path of a *Connection* is not considered as basic layout information. All *Connections* that are connected to a displayed *ComponentPort* are considered to be display on the diagram represented by *ConnectionViewSpecification*.

1.10.6 Wiring Specification

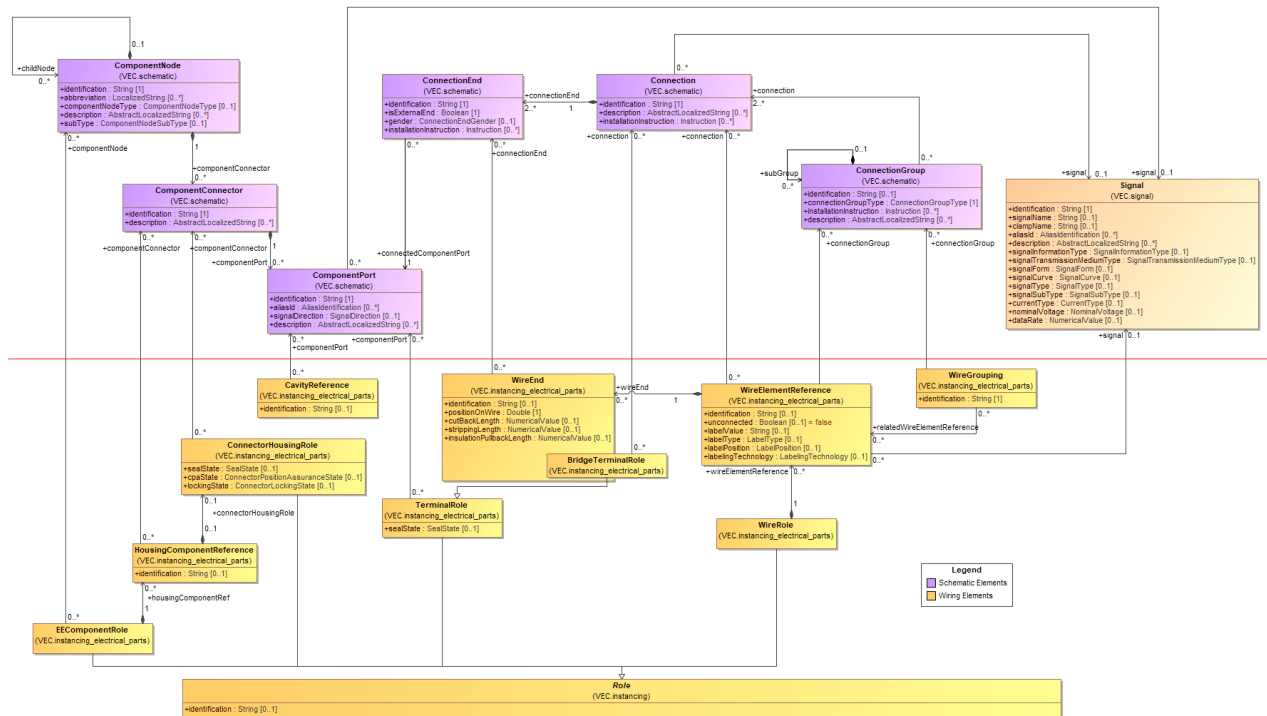


Figure 90: Wiring Specification

The VEC model breaks down the information that is usually included on an electrical system wiring plan into three aspects: part usage information, contacting information and mating information. As part usage information is for instance also relevant in the geometry respectively part placement context and as contacting and mating information is also relevant in the KBL context (the final definition of wiring harnesses and harness modules), the VEC model does not define an integrated *WiringSpecification* class for all these aspects. Instead, the VEC model defines for each of the three mentioned aspects a separate specification class, which can be reused when necessary (see also "Instances of Components").

EE-components in a wiring diagram are intended to be described as *PartUsages* with an *EEComponentRole*. Each of these *PartUsages* can reference an *EEComponentSpecification* in order to describe relevant technical properties. The *EEComponentRole* can specify a *TerminalRole* for each EE-component pin. This can act as the basis for a mating definition between the EE-component pins and harness connector pins.

Harness connectors in a wiring diagram are intended to be described as *PartUsages* with a *ConnectorHousingRole*. Each of these *PartUsages* can reference a *ConnectorHousingSpecification* in order to describe relevant technical properties.

Harness connector pins in a wiring diagram are intended to be described as *PartUsages* with a *PluggableTerminalRole*. Each of these *PartUsages* can reference a *PluggableTerminalSpecification* in order to describe relevant technical properties. The relationship between harness connector and harness connector pin is intended to be described by a *ContactingSpecification* which defines an appropriate *CavityMounting*.

Wiring connections are intended to be described as *PartUsages* with a *WireRole* in combination with a *ContactingSpecification* which defines an appropriate *WireMounting*. Each of these *PartUsages* can reference a *WireSpecification* in order to describe relevant technical properties.

The *PartUsages* with *EEComponentRole* respectively *ConnectorHousingRole*, *TerminalRole* or *WireRole* can each reference their specific corresponding element within a *ConnectionSpecification* in order to express a refinement-relationship.

1.10.7 Contacting Specification

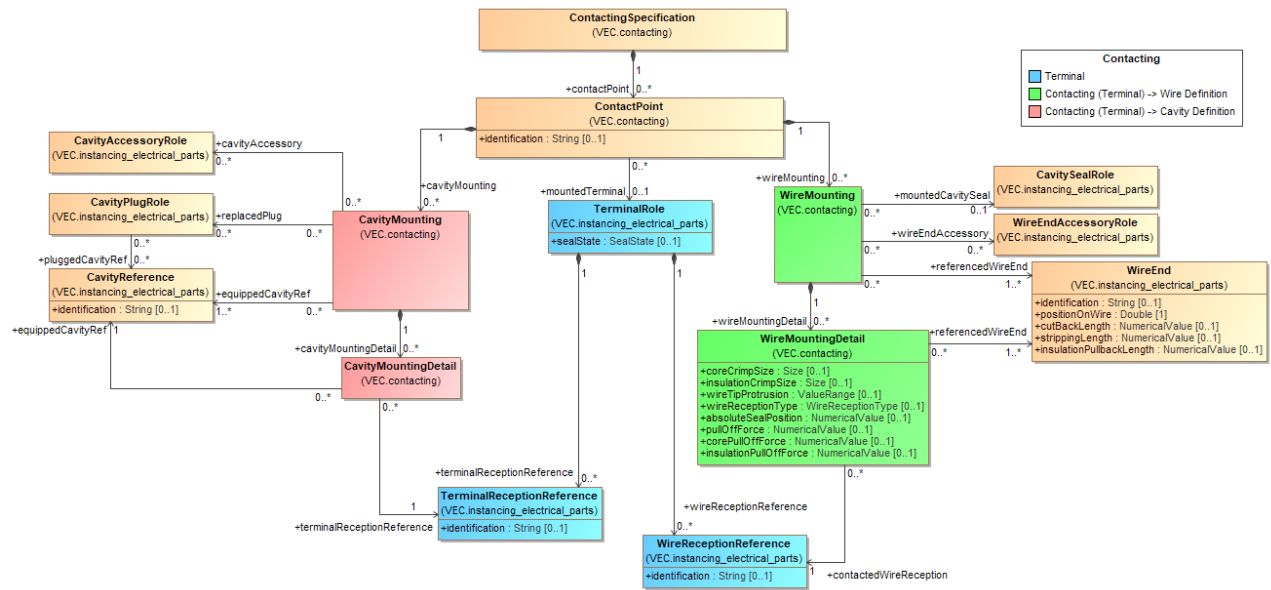


Figure 91: Contacting Specification

A *ContactingSpecification* is used to describe contacting information in several use cases: It can be used to describe both the more abstract contacting information which is contained on an electrical system wiring plan and the very concrete contacting of an assembly / module / harness configuration or even a 150% harness description.

A *ContactingSpecification* is a container for various *ContactPoints*. Each *ContactPoint* can specify several *CavityMountings* and *WireMountings* and will in most cases reference a *TerminalRole*.

A *WireMounting* references one or more *WireEnds* stating they are contacted with the *TerminalRole* referenced by the *ContactPoint*. In cases of ambiguity a *WireMounting* can specify various *WireMountingDetails* in order to describe which *WireEnd* is contacted to which *WireReception* of the referenced *TerminalRole*. Finally, a *WireMounting* can reference a *CavitySealRole*.

A *CavityMounting* references one or more *CavityReferences* expressing they are equipped by the *TerminalRole* referenced by the *ContactPoint*. In cases of ambiguity a *CavityMounting* can specify various *CavityMountingDetails* in order to describe which *TerminalReception* is mounted in which *CavityReference*. Finally, a *CavityMounting* can reference various *CavityPlugRoles* stating the *CavityMounting* replaces those *CavityPlugs*.

Note: *ContactPoints* are *ConfigurableElements* and therefore can reference a *VariantConfiguration*.

Note: There may be use cases where a *ContactingSpecification* is wanted to describe an incomplete contacting. An example might be an assembly or module with wires together with mounted terminals but without a mounting of the terminals into a connector housing. The missing contacting information (the mounting of the terminals into a connector housing) may be subject of other *ContactingSpecifications* respectively *ContactPoints* which may be specified configuration dependent.

1.10.9 Wire Grouping Specification

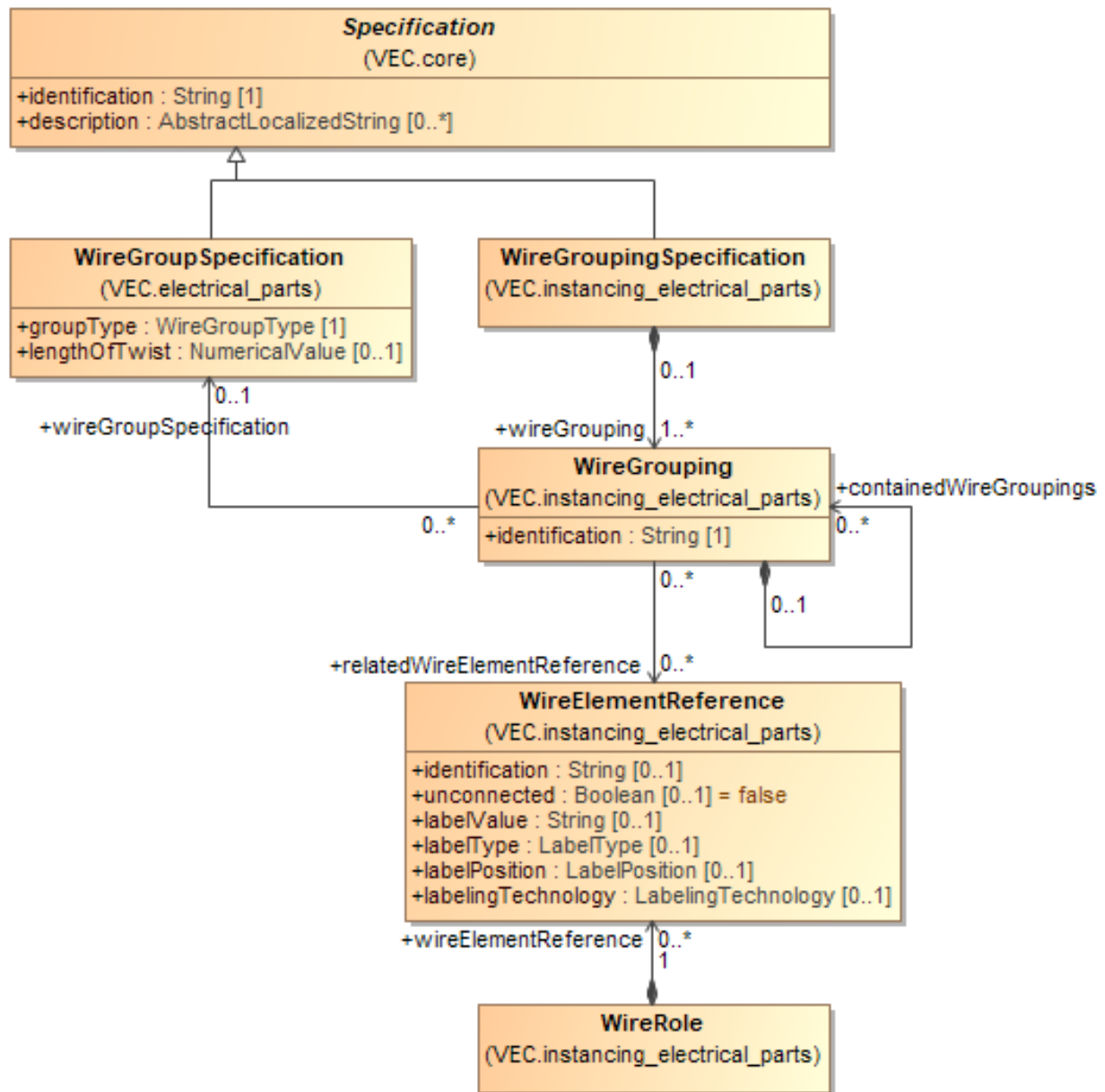


Figure 93: Wire Grouping Specification

A *WireGroupingSpecification* is used to describe grouping requirements (e.g. twisted pair) in a certain context of usage (e.g. in the context of a 150% harness description). For this, a *WireGroupingSpecification* is a container for various *WireGroupings*. For more details, please refer to the description of the *WireGrouping*.

1.10.10 Wire Tuple Termination

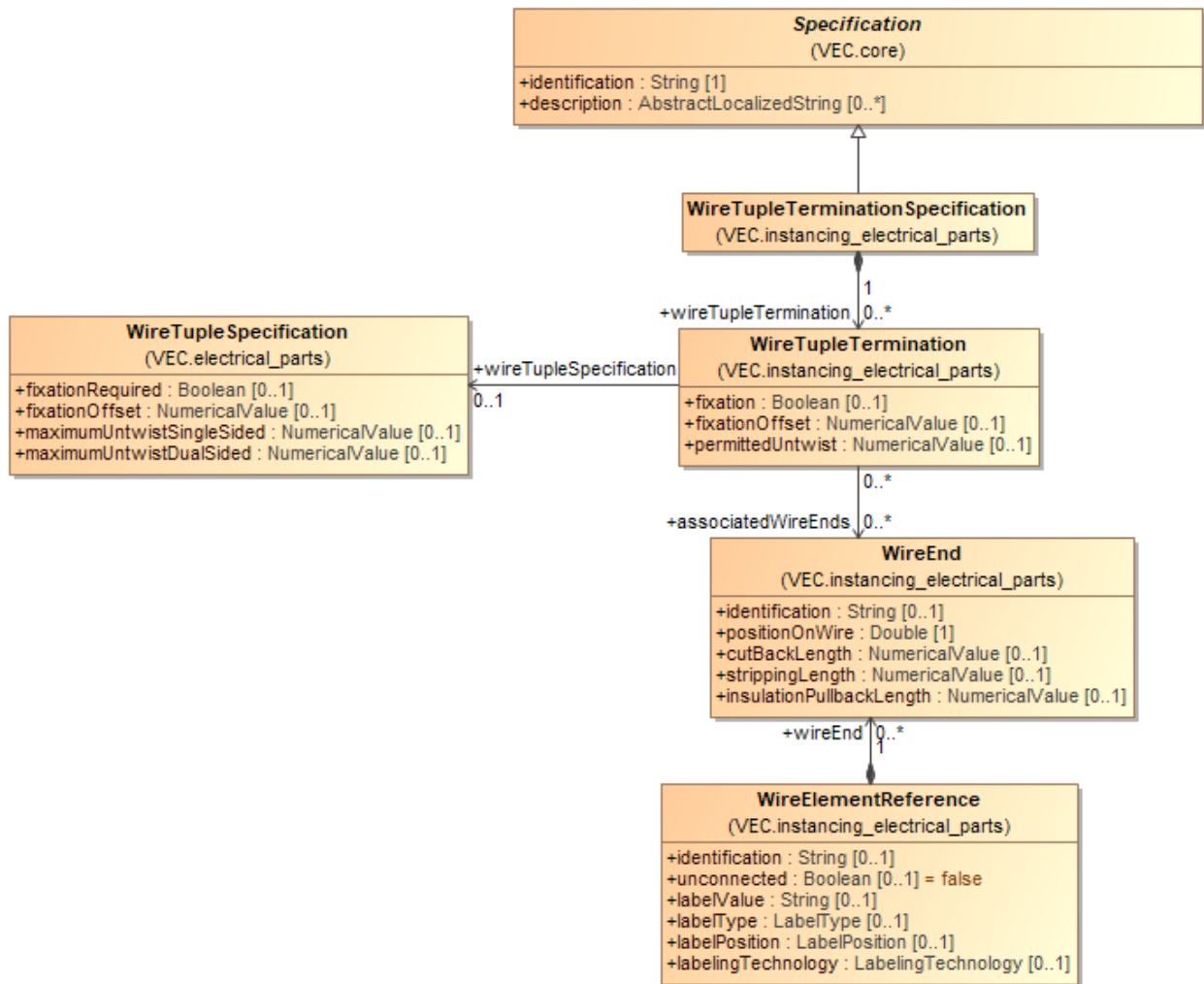


Figure 94: Wire Tuple Termination

A *Specification* to define requirements for the handling of wire / wire element tuples (pairs, triple...) with the focus on twisting / untwisting. The *WireTupleSpecification* defines the technical properties of the untwisting from a part master data perspective (see also the diagram "Wire Tuple" in "Component Characteristics"). The *WireTupleTermination* defines properties for a specific end of a specific wire tuple (a concrete usage).

1.10.11 Pin Wire Mapping

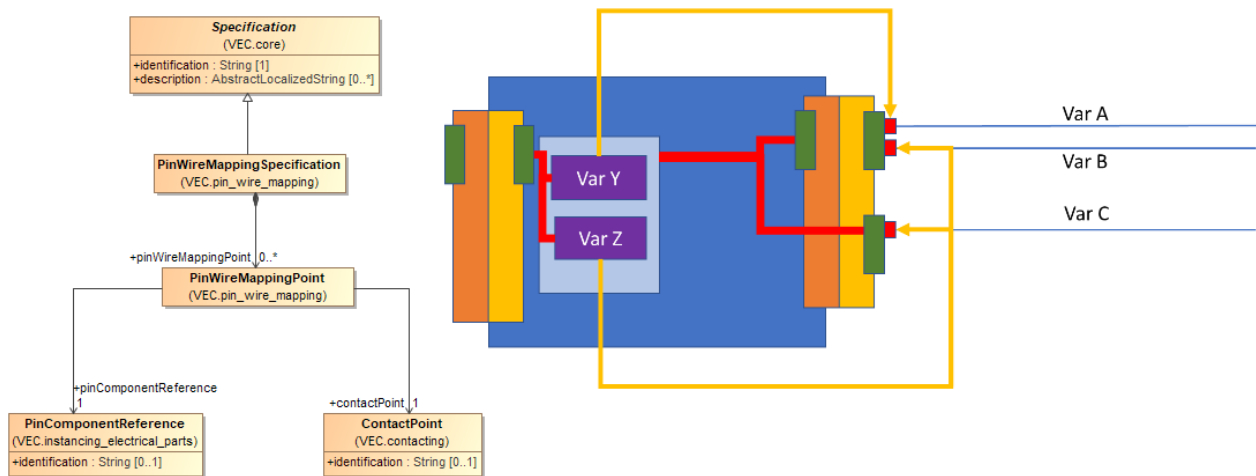


Figure 95: Pin Wire Mapping

A *PinWireMappingSpecification* can be used to create **variance free** mappings between a wire (represented by the contact point) and the pin of an E/E component. This is a possibility to create a shortcut in the model between a wire and its connected E/E-component (e.g. a fuse) that might be only indirectly connected to a wire (e.g. via a fuse and relay carrier). This is a relevant information for e.g. the validation of fusing.

In a specific 100% variant, this is an information that could be derived from the data provided by the VEC, if available. Informally spoken, the way in the model would be from the *WireEnd* over to the *ContactPoint* and from there to the cavity of the harness connector. From the harness connector to the fuse and relay carrier *PinComponentReference*, via the *CouplingPoint*. From the *PinComponentReference*, via the *InternalComponentConnection* to the slot of the fuse and then via another *CouplingPoint* to the fuse.

However, in a 150% scenario, even if all component assignments to modules are known, this information cannot be derived unambiguously without extensive knowledge of the variant management mechanisms. This is illustrated with the graphic representation in the diagram. There are two wires (controlled by Variant A & B) attached to a single cavity (green). Via an internal connection of the carrier, they are connected to a single component slot (light blue). This slot contains two different fuses (controlled by variant Y & Z). Without knowledge of the conditions between the variant A, B, Y & Z it is impossible to say what fuse is responsible for the protection of which wire. That knowledge might not be available at all times, for all partners in the process.

The *PinWireMappingPoint* provides a possibility to define this information in the VEC (the golden / yellow arrows in the illustration).

1.10.12 Signal Graphs

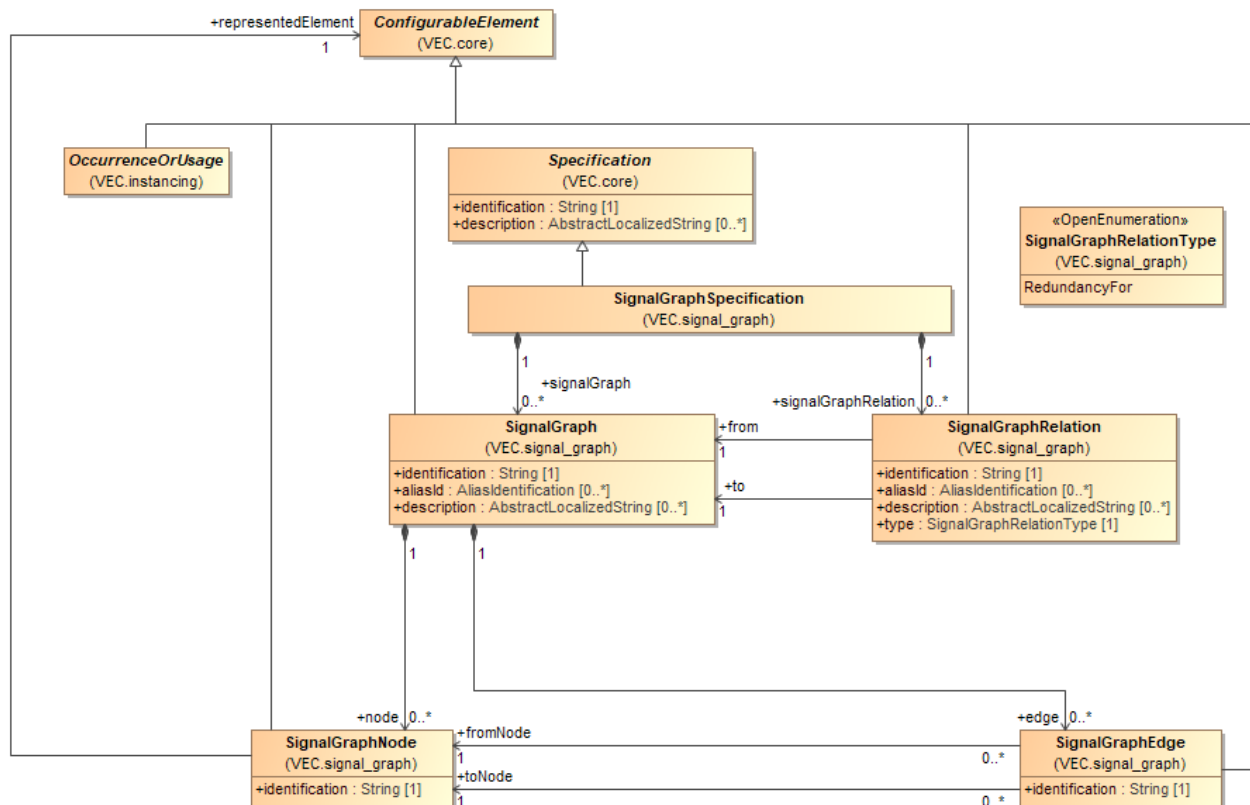


Figure 96: Signal Graphs

A *SignalGraph* is a directed graph of elements in the vehicle network that participate in the transmission of a signal / function / power / ground etc. A *SignalGraph* is expected to be a unilaterally connected graph at least, meaning it contains a directed path from *u* to *v* or a directed path from *v* to *u* for every pair *u* and *v* of *SignalGraphNode*s in the *SignalGraph*.

A *SignalGraph* might be the result of a complex calculation based on existing information (e.g. a power supply or grounding graph) or it might user defined (e.g. to apply specific requirements to a chain of elements). Relations between *SignalGraph*s can be expressed with a *SignalGraphRelation*, for example if one graph is the redundant implementation of another one.

Note: The "signal" concept of a *SignalGraph* is much broader than that of the *Signal* class. Due to the lack of a better term, the same term was used for slightly different concepts. For example, a *SignalGraph* might be used to represent all possible elements that are involved in the power supply for a specific component. This involves all elements on the way from the component to the battery. However, the electrical signal (*Signal*-class) will be named differently for the supply and distribution sides of a power distributor / fuse box.

1.11 External Mapping

The external mapping in the VEC defines a mechanism to provide a link between the content data of the VEC and external data sources (e.g. SVG images, JT models, requirements, ...). The only requirement on the external data source for this mapping approach is, that the contained data is in some way structured and that the elements are identifiable with a unique key.

The reasons for embedding the mapping information in the VEC are:

- The mapping can be easily read and recreated together with VEC and references to VEC elements can be followed in a simple way since the mapping can reference the elements directly within a single file (IDREF).

- No additional format definition is necessary.

1.11.1 External Mapping

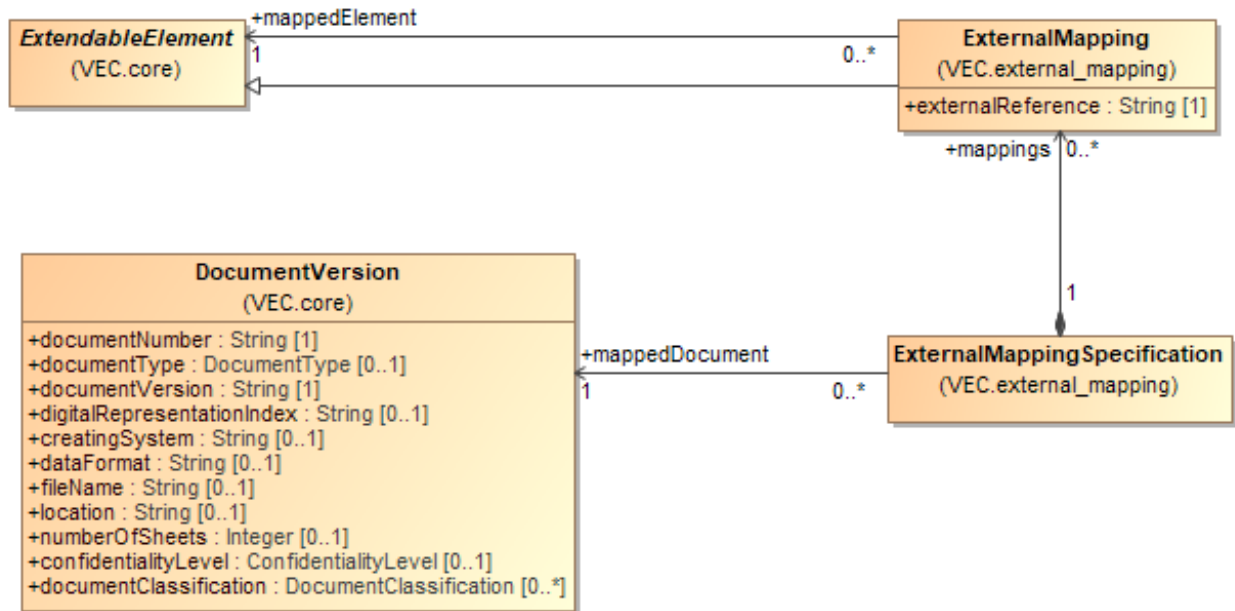


Figure 97: External Mapping

The diagram displays the extensions in the VEC Modell necessary to support a mapping to external sources. The extension consists of two classes *ExternalMappingSpecification* and *ExternalMapping*. An *ExternalMappingSpecification* defines a mapping for a single *DocumentVersion* specified by the association *mappedDocument*. The referenced *DocumentVersion* itself identifies by its attributes the external document which should be mapped (e.g. an SVG-File, a requirements-document).

The mapping of the individual elements in the VEC is defined with the class *ExternalMapping*. It points to an *ExtendableElement* (which can be almost any VEC-Element) as *mappedElement*. The attribute *externalReference* contains the unique key of the same element in the context of the mapped *DocumentVersion*.

The reason for defining the *ExternalMapping* elements within a separate *Specification* is that the *ExternalMappingSpecification* can be placed in the VEC within its own *DocumentVersion*. The creation of mapping does not require a modification and thus a versioning of the mapped content in the VEC.

2 VEC Classes

2.1 Module assignment_groups

2.1.1 Class AssignmentGroup

An *AssignmentGroup* is a concept that allows the clustering of arbitrary elements in ways that are orthogonal to hierarchical and semantic structure of the VEC.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the AssignmentGroup. The identification is guaranteed to be unique within the specification.
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the <i>AssignmentGroup</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
assignmentGroup	0..*	Y	AssignmentGroup Specification	1	Contains the AssignmentGroups that are defined by this AssignmentGroupSpecification.
associatedAssignmentGroups	0..*	N	ConfigurableElement	0..*	NA

2.1.2 Class AssignmentGroupSpecification

Specification that contains *AssignmentGroups*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
assignmentGroup	AssignmentGroup	0..*	Y	N	1	Y	Contains the AssignmentGroups that are defined by this AssignmentGroupSpecification.

2.1.3 Class DocumentRelatedAssignmentGroup

A DocumentRelatedAssignmentGroup allows the creation of traceability links to elements in a *DocumentVersion* for a set of VEC objects. The semantic of the traceability link is defined by the *DocumentRelationType*.

General Information

Base Classifier	AssignmentGroup
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	DocumentRelationType	0..1	NA
relatedIdentification	String	0..1	If this group relates to a specific element in the <i>relatedDocumentVersion</i> the identifier of the element is defined in this attribute (e.g. a requirements number).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
relatedDocumentVersion	DocumentVersion	1	Y	N	0..*	N	References the <i>DocumentVersion</i> to which this group relates.
relatedSheetOrChapter	SheetOrChapter	0..1	Y	N	0..*	N	Allows a more specific relationship to a <i>SheetOrChapter</i> within the <i>relatedDocumentVersion</i> .

2.1.4 Class FunctionalAssignmentGroup

The *FunctionalAssignmentGroup* clusters elements that contribute to a specific function or a functional aspect. With such a group, certain functional requirements can be associated.

General Information

Base Classifier	AssignmentGroup
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
functionalRequirements	FunctionalRequirement	0..*	Functional requirements that apply to the members of this group.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
containedGroups	0..*	N	FunctionalStructureNode	0..*	NA

2.1.5 Class FunctionalRequirement

Allows the definition of functional requirements in an enumerable way (e.g. conformance to a certain ASIL level). Attributes of this type have the multiplicity of [0..*]. The following restrictions apply:

- For a combination of type & referenceSystem only a single value is allowed. For single type
- For a specific type and different references systems, multiple values are allowed. However, they must express the same semantic value.
- For different types multiple values are allowed.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	FunctionalRequirementType	0..1	The type defines to which category a requirement belongs (e.g. Functional Safety).
referenceSystem	String	1	The reference system identifies the system in which the values are defined (e.g. ISO26262)
value	String	1	The value that represents the functional requirement (e.g. ASIL D).

2.1.6 Class FunctionalStructureNode

FunctionalStructureNodes can be used to define a hierarchical structure on *FunctionalAssignmentGroups*. Every *FunctionalStructureNode* can reference *FunctionalAssignmentGroups* and *FunctionalStructureNodes* as children.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>FunctionalStructureNode</i> .
aliasId	AliasIdentification	0..*	Room to specify additional identifiers for the <i>FunctionalStructureNode</i> .
description	AbstractLocalizedString	0..*	On optional human readable description of the <i>FunctionalStructureNode</i> .
abbreviation	LocalizedString	0..1	Room for a human readable short name, title etc. of the <i>FunctionalStructureNode</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
childNodes	FunctionalStructureNode	0..*	Y	N	0..1	Y	NA
containedGroups	FunctionalAssignmentGroup	0..*	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
childNodes	0..*	Y	FunctionalStructureNode	0..1	NA
rootNode	1	Y	FunctionalStructureSpecification	0..1	

2.1.7 Class FunctionalStructureSpecification

Specification to define any hierarchical structure on functional assignment groups (e.g. by the means of a functional organization). The hierarchy starts with a single root node.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
rootNode	FunctionalStructureNode	1	Y	N	0..1	Y	

2.1.8 Enumeration DocumentRelationType

NA

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
ResponsibilityLink	NA
RequirementsLink	NA

2.1.9 Enumeration FunctionalRequirementType

NA

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
FunctionalSafety	Functional safety requirements e.g. ASIL Level.
Crash	Crash requirements e.g. post-crash functionality.
Legal	Legal Requirements.
Function	Requirements from a functional point of view (e.g. standby current).

2.2 Module contacting

2.2.1 Class CavityMounting

A CavityMounting defines the cavities (CavityReference) where the contacted elements (Terminal) will be mounted.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavityAccessory	CavityAccessoryRole	0..*	Y	N	0..*	N	NA
cavityMountingDetail	CavityMountingDetail	0..*	Y	N	1	Y	Specifies the CavityMountingDetails, if a detailed description of the relationships between Cavities and TerminalReceptions is needed.
equippedCavityRef	CavityReference	1..*	Y	N	0..*	N	References the cavities that are used for the cavity mounting.
replacedPlug	CavityPlugRole	0..*	Y	N	0..*	N	References the cavity plugs that are obsolete if the cavity mounting is realized.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavityMounting	0..*	Y	ContactPoint	1	Defines the mounting to a cavity of the terminal associated with the ContactPoint. The cardinality is 0..* in order to allow a variant dependent cavity mounting. In such a scenario a cavity mounting is valid in a configuration if all addressed cavities and therefore the associated connector housing is available.

2.2.2 Class CavityMountingDetail

With a CavityMountingDetail it is possible to describe a detailed cavity mounting.

This is needed if the information which terminal reception is mounted into which cavity is important. There are cases where this information can be relevant (e.g. bridge contacts with an asymmetric wire mounting).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
equippedCavityRef	CavityReference	1	Y	N	0..*	N	References the cavity that is used for the detailed description of the cavity mounting.
terminalReceptionReference	TerminalReceptionReference	1	Y	N	0..*	N	References the TerminalReception that is used for the detailed description of the cavity mounting.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavityMountingDetail	0..*	Y	CavityMounting	1	Specifies the CavityMountingDetails, if a detailed description of the relationships between Cavities and TerminalReceptions is needed.

2.2.3 Class ContactingSpecification

Specification for the description of the contacting. A contacting defines the relationships between Terminals, Seals, Plugs, Cavities and Wires.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
contactPoint	ContactPoint	0..*	Y	N	1	Y	Specifies the ContactPoints defined by the ContactingSpecification.

2.2.4 Class ContactPoint

A contact point defines the relationship between Terminals, Seals, Plugs, Cavities and Wires. It specifies a single contacting variant. This means that the contacting is manufactured, as specified by the *ContactPoint*. Either all participants (Cavities, Terminals, Seals, Wires) set into a relationship by the *ContactPoint* exist in a specific harness or none. There is no requirement, to filter the participants of a contacting situation with information derived from VariantConfigurations or assembly / module associations in order to create a manufacturing variant.

The *ContactPoint* represents a single potential. Consequently, all cavities and wires referencing / being referenced by a *ContactPoint* are short-circuited and have the same potential (even if the signals on the wires are named differently. If a contacting of a terminal has more than one potential (e.g. a coax-contact) one contact point for each potential is needed.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the ContactPoint. The identification is guaranteed to be unique within the ContactingSpecification.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavityMounting	CavityMounting	0..*	Y	N	1	Y	Defines the mounting to a cavity of the terminal associated with the ContactPoint. The cardinality is 0..* in order to allow a variant dependent cavity mounting. In such a scenario a cavity mounting is valid in a configuration if all addressed cavities and therefore the associated connector housing is available.
mountedTerminal	TerminalRole	0..1	Y	N	0..*	N	References the terminal that is used for contacting defined by the ContactPoint.
wireMounting	WireMounting	0..*	Y	N	1	Y	Specifies the WireMounting defined by ContactPoint. More than one WireMounting is allowed in order to support variance. In concrete configuration the WireMounting with all referenced elements present is used.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
contactPoint	0..*	Y	ContactingSpecification	1	Specifies the ContactPoints defined by the ContactingSpecification.
contactPoint	1	N	PinWireMappingPoint	NA	NA

2.2.5 Class WireMounting

A *WireMounting* defines the mounting of a wire end to terminal. If the contacting is required to be waterproof a cavity seal can be mounted additionally.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
mountedCavitySeal	CavitySealRole	0..1	Y	N	0..*	N	References the cavity seal that is used for the wire mounting.

referencedWireEnd	WireEnd	1..*	Y	N	0..*	N	References the wire ends that are used for the wire mounting. The minimum cardinality is one, because a wire mounting without wire end makes no sense. The maximum cardinality is * in order to support multi crimps.
wireEndAccessory	WireEndAccessoryRole	0..*	Y	N	0..*	N	NA
wireMountingDetail	WireMountingDetail	0..*	Y	N	1	Y	Specifies the WireMountingDetails, if a detailed description of the relationships between WireEnds and WireReceptions is needed.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
wireMounting	0..*	Y	ContactPoint	1	Specifies the WireMounting defined by ContactPoint. More than one WireMounting is allowed in order to support variance. In concrete configuration the WireMounting with all referenced elements present is used.

2.2.6 Class WireMountingDetail

With a WireMountingDetail it is possible to describe a detailed wire mounting.

This is needed if the information which wire end is mounted onto which wire reception is important (e.g. coax contacts).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
coreCrimpSize	Size	0..1	Defines the required core crimp size (width & height) of this specific wire mounting situation.
insulationCrimpSize	Size	0..1	Defines the required insulation crimp size (width & height) of this specific wire mounting situation.
wireTipProtrusion	ValueRange	0..1	Defines the valid length range in which the protrusion of the conductor from the conductor crimp must be located.
wireReceptionType	WireReceptionType	0..1	Defines the joining method to be applied for this <i>WireMountingDetail</i> . This is particularly relevant for wire receptions that support multiple joining methods (see <i>WireReceptionSpecification.WireReceptionType</i>).
absoluteSealPosition	NumericalValue	0..1	The <i>absoluteSealPosition</i> defines the distance between the conductor tip and edge of the <i>CavitySeal</i> facing the conductor tip.
pullOffForce	NumericalValue	0..1	The minimum force, of this specific wire mounting situation, that the completed crimp should withstand when the wire is being pulled off the terminal. Above that force, the crimp can be expected to be destroyed at any time. This value applies when both core & insulation crimp are functional.
corePullOffForce	NumericalValue	0..1	The minimum force, of this specific wire mounting situation, that the completed crimp should withstand when the wire is being pulled off the terminal. Above that force, the crimp can be expected to be destroyed at any time. This value applies when only the core crimp is present or functional (insulation crimp is either not present or disabled).

insulationPullOffForce	NumericalValue	0..1	The minimum force, of this specific wire mounting situation, that the completed crimp should withstand when the wire is being pulled off the terminal. Above that force, the crimp can be expected to be destroyed at any time. This value applies when only the insulation crimp is functional.
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Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
contactedWireReception	WireReceptionReference	1	Y	N	0..*	N	References the WireReception that is used for the WireMounting.
referencedWireEnd	WireEnd	1..*	Y	N	0..*	N	References the WireEnds that are mounted to referenced WireReception. A cardinality of more than one is allowed in order support parallel connectors, where multiple wire ends are placed on one side of the connector (one wire reception) and the other wire ends are placed on the other side of the connector (the other wire reception).

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
wireMountingDetail	0..*	Y	WireMounting	1	Specifies the WireMountingDetails, if a detailed description of the relationships between WireEnds and WireReceptions is needed.

2.3 Module core

2.3.1 Class AbstractLocalizedString

Abstract super-class for Localized text values.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	true

Attributes

Name	Type	Mult	Comment
languageCode	LanguageCode	1	References the corresponding languageCode of the value.
value	String	1	The value of the LocalizedString in language defined by the languageCode.

2.3.2 Class AliasIdentification

Class for the definition of alias identifications. Alias identifications are additional identifications for VEC-elements, which are valid in a certain scope (e.g. PPS-ID's).

General Information

Base Classifier	NA
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Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identificationValue	String	1	Specifies the identification value.
type	AliasIdentificationType	0..1	Defines the type (or the role) of the AliasIdentification. Defined literals are contained in an OpenEnumeration.
scope	String	0..1	The scope in which the AliasIdentification is valid / or the issuer of the alias id. This could be for example a certain process, a company or an IT-System.
description	AbstractLocalizedString	0..*	On optional human readable description of the AliasIdentification.

2.3.3 Class BaselineSpecification

A *BaselineSpecification* defines a set of *ItemVersions* (*Document-* and *PartVersions*) that relate to each other in a certain way e.g. all parts and documents in their specific versions that contributed to a specific manufactured result.

Baselines are a standard mechanism to support change, release and configuration management.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
state	BaselineState	0..1	Defines the state of the baseline itself (e.g. if it is finalized or work in progress).
content	BaselineContent	0..1	Defines the state of the content of the baseline in regard of its defined scope.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
validVersions	ItemVersion	0..*	Y	N	0..*	N	References the <i>ItemVersions</i> that are the content of the baseline.

2.3.4 Class ConfigurableElement

Abstract base class for all elements which can be configured with a VariantConfiguration.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
applicationConstraint	ApplicationConstraint	0..*	Y	N	NA	N	<u>Deprecated since 2.0.0:</u> <u>Reason: Replaced with ConfigurationConstraint.</u> References the application constraints that apply to the ConfigurableElement.
associatedAssignmentGroups	AssignmentGroup	0..*	Y	N	0..*	N	NA
configInfo	VariantConfiguration	0..1	Y	N	0..*	N	<u>Deprecated since 2.0.0:</u> <u>Reason: Replaced with ConfigurationConstraint.</u> References the configuration information that applies to the ConfigurableElement.
configurationConstraint	ConfigurationConstraint	0..*	Y	N	0..1	Y	<u>Deprecated since 2.0.2:</u> <u>Reason: The relationship has been inverted, allowing external configuration of ConfigurableElements. The ConfigurationConstraint is now contained in its own Specification.</u> Contains or more <i>ConfigurationConstraints</i> that apply to this <i>ConfigurableElement</i> . This is explained in more details in the recommendation chapter / model diagram "Variants".

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
constrainedElements	1..*	N	ConfigurationConstraint	0..*	Introduced with VEC V2.0.2. References the <i>ConfigurableElements</i> that are constrained by this <i>ConfigurationConstraint</i> . This association shall only be used in combination with a containment in a <i>ConfigurationConstraintSpecification</i> . When using the deprecated containment in the <i>ConfigurableElement</i> it shall not be used.
representedElement	1	N	SignalGraphNode	NA	References the <i>ConfigurableElement</i> that is represented by a <i>SignalGraphNode</i> . This is expected to be some physical element of the vehicle network, which participates in signal transmission. In many cases this will be an <i>OccurrenceOrUsage</i> , however in some cases sub elements of the <i>OccurrenceOrUsage</i> might be relevant, e.g. a <i>PinComponentReference</i> or a <i>WireElementReference</i> .

2.3.5 Class ConfigurationConstraint

Represents a *ConfigurationConstraint* ("When does this element exist?") for a specific application (*ApplicationConstraint*) and a "configuration universe" (*VariantConfiguration.configurationType*)

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA

Is Abstract	false
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Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the <i>ConfigurationConstraint</i> . The identification is guaranteed to be unique within the <i>ConfigurableElement</i> and does not change over the time.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
applicationConstraint	ApplicationConstraint	0..*	Y	N	NA	N	References the application constraint that applies to the <i>ConfigurationConstraint</i> .
configInfo	VariantConfiguration	0..1	Y	N	NA	N	References the configuration information that applies to the <i>ConfigurationConstraint</i> .
constrainedElements	ConfigurableElement	1..*	Y	N	0..*	N	Introduced with VEC V2.0.2. References the <i>ConfigurableElements</i> that are constrained by this <i>ConfigurationConstraint</i> . This association shall only be used in combination with a containment in a <i>ConfigurationConstraintSpecification</i> . When using the deprecated containment in the <i>ConfigurableElement</i> it shall not be used.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
configurationConstraint	0..*	Y	ConfigurableElement	0..1	Contains or more <i>ConfigurationConstraints</i> that apply to this <i>ConfigurableElement</i> . This is explained in more details in the recommendation chapter / model diagram "Variants".
configurationConstraint	1..*	Y	ConfigurationConstraintSpecification	0..1	Specifies the individual <i>ConfigurationConstraint</i> defined in the <i>ConfigurationConstraintSpecification</i> .

2.3.6 Class ConfigurationConstraintSpecification

Specification container for the definition of *ConfigurationConstraints*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

configurationConstraint	ConfigurationConstraint	1..*	Y	N	0..1	Y	Specifies the individual <i>ConfigurationConstraint</i> defined in the <i>ConfigurationConstraintSpecification</i> .
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2.3.7 Class DocumentClassification

A *DocumentClassification* is used to categorize and classify a document in the context of a document management system. It helps to maintain consistency and compliance by ensuring that documents are appropriately handled according to their classification. Classifications can be used for example to define retention periods or disposal dates. A classification is defined by a *Key* in a corresponding *ReferenceSystem*.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
key	String	1	The <i>Key</i> of the <i>DocumentClassification</i> in the corresponding classification reference system.
referenceSystem	String	1	The identification of the classification reference system, which is defining possible values and the semantic of <i>DocumentClassification</i> keys.
description	LocalizedString	0..*	On optional human readable description of the classification (e.g. the name or clear text, for example "max. 99 Years").

2.3.8 Class DocumentVersion

A *DocumentVersion* is a unique identifier for a piece of information in a company context. The *DocumentVersion* is one of the three anchors for PDM information in the VEC. All technical information about a *PartVersion* is contained in one or more documents. The documents are containing the actual content of the VEC since all *Specifications* are an element of a document.

General Information

Base Classifier	ItemVersion
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
documentNumber	String	1	The documentNumber is the major identifier of a <i>DocumentVersion</i> . The format is user defined and respectively company specific.
documentType	DocumentType	0..1	The type of the document, which is defined in an <i>OpenEnumeration</i> and gives a hint about the content of the document. Values for typical types of documents in the process are predefined (e.g. a part master document for the specification of a <i>PartVersion</i>). At later point, further constraint might be attached to <i>documentType</i> defining a minimum content for certain types of documents.
documentVersion	String	1	The documentVersion specifies the version index of a document (see also documentNumber).
digitalRepresentationIndex	String	0..1	An arbitrary change index that indicates if the digital representation (the content in VEC) of this <i>DocumentVersion</i> has been changed /

			regenerated. This can be for example an index, a timestamp or a checksum. This allows the detection of changes in the content, even when the DocumentNumber & DocumentVersion is the same. For a more detailed explanation in the context see "Parts & Documents". KBLFRM-837.
creatingSystem	String	0..1	The creatingSystem specifies the computer application or the machine which is used to create the document.
dataFormat	String	0..1	The dataFormat specifies the convention that was used to structure the information in the document. This is useful if the DocumentVersion is a pointer to an external document, which is not contained in the VEC or if the content of this DocumentVersion was automatically generated by the extraction of the information out of the original document.
fileName	String	0..1	The name of the file as it appears in the VEC-Package, including the folder structure (fully qualified name) that contains this <i>DocumentVersion</i> . If this DocumentVersion is a link to an external document (e.g. a ComponentSymbol), then the fileName attribute points to the file containing the original document. The usage of this attribute is only valid, if the original document is distributed along with the VEC-file in a VEC-Package. It must not point to any file location which is not part of the VEC-Package (e.g. a File on a central server file share). The <i>fileName</i> is relative to the VEC-Package root. It MUST NOT contain a drive or device letter, or a leading slash. All slashes MUST be forward slashes '/' (UNIX-style).
location	String	0..1	The location is a possibility to provide a reference to the source location of the <i>DocumentVersion</i> (e.g. a document management system or an archive system) where the original document can be found. The location shall be provided either as an URN or URL.
numberOfSheets	Integer	0..1	The number of sheets contained in the document.
confidentialityLevel	ConfidentialityLevel	0..1	The <i>ConfidentialityLevel</i> of this <i>DocumentVersion</i> .
documentClassification	DocumentClassification	0..*	Defines the <i>DocumentClassifications</i> of this <i>DocumentVersion</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
itemEquivalence	ItemEquivalence	0..*	Y	N	1	Y	Specifies ItemEquivalences defined by the DocumentVersion.
referencedPart	PartVersion	0..*	Y	N	0..*	N	The association is an informative link which PartVersions are described by the DocumentVersion.
relatedDocument	DocumentVersion	0..*	Y	N	0..*	N	The association is an informative link which DocumentVersion are related to each other (e.g. by derivation, A Harness-Drawing is related to a 3D-Model).
sheetOrChapter	SheetOrChapter	0..*	Y	N	1	Y	Specifies SheetOrChapters defined in this DocumentVersion. These are especially useful if the DocumentVersion represents an external reference.
specification	Specification	0..*	Y	N	0..1	Y	Specifies the Specifications contained in the DocumentVersion. All structured, technical information in the VEC is described by such Specifications.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
documentVersion	0..*	Y	VecContent	1	Specifies the DocumentVersions contained in the VEC-file.
documentVersion	1	N	RequirementsConformanceStatement	NA	References the <i>DocumentVersion</i> that contains the requirements to which a conformance statement shall be expressed.
mappedDocument	1	N	ExternalMappingSpecification	0..*	Reference to the <i>DocumentVersion</i> that represents the external data source that connected to the VEC content by the <i>ExternalMappingSpecification</i> .
referencedDocument	1	N	DocumentBasedInstruction	0..*	References the DocumentVersion that is used as an Instruction.
referencedExternalDocuments	0..*	N	ExtendableElement	0..*	This association allows all <i>ExtendableElements</i> in the VEC to reference <i>DocumentVersions</i> as "external reference". This association shall be used for the extension of elements in the VEC with information that cannot be represented in the VEC in an appropriate way but can be expressed in some external format (e.g. a specific symbol for a <i>ComponentNode</i>). <i>DocumentVersions</i> referenced by this association shall not contain any <i>Specifications</i>. This association is no replacement for associations with a more precise semantic like the <i>DocumentBasedInstruction</i> or the associations between <i>PartVersion</i> and <i>DocumentVersion</i>.
relatedDocument	0..*	N	DocumentVersion	0..*	The association is an informative link which DocumentVersion are related to each other (e.g. by derivation, A Harness-Drawing is related to a 3D-Model).
relatedDocumentVersion	1	N	DocumentRelatedAssignmentGroup	0..*	References the <i>DocumentVersion</i> to which this group relates.

2.3.9 Class ExtendableElement

Abstract base class for extendable elements. Extendable elements have the possibility to define non-standard custom properties. The *ExtendableElement* is also used as base class for all higher-level entities in the VEC (in contrast to a *StructuredPrimitive*). So, in general, referencing *ExtendableElement* should be possible.

General Information

Base Classifier	NA
Applied Stereotype	NA
Is Abstract	true

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
customProperty	CustomProperty	0..*	Y	N	1	Y	Specifies the CustomProperties of the ExtendableElement.

referencedExternal Documents	DocumentVersion	0..*	Y	N	0..*	N	This association allows all <i>ExtendableElements</i> in the VEC to reference <i>DocumentVersions</i> as "external reference". This association shall be used for the extension of elements in the VEC with information that cannot be represented in the VEC in an appropriate way but can be expressed in some external format (e.g. a specific symbol for a <i>ComponentNode</i>). <i>DocumentVersions</i> referenced by this association shall not contain any <i>Specifications</i>. This association is no replacement for associations with a more precise semantic like the <i>DocumentBasedInstruction</i> or the associations between <i>PartVersion</i> and <i>DocumentVersion</i>.
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Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
mappedElement	1	N	ExternalMapping	0..*	NA
template	1	N	Reusage	0..*	The <i>template</i> references a model element that is used as "source" of information.
usage	1	N	Reusage	0..*	The <i>usage</i> references a model element that is the reuse of the <i>template</i> . The types of elements referenced by <i>usage</i> and <i>template</i> shall be the same.

2.3.10 Class ItemVersion

Abstract super-class for physical objects (e.g. a Terminal), virtual objects (e.g. a 150% Harness) as well as documents (e.g. a wiring diagram). In difference to AP 212 the VEC makes it only possible to describe/exchange information about Versions since Master-Objects cannot exist without one or more Versions.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
abbreviation	LocalizedString	0..*	Room for a short name of the Item. In case of a document the attribute is wanted to contain its title.
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the ItemVersion. e.g. Buchsengehäuse 26-polig
companyName	String	1	Defines the publishing company of the ItemVersion. The companyName is part of the main identifier of an ItemVersion together with the corresponding number (partNumber or documentNumber) and version (partVersion or documentVersion).
processingInstruction	Instruction	0..*	Processing instructions for the application of the part or the document.
changeRestriction	ChangeRestrictionType	0..*	This attribute defines if the <i>ItemVersion</i> is subject to some kind of change restriction. For example, some parts or document might not be changeable anymore after a certain stage in the homologation

			process for a product. The <i>ChangeRestrictionType</i> defines the reason for the <i>changeRestriction</i> .
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Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
approval	Approval	0..*	Y	N	1	Y	Specifies the approval information of the <i>ItemVersion</i> .
changeDescription	ChangeDescription	0..*	Y	N	0..1	Y	Specifies the change history of the <i>ItemVersion</i> .
contract	Contract	0..*	Y	N	0..*	N	References the contracts that apply to an <i>ItemVersion</i> .
copyrightInformation	CopyrightInformation	0..1	Y	N	0..*	N	References the <i>CopyrightInformation</i> that is in effect for this <i>ItemVersion</i> . If no <i>CopyrightInformation</i> is referenced by the <i>ItemVersion</i> , the <i>CopyrightInformation</i> that is referenced by the <i>VecContent</i> (if defined) shall be considered as in effect for this <i>ItemVersion</i> .
creation	Creation	0..1	Y	N	1	Y	Specifies the information about the creation of the <i>ItemVersion</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
item	2..*	N	ItemEquivalence	0..*	References all <i>ItemVersion</i> that are considered to be equivalent by the <i>ItemEquivalence</i> . A single <i>ItemEquivalence</i> shall only reference <i>ItemVersions</i> of the same class (either <i>DocumentVersions</i> or <i>PartVersions</i>).
predecessorVersion	1	N	ItemHistoryEntry	0..*	References the <i>ItemVersion</i> that is the predecessor in the <i>ItemHistoryEntry</i> .
successorVersion	1	N	ItemHistoryEntry	0..*	References the <i>ItemVersion</i> that is the successor in the <i>ItemHistoryEntry</i> .
validVersions	0..*	N	BaselineSpecification	0..*	References the <i>ItemVersions</i> that are the content of the baseline.

2.3.11 Class LocalizedString

Allows the internationalization of text contents. Attributes of the type *LocalizedString* normally have the multiplicity [0..*]. This means that such an attribute can have multiple values for different locales. It must not have multiple values for the same locale.

General Information

Base Classifier	AbstractLocalizedString
Applied Stereotype	StructuredPrimitive
Is Abstract	false

2.3.12 Class LocalizedTypedString

Allows the internationalization of text contents in a typed way. Attributes of the type *LocalizedTypedString* normally have the multiplicity [0..*]. This means that such an attribute can have multiple values for different locales and types. It must not have multiple values for the same locale and type.

General Information

Base Classifier	AbstractLocalizedString
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	LocalizedStringType	1	Defines the <i>type</i> of the <i>LocalizedString</i> . This allows the definition of a more detailed semantic than the semantic of the attribute itself with the type <i>LocalizedString</i> . Agreed type values are defined in an OpenEnumeration.

2.3.13 Class PartOrUsageRelatedSpecification

Base class for all specifications which are describing a *PartVersion* or a *PartUsage*. A *PartOrUsageRelatedSpecification* specifies a certain aspect of the described part or usage (e.g. general technical part information, connector housing aspects or wire aspects).

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
specialPartType	String	0..1	The specialPartType allows the specification of subclassifications for a PartOrUsageRelatedSpecification (e.g. different types of connector housings).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
describedPart	PartVersion	0..*	Y	N	0..*	N	References the PartVersion(s) to which the information defined in this specification applies. Example: If the PartOrUsageRelatedSpecification is a GeneralTechnicalPartSpecification and it defines that the color is "green" then all PartVersion referenced by this association are "green".

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
partOrUsageRelatedSpecification	0..*	N	PartUsage	0..*	References the PartOrUsageRelatedSpecification(s) that describe the PartOrUsageRelatedSpecification. KBLFRM-399

specification	1	N	SpecificRole	0..*	References the <i>PartOrUsageRelatedSpecification</i> that is instantiated by this <i>SpecificRole</i> .
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2.3.14 Class PartSubstitutionSpecification

A *PartSubstitutionSpecification* defines a set of *PartVersions* that can be used alternatively, due to an incomplete specification for the 150% product description. For a concrete wiring harness only on valid *PartVersion* remains. The selection logic for valid *PartVersions* is not included in the VEC.

A *PartSubstitutionSpecification* can be used for example tubes or ring terminals, where a part of the specification is known at design time, but not yet the complete specification. For tubes for example the tube diameter is not known at design time, since it depends on the bundle diameter of a specific configuration.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
alternativePartVersions	PartVersion	0..*	Y	N	NA	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
partSubstitution	0..1	N	PartUsage	NA	NA

2.3.15 Class PartVersion

A *PartVersion* is unique identifier for a part in a company context. A part can be any components within a vehicle. The *PartVersion* is one of the three anchors for PDM information in the VEC. All technical information about a *PartVersion* is contained in one or more documents. These describing elements are normally referencing to the *PartVersion*.

General Information

Base Classifier	ItemVersion
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
partNumber	String	1	The partNumber is the major identifier of a PartVersion. The format is user defined and respectively company specific. For all VEC-documents a PartVersion-instance can be trusted to be identical if the combination of partNumber, partVersion and companyName is identical.
partVersion	String	1	The partVersion specifies the version index of a part (see also partNumber).
primaryPartType	PrimaryPartType	1	The primary type of the part defines the type of the part (e.g. <i>ConnectorHousing</i> , <i>Fixing</i> , etc.) Since the VEC supports dual use

			parts (e.g. <i>Fixing</i> & <i>WireProtection</i>) the primary part type is necessary to define which specification associated to part is the primary character of the part. Therefore, all primary part types correspond to a <i>PartOrUsageRelatedSpecification</i> and a <i>Role</i> (e.g. <i>ConnectorHousing</i> --> <i>ConnectorHousingSpecification</i> --> <i>ConnectorHousingRole</i>).
isPreferredPart	String	0..1	Flags a part as "preferred" by the means of being a preferred part out of a group of parts with identical technical properties. The preferred part should be used, if the other properties of a couple of parts do not allow a distinct decision. Even though the name suggests a Boolean value, this attribute has the type "string". The way in which preference is expressed varies greatly between the process and usually it is a multi-value flag and not just a yes/no answer.
preferredUseCase	LocalizedString	0..*	Contains a part description from the point of view of the preferred use case. Defines the function for which the part was initially designed. (e.g. "Grommet for Hatch", "...passenger compartment"). This can be an important information for searching and selecting parts.
aliasId	AliasIdentification	0..*	Room to specify additional identifiers for the PartVersion. This field must not be used for alternative PartNumbers. It is intended for identifiers others than PartNumbers, such as human readable identifiers printed on the part e.g. a number of fuse or a relay. Therefore, it does not have to be strictly unique.
nature	PartNature	0..1	The <i>nature</i> specifies how the <i>PartVersion</i> can be used in the different processes or the significance of the <i>PartVersion</i> in the process. The <i>nature</i> of a <i>PartVersion</i> is normally inherent and does not change. If the <i>nature</i> the underlying part changes a new (other) <i>partNumber</i> is assigned to the part and respectively a new <i>PartVersion</i> is created. It used to differentiate for example normal (productive) part numbers from preliminary (prototypic) part numbers.
partNumberType	PartNumberType	0..1	Specifies the classification of the part number associated with this <i>PartVersion</i> , indicating its semantic role within the BOM methodology of the issuing process (see <i>PartNumberType</i> for more details).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
project	Project	0..1	Y	N	0..*	N	References the project that develops the PartVersion.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
A	1	N	Mapping	0..*	NA
accessoryPart	1..*	N	PartRelation	0..*	References the PartVersions that are related by the PartRelation.
alternativePartVersions	0..*	N	PartSubstitutionSpecification	NA	NA
B	1	N	Mapping	0..*	NA

constrainedParts	0..*	N	UsageConstraintSpecification	0..*	References the <i>PartVersions</i> to which this <i>UsageConstraintSpecification</i> applies.
describedPart	0..*	N	PartOrUsageRelatedSpecification	0..*	References the <i>PartVersion</i> (s) to which the information defined in this specification applies. Example: If the <i>PartOrUsageRelatedSpecification</i> is a <i>GeneralTechnicalPartSpecification</i> and it defines that the color is "green" then all <i>PartVersion</i> referenced by this association are "green".
firstTerminal	1	N	TerminalPairing	0..*	References the first terminal of the <i>TerminalPairing</i> .
part	0..1	N	PartOccurrence	0..*	References the <i>PartVersion</i> that is instantiated by this <i>PartOccurrence</i> .
partVersion	0..*	Y	VecContent	1	Specifies the <i>PartVersions</i> contained in the VEC-file.
referencedPart	0..*	N	DocumentVersion	0..*	The association is an informative link which <i>PartVersions</i> are described by the <i>DocumentVersion</i> .
referencedPart	0..*	N	SheetOrChapter	0..*	The association is an informative link which <i>PartVersions</i> are described by the <i>SheetOrChapter</i> .
secondTerminal	1	N	TerminalPairing	0..*	References the second terminal of the <i>TerminalPairing</i> (first and second does not imply any specific order).
validForMatingTerminal	0..*	N	TerminalCurrentInformation	0..*	<i>PartVersions</i> of mating terminals for which this <i>TerminalCurrentInformation</i> is valid. This is a relevant information, because the thermal absorption of the counterpart is a relevant influence factor for possible current.

2.3.16 Class ResourceVersion

A *ResourceVersion* is unique identifier for a resource in a company context. Resources are elements used to produce, handle or service a harness, e.g. tools or machines, which do not end up in the product (see *PartVersion*). The *ResourceVersion* is one of the three anchors for PDM information in the VEC.

General Information

Base Classifier	ItemVersion
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
resourceNumber	String	1	The <i>ResourceNumber</i> is the major identifier of a <i>ResourceVersion</i> . The format is user defined and respectively company specific. For all VEC-documents a <i>ResourceVersion-instance</i> can be trusted to be identical if the combination of <i>ResourceNumber</i> , <i>ResourceVersion</i> and <i>CompanyName</i> is identical.
resourceVersion	String	1	The <i>ResourceVersion</i> specifies the version index of a resource (see also <i>ResourceNumber</i>).
primaryResourceType	PrimaryResourceType	1	The primary type of the resource defines the type of the resource. In a future version of the VEC the <i>PrimaryResourceType</i> might correspond to <i>Specifications</i> describing the <i>ResourceVersion</i> (see <i>PrimaryPartType</i>).

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment

extractionTool	0..*	N	PluggableTerminal Specification	0..*	NA
machineCrimpTool	0..*	N	CrimpDetail	0..*	A list of tools that are permitted for machine crimping processes.
manualCrimpTool	0..*	N	CrimpDetail	0..*	A list of tools that are permitted for manual crimping processes.
resourceVersion	0..*	Y	VecContent	1	NA

2.3.17 Class RoutableElement

A RoutableElement is an element that can be routed, which mean it is possible to assign it to a Path in the Topology.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	true

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
routedElement	1	N	Routing	0..*	Specifies the Element that is routed.

2.3.18 Class SheetOrChapter

Documents can be structured into sheets or chapters. Since it is possible, that one document describes a couple of parts it is necessary to be able to specify which sheet, or chapter contains the description of which part. (see KBLFRM-308)

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	The identification of the sheet or chapter within the document. This value must be distinct within the context of a document.
sheetNumber	String	0..1	The sheetNumber is the major identifier of a SheetOrChapter. The format is user defined and respectively company specific. This field has to be used if a SheetOrChapter has its own "DocumentNumber".
sheetVersion	String	0..1	The sheetVersion specifies the version index of a sheet (see also sheetNumber)
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the SheetOrChapter.
sheetFormat	String	0..1	Defines the format of the SheetOrChapter. This is an alias for a specific sheet / paper size (e.g. DIN A4). If both attributes are defined the <i>SheetSize</i> takes precedence.
sheetSize	Size	0..1	Defines the size of the sheet with numerical values. This is the size of the "paper", the actual drawing area might be smaller for example due to drawing frames.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
changeDescription	ChangeDescription	0..*	Y	N	0..1	Y	Specifies the change history of the SheetOrChapter.
referencedPart	PartVersion	0..*	Y	N	0..*	N	The association is an informative link which PartVersions are described by the SheetOrChapter.
specification	Specification	0..*	Y	N	0..1	Y	Specifies the Specifications contained in the SheetOrChapter. All structured, technical information in the VEC is described by such Specifications.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
referencedSheetOrChapter	0..1	N	DocumentBasedInstruction	0..*	References the SheetOrChapter that is used as an Instruction.
relatedSheetOrChapter	0..1	N	DocumentRelatedAssignmentGroup	0..*	Allows a more specific relationship to a <i>SheetOrChapter</i> within the <i>relatedDocumentVersion</i> .
sheetOrChapter	0..*	Y	DocumentVersion	1	Specifies SheetOrChapters defined in this DocumentVersion. These are especially useful if the DocumentVersion represents an external reference.

2.3.19 Class Specification

Abstract super-class for all specifications. Every technical information exchanged with the VEC is contained in the different specializations of a specification.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the specification. The identification is guaranteed to be unique within the document containing the specification. For all VEC-documents a Specification-instance can be trusted to be identical if the DocumentVersion-instance is the same (see DocumentVersion) and the identification of the Specification is the same.
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the specification.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
specification	0..*	Y	SheetOrChapter	0..1	Specifies the Specifications contained in the SheetOrChapter. All structured, technical

					information in the VEC is described by such Specifications.
specification	0..*	Y	DocumentVersion	0..1	Specifies the Specifications contained in the DocumentVersion. All structured, technical information in the VEC is described by such Specifications.

2.3.20 Class VecContent

The VecContent is the XML-Root node for any VEC-Document.

Note about *GeneratingSystemName* and *GeneratingSystemVersion*: Both together are intended to provide meta information to identify the generating system and its version for reasons of traceability (system is used as general term and can also be a system landscape, a highly modularized micro service architecture or a tool chain). It is not intended to provide detailed information about the composition or configuration of a system and its modules. Especially in a scenario where a system is highly modularized (e.g. a micro service architecture) any detailed information would be subject to very individual interpretation.

Therefore, the VEC provides only the means to identify a specific system configuration, not to describe it. The *GeneratingSystemName* allows identification in space (who did it), and the *GeneratingSystemVersion* provides identification in time (when was it done, with which version of the configuration). The generator of the dataset is responsible to provide meaningful identifications for both, that satisfy the traceability requirements in the context.

The following scenarios are explicitly allowed:

1. Single System: Single identifiers are used which are mapping directly to a system / system version (e.g. "Component DB" / "V7.1.0")
2. Combined System: The identifiers are a combination of the participating modules (e.g. "Design System X+Userware ACME+DBExport" / "V3+V2.7+V1.2.3").
3. Baseline / System Configuration: The identifiers are pointing to a baseline in the configuration management of a system landscape. The systems and their versions are tracked there, outside the VEC (e.g. "Harness Tool Chain" / "V12-2023-5").

General Information

Base Classifier	ExtendableElement
Applied Stereotype	xml-root
Is Abstract	false

Attributes

Name	Type	Mult	Comment
vecVersion	VecVersion	1	Specifies the version of the VEC used for the file.
generatingSystemName	String	0..1	Specifies the name of the system that has generated the VEC-file (see note in class documentation).
dateOfCreation	Date	0..1	Specifies the date of creation of the VEC-file.
generatingSystemVersion	String	0..1	Specifies the version of the system that has generated the VEC-file (see note in class documentation).

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
contract	Contract	0..*	Y	N	1	Y	Specifies the contracts used in the VEC-file.
copyrightInformation	CopyrightInformation	0..*	Y	N	1	Y	Specifies the CopyrightInformation used in the VEC-file.
documentVersion	DocumentVersion	0..*	Y	N	1	Y	Specifies the DocumentVersions contained in the VEC-file.
itemHistoryEntry	ItemHistoryEntry	0..*	Y	N	1	Y	Specifies the ItemVersionHistoryEntries for ItemVersions contained in the VEC-file.
partVersion	PartVersion	0..*	Y	N	1	Y	Specifies the PartVersions contained in the VEC-file.
project	Project	0..*	Y	N	1	Y	Specifies the Projects used in the VEC-file.
quantityKind	QuantityKind	0..*	Y	N	1	Y	Specifies the <i>QuantityKind</i> used in the VEC-file.
resourceVersion	ResourceVersion	0..*	Y	N	1	Y	NA
standardCopyrightInformation	CopyrightInformation	0..1	Y	N	0..*	N	References the <i>CopyrightInformation</i> that is in effect for the complete content of this <i>VecContent</i> . It is applied to all <i>ItemVersions</i> that do not references their own individual <i>CopyrightInformation</i> .
unit	Unit	0..*	Y	N	1	Y	Specifies the Units used in the VEC-file.

2.3.21 Enumeration AliasIdentificationType

NA

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
UUID	Defines that this AliasIdentification represents a "Universally Unique Identifier". Although a UUID is technical definition of a 128-Bit number, the primary relevance of this type is not its technical representation, but it's other properties. An AliasIdentification with the type "UUID" of an element is its unique identifier, which is constant over time, never changes and is never reused for other elements. Such an AliasIdentification can be used to trace elements through different systems, companies and processes.
DiagnosticAddress	A <i>DiagnosticAddress</i> identifies a vehicle network participant during diagnosis. This is normally used on <i>NetworkPorts</i> , <i>ComponentPorts</i> and <i>PinComponentBehaviours</i> .
SoftwareAddress	A <i>SoftwareAddress</i> identifies a vehicle network participant during regular operation. This is normally used on <i>NetworkPorts</i> , <i>ComponentPorts</i> and <i>PinComponentBehaviours</i> .

2.3.22 Enumeration BaselineContent

Enumeration to define the state of the content of the baseline in regard of its defined scope.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
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Partial	A partial baseline does not contain all elements of the final scope of the baseline (e.g. it is a baseline created during the development process).
Complete	A complete baseline contains <u>all</u> elements of the final scope of the baseline.

2.3.23 Enumeration BaselineState

Enumeration the define the valid states of a baseline.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Draft	Draft means that the baseline is not finalized yet and new ItemVersions can be added without the necessity to create a new version of the baseline itself.
Frozen	Frozen means that the baseline is finalized and new ItemVersions <u>must not</u> be added without creating a new version of the baseline itself.

2.3.24 Enumeration ChangeRestrictionType

The *ChangeRestrictionType* defines the reason for a *changeRestriction* (see *ItemVersion.changeRestriction* for more details).

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Legal	The <i>ItemVersion</i> falls under legal and other regulatory aspects (e.g. it defines type approval characteristics).
Safety	The <i>ItemVersion</i> influences safety critical elements, and their changes are underlying specific restrictions.
Function	A change of the <i>ItemVersion</i> might create a heavy impact on the function of the product, e.g. because of critical interdependencies like a bearing fit.

2.3.25 Enumeration ConfidentialityLevel

Defines different levels of confidentiality for *DocumentVersions*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Public	NA
Internal	NA
Confidential	NA
Secret	NA

2.3.26 Enumeration DocumentType

Defines the predefined *DocumentTypes* of a *PartVersion*. A certain *DocumentType* has normally a typical set of information that is defined within its scope. E.g. a part master document contains *Specifications* that are used for the description of a defined *PartVersion*.

The content and the degree of information a DocumentType may vary in the different processes.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
BaselineDefinition	The <i>DocumentVersion</i> represents the definition of a baseline (see <i>BaselineSpecification</i>).
PartMaster	Documents of this type describe the master data of a part / component (e.g. a <i>ConnectorHousing</i> , a <i>Terminal</i> , a <i>Wire</i>).
PartMasterTemplate	Documents of this type describe reusable snippets of component master data (e.g. a standard cavities or wire receptions, wires without color or interface definition for E/E components).
NetworkArchitecture	
SystemSchematic	
SchematicProject	The <i>SchematicProject</i> is a bracket around multiple <i>SystemSchematics</i> which are by itself often versioned documents.
WiringDescription	
HarnessDescription	Documents of this type contain the description of a <i>Harness</i> (which is for example the scope of the KBL).
InstallationDescription	
ManufacturingDescription	
RequirementsDescription	A requirements description is some kind of document that contains requirements for a certain part or a group of parts, their design or their properties. A requirements description can be a requirements specification (e.g. REQ-IF) or any other document containing requirements (e.g. a norm).
ChangeDescription	
GraphicsSymbol	Documents of this type represent a graphical symbol (2D), which is normally an external file (e.g. an SVG, PNG). The use case for this document type are symbols that are used in 2D harness and form board drawings.
GeometryModel	Documents of this type represent a geometry model (3D) of a component or part, normally as a reference to an external file (e.g. a JT File or some native file of a 3D modelling tool). <i>GeometryModels</i> are original CAD Models and not simplified. Typical use cases are for example the models of components (e.g. connectors) used in a 3D model or the 3D visualization of a Harness.
GeometryModelSimplified	Documents of this type represent a geometry model (3D) of a component or part, normally as a reference to an external file (e.g. a JT File or some native file of a 3D modelling tool). Simplified means, that those models contain only details necessary for representation in wiring harness and not every detail (e.g. for production of the component itself). For example, a simplified connector model might not contain cavity details. However, a simplified model could contain additional (meta-)information necessary in the harness development process (e.g. segment connection points).
GeometryModelOccurrence	Documents of this type represent a geometry model (3D) of a component or part, normally as a reference to an external file (e.g. a JT File or some native file of a 3D modelling tool). "Occurrence" means, that the model represents the component in a specific usage situation, meaning in a specific shape and position (e.g. a deformed grommet).

GeometryTemplate	A <i>GeometryTemplate</i> is a <i>GeometryModelSimplified</i> who's actual appearance can be modified with a set of parameters, e.g. a tube can have a variable length, diameter and deformation. A specific set of parameters is defined in <i>GeometryParameters</i> .
GeometryParameters	An actual set of parameters for a <i>GeometryTemplate</i> .
MasterDataDefinition	Documents of this type contain master data definitions e.g. list valid <i>UsageNodes</i> or list valid <i>Signals</i> .
HarnessCoupling	<i>DocumentVersions</i> of this type define the coupling information of wiring harnesses in a vehicle network. The <i>DocumentVersion</i> contains the necessary <i>CouplingSpecifications</i> and the <i>PartUsages</i> for the coupling devices.
SchematicSymbol	Documents of this type represent a graphical symbol (2D), which is normally an external file (e.g. an SVG, PNG). The use case for this document type are symbols that are used in schematic diagrams.
NetworkSymbol	Documents of this type represent a graphical symbol (2D), which is normally an external file (e.g. an SVG, PNG). The use case for this document type are symbols that are used in network / architecture diagrams.
DeviationTable	Documents that define allowed deviations from specified part numbers.
ComponentDrawing	<i>DocumentVersion</i> of this type represent the drawing of a component (Deutsch: Einzelteilezeichnung). Those <i>DocumentVersion</i> are normally used as external reference to the document containing the graphical representation of the component and do not contain <i>Specifications</i> . <i>DocumentVersion</i> describing a component in terms of the VEC (with <i>Specifications</i>) shall have the <i>DocumentType PartMaster</i> .
ProcessingInstruction	Documents of this type specify the processing / handling of a part during production (e.g. a crimp specification for terminals).
MatingInterface	Documents of this type are a specification of the geometrical interface for a connector family. All connectors satisfying this interface are plugging compatible with the corresponding component connector, provided that a matching coding is selected.

2.3.27 Enumeration LanguageCode

Enumeration for the definition of ISO language codes. (see KBLFRM-317)

General Information

Applied Stereotype	ClosedEnumeration
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Enumeration Literals

Name	Comment
Aa	Afar
Ab	Abkhazian
Af	NA
Am	NA
Ar	NA
As	NA
Ay	NA
Az	NA
Ba	NA
Be	NA
Bg	NA
Bh	NA

Bi	NA
Bn	NA
Bo	NA
Br	NA
Ca	NA
Co	NA
Cs	NA
Cy	NA
Da	NA
De	NA
Dz	NA
El	NA
En	NA
Eo	NA
Es	NA
Et	NA
Eu	NA
Fa	NA
Fi	NA
Fj	NA
Fo	NA
Fr	NA
Fy	NA
Ga	NA
Gd	NA
Gl	NA
Gn	NA
Gu	NA
Ha	NA
Hi	NA
He	NA
Hr	NA
Hu	NA
Hy	NA
Ia	NA
Id	NA
Ie	NA

Ik	NA
In	NA
Is	NA
It	NA
Iu	NA
Iw	NA
Ja	NA
Ji	NA
Jw	NA
Ka	NA
Kk	NA
Kl	NA
Km	NA
Kn	NA
Ko	NA
Ks	NA
Ku	NA
Ky	NA
La	NA
Ln	NA
Lo	NA
Lt	NA
Lv	NA
Mg	NA
Mi	NA
Mk	NA
MI	NA
Mn	NA
Mo	NA
Mr	NA
Ms	NA
Mt	NA
My	NA
Na	NA
Ne	NA
NI	NA
No	NA

Oc	NA
Om	NA
Or	NA
Pa	NA
Pl	NA
Ps	NA
Pt	NA
Qu	NA
Rm	NA
Rn	NA
Ro	NA
Ru	NA
Rw	NA
Sa	NA
Sd	NA
Sg	NA
Sh	NA
Si	NA
Sk	NA
Sl	NA
Sm	NA
Sn	NA
So	NA
Sq	NA
Sr	NA
Ss	NA
St	NA
Su	NA
Sv	NA
Sw	NA
Ta	NA
Te	NA
Tg	NA
Th	NA
Zi	NA
Tk	NA
Tl	NA

Tn	NA
To	NA
Tr	NA
Ts	NA
Tt	NA
Tw	NA
Ug	NA
Uk	NA
Ur	NA
Uz	NA
Vi	NA
Vo	NA
Wo	NA
Xh	NA
Yi	NA
Yo	NA
Za	NA
Zh	Chinese
Zu	Zulu

2.3.28 Enumeration LocalizedTypedStringType

NA

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Title	The LocalizedTypedString represents a Title in accordance with ISO 7200.
SupplementaryTitle	The LocalizedTypedString represents a SupplementaryTitle in accordance with ISO 7200.
MainDescription	The primary description of an element, possibly inherited from a parent element (e.g. a <i>ComponentNode</i> might inherit its description from the assigned <i>UsageNode</i>).
AdditionalDescription	An additional description that is an extension or a detailing of the <i>MainDescription</i> .

2.3.29 Enumeration PartNature

OpenEnumeration that defines the nature of a *PartVersion*. The nature specifies how the *PartVersion* can be used in the different processes or the significance of the *PartVersion* in the process.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Preliminary	<i>Preliminary PartVersions</i> represent part numbers which are used as place holders for parts that are not yet developed, or which are representing prototype parts that are not allowed in serial production.
Productive	<i>Productive PartVersions</i> represent regular part numbers that are used in serial production.
CustomerOrder	<i>CustomerOrder PartVersions</i> represent part numbers that are <u>not</u> regularly used in serial production. They are only used to fulfil special customer orders (e.g. an adapter connector for equipment of a special purpose vehicle).

2.3.30 Enumeration PartNumberType

Specifies the classification of a part number associated, indicating its semantic role within the BOM methodology of the issuing process. This enables the distinction between different types of part numbers, such as those referencing actual physical parts and those used to represent virtual constructs or implementation logic. In some cases, positions in the bill of material are created to express specific aspects of the product structure, even though they do not (directly) refer to a concrete part.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
RealPart	The part number identifies a physically existing part. It uniquely refers to a concrete, realizable item in the finished product.
VirtualPart	The part number identifies a virtual part, which is used for structuring or modeling purposes. It does not refer to a physically existing component in the final product.
TablePart	The part number identifies a table or list of possible part numbers. A specific real part must be selected from this table during subsequent steps of the development or configuration process (e.g. an appropriate ring terminal for a specific configuration of a harness).
ImplementationRegulation	The part number refers to a rule or instruction that governs how a part or system must be implemented. It does not directly identify a physical or virtual part but rather describes a processing step, constraints or requirements (e.g. an ultra-sonic splice).

2.3.31 Enumeration PrimaryPartType

The primary type of the part defines the type of the part (e.g. ConnectorHousing, Fixing, etc.) Since the VEC supports dual use parts (e.g. Fixing & WireProtection) the primary part type is necessary to define which specification associated to part is the primary character of the part. Therefore, all primary part types correspond to a PartOrUsageRelatedSpecification (e.g. ConnectorHousing --> ConnectorHousingSpecification).

The primary part type '*Other*' is used if the PartVersion is not further specified by the VEC, which means it has no PartOrUsageRelatedSpecification, only a GeneralTechnicalPartSpecification or a direct instance of PartOrUsageRelatedSpecification.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Antenna	NA
Battery	NA
BoltMountedFixing	NA

BoltTerminal	NA
BridgeTerminal	NA
CableDuct	NA
CableSeal	NA
CableSealPlug	NA
CableTie	NA
Capacitor	NA
CavityAccessory	NA
CavityPlug	NA
CavitySeal	NA
Coating	NA
ConnectorHousing	NA
ConnectorHousingCap	NA
ConnectorHousingCover	NA
CorrugatedPipe	NA
Diode	NA
EdgeMountedFixing	NA
EEDComponent	NA
Ferrite	NA
Fitting	NA
Fixing	NA
Fuse	NA
Grommet	NA
HoleMountedFixing	NA
MultiCavityPlug	NA
MultiCavitySeal	NA
MultiFuse	NA
Other	The <i>PrimaryPartType</i> "Other" is used for parts that are described by a direct instance of <i>PartOrUsageRelatedSpecification</i> . These are parts that do not have a specific classification in the VEC and can be described with a <i>PartOrUsageRelatedSpecification</i> and <i>CustomProperties</i> . The corresponding <i>Role</i> is the <i>SpecificRole</i> .
OpenWireEndTerminal	NA
OpenWireEnd	<u>Deprecated since 2.1.0:</u> <u>Reason: The literal is violating VEC conventions. OpenWireEndTerminal shall be used.</u> NA
PartStructure	The <i>PrimaryPartType</i> "PartStructure" has an inconsistency with VEC conventions for historical reasons, which is kept for backwards compatibility. The corresponding <i>PartOrUsageRelatedSpecification</i> is the <i>PartStructureSpecification</i> . However, the corresponding <i>Role</i> is the <i>PartWithSubComponentsRole</i> .
PluggableTerminal	NA

PotentialDistributor	NA
Relay	NA
RingTerminal	NA
ShrinkableTube	NA
SpliceTerminal	NA
Stripe	NA
Tape	NA
Terminal	NA
TerminalProtection	NA
Tube	NA
Wire	NA
WireEndAccessory	NA
WireProtection	NA

2.3.32 Enumeration PrimaryResourceType

The primary type of the resource defines the type of the resource. In a future version of the VEC the *PrimaryResourceType* might correspond to *Specifications* describing the *ResourceVersion* (see *PrimaryPartType*).

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
CrimpTool	NA
ExtractionTool	A tool for safely extracting a terminal from a cavity without causing any damage.

2.3.33 Enumeration VecVersion

NA

General Information

Applied Stereotype	ClosedEnumeration
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Enumeration Literals

Name	Comment
1.1.1	NA
1.1.2	NA
1.1.3	NA
1.2.0	NA
1.2.1	NA
1.2.2	NA
2.0.0	NA

2.0.1		NA
2.0.2		NA
2.1.0		NA
2.2.0		NA

2.4 Module coupling

2.4.1 Class CavityCoupling

A *CavityCoupling* defines the mapping between two cavities of the *ConnectorHousingRoles* associated with the *CouplingPoint*.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	NA

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
firstCavity	CavityReference	1	Y	N	0..*	N	NA
secondCavity	CavityReference	1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavityCoupling	0..*	Y	SlotCoupling	1	NA

2.4.2 Class CouplingPoint

A *CouplingPoint* defines a single coupling. If a coupling takes place, all sub elements are connected. If the coupling is disconnected, all sub elements are disconnected.

If a coupling occurs between two connectors, and not just between two terminals, the *CouplingPoint* references the respective *ConnectorHousingRoles*.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	NA

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
firstConnector	ConnectorHousingRole	0..1	Y	N	0..*	N	NA
matingPoint	MatingPoint	0..*	Y	N	1	Y	Specifies the <i>MatingPoints</i> defined within the <i>CouplingPoint</i> .
secondConnector	ConnectorHousingRole	0..1	Y	N	0..*	N	NA
slotCoupling	SlotCoupling	0..*	Y	N	1	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
couplingPoint	0..*	Y	CouplingSpecification	1	NA

2.4.3 Class CouplingSpecification

Specification for the description of a Coupling. A coupling allows the mapping between independent harness sections or EECComponents of the electrical system. This is done by the mapping of either-or both, ConnectorHousingRoles & TerminalRoles of one side to the other.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
couplingPoint	CouplingPoint	0..*	Y	N	1	Y	NA

2.4.4 Class MatingDetail

If the mating of the two terminals is not unambiguously, a MatingDetail can specify the TerminalReceptions that are mated.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA

Is Abstract	false
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Attributes

Name	Type	Mult	Comment
identification	String	1	NA

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connection	Connection	0..*	Y	N	0..*	N	References the <i>Connection</i> that is realized by this <i>MatingPointDetail</i> . For example, when a connection is realized by directly plugging or screwing two E/E components together. The definition at level of the <i>MatingDetail</i> might be required if the <i>TerminalRole</i> of the <i>MatingPoint</i> carries multiple different potentials (e.g. Coax).
firstTerminalReception	TerminalReceptionReference	1	Y	N	0..*	N	References the first terminal reception that is mated.
secondTerminalReception	TerminalReceptionReference	1	Y	N	0..*	N	References the second terminal reception that is mated.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
matingDetail	0..*	Y	MatingPoint	1	Specifies the MatingDetails, if a detailed description of the relationships between TerminalReceptions and TerminalReceptions is needed.

2.4.5 Class MatingPoint

A MatingPoint defines the Mating of two terminals. This normally occurs when two inliners are connected. Then terminals of one side (female) are mated with terminals of the other side (male).

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the <i>MatingPoint</i> . The identification is guaranteed to be unique within a <i>CouplingPoint</i> .

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connection	Connection	0..*	Y	N	0..*	N	References the <i>Connection</i> that is realized by this <i>MatingPoint</i> . For example, when a connection is realized by directly plugging or screwing two E/E components together.
firstTerminalRole	TerminalRole	1	Y	N	0..*	N	References the first terminal that is mated.
matingDetail	MatingDetail	0..*	Y	N	1	Y	Specifies the MatingDetails, if a detailed description of the relationships between TerminalReceptions and TerminalReceptions is needed.
secondTerminalRole	TerminalRole	1	Y	N	0..*	N	References the second terminal that is mated.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
matingPoint	0..*	Y	CouplingPoint	1	Specifies the <i>MatingPoints</i> defined within the <i>CouplingPoint</i> .

2.4.6 Class SlotCoupling

A *SlotCoupling* defines the mapping between two slots of the *ConnectorHousingRoles* associated with the *CouplingPoint*.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	NA

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavityCoupling	CavityCoupling	0..*	Y	N	1	Y	NA
firstSlot	AbstractSlotReference	1	Y	N	0..*	N	NA
secondSlot	AbstractSlotReference	1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
slotCoupling	0..*	Y	CouplingPoint	1	NA

2.5 Module custom_properties

2.5.1 Class BooleanValueProperty

A custom property with a Boolean value.

General Information

Base Classifier	CustomProperty
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
value	Boolean	1	Defines the value of the CustomProperty.

2.5.2 Class ComplexProperty

A custom property that represents a tuple of values.

General Information

Base Classifier	CustomProperty
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
customProperty	CustomProperty	0..*	Y	N	1	Y	The customProperties that represent the individual values of the complex property.

2.5.3 Class CustomProperty

Abstract base class for custom properties. Basically, a custom property is key / value pair. The key (propertyType) defines the meaning of the value. A custom property can either be a simple value (string), a numerical value or a value range.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	true

Attributes

Name	Type	Mult	Comment
propertyType	String	1	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment

customProperty	0..*	Y	ComplexProperty	1	The customProperties that represent the individual values of the complex property.
customProperty	0..*	Y	ExtendableElement	1	Specifies the CustomProperties of the ExtendableElement.

2.5.4 Class DateValueProperty

A custom property with a date value.

General Information

Base Classifier	CustomProperty
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
value	Date	1	Defines the value of the CustomProperty.

2.5.5 Class DoubleValueProperty

A custom property with a double value.

General Information

Base Classifier	CustomProperty
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
value	Double	1	Defines the value of the CustomProperty.

2.5.6 Class IntegerValueProperty

A custom property with an integer value.

General Information

Base Classifier	CustomProperty
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
value	Integer	1	Defines the value of the CustomProperty.

2.5.7 Class LocalizedStringProperty

NA

General Information

Base Classifier	CustomProperty
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Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
value	LocalizedString	1	Defines the value of the <i>CustomProperty</i> .

2.5.8 Class MultiDimensionalValueProperty

NA

General Information

Base Classifier	CustomProperty
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
value	MultiDimensionalValue	1	NA

2.5.9 Class NumericalValueProperty

A custom property with a numerical value. (see KBLFRM-319)

General Information

Base Classifier	CustomProperty
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
value	NumericalValue	1	Defines the value of the <i>CustomProperty</i> .

2.5.10 Class SimpleValueProperty

A custom property with a simple value (string).

General Information

Base Classifier	CustomProperty
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
value	String	1	Defines the value of the <i>CustomProperty</i> .

2.5.11 Class ValueRangeProperty

A custom property with a value range. (see KBLFRM-319)

Documentation

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General Information

Base Classifier	CustomProperty
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
value	ValueRange	1	Defines the value of the CustomProperty.

2.6 Module electrical_parts

2.6.1 Class AbstractSlot

An *AbstractSlot* represents a geometrical location in a connector housing that can **contain or group one or more cavities**. This grouping can happen in two different ways:

- **Directly**, via a *Slot*, where the slot is an **inseparable part of the connector housing**, formed during the manufacturing process.
- **Indirectly**, via a *ModularSlot*, which serves as a **placeholder** where other connector housings (modules) can be inserted during assembly.

The *AbstractSlot* is primarily relevant on the **mating side** of the connector. It enables a logical and geometrical grouping of cavities that may affect how mating is performed.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
slotNumber	String	0..1	Specifies the number of the slot. This must be unique within a ConnectorHousingSpecification.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
coding	Coding	0..1	Y	N	0..1	Y	Defines coding of the slot that is satisfied by the Slot.
slotSpecification	SlotSpecification	0..1	Y	N	0..*	N	References the SlotSpecification that is satisfied by the slot.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
referencedSlot	1	N	AbstractSlotReference	0..*	Points to the slot referenced by the slot reference.

slot	0..*	Y	ConnectorHousing Specification	1	Specifies the slots forming the ConnectorHousing.
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2.6.2 Class AntennaSpecification

Specification of the electrological aspects of an antenna.

General Information

Base Classifier	EEComponentSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
fMin	NumericalValue	0..1	Specifies the minimum operating frequency of the antenna.
fMax	NumericalValue	0..1	Specifies the maximum operating frequency of the antenna.
impedance	NumericalValue	0..1	Specifies the impedance of the antenna.

2.6.3 Class BatterySpecification

Specification of the electrological aspects of a battery.

General Information

Base Classifier	EEComponentSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
u	NumericalValue	0..1	Specifies the nominal voltage of the battery.
i	NumericalValue	0..1	Specifies the current the battery provides.
iCool	NumericalValue	0..1	Specifies the battery's current in cool state.
capacity	NumericalValue	0..1	Specifies the power capacity of the battery.

2.6.4 Class BoltTerminalSpecification

Specification for the definition of bolt terminals. These are the counterparts to ring terminals.

General Information

Base Classifier	TerminalSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
boltDiameter	NumericalValue	0..1	Specifies the diameter of the bolt in a numerical way.
boltHeight	NumericalValue	0..1	Specifies the height of the bolt (the height of the part to which ring terminals can be attached).

boltNominalSize	TerminalBoltNominalSize	0..1	Defines the size (diameter) of the bolt in a nominal way (e.g. "M8").
boltType	TerminalBoltType	0..1	Specifies the type of the bolt.
torsionProtection	Boolean	0..1	Specifies if the bolt provides torsion protected or not.
maxTerminalCount	Integer	0..1	Defines the maximum number of (ring) terminals that can be attached to this bolt at the same time.

2.6.5 Class BridgeTerminalSpecification

A bridge terminal is a part that behaves like terminal but has no *WireReceptions*. It is used to create short circuit between different pins in a connector. In its use, it can realize a schematic connection on its own and independently of other components.

General Information

Base Classifier	TerminalSpecification
Applied Stereotype	NA
Is Abstract	false

2.6.6 Class CableSealEntry

Specifies an opening in the cable seal.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Provides an identifier for the <i>CableSealEntry</i> . The <i>Identification</i> shall be unique within a <i>CableSealSpecification</i> .
wireElementOutsideDiameter	ValueRange	0..1	Specifies a range of valid wire diameters to which the <i>CableSealEntry</i> fits.
insideDiameter	NumericalValue	0..1	Defines the inside diameter in the relaxed state for the opening in the cable seal.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
position	Cavity	0..1	Y	N	0..*	N	Specifies the <i>Cavity</i> in the <i>CavityLayout</i> of the <i>CableSealSpecification</i> where this <i>CableSealEntry</i> is positioned.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cableSealEntry	1..*	Y	CableSealSpecification	NA	NA

cableSealEntry	1	N	CableSealEntryReference	NA	NA
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2.6.7 Class CableSealPlugSpecification

A cable seal plug is a plug that is used to seal entries of a cable seal. A cable seal plug can seal multiple *CableSealEntry* at the same time. A cable seal plug can have additional feature/functions at the same time. For example, it might also serve as a seal retainer in connector housing.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavityLayout	CavityLayout	0..1	Y	N	0..1	Y	NA
sealingPin	SealingPin	1..*	Y	N	NA	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cableSealPlugSpecification	1	N	CableSealPlugRole	NA	NA

2.6.8 Class CableSealSpecification

Specification for seals that are associated with one or more wires and not with specific cavity or crimp. A *CavitySeal* is a watertight non-electrical object to fill a populated Cavity. Cable seals are designed to seals specific areas (covering one or more wires) in a connector or a grommet. A cable seal is used on a cable outside the crimp. A *CableSealSpecification* defines one or more *CableSealEntries*. Cable seals can have more than one entry with different wire sizes.

There are two variants of cable seals:

1. The cable seal has openings for all possible assignments. Openings that are unpopulated in a specific configuration must be sealed with a cable seal plug. A cable seal plug can seal one or multiple openings at the same time and can also provide additional functions, like serving as seal retainer.
2. The cable seal is available in different versions, each with different openings of the theoretically possible ones that are actually open.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cableSealEntry	CableSealEntry	1..*	Y	N	NA	Y	NA
cavityLayout	CavityLayout	0..1	Y	N	0..1	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cableSealSpecification	1	N	CableSealRole	NA	NA

2.6.9 Class CapacitorSpecification

Specification of the electrological aspects of a capacitor.

General Information

Base Classifier	EEComponentSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
capacity	NumericalValue	0..1	NA
impedance	NumericalValue	0..1	NA
uMax	NumericalValue	0..1	The breakdown voltage of the capacitor.

2.6.10 Class Cavity

A cavity is a defined space in a connector housing for location of an electrical terminal or cavity plug or seal. A cavity may also be empty.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
available	Boolean	0..1	Defines whether the cavity is available for contacting. If the cavity is not available, it means that it is completely closed.
cavityNumber	String	0..1	Provides an identifier for the cavity. The cavity number needs to be unique within a <i>Slot</i> .
hasIntegratedTerminal	Boolean	0..1	Defines whether the cavity has an integrated terminal (for example an IDC cavity) or if an additional terminal is required. If this attribute is "true", the cavity can reference a <i>TerminalSpecification</i> as <i>integratedTerminalSpecification</i> in order to specify the integrated terminal.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavitySpecification	CavitySpecification	0..1	Y	N	0..*	N	References the CavitySpecification that is satisfied by the cavity.
integratedTerminalSpecification	TerminalSpecification	0..1	Y	N	NA	N	Specifies the terminal, if the cavity has an integrated terminal (e.g. an IDC).
positionDetail	CavityPositionDetail	0..1	Y	N	1	Y	Optional details for geometric properties of the cavity in the connector.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavity	1..*	Y	Slot	1	Specifies the Cavities forming the Slot.
cavity	1	N	CavityAddOn	0..*	NA
openCavities	1..*	N	OpenCavitiesAssignment	0..*	Specifies the cavities that are open.
position	0..1	N	SealingPin	0..*	Specifies the <i>Cavity</i> in the <i>CavityLayout</i> of the <i>CableSealPlugSpecification</i> where this <i>SealingPin</i> is positioned.
position	0..1	N	CableSealEntry	0..*	Specifies the <i>Cavity</i> in the <i>CavityLayout</i> of the <i>CableSealSpecification</i> where this <i>CableSealEntry</i> is positioned.
reachableCavities	0..*	N	SegmentConnectionPoint	0..*	Specifies the <i>Cavities</i> that are reachable with wires through this <i>SegmentConnectionPoint</i> .
referencedCavity	0..1	N	PinComponent	0..*	Defines the cavity in the corresponding ConnectorHousingSpecification of the HousingComponent where the PinComponent is located. (see KBLFRM-300)
referencedCavity	1	N	CavityReference	0..*	Points to the cavity referenced by the cavity reference.
sealedCavities	1..*	N	SealedCavitiesAssignment	0..*	Specifies the Cavities that are sealed.
virtualCavity	0..*	Y	CavityLayout	1	NA

2.6.11 Class CavityAccessorySpecification

A *CavityAccessory* is a non-electrical part used in a cavity with no sealing properties (e.g. a wire fixation).

General Information

Base Classifier	CavityPartSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
wireElementOutsideDiameter	ValueRange	0..1	Specifies a range of valid wire diameters to which the cavity accessory fits.

wireReceptionType	WireReceptionType	0..1	Specifies the wireReceptionType to which the cavity accessory fits.
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Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavityAccessorySpecification	1	N	CavityAccessoryRole	0..*	NA

2.6.12 Class CavityAddOn

Specifies the wire addon needed to reach a *Cavity* from a specific *SegmentConnectionPoint*. More specifically, it is the add-on required to calculate the cutting length of a wire. It is not a measured distance between the segment connection point and the insertion point of the cavity. For each *SegmentConnectionPoint* there shall be no more than one add-on value per cavity and type.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult.	Comment
wireAddOn	NumericalValue	1	Specifies the wire length add on needed for the cavity.
type	WireAddOnType	1	Defines the type of the add-on (see <i>CavityAddOn</i>).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavity	Cavity	1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavityAddOns	0..*	Y	SegmentConnectionPoint	1	NA

2.6.13 Class CavityLayout

A *CavityLayout* represents a specific configuration pattern for a component with cavities. It defines a set of *virtualCavities* that serve as placeholders for the actual cavities provided by component that satisfies this layout (e.g. a contact module).

It acts as a structural template, such as single-pole, five-pole, or multi-row configurations, that allows the definition of properties (e.g. a cavity mapping) for all components that satisfy this layout, instead of specifying those for each component individually (e.g. a cavity mapping for a contact modules within a specific modular slot of a collector housing). Its purpose is to create a layer of indirection between.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA

Is Abstract	false
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Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>ModularSlotLayout</i> . The identification is guaranteed to be unique within the <i>ModularSlot</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
virtualCavity	Cavity	0..*	Y	N	1	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavityLayout	0..1	Y	CableSealSpecification	0..1	NA
cavityLayout	0..1	Y	CableSealPlugSpecification	0..1	NA
layout	0..*	Y	ModularSlot	0..1	NA
validLayoutCombination	0..*	N	ModularSlotConfiguration	0..*	NA

2.6.14 Class CavityPartSpecification

The *CavityPartSpecification* is an abstract class for common properties of non-electrical parts that are used in *Cavities*.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
cavityDimension	Size	0..*	Specifies a valid cavity dimension to which the cavity part fits. The dimension defines the size of the sealing area of the cavity (crimp end), not in the contacting area (box end). Note: CavityDimension is of type Size which is defined as x & y with type NumericalValue. NumericalValue can define tolerances. So, a cavity dimension is not necessarily a single fixed value.
hardness	NumericalValue	0..1	Specifies the hardness of the cavity seal.
geometry	SealingGeometry	0..1	Defines the geometry of the cavity sealing. This attribute is defined as an OpenEnumeration.
compatibleTerminalType	TerminalType	0..*	Defines a list of terminal types that are compatible to this CavitySealSpecification. This defines the compatible cavities as well, since a plug is normally used when no terminals are present.

compatibleCavityGeometry	CavityGeometry	0..*	Defines a list of <i>CavityGeometries</i> that are compatible with this cavity part.
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2.6.15 Class CavityPlugSpecification

Specification for the definition of cavity seals. A cavity plug is a watertight non-electrical object to fill an empty cavity.

General Information

Base Classifier	CavityPartSpecification
Applied Stereotype	NA
Is Abstract	false

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavityPlugSpecification	1	N	CavityPlugRole	0..*	References the <i>CavityPlugSpecification</i> that is instantiated by this <i>CavityPlugRole</i> .

2.6.16 Class CavityPositionDetail

CavityPositionDetail is an aggregator class for all geometric properties of a cavity within the connector (e.g. rotation, position).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
rotation	NumericalValue	0..1	The rotation of the cavity is defined as the angle between the primary locking of the cavity and the Y-axis of the connector housing coordinate system.
cavitySealingOffset	NumericalValue	0..1	Defines the offset (length) by which the starting point of the sealing area is displaced <i>insertionPosition</i> in insertion direction.
insertionDepth	NumericalValue	0..1	The <i>insertionDepth</i> of a cavity is the length of the insertion path for a terminal. The distance a terminal must travel from the <i>InsertionPosition</i> to its final position.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
baseUnit	Unit	1	Y	N	0..*	N	The <i>Unit</i> in which all coordinates (e.g. cartesian points) of this <i>CavityPositionDetail</i> are defined. Shall be a unit of length (e.g. millimetre).
insertionPosition	CartesianPoint3D	0..1	Y	N	1	Y	The position of the center of the cavity in the entrance surface of the connector during the insertion process.

insertionVector	CartesianVector3 D	0..1	Y	N	NA	Y	A unit vector that points in the direction in which a terminal is inserted from the <i>InsertionPosition</i> into the connector.
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Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
positionDetail	0..1	Y	Cavity	1	Optional details for geometric properties of the cavity in the connector.

2.6.17 Class CavitySealSpecification

Specification for the definition of cavity seals. A *CavitySeal* is a watertight non-electrical object to fill a populated Cavity. A *CavitySeal* is normally used at a *WireEnd* within the crimp of a terminal.

General Information

Base Classifier	CavityPartSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
wireElementOutsideDiameter	ValueRange	0..1	Specifies a range of valid wire diameters to which the cavity seal fits.
wireReceptionType	WireReceptionType	0..1	Specifies the wireReceptionType to which the cavity seal fits.
insideDiameter	NumericalValue	0..1	Defines the inside diameter in the relaxed state for a cavity seal.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavitySealSpecification	1	N	CavitySealRole	0..*	References the <i>CavitySealSpecification</i> that is instantiated by this <i>CavitySealRole</i> .

2.6.18 Class CavitySpecification

Specification for the definition of cavities.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
angle	NumericalValue	0..2	Specifies the angle against two planes of the connector housing a terminal used in this cavity can be buckled.
cavityDesign	String	0..1	<u>Deprecated since 1.1.4:</u> <u>Reason: Replaced by the more meaningful mechanism TerminalTypes.</u>

geometry	CavityGeometry	0..1	Defines the geometry of a cavity in the sealing area (crimp end).
cavityDimension	Size	0..1	Specifies the dimension of the cavity in the sealing area of the cavity (crimp end), not in the contacting area (box end). The <i>Height</i> of a cavity is defined as the dimension of cavity in the direction of the primary locking. The <i>Width</i> is the dimension orthogonal to the <i>Height</i> . For cavities with <i>Geometry = Circular</i> the <i>Size.Width</i> shall be equal to <i>Size.Height</i> .
minWireElementOutsideDiameter	NumericalValue	0..1	Specifies the minimum diameter a wire is allowed to have to fit into the cavity. This definition is necessary, since wires that are too small might cause movements and in acceptable torsion forces on terminals.
maxWireElementOutsideDiameter	NumericalValue	0..1	Specifies the maximum diameter a wire is allowed to have to fit into the cavity.
primaryLockingType	PrimaryLockingType	0..1	Specifies if the cavity has a primary locking and of what type it is.
sealable	Boolean	0..1	Specifies if the cavity is sealable.
compatibleTerminalType	TerminalType	0..*	Defines a list of terminal types that are compatible to this CavitySpecification.
cavitySealingLength	NumericalValue	0..1	Defines the length of the sealing area in the cavity.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavitySpecification	0..1	N	Cavity	0..*	References the CavitySpecification that is satisfied by the cavity.

2.6.19 Class CoatingSpecification

A CoatingSpecification is a special *WireEndAccessorySpecifications* are describing parts / procedures with which the wire end is treated before the actual terminal is attached to the *WireEnd* (e.g. a tin coating of the wire end). Properties of the procedure are defined by the referenced *WireReceptionSpecification*, e.g. the coating material is defined by the *GeneralTechnicalPartSpecification.MaterialInformation*.

General Information

Base Classifier	WireEndAccessorySpecification
Applied Stereotype	NA
Is Abstract	false

2.6.20 Class Coding

Specifies the coding of a slot or a connector housing.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
coding	CodingName	1	Specifies the name of the coding.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
coding	0..1	Y	ConnectorHousing Specification	0..1	Defines coding of the connector housing that is satisfied by the connector housing.
coding	0..1	Y	AbstractSlot	0..1	Defines coding of the slot that is satisfied by the Slot.

2.6.21 Class ConductorCurrentInformation

The *ConductorCurrentInformation* specifies the maximum current for which a conductor is approved. As the maximum current is dependent from the voltage and the environment temperature it is modelled as a class and not only as an attribute.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
maxCurrent	NumericalValue	1	The maximum current value.
environmentTemperature	NumericalValue	1	The environment temperature for which this maximum current value is applicable.
voltage	NumericalValue	1	The voltage for which this maximum current value is applicable.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
currentInformation	0..*	Y	ConductorSpecification	1	Specifies the current information of the conductor. These are the maximum currents for which the conductor is approved.

2.6.22 Class ConductorMaterial

ConductorMaterial is a helper class to specify *validConductorMaterials*. This is necessary, since all attributes of the type *Material* have a multiplicity of * with the semantics that it always defines one *Material* with the possibility to define it in different reference systems.

Since a wire reception can have more than one *validConductorMaterial* this container class is necessary to keep the semantics clear.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
material	Material	1..*	Specifies the material.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
validConductorMaterial	0..*	Y	TerminalProtectionSpecification	1	Specifies the materials of a conductor, which are valid to use with this <i>TerminalProtectionSpecification</i> . This material shall be matched against the <i>ConductorSpecification.material</i> .
validConductorMaterials	0..*	Y	WireReceptionSpecification	1	Specifies the materials of a conductor, which are valid to use with this <i>WireReceptionSpecification</i> . This material shall be matched against the <i>ConductorSpecification.material</i> .

2.6.23 Class ConductorSpecification

Specification for the definition of conducting properties of a WireElement.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
wireType	WireType	0..*	Defines the type of the conductor. A <i>WireType</i> is a key in a reference system for the definition of wires. Some reference systems define independent type fragment for insulations and conductors and the <i>WireType</i> of a <i>WireElement</i> is the combination of both. A conductor must not have more than one type. This attribute allows more than one value for the reason, that the same type can be expressed in multiple reference systems.
crossSectionArea	NumericalValue	0..1	Specifies the cross-section area of the conductor (e.g. 0,5 mm ²). The cross-section area is a nominal value, which refers to the conducting properties of the conductor normalized to the properties of a full material core.
massInformation	MassInformation	0..1	Specifies the mass information of the conductor. In most cases this mass information is known as copper weight and is normally specified as mass per length (e.g. gram per meter).
material	Material	0..*	Specifies the material of the conductor.
resistance	NumericalValue	0..1	Specifies the electrical resistance of the conductor. In most cases this is specified as resistance per length (e.g. Ohm per meter).
structure	ConductorStructure	0..1	Specifies the structure of the conductor according to ISO 6722-1 for ConductorSpecifications with Type = Stranded. <i>Should only be used for ConductorSpecifications with type=Stranded.</i>
type	ConductorType	0..1	Specifies the type of the conductor, e.g. if it is rigid or stranded.
numberOfStrands	NumericalValue	0..1	Specifies the number of individual strands used to construct the conductor. If the conductor is solid than the number of strands would be one. This value indicates the total number of strands forming the conductor. While typically an integer, this attribute is intentionally

			modelled as a <i>NumericalValue</i> to support tolerances, as found in technical data sheets — especially for high-voltage / large cross-section cables, where manufacturers may specify strand count as a range or with a deviation (e.g., 1.070 ± 5%). Usage Notes: • If a precise strand count is known and no variation applies, a <i>NumericalValue</i> without tolerance (e.g., 19) should be used. • If the strand count includes a tolerance, the tolerance properties of <i>NumericalValue</i> should be utilized. • The <i>Unit</i> used should be set to <i>Piece</i>.
platingMaterial	Material	0..*	Specifies the plating material of the conductor.
strandDiameter	NumericalValue	0..1	Specifies the diameter of a single strand in the conductor.
voltageRange	NumericalValue	0..1	<u>Deprecated since 2.2.0:</u> <u>Reason: Replaced by</u> <u>GeneralTechnicalPartSpecification.VoltageRating.</u> Specifies the voltageRange for which the conductor is approved.
transmissionMedium	SignalTransmissionMediumType	0..1	Specifies the type of the transmission medium for the conductor.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
currentInformation	ConductorCurrentInformation	0..*	Y	N	1	Y	Specifies the current information of the conductor. These are the maximum currents for which the conductor is approved.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
appliesTo	1	N	CoreCrimpDetail	NA	NA
conductorSpecification	0..1	N	InternalComponentConnection	0..*	NA
conductorSpecification	0..1	N	WireElementSpecification	0..*	If the <i>WireElement</i> has a core then the specification of the core is referenced here.
recommendedConductorSpecification	0..1	N	Signal	NA	Defines a recommended Specification for the cores that implement this signal.
referencedCoreSpecification	0..1	N	TerminalPairing	0..*	References the CoreSpecification that is used on both sides of the ContactSystem.

2.6.24 Class ConnectorHousingCapSpecification

Specification for the definition of caps (backshells) of connectors. Different caps can add additional wire length addons to a connector housing.

A 'cap' which already defines the number of cavities, coding etc. is in the VEC defined by *ConnectorHousingSpecification* with *ModularSlots* and not by a *ConnectorHousingCapSpecification*.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
wireAddOn	NumericalValue	0..1	Specifies the wire length add on needed for the cap.
outletDirection	ConnectorOutletDirection	0..1	Defines the <i>OutletDirection</i> for wires. This attribute is defined as an <i>OpenEnumeration</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
connectorHousingCapSpecification	1	N	ConnectorHousingCapRole	0..*	References the <i>ConnectorHousingCapSpecification</i> that is instantiated by this <i>ConnectorHousingCapRole</i> .

2.6.25 Class ConnectorHousingCoverSpecification

Specification for the definition of covers for connectors. A *ConnectorHousingCover* is used to protect the contacting side (front) of the connector during transport of the harness or in the vehicle, if the connector remains unconnected (e.g. in case of "give-away" features or retrofit options).

There are two types of covers, on the one hand components whose sole purpose is to be used as a cover and on the other hand normal harness connectors which can also be used as covers, if necessary sealed with additional plugs. The former have the *PrimaryPartType*="ConnectorCover" and are only specified by a *ConnectorHousingCoverSpecification*, the latter have the *PrimaryPartType*="ConnectorHousing" and have an additional *ConnectorHousingCoverSpecification*.

The relation between a connect and its covers is established in part master data with the *PartRelation*.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
connectorHousingCoverSpecification	1	N	ConnectorHousingCoverRole	0..*	NA

2.6.26 Class ConnectorHousingSpecification

Specification for the definition of connector housings. A connector housing consists of a one or more slots. In the means of the VEC, a connector housing can be a conventional connector housing, a contact module or a connector shell.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
averageWireLengthAddOn	NumericalValue	0..1	Specifies the average wire length add on for this connector.
voltageRange	ValueRange	0..1	<u>Deprecated since 2.2.0:</u> <u>Reason:</u> Replaced by <i>GeneralTechnicalPartSpecification.VoltageRating</i> . Specifies the allowed voltage range for the connector housing.
coupleable	Boolean	0..1	Defines whether the connector is coupleable or not. Connectors that are coupleable can be used in an inline position. Connectors that are not coupleable can be connected only to an ECU or something similar.
connectorPositionAssurance	Boolean	0..1	If <i>true</i> the <i>ConnectorHousing</i> has a connector position assurance (CPA). A CPA is some sort of feature of a connector, which ensures that the connector in its correctly assembled position with its mating part.
connectorPositionAssuranceType	ConnectorPositionAssuranceType	0..1	Defines the type of the connector position assurance. This is only valid of <i>connectorPositionAssurance = true</i> .
outletDirection	ConnectorOutletDirection	0..1	Defines the <i>OutletDirection</i> for wires. This attribute is defined as an <i>OpenEnumeration</i> .
lockingType	ConnectorLockingType	0..1	Defines the type of locking of this connector against its counterpart.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
coding	Coding	0..1	Y	N	0..1	Y	Defines coding of the connector housing that is satisfied by the connector housing.
modularSlotConfiguration	ModularSlotConfiguration	0..*	Y	N	1	Y	NA
referenceSurfaceDefinition	ReferenceSurfaceDefinition	0..1	Y	N	1	Y	Contains the information about reference surfaces of the connector.
segmentConnectionPoint	SegmentConnectionPoint	0..*	Y	N	1	Y	Specifies the <i>SegmentConnectionPoints</i> the connector housing.
slot	AbstractSlot	0..*	Y	N	1	Y	Specifies the slots forming the <i>ConnectorHousing</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
connectorHousingSpecification	1	N	ConnectorHousingRole	0..*	References the <i>ConnectorHousingSpecification</i> that is instantiated by this <i>ConnectorHousingRole</i> .

housingSpecification	0..1	N	HousingComponent	0..*	References the ConnectorHousingSpecification that is describing the connector interface of the HousingComponent (e.g. Slots, Cavities, Design, Coding).
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2.6.27 Class CoreCrimpDetail

The *CoreCrimpDetail* defines crimp values for a conductor crimp (wire crimp). The selection criteria to which the *CoreCrimpDetail* applies are defined by the referenced *ConductorSpecification*.

General Information

Base Classifier	CrimpDetail
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
corePullOffForce	NumericalValue	0..1	The minimum force that the completed crimp should withstand when wire is being pulled off the terminal. Above that force, the crimp can be expected to be destroyed at any time. This value applies when only the core crimp is present or functional (insulation crimp is either not present or disabled).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
appliesTo	ConductorSpecification	1	Y	N	NA	N	NA
insulationCrimpDetails	InsulationCrimpDetail	0..*	Y	N	1	Y	Defines the different <i>InsulationCrimpDetails</i> that are valid combinations for this <i>CoreCrimpDetail</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
coreCrimpDetails	0..*	Y	WireReceptionSpecification	1	Defines the <i>CrimpDetails</i> for this WireReception.

2.6.28 Class CoreSpecification

Defines the properties of a circular conductor (core) which are specific for them.

General Information

Base Classifier	ConductorSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
outsideDiameter	NumericalValue	0..1	Specifies the outside diameter of the core.

2.6.29 Class CrimpDetail

Abstract base class for the definition of *CrimpDetail*-information. See *CoreCrimpDetail* & *InsulationCrimpDetail*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
size	Size	0..1	Defines the expected size of the crimp. The <i>height</i> is measured in direction of the crimp opening. The <i>width</i> is measured in orthogonal to the <i>height</i> and orthogonal to main axis of the terminal (<i>TerminalSpecification.OverallLength</i>).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
machineCrimpTool	ResourceVersion	0..*	Y	N	0..*	N	A list of tools that are permitted for machine crimping processes.
manualCrimpTool	ResourceVersion	0..*	Y	N	0..*	N	A list of tools that are permitted for manual crimping processes.

2.6.30 Class DiodeSpecification

Specification of the electrological aspects of a diode.

General Information

Base Classifier	EEComponentSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
thresholdVoltage	NumericalValue	0..1	Voltage at which the diode starts conducting in forward direction.
breakDownVoltage	NumericalValue	0..1	Voltage at which the diode starts conducting in reverse direction.
iMax	NumericalValue	0..1	Specifies the maximum electric current tolerated by the diode.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
anode	PinComponent	0..1	Y	N	0..*	N	NA

cathode	PinComponent	0..1	Y	N	0..*	N	NA
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2.6.31 Class EEComponentSpecification

Base-class for the specification of electrological components, which are connected to the harness. Usually, electrological components are not part of the harness e.g. a fuse, a switch or a control device. All EEComponents can have one or more HousingComponents which are possible interfaces for the connection to a harness.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
powerConsumption	PowerConsumption	0..*	Specifies the <i>PowerConsumptions</i> of this <i>EEComponentSpecification</i> . This refers to the internal power consumption of this E/E-component, not the power-consumption that might occur indirectly over other E/E-components that are supplied through this component.
voltageRating	NumericalValue	0..1	<i>Deprecated since 2.2.0:</i> <i>Reason:</i> Replaced by <i>GeneralTechnicalPartSpecification.VoltageRating</i> . The <i>voltageRating</i> of an <i>EEComponentSpecification</i> indicates the maximum voltage of the circuit for which the E/E component (e.g. a fuse) is designed to operate safely.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentNode	ComponentNode	0..1	Y	N	0..*	N	The <i>ComponentNode</i> that defines the internal connectivity of this <i>EEComponentSpecification</i> .
connections	InternalComponentConnection	0..*	Y	N	1	Y	NA
extensionSlots	ExtensionSlot	0..*	Y	N	1	Y	Specifies the available ExtensionSlots of the EEComponent.
housingComponent	HousingComponent	0..*	Y	N	1	Y	Specifies the available connector interfaces of the EEComponent.
states	SwitchingState	0..*	Y	N	1	Y	Specifies the available <i>SwitchingStates</i> of the EEComponent.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
EEComponentSpecification	1	N	EEComponentRole	0..*	References the <i>EEComponentSpecification</i> that is instantiated by this <i>EEComponentRole</i> .

2.6.32 Class ExtensionSlot

An *ExtensionSlot* defines a slot within an EE-Component where other EE-Components can be plugged into (modular extension). This is necessary for example for modular power distributions.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Identification of the ExtensionSlot, which must be distinct for all ExtensionSlots of an EEComponent.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
allowedInserts	PartRelation	0..*	Y	N	0..*	N	References the <i>PartRelations</i> that are valid inserts for this <i>ExtensionSlot</i> . This reference points to <i>PartRelations</i> in order to allow referencing indirectly a <i>PartVersion</i> if the description of individual <i>PartVersions</i> is done with one physical VEC file per <i>PartVersion</i> and to allow the expression of optional inserts, choices etc. However, inserts for an <i>ExtensionSlot</i> are always <i>EEComponents</i> by itself. Therefore, the referenced <i>PartVersion</i> shall have a <i>PrimaryPartType</i> = <i>EEComponent</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
extensionSlot	1	N	ExtensionSlotReference	NA	NA
extensionSlots	0..*	Y	EEComponentSpecification	1	Specifies the available ExtensionSlots of the EEComponent.

2.6.33 Class FillerSpecification

Specification for the definition of filler elements in the wire.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
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material	Material	0..*	Specifies the material of the filler.
diameter	NumericalValue	0..1	Specifies the diameter of the filler.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
fillerSpecification	0..1	N	WireElementSpecif ication	0..*	If the <i>WireElement</i> is a filler then the specification of the filler is referenced here.

2.6.34 Class FlatCoreSpecification

Defines the properties of a flat (rectangular) conductor which are specific for them.

General Information

Base Classifier	ConductorSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
size	Size	0..1	Defines the size (width & height) of the flat core.

2.6.35 Class FuseCharacteristic

The *FuseCharacteristic* specializes the *MultiDimensionalValue* for describing fuse behavior. It references two mandatory *dimensions*: one for current (I) and one for tripping time (T). Additional dimensions may be included as needed (e.g., temperature).

General Information

Base Classifier	MultiDimensionalValue
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
time	QuantityKind	1	Y	N	0..1	N	NA
trippingCurrent	QuantityKind	1	Y	N	0..1	N	NA

2.6.36 Class FuseComponent

Deprecated since 2.2:

Reason: The *FuseComponent* was a very specific model element for internal connectivity and has been replaced with the more generic approach of adding a *ConnectionSpecification* to a E/E component description (see "Internal Connectivity").

NA

General Information

Base Classifier	ExtendableElement
Applied Stereotype	Deprecated
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Identification of the FuseComponent, which must be distinct for all FuseComponents of an MultiFuseSpecification.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connectedPins	PinComponent	0..2	Y	N	0..*	N	NA
fuseSpecification	FuseSpecification	0..1	Y	N	NA	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
fuseComponents	0..*	Y	MultiFuseSpecification	1	NA

2.6.37 Class FuseSpecification

Specification of the electrological aspects of a fuse.

General Information

Base Classifier	EEComponentSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
fuseType	FuseType	0..1	The geometric type of a fuse.
iMax	NumericalValue	0..1	Specifies the maximum electric current tolerated by the fuse.
characteristic	FuseCharacteristic	0..1	The characteristics of a fuse design refers to how rapidly the fuse responds to various current overloads.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
fuseSpecification	0..1	N	FuseComponent	NA	NA

2.6.38 Class GeneralTechnicalPartSpecification

Specification for the definition of common properties for technical parts.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
colorInformation	Color	0..*	Specifies the color of the part.
massInformation	MassInformation	0..*	Specifies the overall mass of the part per Unit of quantity (e.g. g/piece or g/meter...)
materialInformation	Material	0..*	Specifies the material of a part with the syntax and nomenclature of an arbitrary reference system. Multiple <i>Material</i> entries in this attribute are only allowed to express the same value in the different reference systems. For parts that are composed of different materials, e.g. wiring harness modules, these materials and their weight proportions can be defined via the <i>MaterialComposition</i> .
materialComposition	MaterialComposition	0..*	Parts and components are often composed from a material-mix. The <i>MaterialComposition</i> allows the definition of the various <i>Materials</i> in a part and their mass. For example, the proportion of copper or other precious metals in a component. It is permissible to list only individual relevant components. A complete listing of all components is not required.
robustnessProperties	RobustnessProperties	0..*	Specifies the robustness of a part.
temperatureInformation	TemperatureInformation	0..*	Specifies valid temperatures for a part.
fitRate	Double	0..1	The Failures In Time (FIT) rate of a device is the number of failures that can be expected in one billion (10 ⁹) device-hours of operation.[14] (E.g. 1000 devices for 1 million hours, or 1 million devices for 1000 hours each, or some other combination.) (see https://en.wikipedia.org/wiki/Failure_rate#Units)
unspecifiedAccessoryPermitted	Boolean	0..1	Defines whether accessories which are not explicitly defined by a <i>PartRelation</i> may be used with instances of this part. If this attribute is not specified the default value is <i>true</i> .
voltageRating	NumericalValue	0..1	The <i>voltageRating</i> of a component indicates the maximum voltage for which the component (e.g. a fuse, connector housing,...) is designed to operate safely.
voltageLevelCompliance	NominalVoltage	0..*	Defines the voltage levels (in terms of <i>NominalVoltage</i>) for which the part is approved.
applicationType	ApplicationType	0..*	Defines the valid application types (or fields of application) for the <i>PartVersion</i> described by this <i>GeneralTechnicalPartSpecification</i>. An application type usually implies a wide variety of aspects, such as process requirements, specifications, applicable standards, procedures to be followed etc. <u>Differentiation from <i>VoltageComplianceLevel</i>:</u> Although there is an overlap between the literals of the <i>NominalVoltage</i> and the <i>ApplicationType</i> the statements expressed with the <i>applicationType</i> and the <i>voltageComplianceLevel</i> are different. The <i>voltageComplianceLevel</i> only refers to a statement about the voltage compatibility of a part, but does not include the other aspects, e.g. process requirements. For example, a connector or cable might withstand high voltage levels (<i>voltageLevelCompliance</i>) but might not be allowed for high voltage applications in a vehicle due to an insufficient color coding.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
boundingBox	BoundingBox	0..1	Y	N	NA	Y	Defines the bounding box of the part.
partRelation	PartRelation	0..*	Y	N	1	Y	Specifies possible relations (accessories) of the specified part with other PartVersion (e.g. caps, clips).

2.6.39 Class HoleTerminalSpecification

Specification for the definition of hole terminals. These are used for example as attachment points of ring terminals at the car body.

General Information

Base Classifier	TerminalSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
holeDiameter	NumericalValue	0..1	Specifies the diameter of the hole in a numerical way.
holeDepth	NumericalValue	0..1	Specifies the depth of the hole.
holeNominalSize	TerminalBoltNominalSize	0..1	Defines the size (diameter) of the hole in a nominal way (e.g. "M8").
holeType	TerminalBoltType	0..1	Specifies the type of the hole.
torsionProtection	Boolean	0..1	Specifies if the hole provides torsion protected or not.
maxTerminalCount	Integer	0..1	Defines the maximum number of (ring) terminals that can be attached to this hole at the same time.

2.6.40 Class HousingComponent

A HousingComponent describes the interface of an EECComponent with which it can be connected to another EECComponent or a harness. The characteristics of the interface can be described with a referenced ConnectorHousingSpecification. (see KBLFRM-300)

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Identification of the HousingComponent, which must be distinct for all HousingComponents of an EECComponent.
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the <i>HousingComponent</i> .

compatibleTypes	HousingComponent Type	0..*	The values of this attribute define the <i>HousingComponentTypes</i> that are valid to be associated with this <i>HousingComponent</i> . In other word, if this <i>HousingComponent</i> can be associated with a relay, a fuse, a connector housing of a harness. The values are matching the <i>PrimaryPartType</i> of the <i>PartVersion</i> of the component that should be associated (plugged) into this housing component.
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Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentConnector	ComponentConnector	0..1	Y	N	0..*	N	The <i>ComponentConnector</i> that represents this <i>HousingComponent</i> in the internal connectivity of the <i>EEComponentSpecification</i> .
housingSpecification	ConnectorHousingSpecification	0..1	Y	N	0..*	N	References the <i>ConnectorHousingSpecification</i> that is describing the connector interface of the <i>HousingComponent</i> (e.g. Slots, Cavities, Design, Coding).
pinComponent	PinComponent	0..*	Y	N	1	Y	Specifies the <i>PinComponents</i> of <i>HousingComponent</i> . (see KBLFRM-300)
segmentConnectionPoint	SegmentConnectionPointHC	0..*	Y	N	1	Y	Specifies the <i>SegmentConnectionPoints</i> the <i>HousingComponent</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
housingComponent	0..*	Y	EEComponentSpecification	1	Specifies the available connector interfaces of the <i>EEComponent</i> .
housingComponent	1	N	HousingComponentReference	0..*	Points to the <i>HousingComponent</i> referenced by the <i>HousingComponent</i> reference. (KBLFRM-401)

2.6.41 Class InsulationCrimpDetail

The *InsulationCrimpDetail* defines crimp values for an insulation crimp. The selection criteria to which the *InsulationCrimpDetail* applies are defined by the referenced *InsulationSpecification* and the *ConductorSpecification* referenced by the containing *CoreCrimpDetail*.

General Information

Base Classifier	CrimpDetail
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
pullOffForce	NumericalValue	0..1	The minimum force that the completed crimp should withstand when wire is being pulled off the terminal. Above that force, the crimp can be expected to be destroyed at any time. This value applies when core & insulation crimp are functional.

insulationPullOffForce	NumericalValue	0..1	The minimum force that the completed crimp should withstand when wire is being pulled off the terminal. Above that force, the crimp can be expected to be destroyed at any time. This value applies when only the insulation crimp is functional.
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Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
appliesTo	InsulationSpecifica tion	1	Y	N	NA	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
insulationCrimpDet ails	0..*	Y	CoreCrimpDetail	1	Defines the different <i>InsulationCrimpDetails</i> that are valid combinations for this <i>CoreCrimpDetail</i> .

2.6.42 Class InsulationSpecification

Specification for the definition of insulation properties of a WireElement.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
wireType	WireType	0..*	Defines the type of the insulation. A <i>WireType</i> is a key in a reference system for the definition of wires. Some reference systems define independent type fragments for insulations and conductors and the <i>WireType</i> of a <i>WireElement</i> is the combination of both. An insulation must not have more than one type. This attribute allows more than one value for the reason, that the same type can be expressed in multiple reference systems.
baseColor	Color	0..*	Specifies the base color of the insulation.
firstIdentificationColor	Color	0..*	Specifies the first identification color of the insulation.
secondIdentificationColor	Color	0..*	Specifies the second identification color of the insulation.
labelIdentificationColor	Color	0..*	Specifies the color of a label printed on the insulation of the wire.
labelIdentificationType	LabelType	0..1	Specifies the type of a label printed, stamped or integrated on or in the insulation of the wire (e.g. alpha numerical, bar code).
labelIdentificationValue	String	0..1	Specifies the value of a label printed on the insulation of the wire. This value applies to all wire elements that satisfy this specification, so all instances share the same label value (e.g. a wire type or a core number in a multi core wire). If instances of a wire require individual labelling it has to be defined on the <i>WireElementReference</i> .
material	Material	0..*	Specifies the material of the insulation.

thickness	NumericalValue	0..1	Specifies the thickness of the insulation.
labelingTechnology	LabelingTechnology	0..1	Specifies used labeling technology if the label is already defined in this specification.
allowedLabelingTechnology	LabelingTechnology	0..*	Specifies the allowed labeling technologies if the insulation is not yet labeled, but support labeling in its usage.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
appliesTo	1	N	InsulationCrimpDetail	NA	NA
insulationSpecification	0..1	N	WireElementSpecification	0..*	If the <i>WireElement</i> has an insulation then the specification of the insulation is referenced here.
recommendedInsulationSpecification	0..1	N	Signal	NA	Defines a recommended Specification for the insulation (e.g. the color) that implements this signal.

2.6.43 Class InternalComponentConnection*Deprecated since 2.2:**Reason:*

The *InternalComponentConnection* has been replaced with the more generic approach of adding a *ConnectionSpecification* to a E/E component description (see "Internal Connectivity").

An *InternalComponentConnection* defines a conductive connection between *PinComponents* within an *EEComponent*. Such a connection can be statically permanent or dynamically switch (e.g. by a relais). This behavior can be defined with a *SwitchingState*.

The electrical behavior of an *InternalComponentConnection* can be further specified by a referenced *ConductorSpecification*.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	Deprecated
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Identification of the <i>InternalComponentConnection</i> , which must be distinct for all <i>InternalComponentConnection</i> of an <i>EEComponent</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
conductorSpecification	ConductorSpecification	0..1	Y	N	0..*	N	NA
pins	PinComponent	2..*	Y	N	0..*	N	NA

Incoming Relations

This End	Other End	General
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Role	Mult.	Agg.	Type	Mult.	Comment
connections	0..*	Y	EEComponentSpecification	1	NA
switchedConnections	0..*	N	SwitchingState	0..*	Specifies the <i>InternalComponentConnections</i> that are switched by this <i>SwitchingState</i> .

2.6.44 Class InternalTerminalConnection

An InternalTerminalConnection represents an electrical connection within a terminal. For standard terminals all receptions (wire- and terminal-receptions) have an electrical connection. For non-standard terminals (e.g. coax) only some receptions have an electrical connection. The InternalTerminalConnection is modelled as a separate class and not as relationship between wire- and terminal-reception, since it is possible that a terminal has only one kind of reception (e.g. a parallel connector, a cavity bridge). (see KBLFRM-302)

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the InternalTerminalConnection. The identification is guaranteed to be unique within the TerminalSpecification (this might be for example an internal connection number).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
terminalReception	TerminalReception	0..*	Y	N	0..1	N	References the TerminalReceptions that participate in the InternalTerminalConnection.
wireReception	WireReception	0..*	Y	N	0..1	N	References the WireReceptions that participate in the InternalTerminalConnection.

Incoming Relations

This End			Other End			General
Role	Mult.	Agg.	Type	Mult.		Comment
internalTerminalConnection	0..*	Y	TerminalSpecification	1		Specifies the InternalTerminalConnections of the terminal.

2.6.45 Class LabelingSpecification

A component (of any type) can be labeled in many ways either during the manufacturing process, for example a label that is imprinted on the component during the general manufacturing process (e.g. the part number), or in its concrete use (e.g. with wire number or a usage node id). The LabelingSpecification defines general labeling properties of the component. Labeling properties in its concrete use are defined with the corresponding role.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
labelType	LabelType	0..1	Specifies the type of a label printed, stamped or integrated on or in component (e.g. alpha numerical, bar code).
labelValue	String	0..1	Specifies the value of a label printed on the component. This value applies to all components that satisfy this specification, so all instances share the same label value (e.g. a part number). If instances of a component require individual labeling it has to be defined on the LabelingRole.
labelingTechnology	LabelingTechnology	0..1	Specifies the used labeling technology if the label is already defined in this specification.
allowedLabelingTechnology	LabelingTechnology	0..*	Specifies the allowed labeling technologies if the component is not yet labeled, but support labeling in its usage.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
labelingSpecification	1	N	LabelingRole	0..*	NA

2.6.46 Class ModularSlot

A *ModularSlot* is a specialization of *AbstractSlot*. Unlike a *Slot*, it does not contain cavities itself but serves as a mounting location for other connector housings (modules) during assembly. This enables flexible, modular assembly and potentially reusable mating strategies.

General Information

Base Classifier	AbstractSlot
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
optional	Boolean	0..1	Specifies whether the allocation of the ModularSlot is optional or not (can a usage of the connector housing leave this ModularSlot empty).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
allowedInserts	PartRelation	0..*	Y	N	0..*	N	References the <i>PartRelations</i> that are valid inserts for this <i>ModularSlot</i> . This reference points to <i>PartRelations</i> in order to allow referencing indirectly a <i>PartVersion</i> if the description of individual <i>PartVersions</i> is done with one physical VEC file per <i>PartVersion</i> and to allow the expression of optional inserts, choices etc. However, inserts for a <i>ModularSlot</i> are always ConnectorHousings by itself. Therefore, the

							referenced <i>PartVersion</i> shall have a <i>PrimaryPartType</i> = <i>ConnectorHousing</i>
layout	CavityLayout	0..*	Y	N	0..1	Y	NA

Incoming Relations

This End			Other End			General	
Role	Mult.	Agg.	Type	Mult.		Comment	
slot	1	N	ModularSlotAddOn	0..*		NA	

2.6.47 Class ModularSlotAddOn

Specifies the wire addon needed to reach a *ModularSlot* from a specific *SegmentConnectionPoint*. The addon needed to reach the cavities of the insert(s) from this point is defined by the *ConnectorHousingSpecification* of the insert.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
wireAddOn	NumericalValue	1	Specifies the wire length add on needed for the modular slot.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
slot	ModularSlot	1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End			General	
Role	Mult.	Agg.	Type	Mult.		Comment	
ModularSlotAddOns	0..*	Y	SegmentConnectionPoint	1		NA	

2.6.48 Class ModularSlotConfiguration

A *ModularSlotConfiguration* defines a valid combination of *ModularSlotLayouts* for a set of modular slots within a connector housing.

This class enables the modeling of complex interdependencies between modular slots—particularly when the validity of one layout depends on the configuration of other slots. It is essential for ensuring consistency in use cases requiring continuous cavity numbering or where spatial constraints affect insert placement.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>ModularSlotConfiguration</i> . The identification is guaranteed to be unique within the <i>ConnectorHousingSpecification</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
validLayoutCombination	CavityLayout	0..*	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
modularSlotConfiguration	0..*	Y	ConnectorHousingSpecification	1	NA

2.6.49 Class MultiCavityPlugSpecification*Deprecated since 2.2.0:*

Reason: Was replaced by the more powerful concept of CablePlugs (see documentation of *MultiCavitySealSpecification* for a detailed explanation).

Specification for the definition of cavity plugs that apply to more than one cavity. A cavity plug is a watertight non-electrical object to fill an empty cavity. MultiCavityPlugs are formed to fit into one connector / slot and to seal more than one cavity at once. Normally there are different variants of these MultiCavityPlugs that can seal a connector in different pinning scenarios. The cavities that are plugged by a MultiCavityPlug are defined with a SealedCavitiesAssignment.

General Information

Base Classifier	CavityPlugSpecification
Applied Stereotype	Deprecated
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
assignment	SealedCavitiesAssignment	0..*	Y	N	1	Y	References the <i>SealedCavitiesAssignments</i> that are valid for this <i>MultiCavityPlug</i> . One individual <i>SealedCavitiesAssignment</i> is used for each connector housing that matches this <i>MultiCavityPlug</i> .

2.6.50 Class MultiCavitySealSpecification*Deprecated since 2.2.0:*

Reason: Was replaced by the more powerful concept of CableSeals (see documentation of *MultiCavitySealSpecification*).

Specification for cavity seals that apply to more than one cavity. A *CavitySeal* is a watertight non-electrical object to fill a populated Cavity. *MultiCavitySeals* are formed to fit into one connector / slot and to seal more than one cavity at once.

There are two types of *MultiCavitySeals*:

1. In first type the *MultiCavitySeal* has an opening for all cavities of the connector. Each opening can be filled either with a wire (without an individual seal) or with a *CavityPlug* (e.g. a synthetic pin) or a *MultiCavityPlug*.
2. In the second type, the *MultiCavitySeal* has a specific configuration of openings for some cavities of the connector. These *MultiCavitySeals* are sealing all cavities with an opening and a wire in it and all cavities where no opening in the *MultiCavitySeal* exists. For each opening that has no wire it an additional *CavityPlug* is needed.

The cavities that are left open by a *MultiCavitySeal* are defined with a *SealedCavitiesAssignment*.

Note: With Version 2.2.0 The *MultiCavitySealSpecification* (and all associated concepts like the *MultiCavityPlugSpecification* and the corresponding Roles) has been replaced by the *CableSealSpecification*. The *MultiCavitySealSpecification* has multiple design flaws that became obvious, when more complex connector assemblies were taken into account (e.g. modern high-voltage or high-frequency connector systems). The main reason for deprecation of this concept were:

1. The *MultiCavitySealSpecification* inherits from *CavitySealSpecification*. Therefore, all openings in a *MultiCavitySeal* have to share the same properties.
2. A *ContactPoint* is required to introduce an occurrence of a *MultiCavitySeal*. However, such seals are often placed at certain position on the wire and are not associated directly with specific *ContactPoint*. They also might be used on multi-core wires before those are split up into individual wire elements (that are contracted individually).
3. It is not possible to assign a wire to a specific opening in the seal, which means an incomplete product definition.
4. Such seals are often used in addition to a conventional cavity seal. However, the *ContactPoint* only permits one *CavitySeal* per contacting situation. This is true for conventional cavity seals that are associated with a specific crimp. This is not true for cable seals where multiple seals can be added to the wire.

General Information

Base Classifier	CavitySealSpecification
Applied Stereotype	Deprecated
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
assignment	<i>OpenCavitiesAssignment</i>	0..*	Y	N	1	Y	References the <i>OpenCavitiesAssignments</i> that are valid for this <i>MultiCavitySeal</i> . One individual <i>OpenCavitiesAssignment</i> is used for each connector housing that matches this <i>MultiCavitySeal</i> .

2.6.51 Class *MultiFuseSpecification*

NA

General Information

Base Classifier	EEComponentSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
fuseType	MultiFuseType	0..1	Defines the manufacturing types of a multi-fuse.
iMaxTotal	NumericalValue	0..1	Specifies the maximum electric current tolerated by the multi-fuse in total.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
fuseComponents	FuseComponent	0..*	Y	N	1	Y	NA

2.6.52 Class OpenCavitiesAssignment*Deprecated since 2.2.0:**Reason: see [MultiCavitySealSpecification](#)*

An OpenCavitiesAssignment groups the cavities of ONE connector that are open in a MultiCavitySeal. If a MultiCavitySeal fits into more than one connector, than there are as many OpenCavitiesAssignments.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	Deprecated
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
openCavities	Cavity	1..*	Y	N	0..*	N	Specifies the cavities that are open.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
assignment	0..*	Y	MultiCavitySealSpecification	1	References the <i>OpenCavitiesAssignments</i> that are valid for this <i>MultiCavitySeal</i> . One individual <i>OpenCavitiesAssignment</i> is used for each connector housing that matches this <i>MultiCavitySeal</i> .

2.6.53 Class OpenWireEndTerminalSpecification

Specification for the definition of open wire ends. An open wire end is an end of wire located at a specific location in the harness. Open wire ends are not connected to some other component, in contrast to regular connectors or ring terminals. Only one wire is ending at a single open wire end, otherwise it would be a splice.

General Information

Base Classifier	TerminalSpecification
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Applied Stereotype	NA
Is Abstract	false

2.6.54 Class PartRelation

A *PartRelation* defines additional parts (e.g. accessories) for a specific part. These parts are in some way or usage scenario required for the part itself to be used. However, they are not included with the part number and have to be ordered separately. This can be used for example for caps, cable ties etc.

The associated *PartRelations* of a *GeneralTechnicalPartSpecification* represent a configurable bill of material that can/must be ordered together with the part, when it is used. Each *PartRelation* represents an item / line in this bill of material. The semantic by which a *PartRelation* is resolved to *PartVersions* is defined by the *PartRelationType*. If multiple *PartRelations* resolve to the same *PartVersions* the resulting bill of material is the sum of them.

If a *PartRelation* references more than one *accessoryPart* the *PartRelationType* defines the semantic to resolve this reference for a resulting bill of material. If the type is *Mandatory* all referenced *PartVersions* shall be in the resulting bill of material. If the type is *Optional*, the referenced *PartVersions* can be selected by choice into the resulting bill of material. However, the choice applies to all *PartVersions* of one *PartRelation*. For *Mandatory* it is semantically equivalent to have one *PartRelation* referencing N *PartVersions* or to have N *PartRelations*, each referencing one *PartVersion*. The *PartRelationType OneOfAll* defines, that exactly one of the referenced *PartVersions* shall be chosen for the resulting bill of material.

If the same *PartVersion* is referenced multiple times, each reference counts as its own position.

Example: To express that a *PartVersion* shall be used at least three times and with a maximum of 6 times, three mandatory and three optional *PartRelations* to this *PartVersion* would be created.

With these concepts, simple yes/no decisions can be represented. However, there are cases where there are constraints between accessory parts (e.g. if part A, then choice of 2 x B or 1 x C). To express such logic in a static object model is not very feasible and inflexible. For such cases, the *relationType 'Custom'* was introduced. In this case, the relationships and constraints between all referenced *accessoryPart* can be expressed with some custom expression language in the *customRelationExpression* attribute. Even if it is custom, the expression shall only refer to elements that are contained in the *accessoryPart* relation and shall not influence other *PartRelations* of the same *GeneralTechnicalPartSpecification*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
relationType	PartRelationType	1	Specifies the type of the relation.
customRelationExpression	String	0..1	Defines the relationship between the accessory parts in a proprietary expression language. This attribute shall only be used, if the <i>relationType</i> = 'Custom'.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
accessoryPart	PartVersion	1..*	N	N	0..*	N	References the PartVersions that are related by the PartRelation.

Incoming Relations

This End	Other End	General
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Role	Mult.	Agg.	Type	Mult.	Comment
allowedInserts	0..*	N	ExtensionSlot	0..*	References the <i>PartRelations</i> that are valid inserts for this <i>ExtensionSlot</i> . This reference points to <i>PartRelations</i> in order to allow referencing indirectly a <i>PartVersion</i> if the description of individual <i>PartVersions</i> is done with one physical VEC file per <i>PartVersion</i> and to allow the expression of optional inserts, choices etc. However, inserts for an <i>ExtensionSlot</i> are always <i>EComponents</i> by itself. Therefore, the referenced <i>PartVersion</i> shall have a <i>PrimaryPartType</i> = <i>EComponent</i> .
allowedInserts	0..*	N	ModularSlot	0..*	References the <i>PartRelations</i> that are valid inserts for this <i>ModularSlot</i> . This reference points to <i>PartRelations</i> in order to allow referencing indirectly a <i>PartVersion</i> if the description of individual <i>PartVersions</i> is done with one physical VEC file per <i>PartVersion</i> and to allow the expression of optional inserts, choices etc. However, inserts for a <i>ModularSlot</i> are always <i>ConnectorHousings</i> by itself. Therefore, the referenced <i>PartVersion</i> shall have a <i>PrimaryPartType</i> = <i>ConnectorHousing</i> .
fixationAccessory	0..*	Y	WireTupleSpecification	1	Specifies <i>PartRelations</i> that can / have to be used for the fixation.
partRelation	0..*	Y	GeneralTechnicalPartSpecification	1	Specifies possible relations (accessories) of the specified part with other <i>PartVersion</i> (e.g. caps, clips).
supplementaryParts	0..*	N	PlacementPoint	NA	References the <i>PartRelations</i> that specify supplementary parts for this <i>PlacementPoint</i> , e.g. cable ties for attaching the component on the harness.
supplementaryParts	0..*	N	CableLeadThroughOutlet	NA	References the <i>PartRelations</i> that specify supplementary parts for this <i>CableLeadThroughOutlet</i> .
supplementaryParts	0..*	N	Slot	0..*	References the <i>PartRelations</i> that specify supplementary parts for this slot.

2.6.55 Class PinComponent

A *PinComponent* describes a pin of an *EComponent*. A *PinComponent* is part of a *HousingComponent* and is defined by three aspects.

1. Its geometric position in the *HousingComponent*, which is defined by the *referencedCavity*.
2. Its physical electrical properties, which are defined by the referenced *TerminalSpecification*.
3. Its electrical behavior, which is defined configuration dependent by its *PinComponentBehaviour*.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Identification of the <i>PinComponent</i> , which must be distinct for all <i>PinComponent</i> of a <i>HousingComponent</i> .
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the <i>PinComponent</i> .

pinComponentType	PinComponentType	0..1	Specifies the type of a <i>PinComponent</i> .
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Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
behaviour	PinComponentBehavior	0..*	Y	N	1	Y	Specifies the configuration dependent electrical behavior of the <i>PinComponent</i> .
componentPort	ComponentPort	0..1	Y	N	0..*	N	The <i>ComponentPort</i> that represents this <i>PinComponent</i> in the internal connectivity of the <i>EEComponentSpecification</i> .
pinSpecification	TerminalSpecification	0..1	Y	N	0..*	N	References the <i>TerminalSpecification</i> describing the electrical connectivity aspect of the <i>PinComponent</i> . (see KBLFRM-300)
referencedCavity	Cavity	0..1	Y	N	0..*	N	Defines the cavity in the corresponding <i>ConnectorHousingSpecification</i> of the <i>HousingComponent</i> where the <i>PinComponent</i> is located. (see KBLFRM-300)

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
anode	0..1	N	DiodeSpecification	0..*	NA
cathode	0..1	N	DiodeSpecification	0..*	NA
connectedPins	0..2	N	FuseComponent	0..*	NA
pinComponent	0..*	Y	HousingComponent	1	Specifies the <i>PinComponents</i> of <i>HousingComponent</i> . (see KBLFRM-300)
pinComponent	1	N	PinComponentReference	0..*	Points to the <i>PinComponent</i> referenced by the <i>PinComponentReference</i> .
pins	2..*	N	InternalComponentConnection	0..*	NA

2.6.56 Class PinComponentBehavior

A *PinComponentBehavior* specifies the electrical behavior of a *PinComponent*. Since the behavior of a pin is configuration dependent (e.g. the software deployed on an ECU) the *PinComponentBehavior* inherits from *ConfigurableElement*. Therefore, a *PinComponent* can specify multiple *PinComponentBehavior*.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
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identification	String	1	Identification of the <i>PinComponentBehavior</i> which identifies it from a user perspective, and which must be distinct for all <i>PinComponentBehaviors</i> of a <i>PinComponent</i> .
aliasId	AliasIdentification	0..*	Specifies additional identifiers for the element.
signalDirection	SignalDirection	0..1	The direction of the signal on this pin.
pinType	PinType	0..1	Specifies the <i>PinType</i> of the <i>PinComponent</i> .
applianceType	PinApplianceType	0..1	Classifies the appliance of a Pin in terms of the duration of the appliance (see <i>PinApplianceType</i>).
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the <i>PinComponentBehaviour</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
currentInformation	PinCurrentInformation	0..*	Y	N	1	Y	Specifies the current information of the <i>PinComponent</i> in this <i>PinComponentBehavior</i> . Since the current values of a pin can be defined for different types and times it is possible to define multiple <i>PinCurrentInformations</i> for a <i>PinComponentBehavior</i> .
opticalInformation	PinOpticalInformation	0..*	Y	N	1	Y	Specifies the optical information of the pin, if it has the type optical.
signal	Signal	0..1	Y	N	0..*	N	Specifies the <i>Signal</i> associated with the pin in this behavior.
voltageInformation	PinVoltageInformation	0..*	Y	N	1	Y	Specifies the voltage information of the <i>PinComponent</i> in this <i>PinComponentBehavior</i> . Since the voltage values of a pin can be defined for different types and times it is possible to define multiple <i>PinVoltageInformations</i> for a <i>PinComponentBehavior</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
behaviour	0..*	Y	PinComponent	1	Specifies the configuration dependent electrical behavior of the <i>PinComponent</i> .

2.6.57 Class PinCurrentInformation

Allows the definition of currents for a pin of an *EEComponent*. A current can be further specified by a duration. Attributes of the type *PinCurrentInformation* normally have the multiplicity [0..*]. This means that such an attribute can have *PinCurrentInformation* entries for different types and durations. It must not have multiple entries for the same type and duration.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	PinCurrentType	1	Defines the type of the current.
current	NumericalValue	1	The current of the pin.
determinationType	ValueDetermination	0..1	Specifies the determination type of the <i>PinCurrentInformation</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
timing	PinTiming	0..*	Y	N	0..1	Y	Specifies the timing of the <i>PinCurrentInformation</i> .

Incoming Relations

This End			Other End		General	
Role	Mult.	Agg.	Type	Mult.	Comment	
currentInformation	0..*	Y	PinComponentBehavior	1	Specifies the current information of the <i>PinComponent</i> in this <i>PinComponentBehavior</i> . Since the current values of a pin can be defined for different types and times it is possible to define multiple <i>PinCurrentInformations</i> for a <i>PinComponentBehavior</i> .	

2.6.58 Class PinOpticalInformation

Allows the specification of optical information in a *PinComponentBehavior*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
frequency	NumericalValue	1	The frequency of the optical signal.
attenuation	NumericalValue	0..1	The attenuation of the optical pin at the defined frequency.

Incoming Relations

This End			Other End		General	
Role	Mult.	Agg.	Type	Mult.	Comment	
opticalInformation	0..*	Y	PinComponentBehavior	1	Specifies the optical information of the pin, if it has the type optical.	

2.6.59 Class PinTiming

Specifies the timing for a *PinCurrentInformation* or a *PinVoltageInformation*.

General Information

Base Classifier	ExtendableElement
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Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	PinTimingType	1	Specifies the type of the timing.
time	NumericalValue	1	Specifies the time value of the timing.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
timing	0..*	Y	PinVoltageInformation	0..1	Specifies the timing of the <i>PinVoltageInformation</i> .
timing	0..*	Y	PinCurrentInformation	0..1	Specifies the timing of the <i>PinCurrentInformation</i> .

2.6.60 Class PinVoltageInformation

Allows the definition of voltages for a pin of an *EEComponent*. A current can be further specified by a duration. Attributes of the type *PinVoltageInformation* normally have the multiplicity [0..*]. This means that such an attribute can have *PinVoltageInformation* entries for different types and durations. It must not have multiple entries for the same type and duration.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	PinVoltageType	1	Defines the type of the voltage.
voltage	NumericalValue	1	The voltage of the pin.
determinationType	ValueDetermination	0..1	Specifies the determination type of the <i>PinVoltageInformation</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
timing	PinTiming	0..*	Y	N	0..1	Y	Specifies the timing of the <i>PinVoltageInformation</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
voltageInformation	0..*	Y	PinComponentBehavior	1	Specifies the voltage information of the <i>PinComponent</i> in this <i>PinComponentBehavior</i> . Since the voltage values of a pin can be defined for different types and times it is possible to define

					multiple <i>PinVoltageInformations</i> for a <i>PinComponentBehavior</i> .
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2.6.61 Class PluggableTerminalSpecification

Specification for the definition of pluggable terminals.

General Information

Base Classifier	TerminalSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
terminalType	PluggableTerminalType	0..1	Defines the type of the terminal. The type defines constraints about the numbers of wire and terminal receptions and their relations.
maximumInsertionForce	NumericalValue	0..1	The maximum force that may be applied to the terminal during the insertion process of the terminal into the housing

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
extractionTool	ResourceVersion	0..*	Y	N	0..*	N	NA

2.6.62 Class PotentialDistributorSpecification

Specifies the properties of a potential distributor (e.g. a joint connector).

General Information

Base Classifier	EEComponentSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
screwable	Boolean	0..1	Defines if the <i>PotentialDistributor</i> can be screwed by itself to be fixed. This fixation is a non-electrical connection. If the <i>PotentialDistributor</i> can be fixated electrical (like a ring terminal) it shall have one <i>PinComponent</i> with a <i>RingTerminalSpecification</i> .
boltDiameter	NumericalValue	0..1	Specifies the diameter of the bolt for which the potential distributor is designed for.
boltNominalSize	TerminalBoltNominalSize	0..1	Defines the size (diameter) of the bolt for which the potential distributor is designed for in a nominal way (e.g. "M8").
boltType	TerminalBoltType	0..1	Specifies the type of the bolt to which the potential distributor is compatible.

2.6.63 Class PowerConsumption

Defines the power consumption of an *EEComponent*. An *EEComponent* can have multiple different *PowerConsumptions* e.g. standby, maximum. An *EEComponent* can have multiple *PowerConsumptions* but must not have more than one *PowerConsumptions* of the same *type*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	PowerConsumptionType	1	The type of a <i>PowerConsumption</i> . This should be an enumeration of values defined in a conformance class.
value	NumericalValue	1	Specifies the power consumption for this type of <i>PowerConsumption</i> .

2.6.64 Class ReferenceSurfaceDefinition

A connector housing can define reference surfaces that serve as contact surfaces in a work piece holder. Those reference surfaces normally have defined tolerance limits to support accurate determination of cavity positions for example during the insertion process.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
hasSurfaces	Boolean	1	Specifies whether the connector has defined reference surfaces or not.
dimensionX	NumericalValue	0..1	Specifies the distance and tolerance between the middle axis of the lowest alpha numerical cavity (origin of the connector coordinate system) and the reference surface in direction of the X-axis.
dimensionY	NumericalValue	0..1	Specifies the distance and tolerance between the middle axis of the lowest alpha numerical cavity (origin of the connector coordinate system) and the reference surface in direction of the X-axis.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
referenceSurfaceDefinition	0..1	Y	ConnectorHousingSpecification	1	Contains the information about reference surfaces of the connector.

2.6.65 Class RelaySpecification

Specification of the electrological aspects of a relay.

General Information

Base Classifier	EEComponentSpecification
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Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
iMax	NumericalValue	0..1	Specifies the maximum current allowed for the relais.
relaisType	RelaisType	0..1	The type of the relay (switching behavior). This is an OpenEnumeration, for values see <i>RelaisType</i> .
lowNoise	Boolean	0..1	Defines if the relais switch with low noise / silently or not.
applianceType	RelaisApplianceType	0..1	Specifies the appliance type of a relais.

2.6.66 Class RingTerminalSpecification

Specification for the definition of ring terminals. These are the counterparts to bolt terminals.

General Information

Base Classifier	TerminalSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
boltDiameter	NumericalValue	0..1	Specifies the diameter of the bolt for which the ring terminal is designed for in a numerical way.
boltNominalSize	TerminalBoltNominalSize	0..1	Defines the size (diameter) of the bolt for which the ring terminal is designed for in a nominal way (e.g. "M8").
thickness	NumericalValue	0..1	Specifies the thickness of the contact surface of the ring terminal.
boltType	TerminalBoltType	0..1	Specifies the type of the bolt to which the ring terminal is compatible.
outsideDimension	NumericalValue	0..1	Specifies the diameter of the circle around the center of the bolt which passes through the farthest outside point of the contact surface of the ring terminal.
torsionProtection	Boolean	0..1	Specifies if the ring terminal is torsion protected or not. (see KBLFRM-311)
nutIntegrated	Boolean	0..1	Defines if the ring terminal has nut integrated for the fixation on the corresponding bolt terminal.

2.6.67 Class SealedCavitiesAssignment

Deprecated since 2.2.0:

Reason: see [MultiCavityPlugSpecification](#)

A *SealedCavitiesAssignment* groups the cavities of ONE connector that are sealed by a MultiCavityPlug. If a MultiCavityPlug fits into more than one connector, than there are as many *SealedCavitiesAssignments*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	Deprecated
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
sealedCavities	Cavity	1..*	Y	N	0..*	N	Specifies the Cavities that are sealed.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
assignment	0..*	Y	MultiCavityPlugSpecification	1	References the <i>SealedCavitiesAssignments</i> that are valid for this <i>MultiCavityPlug</i> . One individual <i>SealedCavitiesAssignment</i> is used for each connector housing that matches this <i>MultiCavityPlug</i> .

2.6.68 Class SealingPin

Specifies a pin to seal a *CableSealEntry*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Provides an identifier for the <i>SealingPin</i> . The <i>Identification</i> shall be unique within a <i>CableSealPlugSpecification</i> .
outsideDiameter	NumericalValue	0..1	Defines the outside diameter in the relaxed state for the <i>SealingPin</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
position	Cavity	0..1	Y	N	0..*	N	Specifies the <i>Cavity</i> in the <i>CavityLayout</i> of the <i>CableSealPlugSpecification</i> where this <i>SealingPin</i> is positioned.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
sealingPin	1..*	Y	CableSealPlugSpecification	NA	NA
sealingPin	1	N	SealingPinReference	NA	NA

2.6.69 Class SegmentConnectionPoint

Specifies a point where the connector can be attached to the topology (sometimes called bundle position point or insertion point).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the SegmentConnectionPoint. The identification is guaranteed to be unique within the ConnectorHousingSpecification.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavityAddOns	CavityAddOn	0..*	Y	N	1	Y	NA
ModularSlotAddOns	ModularSlotAddOn	0..*	Y	N	1	Y	NA
placementPoint	PlacementPoint	0..1	Y	N	0..*	N	Specifies the <i>PlacementPoint</i> that represents this <i>SegmentConnectionPoint</i> in a <i>PlaceableElementSpecification</i> .
reachableCavities	Cavity	0..*	Y	N	0..*	N	Specifies the <i>Cavities</i> that are reachable with wires through this <i>SegmentConnectionPoint</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
connectorSegmentConnectionPoint	0..1	N	SegmentConnectionPointHC	0..*	References the <i>SegmentConnectionPoint</i> of the <i>ConnectorHousingSpecification</i> that is used to "enter" the connector housing.
segmentConnectionPoint	0..*	Y	ConnectorHousingSpecification	1	Specifies the <i>SegmentConnectionPoints</i> the connector housing.

2.6.70 Class SegmentConnectionPointHC

Specifies a point where the *HousingComponent* can be attached to the topology [if required] (sometimes called bundle position point or insertion point).

Note: The suffix HC is justified by the fact that class names must be unique within the model and the name SegmentConnectionPoint is already used within the ConnectorHousingSpecification. However, the represented semantics of the two connection points are so similar that the use of actually different names for both would rather have led to confusion. So SegmentConnectionPointHC actually is shorthand for "SegmentConnectionPoint within HousingComponent".

General Information

Base Classifier	ExtendableElement
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Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	NA
wireAddOn	NumericalValue	0..1	Defines the wire add-on that is required to reach the referenced SegmentConnectionPoint from this SegmentConnectionPointHC with which the individual cavity add-ons are defined.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connectorSegmentConnectionPoint	SegmentConnectionPoint	0..1	Y	N	0..*	N	References the <i>SegmentConnectionPoint</i> of the <i>ConnectorHousingSpecification</i> that is used to "enter" the connector housing.
placementPoint	PlacementPoint	0..1	Y	N	0..*	N	Specifies the <i>PlacementPoint</i> that represents this <i>SegmentConnectionPoint</i> in a <i>PlaceableElementSpecification</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
segmentConnectionPoint	0..*	Y	HousingComponent	1	Specifies the <i>SegmentConnectionPoints</i> the <i>HousingComponent</i> .

2.6.71 Class ShieldSpecification

Specifies the properties of a shield.

General Information

Base Classifier	ConductorSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
opticalTissueDensity	NumericalValue	0..1	<u>Deprecated since 2.1.0:</u> <i>Reason: Attribute had incorrectly the data NumericalValue, a value with unit, which is not correct for a value representing a ratio. Use OpticalCoverage instead.</i> Defines the optical tissue density of the strands of a shield. This is defined as a value in percentage. <i>Should only be used for ShieldSpecifications with type=Braided.</i>
opticalCoverage	Double	0..1	Defines the optical coverage of the strands of a shield. This value is defined as the ratio of overall area to the area covered by the

			shield. A value of 0.0 would mean no cover at all, a value of 1.0 means a gapless covering by the shield. <i>Should only be used for ShieldSpecifications with type=Braided.</i>
windingType	FoilShieldWindingType	0..1	Defines the type of winding of the shield. <i>Should only be used for ShieldSpecifications with type=Foil.</i>

2.6.72 Class Slot

A *Slot* is a specific type of *AbstractSlot* that represents a group of cavities **integrated directly** into a connector housing. The design of a slot is described by a *SlotSpecification*.

Slots serve two primary purposes:

1. **Cavity grouping within a housing:** Each slot contains or groups cavities with specific properties and geometric placement.
2. **Definition of mating-side structure:** A slot represents a logical and physical mating region of a connector. This makes it a fundamental unit for modular mating. For example, a connector with three slots can be mated with another connector that has only one slot, occupying just one slot on the multi-slot connector. This supports asymmetric or partial mating scenarios.

General Information

Base Classifier	AbstractSlot
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
colorInformation	Color	0..*	Specifies the color of the slot.
sealingType	SlotSealingType	0..1	Specifies the type of the sealing of the slot, if sealed. The values are defined in an <i>OpenEnumeration</i> .
permittedTerminalSupplierCompanyNames	String	0..*	If this attribute is defined, it is only permitted to use terminals of one of the listed terminal suppliers. The used company name shall be same as the one used as <i>PartVersion.companyName</i> for part numbers of this supplier.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavity	Cavity	1..*	Y	N	1	Y	Specifies the Cavities forming the Slot.
supplementaryParts	PartRelation	0..*	Y	N	0..*	N	References the <i>PartRelations</i> that specify supplementary parts for this slot.

2.6.73 Class SlotLayout

For regularly laid out slots the slot layout describes the positions of the cavities

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
gridX	NumericalValue	0..1	The horizontal distance between the center points of two cavities.
gridY	NumericalValue	0..1	The vertical distance between the center points of two cavities.
rowCount	Integer	0..1	The number of cavity rows of the slot.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
slotLayout	0..1	Y	SlotSpecification	1	References the layout associated with this slot.

2.6.74 Class SlotSpecification

Specification for the definition of slots.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
emvProtectionRequired	Boolean	0..1	Specifies whether the slot needs EMV protection.
gender	SlotGender	0..1	Specifies the gender of the slot. Valid values are defined in an open enumeration.
layoutType	SlotLayoutType	0..1	Specifies the layout of the slot from a mechanical / geometrical point of view.
secondaryLocking	Boolean	0..1	Specifies whether the slot supports "secondary locking", which refers to a locking or a lock reinforcement of the terminals in the connector, in addition to the primary locking of terminal itself (e.g. lances on the terminal). This is also often referred to as TPA (Terminal Position Assurance).
secondaryLockingType	SecondaryLockingType	0..1	Defines the type of the secondary locking, if the slot features a secondary locking (<i>secondaryLocking=true</i>).
numberOfCavities	Integer	1	The possible number of cavities in the layout defined by the SlotSpecification. This includes all cavities in the layout. The actual Slot can define specific cavities in the layout as "not available".

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
slotLayout	SlotLayout	0..1	Y	N	1	Y	References the layout associated with this slot.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
slotSpecification	0..1	N	AbstractSlot	0..*	References the SlotSpecification that is satisfied by the slot.

2.6.75 Class SpliceTerminalSpecification

Specification for the definition of splice terminals.

General Information

Base Classifier	TerminalSpecification
Applied Stereotype	NA
Is Abstract	false

2.6.76 Class SwitchingState

A *SwitchingState* defines a certain static state of an *EEComponent*. Under which conditions or when such a state applies, should be described in an external description model.

A *SwitchingState* references a collection of *InternalComponentConnections*, with the semantic that these connections exist (and only these connections) when the switching state is active.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Identification of the SwitchingState, which must be distinct for all SwitchingStates of an EEComponent.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
switchedConnections	InternalComponentConnection	0..*	Y	N	0..*	N	Specifies the <i>InternalComponentConnections</i> that are switched by this <i>SwitchingState</i> .
switchedLogicalConnections	Connection	1..*	Y	N	NA	N	Specifies the <i>Connections</i> in the internal connectivity of the <i>EEComponentSpecification</i> , that are switched by this <i>SwitchingState</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
states	0..*	Y	EEComponentSpecification	1	Specifies the available <i>SwitchingStates</i> of the EEComponent.

2.6.77 Class TerminalCurrentInformation

Allows the definition of valid current ranges for a terminal. A current range is always defined for a *coreCrossSectionArea* and a *nominalVoltage*. Attributes of the type *TerminalCurrentInformation* normally have the multiplicity [0..*]. This means that such an attribute can have *TerminalCurrentInformation* entries for different *coreCrossSectionAreas*, *nominalVoltages*, *environmentTemperature* and mating terminals. It must not have multiple entries for the same values.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
currentRange	ValueRange	1	Specifies the valid current range for the terminal. It is defined as a value range, because there are terminals where minimum current is needed to avoid corrosion.
nominalVoltage	NumericalValue	0..1	Specifies the nominalVoltage for which the CurrentRangeInformation is valid.
coreCrossSectionArea	NumericalValue	1	Specifies the coreCrossSectionArea for which the CurrentRangeInformation is valid. The coreCrossSectionArea is a relevant information, because the thermal absorption of the core depends on the cross-section area and thus is an influence factor for the valid current range.
environmentTemperature	NumericalValue	0..1	The environment temperature for which this maximum current value is applicable.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
validForMatingTerminal	PartVersion	0..*	Y	N	0..*	N	<i>PartVersions</i> of mating terminals for which this <i>TerminalCurrentInformation</i> is valid. This is a relevant information, because the thermal absorption of the counterpart is a relevant influence factor for possible current.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
currentInformation	0..*	Y	TerminalSpecification	1	Specifies the <i>TerminalCurrentInformation</i> that is applicable for the terminal.

2.6.78 Class TerminalDistanceInformation

Allows the definition of valid minimum distances for a terminal. A minimum distance is always a *NumericalValue* defined for a *TerminalDistanceType*. Attributes of the type *TerminalDistanceInformation* normally have the multiplicity [0..*]. This means that such an attribute can have *TerminalDistanceInformation* entries for different *TerminalDistanceType*. It must not have multiple entries for the same *TerminalDistanceType*.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult.	Comment
type	TerminalDistanceType	1	The type of the <i>TerminalDistanceInformation</i> .
value	NumericalValue	1	Defines the valid minimum distance.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
minimumDistance	0..*	Y	TerminalSpecification	0..1	A terminal can have minimum distance requirements to other elements in the harness. These are defined with the <i>TerminalDistanceInformation</i> .

2.6.79 Class TerminalProtectionSpecification

A *TerminalProtection* is an insulation or sealing that is applied after the joining process between the terminal and the wire, typically in the contact area of the WireReception. Examples include caps for splices or heat shrink tubing for ring terminals.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult.	Comment
insideLength	NumericalValue	0..1	The length on the inside of the terminal protection. For a splice cap, for example, this is the measurement how far wires can be inserted.
insideDiameter	NumericalValue	0..1	The inside diameter of the terminal protection.
outsideLength	NumericalValue	0..1	The length of the terminal protection on the outside.
outsideDiameter	NumericalValue	0..1	The outside diameter of the terminal protection.
wallThickness	NumericalValue	0..1	The wall thickness of the terminal protection. Usually, this should be $(outsideDiameter - insideDiameter) / 2$.
shrinkable	Boolean	0..1	Specifies if the terminal protection is heat shrinkable.
shrinkingFactor	Double	0..1	Defines the factor of shrinking for the terminal protection.
maximumLongitudinalShrinkage	NumericalValue	0..1	Defines the shrinkage in longitudinal direction.

adhesivePresent	Boolean	0..1	Defines if an adhesive / sealing is already present in the terminal protection.
adhesive	Material	0..*	Defines the material used as adhesive / sealing, if present.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
validConductorMaterial	ConductorMaterial	0..*	Y	N	1	Y	Specifies the materials of a conductor, which are valid to use with this <i>TerminalProtectionSpecification</i> . This material shall be matched against the <i>ConductorSpecification.material</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
terminalProtectionSpecification	1	N	TerminalProtectionRole	0..*	NA

2.6.80 Class TerminalReception

A TerminalReception is the area of a terminal where the contacting with another terminal (e.g. between a connector housing and a control unit) takes place. Normally the terminal reception is placed in a cavity of a connector housing.

This class represents such an area on the terminal.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the TerminalReception. The identification is guaranteed to be unique within the TerminalSpecification (this might be for example a reception number).
gender	TerminalReceptionGender	0..1	Specifies the gender of the TerminalReception. The gender is included in the TerminalReception class in order to be able to refer the same TerminalReceptionSpecification gender independent.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
terminalReceptionSpecification	TerminalReceptionSpecification	0..1	Y	N	0..*	N	References the TerminalReceptionSpecification that specifies the TerminalReception.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
terminalReception	0..*	Y	TerminalSpecification	1	Specifies the TerminalReceptions of the terminal described by the TerminalSpecification.
terminalReception	1	N	TerminalReceptionReference	0..*	References the <i>TerminalReception</i> that is instanced by this <i>TerminalReceptionReference</i> .
terminalReception	0..*	N	InternalTerminalConnection	0..1	References the TerminalReceptions that participate in the InternalTerminalConnection.

2.6.81 Class TerminalReceptionSpecification

Specification for the definition of terminal receptions. A TerminalReception is the area of a terminal where the contacting with another terminal (e.g. between a connector housing and a control unit) takes place. Normally the terminal reception is placed in a cavity of a connector housing.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
cavityDesign	String	0..1	<i>Deprecated since 1.1.4:</i> <i>Reason: Has been replaced by the more meaningful mechanism with TerminalTypes.</i>
platingMaterial	Material	0..*	Specifies the plating material of the terminal reception.
primaryLockingType	PrimaryLockingType	0..1	Specifies if the terminal reception has a primary locking and of what type it is.
pullOutForce	NumericalValue	0..1	The force that the terminal can withstand when being pulled out of the housing. Above that force, the terminal should be expected to be pulled out of the housing (destructive).
testingPullOutForce	NumericalValue	0..1	The force that must be applied when the terminal is pulled back out of the housing (without further movement) in order to be able to assume that the terminal is securely latched in the housing (check that the terminal is correctly seated in the connector) and that an insertion process was successful.
contactRangeLength	NumericalValue	0..1	Specifies the length of the contact area (see diagram "Terminal Dimensions").

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
terminalType	TerminalType	0..1	Y	N	0..1	Y	Specifies the terminal type that is associated with the terminal reception.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
terminalReceptionSpecification	0..1	N	TerminalReception	0..*	References the TerminalReceptionSpecification that specifies the TerminalReception.

2.6.82 Class TerminalSpecification

Specification for the definition of terminals. A terminal can own multiple WireReceptions & TerminalReceptions.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
voltageRange	ValueRange	0..1	<u>Deprecated since 2.2.0:</u> <u>Reason:</u> Replaced by <i>GeneralTechnicalPartSpecification.VoltageRating</i> . Specifies the allowed voltage range for the connector housing.
sealingType	TerminalSealingType	0..1	Defines the <i>SealingType</i> of the Terminal. This type always refers to the sealing of the terminal itself. However, even a terminal which is not sealable can be used in sealed locations with additional measures (e.g. on the slot).
connectionALength	NumericalValue	0..1	Specifies the length of the terminal between the contact area (terminal reception) and the crimp are (wire reception, see diagram "Terminal Dimensions").
overallLength	NumericalValue	0..1	Specifies the overall length the terminal (see diagram "Terminal Dimensions").
transmissionMedium	SignalTransmissionMediumType	0..1	Specifies the type of the transmission medium for the terminal.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
currentInformation	TerminalCurrentInformation	0..*	Y	N	1	Y	Specifies the <i>TerminalCurrentInformation</i> that is applicable for the terminal.
internalTerminalConnection	InternalTerminalConnection	0..*	Y	N	1	Y	Specifies the InternalTerminalConnections of the terminal.
minimumDistance	TerminalDistanceInformation	0..*	Y	N	0..1	Y	A terminal can have minimum distance requirements to other elements in the harness. These are defined with the <i>TerminalDistanceInformation</i> .
terminalReception	TerminalReception	0..*	Y	N	1	Y	Specifies the TerminalReceptions of the terminal described by the TerminalSpecification.
wireReception	WireReception	0..*	Y	N	1	Y	Specifies the WireReceptions of the terminal described by the TerminalSpecification.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
integratedTerminalSpecification	0..1	N	Cavity	NA	Specifies the terminal, if the cavity has an integrated terminal (e.g. an IDC).
pinSpecification	0..1	N	PinComponent	0..*	References the <i>TerminalSpecification</i> describing the electrical connectivity aspect of the <i>PinComponent</i> . (see KBLFRM-300)
terminalSpecification	1	N	TerminalRole	0..*	References the <i>TerminalSpecification</i> that is instanced by this <i>TerminalRole</i> .

2.6.83 Class TerminalType

Defines the type (system) of a terminal. The type is specified by a combination of a name for the system and an optional nominalSize.

In some processes the terminal type is referred as cavity system, because the system of cavities, terminals, seals, plugs and other cavity accessories must match / be compatible.

However, since a terminal has only one since type and a cavity can be compatible to more than one it is named *TerminalType* in the VEC.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
name	String	1	Specifies the type of the cavity (e.g. MQS, DFK, ...).
nominalSize	String	0..1	Specifies the nominal size of terminals that fit into the cavity. (e.g. 2x4).

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
terminalType	0..1	Y	TerminalReceptionSpecification	0..1	Specifies the terminal type that is associated with the terminal reception.

2.6.84 Class WireElement

A *WireElement* specifies a *WireElementSpecification* in the context of a *WireSpecification*. This is necessary to define a unique *identification* of a *WireElementSpecification* in the context of a *WireSpecification*. Additionally, the *WireElement* serves as anchor for the instancing of a wire (*WireElementReference*) as the *WireElementSpecifications* are not uniquely related to a *WireSpecification* for reasons of reusability.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
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identification	String	1	The identification of the WireElement. The identification is guaranteed to be unique within a wire and immutable. The identification is guaranteed to be the same for the same WireElement over different VEC documents.
index	Integer	0..1	The <i>index</i> defines an order on the <i>WireElements</i> that are <i>SubWireElements</i> of the same <i>WireElement</i> and that are on the same <i>layer</i> . If defined for an individual <i>WireElement</i> , the <i>index</i> shall be defined for all <i>SubWireElements</i> within the same <i>WireElement</i> . The order defined for a specific wire can be interpreted either clockwise or counterclockwise, or from left to right and vice versa, depending on the respective direction of view.
layer	Integer	0..1	The layer defines the layering of <i>WireElements</i> that are <i>subWireElements</i> of the same <i>WireElement</i> . <i>WireElements</i> with lower <i>layer</i> values are positioned further inward. If defined for an individual <i>WireElement</i> , the <i>layer</i> shall be defined for all <i>SubWireElements</i> within the same <i>WireElement</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
subWireElement	WireElement	0..*	Y	N	0..1	Y	Defines the <i>subWireElements</i> of this <i>WireElement</i> . The <i>subWireElements</i> shall be consistent with the <i>subWireElementSpecifications</i> of the <i>WireElementSpecification</i> referenced by this <i>WireElement</i> and shall resemble that hierarchy.
wireElementSpecification	WireElementSpecification	1	Y	N	0..*	N	Reference the <i>WireElementSpecification</i> that is represented by the <i>WireElement</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
referencedWireElement	1	N	WireElementReference	0..*	References the WireElement that is represented by the WireElementReference.
subWireElement	0..*	Y	WireElement	0..1	Defines the <i>subWireElements</i> of this <i>WireElement</i> . The <i>subWireElements</i> shall be consistent with the <i>subWireElementSpecifications</i> of the <i>WireElementSpecification</i> referenced by this <i>WireElement</i> and shall resemble that hierarchy.
wireElement	1	Y	WireSpecification	0..1	Specifies the <i>WireElement</i> that represents the root of the <i>WireSpecification</i> .

2.6.85 Class WireElementSpecification

A WireElementSpecification is the basic element to describe a wire in the VEC. A WireElementSpecification can be atomic or composed recursively out of other WireElementSpecifications. A WireElementSpecification can reference an InsulationSpecification, if it has an insulation, a CoreSpecification, if it has a core or a WireGroupSpecification if it is a grouping of other WireElementSpecifications in the Wire (e.g. a multi-core wire with twisted pairs).

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	WireType	0..*	Defines the type of the wire. A wire must not have more than one type. This attribute allows more than one value for the reason, that the same type can be expressed in multiple reference systems.
minBendRadiusDynamic	NumericalValue	0..1	Specifies the minimum bend radius for wire element, if it is used in a dynamic environment, where it is bended repeatedly (e.g. in grommet of the back door). (see KLBFRM-311)
minBendRadiusStatic	NumericalValue	0..1	Specifies the minimum bend radius for wire element, if it is used in a static environment, where it is bended once during installation. After that it remains unchanged in its bended position during usage. (see KLBFRM-311)
outsideDiameter	NumericalValue	0..1	Specifies the outside diameter of the WireElement.
suitedForDynamicUse	Boolean	0..1	Specifies if it is allowed to use the WireElement in a dynamic environment. (see KLBFRM-311)
impedance	NumericalValue	0..1	Defines the impedance of this wireElement. Typically used for WireElements that have subWireElements e.g. twisted pair or coax wires.
size	Size	0..1	Defines the size of a WireElement if it has not the shape of circle. If it has the shape of a circle the size is normally defined by its outside diameter.
validWireReceptionTypes	WireReceptionType	0..*	Defines the <i>WireReceptionTypes</i> that are allowed for joining with the specified <i>WireElement</i> .
gridSpacing	NumericalValue	0..1	Defines the grid spacing. The grid spacing is the distance between the centers of two adjacent sub wire elements. This attribute is only valid for <i>WireElementSpecifications</i> that have <i>SubWireElementSpecifications</i> .
shape	WireElementShape	0..1	Defines the shape of a <i>WireElement</i> . Circular wire elements are defined by their outsideDiameter, all others are defined by their size.
transmissionMediumType	TransmissionMediumType	0..1	Defines the transmission medium in this wire element.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
conductorSpecification	ConductorSpecification	0..1	Y	N	0..*	N	If the <i>WireElement</i> has a core then the specification of the core is referenced here.
fillerSpecification	FillerSpecification	0..1	Y	N	0..*	N	If the <i>WireElement</i> is a filler then the specification of the filler is referenced here.
insulationSpecification	InsulationSpecification	0..1	Y	N	0..*	N	If the <i>WireElement</i> has an insulation then the specification of the insulation is referenced here.
subWireElementSpecification	WireElementSpecification	0..*	N	N	0..*	N	<u>Deprecated since 2.2.0:</u> <u>Reason: Redundant to WireElement->SubWireElement (exists for historical reasons and reuse of complex wire structures is not recommended).</u> Specifies the contained subWireElements if the WireElement has sub elements. If a <i>WireElementSpecification</i> contains the same

							<i>WireElementSpecification</i> multiple times, then it shall be referenced as often as it occurs in the reality. Otherwise, the <i>WireElementSpecification</i> wouldn't specify a wire element unambiguously, because the representation in the model would be the same, regardless of the number of sub elements. Therefore, this association must not be realized with a "set" semantic.
wireGroupSpecification	WireGroupSpecification	0..1	Y	N	0..*	N	If the <i>WireElementSpecification</i> is representing a wire group, then the specification of the wire group is referenced here. That means as well, that the <i>WireElementSpecification</i> shall have <i>subWireElementSpecifications</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
subWireElementSpecification	0..*	N	WireElementSpecification	0..*	Specifies the contained subWireElements if the WireElement has sub elements. If a <i>WireElementSpecification</i> contains the same <i>WireElementSpecification</i> multiple times, then it shall be referenced as often as it occurs in the reality. Otherwise, the <i>WireElementSpecification</i> wouldn't specify a wire element unambiguously, because the representation in the model would be the same, regardless of the number of sub elements. Therefore, this association must not be realized with a "set" semantic.
wireElementSpecification	0..1	N	WireSpecification	0..*	References the <i>WireElementSpecification</i> that defines the properties of the top-most <i>WireElement</i> .
wireElementSpecification	1	N	WireElement	0..*	Reference the <i>WireElementSpecification</i> that is represented by the <i>WireElement</i> .

2.6.86 Class WireEndAccessorySpecification

WireEndAccessorySpecifications are describing parts that are used for a treatment of a wire end before the actual terminal is attached to the *WireEnd* (e.g. a ferrule).

The actual specification of the properties of the wire end accessory is done by the referenced *WireReceptionSpecification* since this is the *Specification* to describe joining properties of a part with a wire end.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
wireReceptionSpecification	WireReceptionSpecification	0..1	Y	N	0..*	N	References the single <i>WireReceptionSpecification</i> that is specifying the properties of the <i>WireEndAccessory</i> that are related to wire.

Incoming Relations

This End	Other End	General
----------	-----------	---------

Role	Mult.	Agg.	Type	Mult.	Comment
wireEndAccessory Specification	1	N	WireEndAccessory Role	NA	References the <i>WireEndAccessorySpecification</i> that is instantiated by this <i>WireEndAccessoryRole</i> .

2.6.87 Class WireGroupSpecification

Specification for the definition of WireGroups. Wire groups are for example used for the representation of twisted pairs.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
groupType	WireGroupType	1	Specifies the type of the group (e.g. twisted pair, ...).
lengthOfTwist	NumericalValue	0..1	Specifies the length of twist if the wire group is representing a twisted pair.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
wireGroupSpecification	0..1	N	WireGrouping	0..*	References the <i>WireGroupSpecification</i> that applies to the <i>WireGrouping</i> .
wireGroupSpecification	0..1	N	WireElementSpecification	0..*	If the <i>WireElementSpecification</i> is representing a wire group, then the specification of the wire group is referenced here. That means as well, that the <i>WireElementSpecification</i> shall have <i>subWireElementSpecifications</i> .

2.6.88 Class WireReception

A WireReception is the area of a terminal where the contacting with a wire element (e.g. by crimping) takes place. This class represents such an area on the terminal. Its description is done with a WireReceptionSpecification, which is independent from the TerminalSpecification. This allows the reuse of the specification of similar WireReception on different terminals.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the WireReception. The identification is guaranteed to be unique within the TerminalSpecification (this might be for example a reception number).
angle	NumericalValue	0..2	Specifies the angles against two planes of the connector housing a wire reception can be buckled. The planes are defined in a way, that angles of [0.0, 0.0] specify an unbuckled (regular) terminal. The two planes are perpendicular to each other and parallel to the plugging direction of the terminal (the direction of the pin).

			The plane for the first angle is the plane in which the uncrimped wire reception would be, if the terminal has not been buckled. Since most terminals are cut or punched in some way out of a sheet of metal, this plane would be the same as metal sheet before further deformation. For ring terminals, the first plane is perpendicular to the bolt direction. With a viewing direction from the wire reception to the terminal reception and the intended wire position above the terminal, buckling directions up and right are expressed by positive angles, down and left by negative angles.
rotation	NumericalValue	0..1	Defines the rotation of a <i>WireReception</i> relative to the <i>TerminalReception</i> . The rotation of the <i>WireReception</i> is the angle between the normal vector on base plane of the crimp area and the y-axis, which is a rotation around the z-axis (right hand rule, compare illustration in " <i>Terminal Coordinate System</i> ").

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
placementPoint	PlacementPoint	0..1	Y	N	0..*	N	Specifies the <i>PlacementPoint</i> that represents this <i>WireReception</i> in a <i>PlaceableElementSpecification</i> .
wireReceptionSpecification	WireReceptionSpecification	0..1	Y	N	0..*	N	References the <i>WireReceptionSpecification</i> that specifies the <i>WireReception</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
wireReception	0..*	Y	TerminalSpecification	1	Specifies the <i>WireReceptions</i> of the terminal described by the <i>TerminalSpecification</i> .
wireReception	1	N	WireReceptionReference	0..*	References the <i>WireReception</i> that is instantiated by this <i>WireReceptionReference</i> .
wireReception	0..*	N	InternalTerminalConnection	0..1	References the <i>WireReceptions</i> that participate in the <i>InternalTerminalConnection</i> .

2.6.89 Class WireReceptionAddOn

Specifies the wire addon required for this wire reception.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
wireAddOn	NumericalValue	1	Specifies the wire length add on needed for the wire reception.
type	WireAddOnType	1	Defines the type of the add-on.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
addOns	0..*	Y	WireReceptionSpecification	1	NA

2.6.90 Class WireReceptionSpecification

Specification for the definition of wire receptions. A WireReception is the area of a terminal where the contacting with a wire element (e.g. by crimping) takes place.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
coreCrossSectionArea	ValueRange	0..1	Specifies a value range for cross-section areas of cores that are compatible with the wire reception. For wire receptions where a single core is attached to, this value defines the <i>ValueRange</i> of the <i>crossSectionArea</i> of the core. For wire receptions where more than one core is attached to (e.g. ring terminals, splices), this attribute defines the valid value range for the sum of the cross-section areas of all attached cores.
insulationDisplacementLength	NumericalValue	0..1	Specifies the length of the insulation which must be stripped off to fit to this wire reception.
multiContact	Boolean	0..1	Specifies if it is possible to contact more than one core at the wire reception. If this attribute is set to true, the wire reception is allowed for more than one wire, but one wire is possible as well. If set to false, only one wire is allowed.
wireReceptionType	WireReceptionType	0..*	Specifies the type of the wire reception. The <i>WireReceptionType</i> defines how the wire is joined with the wire reception. This attribute is defined in an OpenEnumeration. As there are terminals that support multiple joining techniques, e.g. crimping, soldering and a combination of both, it is possible to define multiple values for this attribute. Joining methods that are a combination of other joining methods are represented as a separate literal. For example, a <i>WireReceptionSpecification</i> with <i>WireReceptionType</i> = <i>[Crimp,Soldering]</i> supports crimping <i>or</i> soldering, but not a combination of both, where as the definition <i>[Crimp,CrimpAndSoldering]</i> would support crimping and crimping with subsequent soldering, but not soldering alone.
wireElementOutsideDiameter	ValueRange	0..1	Specifies a value range for valid diameters a WireElement must have to fit to the wire reception.
platingMaterial	Material	0..*	Specifies the plating material of the wire reception.
sealable	Boolean	0..1	Specifies if the wire reception can be sealed. (see KBLFRM-311)
frontBellMouthLength	NumericalValue	0..1	Specifies the length of the front (facing the terminal reception) bell mouth of the conductor crimp (wire reception, see diagram "Terminal Dimensions").
conductorCrimpLength	NumericalValue	0..1	Specifies the length of the conductor crimp (wire reception, see diagram "Terminal Dimensions").
conductorCrimpShape	CrimpShape	0..1	The shape of the conductor crimp (e.g. F).

conductorCrimpLegHeight	NumericalValue	0..1	The maximum height of the legs above the normal plane of the terminal (uncrimped state).
rearBellMouthLength	NumericalValue	0..1	Specifies the length of the rear bell mouth of the conductor crimp (wire reception, see diagram "Terminal Dimensions").
crimpConnectionLength	NumericalValue	0..1	Specifies the length of the crimp area, conductor + insulation crimp (wire reception, see diagram "Terminal Dimensions").
insulationCrimpLength	NumericalValue	0..1	Specifies the length of the insulation crimp (wire reception, see diagram "Terminal Dimensions").
insulationCrimpShape	CrimpShape	0..1	The shape of the insulation crimp (e.g. O).
insulationCrimpLegHeight	NumericalValue	0..1	The maximum height of the legs above the normal plane of the terminal (uncrimped state).
cutOffTabLength	NumericalValue	0..1	Specifies the length of the cut off tab (wire reception, see diagram "Terminal Dimensions").
connectionBLength	NumericalValue	0..1	Specifies the length of the connection between the conductor and the insulation crimp (wire reception, see diagram "Terminal Dimensions").
sheetThickness	NumericalValue	0..1	The thickness of the material of terminal (metal sheet) in the crimp area.
wireTipProtrusion	ValueRange	0..1	Defines the valid length range in which the protrusion of the conductor from the conductor crimp must be located.
crimpBarrelType	CrimpBarrelType	0..1	The <i>CrimpBarrelType</i> defines the shape of the crimp barrel in its unprocessed state. When the crimp barrel is open the wire can be laid into the terminal from above. If the crimp barrel is closed the wire must be fed in from the front.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
addOns	WireReceptionAddOn	0..*	Y	N	1	Y	NA
coreCrimpDetails	CoreCrimpDetail	0..*	Y	N	1	Y	Defines the <i>CrimpDetails</i> for this <i>WireReception</i> .
validConductorMaterials	ConductorMaterial	0..*	Y	N	1	Y	Specifies the materials of a conductor, which are valid to use with this <i>WireReceptionSpecification</i> . This material shall be matched against the <i>ConductorSpecification.material</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
wireReceptionSpecification	0..1	N	WireEndAccessorySpecification	0..*	References the single <i>WireReceptionSpecification</i> that is specifying the properties of the <i>WireEndAccessory</i> that are related to wire.
wireReceptionSpecification	0..1	N	WireReception	0..*	References the <i>WireReceptionSpecification</i> that specifies the <i>WireReception</i> .

2.6.91 Class WireSpecification

Specification for the definition of a wire. This can either be a complex multicore wire or a normal single core. In the VEC a wire is defined by one *WireElement*, which may be composed from other *WireElements*.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
wireElement	WireElement	1	Y	N	0..1	Y	Specifies the <i>WireElement</i> that represents the root of the <i>WireSpecification</i> .
wireElementSpecification	WireElementSpecification	0..1	Y	N	0..*	N	<u>Deprecated since 2.2.0:</u> <u>Reason: Redundant to WireSpecification->WireElement->WireElementSpecification (exists for historical reasons).</u> References the <i>WireElementSpecification</i> that defines the properties of the top-most <i>WireElement</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
wireSpecification	1	N	WireRole	0..*	References the <i>WireSpecification</i> that is instantiated by this <i>WireRole</i> .

2.6.92 Class WireTupleSpecification

A *Specification* to define requirements for the handling of wire / wire element tuples (pairs, triple,...) with the focus on twisting. More precisely, the allowed untwisting at the wire ends and required measures against it.

This definition is made independent from the *WireGroup*, as it might be defined for the part master data (the wire on the roller). The specifications for untwisting are based on the concrete usage.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
fixationRequired	Boolean	0..1	Specifies if an implementation of this specification in a concrete harness required to define a fixation against untwisting of the wire ends.
fixationOffset	NumericalValue	0..1	The offset from the wire end, at which the fixation has to be placed.
maximumUntwistSingleSided	NumericalValue	0..1	Specifies the maximum untwist for one side.
maximumUntwistDualSided	NumericalValue	0..1	Specifies the maximum untwist for both sides together (A+B).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
fixationAccessory	PartRelation	0..*	Y	N	1	Y	Specifies <i>PartRelations</i> that can / have to be used for the fixation.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
wireTupleRequirements	0..1	N	Signal	NA	Defines a requirement for wire tuples.
wireTupleSpecification	0..1	N	WireTupleTermination	NA	NA

2.6.93 Class WireType

Specifies a wire type. A wire type is always defined by a key value. What wire type is meant by this key value is defined by a standard reference system.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	String	1	Specifies the type of the wire (e.g. FLRY, NYFAZw). Valid values are defined by the wireTypeReferenceSystem.
referenceSystem	String	1	Specifies the reference system for the wire type.

2.6.94 Enumeration ApplicationType

OpenEnumeration defines different fields of application / application types for and within wiring harnesses.

For definitions of 12V to HV5 see *NominalVoltage*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
12V	NA
24V	NA
48V	NA
HV	NA
HV1	NA

HV2	NA
HV3	NA
HV4	NA
HV5	NA
FFC/FPC	Flat conductors, FFC = Flat Flexible Cable, FPC = Flexible Printed Circuitry.
HF	High Frequency
OpticalFibre	NA
Busbar	NA

2.6.95 Enumeration CavityGeometry

Defines valid values of the geometry of cavities in the sealing area (crimp end).

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Square	
Circular	
Oval	
Octagon	NA

2.6.96 Enumeration ConductorStructure

Specifies the structure of the conductor according to ISO 6722-1 for ConductorSpecifications with Type = Stranded.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Symmetric	Symmetric conductor structure in accordance with ISO 6722-1 (Type A)
Asymmetric	Asymmetric conductor structure in accordance with ISO 6722-1 (Type B)
MultiStrand	Multi-Strand conductor structure in accordance with ISO 6722-1 (Type C)

2.6.97 Enumeration ConductorType

Specifies the type of the conductor, e.g. if it is rigid or stranded.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
------	---------

Rigid	Used for conductors that are made of solid material.
Stranded	Used for conductors that are made of multiple individual strands (used for most automotive cores).
Foil	Used for conductors that are a foil (e.g. some shields).
Braided	Used for conductors that are made of multiple individual strands that are braided together (often used for shields).

2.6.98 Enumeration ConnectorLockingType

Defines the valid primary locking types for connectors.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
SnapLock	The plastic of connector has some notches, lances, retainer (or its counterpart) that snap into position when connector is placed.
Slider	NA
Lever	NA
Bolting	NA
Spring	NA
RotarySlider	NA

2.6.99 Enumeration ConnectorOutletDirection

Defines the *OutletDirection* of a connector for wires.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Straight	NA
Angled	NA

2.6.100 Enumeration ConnectorPositionAssuranceType

Defines the valid connector position assurance types for connectors.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Mechanical	The connector features some sort of mechanical device (switch, snapper, slider, spring), that ensure the correct placement and closing of the primary locking of the connector.

Electronical		
ScrewMonitoring		The correct positioning of the connector can be ensured by a screw monitoring.

2.6.101 Enumeration CrimpBarrelType

The *CrimpBarrelType* defines in an *OpenEnumeration* the shape of the crimp barrel in its unprocessed state.

General Information

Applied Stereotype	NA
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Enumeration Literals

Name	Comment
Open	NA
Closed	NA

2.6.102 Enumeration CrimpShape

OpenEnumeration of possible *CrimpShapes* for *Core-* and *InsulationCrimps*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
F	NA
O	NA
OVL	NA

2.6.103 Enumeration FoilShieldWindingType

Defines the type of winding for foil shields. It can be folded around and along the inner wire elements like a cigarette paper or it can be wound around them in a spiral. Both types result in different manufacturing and EMC properties.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Overlap	Spirally wound around the shielded elements with minimal overlap (i.e. without gaps)
Longitudinal	Folded lengthwise around the shielded elements (like a cigarette paper).

2.6.104 Enumeration FuseType

Defines the values for the type a fuse. This is the geometrical type.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Type_SF51	also known as: mega
Type_SF30	also known as: midi
Type_SF	also known as: strip
Type_F	also known as: mini
Type_C	also known as: ato
Type_E	also known as: maxi
Type_A1	also known as: jcase
Type_A1S	also known as: flat
Form_CB15_CatE	also known as: auto
MITOX	NA

2.6.105 Enumeration GrommetMountingType

OpenEnumeration to define different mounting possibilities of a grommet onto a wiring harness.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Moulded	The grommet created in place around the harness with some kind of moulding or foaming manufacturing process.
Slit	The grommet has a longitudinal slit through which the wires can be inserted.
MultiPiece	The grommet consists of multiple parts that are assembled around the harness to create the final appearance.
Threading	The wires must threaded through the grommet.

2.6.106 Enumeration GrommetType

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
ConnectorCover	NA
BendProtection	NA
LeadThrough	NA

2.6.107 Enumeration HousingComponentType

Defines the valid HousingComponentTypes. The values in this Enumeration are matching the relevant *PrimaryPartTypes*. For a description of the values see *PrimaryPartType*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
ConnectorHousing	NA
Fuse	NA
Relay	NA
EEDComponent	NA
Terminal	NA
RingTerminal	NA
MultiFuse	NA

2.6.108 Enumeration LabelingTechnology

OpenEnumeration of standard labeling technologies.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Lasered	NA
Printed	NA
Imprinted	NA
Moulded	NA
PreCoded	In case precoded labels (e.g. RFID).
Codeable	In case of labels that can be coded at a later stage (e.g. RFID).

2.6.109 Enumeration LabelType

OpenEnumeration of different labeling types. This refers to the way the information is encoded.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
AlphaNumerical	NA
BarCode	NA
QRCode	NA
RFID	NA

2.6.110 Enumeration MultiFuseType

OpenEnumeration for the manufacturing types of the multi-fuses.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Stamped	NA
Clinched	NA
OverMolded	NA
Welded	NA

2.6.111 Enumeration PartRelationType

Defines how the set of *accessoryParts* referenced by a *PartRelation* should be interpreted.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Mandatory	<i>Mandatory</i> means that in a usage of a component <u>all</u> referenced <i>accessoryParts</i> must be used.
Optional	<i>Optional</i> means that in a usage of a component <u>some</u> referenced <i>accessoryParts</i> can be used by choice.
OneOfAll	<i>OneOfAll</i> means that in a usage of a component exactly <u>one</u> of the referenced <i>accessoryParts</i> must be selected.
Custom	Custom means, that there is some kind of custom constraints / logic between referenced <i>PartVersions</i> that has to be evaluated to determine the selected accessories.

2.6.112 Enumeration PinApplianceType

Classifies the appliance of a Pin in terms of the duration of the appliance.

E.g. the power supply pin of a power window has "shortTerm" PinApplianceType, in contrast to the head light which is a "long term".

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
ShortTerm	NA
LongTerm	NA

2.6.113 Enumeration PinComponentType

Specifies the type of a *PinComponent*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Switch	NA
Coil	NA
Ground	NA
PowerSupply	<i>PowerSupply</i> defines a <i>PinComponent</i> that is used to supply the component itself with power (in contrast to <i>PowerDistribution</i>). <i>PinComponentBehaviours</i> of <i>PinComponents</i> with this type always have the <i>SignalDirection</i> "In".
PowerDistribution	<i>PowerDistribution</i> defines a <i>PinComponent</i> that is used to distribute power to other components (in contrast to <i>PowerSupply</i>). The semantic of this type depends on the <i>SignalDirection</i> . <i>In</i> means, that this <i>PinComponent</i> is used to supply power to the EECComponent for further distribution to other components. <i>Out</i> means that the <i>PinComponent</i> is a source of power for other EECComponents. It is undefined whether the pin is fused/limited or not.
PowerDistributionLimited	<i>PowerDistributionLimited</i> has the same meaning as <i>PowerDistribution</i> , except that <i>PowerDistributionLimited</i> defines a pin that is limited <u>by an electronic measure</u> within the parent component (eFuse). Limitation with a conventional fuse is <u>not</u> within the semantic of <i>PowerDistributionLimited</i> since conventional fusing can be recognized with the internal connectivity.
Signal	NA
NotConnected	NA

2.6.114 Enumeration PinCurrentType

Defines the different available current types of a pin.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
MinCurrent	NA
MaxCurrent	NA
TypicalCurrent	NA
StandbyCurrent	NA
InitCurrent	NA
BlockCurrent	NA
StartStopCurrent	NA
OverrunCurrent	NA
LeakageCurrent	Defines the current that leaks at this pin.
DeepSleepCurrent	Defines the current on the pin, when the EECComponent is in deep sleep mode.
ClippingCurrent	Current at which the output driver of an ECU turns off or limits the current on an output in (a behavior similar to a fuse).

2.6.115 Enumeration PinTimingType

Defines the different available timing types of a pin.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
InitTime	NA
StartStopTime	NA
OverrunTime	NA
BlockTime	NA
CurrentPeakDistances	NA
Duration	NA

2.6.116 Enumeration PinType

Defines the type of a pin.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Resistive	NA
Inductive	NA
Capacitive	NA
Optical	NA
Filament	NA

2.6.117 Enumeration PinVoltageType

Defines the different available voltage types of a pin.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
MinVoltage	NA
MaxVoltage	NA
TypicalVoltage	NA
StandbyVoltage	NA
InitVoltage	NA

BlockVoltage	NA
StartStopVoltage	NA
OverrunVoltage	NA

2.6.118 Enumeration PluggableTerminalType

Defines valid values for the type of PluggableTerminals. The type defines constraints about the numbers of wire and terminal receptions and their relations.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Standard	NA
Coax	NA
MultiContact	NA
Bridge	NA
PiggyBack	NA

2.6.119 Enumeration PowerConsumptionType

OpenEnumeration for predefined *PowerConsumptionTypes*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Typical	NA
Standby	NA
Minimum	NA
Maximum	NA
Sleep	NA
DeepSleep	NA
StartStop	NA

2.6.120 Enumeration PrimaryLockingType

Defines the valid primary locking types for terminals.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
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Lance	Lance specifies, that the terminal has a lance or lances (syn. tang) to lock itself in the cavity.
Lanceless	Lanceless specifies, that the terminal itself has no lance (syn. tang) to lock itself in the cavity. However, it is locked in the cavity by some mechanical feature of the cavity (e.g. a lance).
None	NA

2.6.121 Enumeration RelaisApplianceType

Specifies the appliance type of a relais.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Soldered	NA
Screwed	NA
Plugged	NA

2.6.122 Enumeration RelaisType

Defines the type of a relais (switching behavior).

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Normally open	
Normally closed	Defines a relais that is <i>closed</i> in its normal state.
Switch	Defines a relais that is a switch.
Bistable	Defines a relais that is <i>bistable</i> .

2.6.123 Enumeration SealingGeometry

Defines valid values of the geometry of a cavity sealing. It defines the cross-section geometry.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Circular	NA
Oval	NA
Square	NA
Segmented	NA

Sectored	NA
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2.6.124 Enumeration SecondaryLockingType

Defines the valid secondary locking types within slots (syn. Terminal Position Assurance).

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
ISL	ISL stands for "Independent Secondary Lock" and means, that the slot as some mechanical slider (or similar) to lock (and ensure) the terminals in their position in the connector. The ISL works independent from the primary locking.
PLR	PLR stands for "Primary Lock Reinforcement" and means, that the slot as some mechanical slider (or similar) to lock (and ensure) the terminals in their position in the connect. However, this system is not independent from the primary locking (see ISL), but by ensuring (reinforcement) that the primary lock is in place and cannot be opened again.

2.6.125 Enumeration SlotGender

Defines the gender of a slot.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Male	NA
Female	NA
Unspecified	NA

2.6.126 Enumeration SlotLayoutType

OpenEnumeration that defines standard slot layouts. A slot layout represents the mechanical / geometrical properties / layout of the slot, not the electrical properties.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
USBTypeA	NA
USBTypeASuperSpeed	NA
USBTypeB	NA
USBTypeBSuperSpeed	NA
USBTypeC	NA
USBMiniA	NA

USBMiniB	NA
USBMiniAB	NA
USBMicroA	NA
USBMicroASuperSpeed	NA
USBMicroB	NA
USBMicroBSuperSpeed	NA
USBMicroAB	NA
USBMicroABSuperSpeed	NA
Type_SF51	Fuse type according to ISO8820, also known as: mega
Type_SF30	Fuse type according to ISO8820, also known as: midi
Type_SF	Fuse type according to ISO8820, also known as: strip
Type_F	Fuse type according to ISO8820, also known as: mini
Type_C	Fuse type according to ISO8820, also known as: ato
Type_E	Fuse type according to ISO8820, also known as: maxi
Type_A1	Fuse type according to ISO8820, also known as: jcase
Type_A1S	Fuse type according to ISO8820, also known as: flat
Form_CB15_CatE	Fuse type according to ISO8820, also known as: auto
MITOX	Fuse type according to ISO8820
Fakra	Fakra (ISO-20860)
RS-453	IEC 60603-11
HSD	High Speed Data Connectors
AMEC	NA
HMTD	High-Speed Modular Twisted-Pair Data

2.6.127 Enumeration SlotSealingType

Defines the possible sealing types for a slot, if the slot shall be sealed in its usage.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
None	The <i>Slot</i> cannot be sealed at all.
SingleSealing	The <i>Slot</i> is sealed with a single <i>Seal</i> per <i>Cavity</i> e.g. a <i>CavitySeal</i> or a <i>CavityPlug</i> .
MultiSealing	The <i>Slot</i> is sealed with a more complex sealing variant, typically sealing multiple cavities with a single seal (e.g. a <i>MultiCavityPlug</i> or a combination of those).
Moulded	The <i>Slot</i> is sealed by moulding it with some sort of sealing compound.

2.6.128 Enumeration TerminalBoltNominalSize

Defines the nominal sizes for threaded bolts as "ISO metric screw thread" or as "Unified Thread Standard" (Imperial).

See also *TerminalBoltType.MetricThread / ImperialThread*.

For the definition of the values see https://en.wikipedia.org/wiki/ISO_metric_screw_thread and https://de.wikipedia.org/wiki/Unified_Thread_Standard.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
M1	NA
M1.2	NA
M1.4	NA
M1.6	NA
M1.8	NA
M2	NA
M2.5	NA
M3	NA
M3.5	NA
M4	NA
M5	NA
M5.5	NA
M6	NA
M7	NA
M8	NA
M10	NA
M12	NA
M14	NA
M16	NA
M18	NA
M20	NA
M22	NA
M24	NA
M27	NA
M30	NA
M33	NA
M36	NA
M39	NA

M42	NA
M45	NA
M48	NA
M52	NA
M56	NA
M60	NA
M64	NA
#0	NA
#1	NA
#2	NA
#3	NA
#4	NA
#5	NA
#6	NA
#8	NA
#10	NA
#12	NA
1/4	NA
5/16	NA
3/8	NA
7/16	NA
1/2	NA
9/16	NA
5/8	NA
3/4	NA
7/8	NA
1	NA

2.6.129 Enumeration TerminalBoltType

Open Enumeration to define different types of bolts for bolt / ring terminals.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
MetricThread	A bolt with a metric scale thread.
ImperialThread	A bolt with an imperial scale thread.
Plain	A plain bolt without any specific features.

Snappable	A bolt with a groove or similar that allows the counterpart to snap on.
-----------	---

2.6.130 Enumeration TerminalDistanceType

OpenEnumeration of possible *TerminalDistanceTypes* for *TerminalDistanceInformation*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Terminal	The distance to the next terminal (e.g. for handling reasons).
UltraSonicConnection	The distance to the next ultra-sonic-welding connection, e.g. a splice or a terminal (on the same wire).

2.6.131 Enumeration TerminalReceptionGender

OpenEnumeration of possible genders for *TerminalReception*

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Female	NA
Male	NA
Unspecified	NA

2.6.132 Enumeration TerminalSealingType

Defines the possible sealing types for a terminal, if the terminal shall be sealed in its usage.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
None	The Terminal cannot be sealed by itself (e.g. a CavitySeal)
Sealed	The Terminal can be sealed and can only be used together with a cavity seal. It is not allowed to be used unsealed.
Neutral	The terminal can be sealed (with a cavity seal), but a seal is not mandatory for this terminal.
Moulded	The terminal can be used in a moulded slot.

2.6.133 Enumeration TransmissionMediumType

OpenEnumeration of the different transmission mediums in wire elements.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Electrical	NA
Optical	NA
Hydraulic	NA
Pneumatic	NA

2.6.134 Enumeration WireAddOnType

Specifies possible values for the *type* of wire add-ons (e.g. *CavityAddOn*).

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Contract	The agreed add-on for any negotiations and calculations.
Production	The add-on length for the use in production environments.

2.6.135 Enumeration WireElementShape

Defines the shape of a wire element.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Circular	NA
Flat	NA
Elliptical	NA

2.6.136 Enumeration WireGroupType

OpenEnumeration for the types of wire groups.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Twisted	NA

2.6.137 Enumeration WireReceptionType

The *WireReceptionType* defines in an *OpenEnumeration* how a wire is joined with a wire reception.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Crimp	NA
Soldering	NA
PlasmaSoldering	NA
SpotResistentWelding	NA
FrictionWelding	NA
UltraSonicWelding	NA
UltraSonicCompaction	UltraSonicCompaction, typically used for aluminum wires.
UniaxialShieldCrimp	A crimping method for coaxial shields, where the crimping process is performed only in the direction of a of single axis.
MultiaxialShieldCrimp	A crimping method for coaxial shields, where the crimping process is performed in the directions of a of multiple axis.
CrimpAndSoldering	A crimp connection with subsequent soldering of the joining area.

2.6.138 Primitive CodingName

NA

General Information

Applied Stereotype	OpenPatternRestriction
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2.7 Module external_mapping

2.7.1 Class ExternalMapping

An *ExternalMapping* is used to relate an *ExtendableElement* in the VEC with an element located in an external data source. The element in the VEC is referenced by the *mappedElement*, the external element is identified by the attribute *externalReference*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
externalReference	String	1	Defines the unique key of the external element. How this key shall be interpreted is dependent from the format of the external data source.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
mappedElement	ExtendableElement	1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
mappings	0..*	Y	ExternalMappingSpecification	1	Specifies the mappings of individual element.

2.7.2 Class ExternalMappingSpecification

An *ExternalMappingSpecification* is used to define a mapping between an external data source (represented by the referenced *mappedDocument*) and the content of a VEC file.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
mappedDocument	DocumentVersion	1	Y	N	0..*	N	Reference to the <i>DocumentVersion</i> that represents the external data source that connected to the VEC content by the <i>ExternalMappingSpecification</i> .
mappings	ExternalMapping	0..*	Y	N	1	Y	Specifies the mappings of individual element.

2.8 Module geo_2d**2.8.1 Class BuildingBlockPositioning2D**

Defines the position of a BuildingBlock2D on a HarnessDrawing.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
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identification	String	0..1	Specifies a unique (optional) identification of the BuildingBlockPositioning2D. The identification is guaranteed to be unique within the HarnessDrawingSpecification2D.
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Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
centerPoint	CartesianPoint2D	0..1	Y	N	0..1	Y	Specifies the center point of the BuildingBlock in the coordinate system of the harness drawing.
referenced2DBuildingBlock	BuildingBlockSpecification2D	1	Y	N	0..*	N	References the building block which is placed on the harness drawing.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
buildingBlockPositionings	1..*	Y	HarnessDrawingSpecification2D	1	Specifies the BuildingBlockPositioning2Ds that are forming the 2D harness drawing.

2.8.2 Class BuildingBlockSpecification2D

Specification for the description of a two-dimensional building block. A building block is a reusable section of a geometry.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
baseUnit	Unit	1	Y	N	0..*	N	NA
boundingBox	CartesianDimension	1	Y	N	0..1	Y	Specifies the size of the area described by the BuildingBlockSpecification2D in Cartesian dimensions.
cartesianPoint	CartesianPoint2D	0..*	Y	N	0..1	Y	Specifies the CartesianPoint2Ds that are used in the BuildingBlockSpecification2D.
geometryNode	GeometryNode2D	0..*	Y	N	1	Y	Specifies the GeometryNode2Ds defined by the BuildingBlockSpecification2D.
geometrySegment	GeometrySegment2D	0..*	Y	N	1	Y	Specifies the GeometrySegment2Ds defined by the BuildingBlockSpecification2D.
placedElementViewItem	OccurrenceOrUsageViewItem2D	0..*	Y	N	1	Y	Specifies the view items for OccurrenceOrUsages on a BuildingBlockSpecification2D.

Incoming Relations

This End	Other End	General
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Role	Mult.	Agg.	Type	Mult.	Comment
referenced2DBuildingBlock	1	N	BuildingBlockPositioning2D	0..*	References the building block which is placed on the harness drawing.

2.8.3 Class CartesianDimension

The CartesianDimension specifies the size of an object in 2D-space.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
height	Double	1	Specifies the height of the object. The unit of this value is the baseUnit of containing BuildingBlockSpecification2D.
width	Double	1	Specifies the width of the object. The unit of this value is the baseUnit of containing BuildingBlockSpecification2D.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
boundingBox	1	Y	BuildingBlockSpecification2D	0..1	Specifies the size of the area described by the BuildingBlockSpecification2D in Cartesian dimensions.

2.8.4 Class CartesianPoint2D

A CartesianPoint2D is a point that is defined by its coordinates in a rectangular two-dimensional Cartesian coordinate system.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
x	Double	1	Specifies the value of x-coordinate of the Cartesian point. The unit of this value is the baseUnit of containing BuildingBlockSpecification2D.
y	Double	1	Specifies the value of y-coordinate of the Cartesian point. The unit of this value is the baseUnit of containing BuildingBlockSpecification2D.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cartesianPoint	1	N	GeometryNode2D	0..*	References the CartesianPoint2D where the GeometryNode2D is located.

cartesianPoint	0..*	Y	NetViewSpecification	0..1	Specifies all the <i>CartesianPoint2Ds</i> that are used within the <i>NetViewSpecification</i> .
cartesianPoint	0..*	Y	ConnectionViewSpecification	0..1	Specifies all the <i>CartesianPoint2Ds</i> that are used within the <i>ConnectionViewSpecification</i> .
cartesianPoint	0..*	Y	BuildingBlockSpecification2D	0..1	Specifies the <i>CartesianPoint2Ds</i> that are used in the <i>BuildingBlockSpecification2D</i> .
centerPoint	0..1	Y	BuildingBlockPositioning2D	0..1	Specifies the center point of the <i>BuildingBlock</i> in the coordinate system of the harness drawing.
controlPoint	0..*	N	PathSegment	0..*	The ordered list of control points through which the <i>PathSegment</i> goes.
origin	1	N	Transformation2D	0..*	References the <i>CartesianPoint2D</i> that is the origin of the <i>Transformation2D</i> .

2.8.5 Class CartesianVector2D

A Cartesian vector in the two-dimensional space.

General Information

Base Classifier	CartesianVector
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
x	Double	1	Specifies the x-coordinate in 2D space.
y	Double	1	Specifies the y-coordinate in 2D space.

2.8.6 Class GeometryNode2D

A *GeometryNode2D* is the geometric representation of a *TopologyNode* in 2D-space.

General Information

Base Classifier	GeometryNode
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cartesianPoint	CartesianPoint2D	1	Y	N	0..*	N	References the <i>CartesianPoint2D</i> where the <i>GeometryNode2D</i> is located.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
endNode	1	N	GeometrySegment2D	0..*	References the <i>GeometryNode2D</i> where the <i>GeometrySegment2D</i> ends.

geometryNode	0..*	Y	BuildingBlockSpecification2D	1	Specifies the GeometryNode2Ds defined by the BuildingBlockSpecification2D.
startNode	1	N	GeometrySegment2D	0..*	References the GeometryNode2D where the GeometrySegment2D starts.

2.8.7 Class GeometrySegment2D

A GeometrySegment2D is the geometric representation of a TopologySegment in 2D-space.

General Information

Base Classifier	GeometrySegment
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
startVector	CartesianVector2D	1	Specifies the start vector of the geometry segment. The start vector is a tangent to the segment at the start position.
endVector	CartesianVector2D	1	Specifies the end vector of the geometry segment. The end vector is a tangent to the segment at the end position.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
endNode	GeometryNode2D	1	Y	N	0..*	N	References the GeometryNode2D where the GeometrySegment2D ends.
pathSegment	PathSegment	0..*	Y	Y	1	Y	Specifies an ordered list of PathSegments that describe the appearance of the GeometrySegment2D. The appearance is described by the concatenation of the PathSegments beginning at the startNode of the GeometrySegment2D.
startNode	GeometryNode2D	1	Y	N	0..*	N	References the GeometryNode2D where the GeometrySegment2D starts.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
geometrySegment	0..*	Y	BuildingBlockSpecification2D	1	Specifies the GeometrySegment2Ds defined by the BuildingBlockSpecification2D.

2.8.8 Class HarnessDrawingSpecification2D

The HarnessDrawingSpecification2D specifies a two-dimensional drawing of a harness. A harness drawing is composed of one or more BuildingBlockSpecification2D which are placed on the drawing.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
buildingBlockPositionings	BuildingBlockPositioning2D	1..*	Y	N	1	Y	Specifies the BuildingBlockPositioning2Ds that are forming the 2D harness drawing.

2.8.9 Class OccurrenceOrUsageViewItem2D

An OccurrenceOrUsageViewItem2D specifies the representation of an OccurrenceOrUsage in a 2DDrawing.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the OccurrenceOrUsageViewItem2D. The identification is guaranteed to be unique within the BuildingBlockSpecification2D.
aliasId	AliasIdentification	0..*	Specifies additional identifiers for the OccurrenceOrUsageViewItem2D.
gridSquare	String	0..1	Specifies the grid square in which the OccurrenceOrUsageViewItem2D is placed (e.g. E/40).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
occurrenceOrUsage	OccurrenceOrUsage	0..*	Y	N	0..*	N	Specifies the <i>OccurrenceOrUsages</i> which are represented by the view item. Important: To use one <i>OccurrenceOrUsageViewItem</i> for multiple <i>OccurrenceOrUsages</i> is only valid, if the referenced items are true alternatives to each other. That means, they must have an identical placement, the geometrical models used for each item must be substitutable and the item must be mutually exclusive to each other.
orientation	Transformation2D	0..1	Y	N	1	Y	Specifies the orientation of the view item.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
placedElementViewItem	0..*	Y	BuildingBlockSpecification2D	1	Specifies the view items for OccurrenceOrUsages on a BuildingBlockSpecification2D.

2.8.10 Class PathSegment

A PathSegment is a part of the 2D presentation of a GeometrySegment2D. The complete presentation of a GeometrySegment2D is built from an ordered list of PathSegments. Each PathSegment has an ordered list of control points through which the path goes. If no curveRadius is specified the control points are connected by a direct straight line. If a curveRadius is specified, the PathSegment can be drawn by a segment of a circle which touches all control points and has the radius specified.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
curveRadius	NumericalValue	0..1	The radius of the curve which describes the appearance of the path segment.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
controlPoint	CartesianPoint2D	0..*	N	Y	0..*	N	The ordered list of control points through which the PathSegment goes.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
pathSegment	0..*	Y	GeometrySegment 2D	1	Specifies an ordered list of PathSegments that describe the appearance of the GeometrySegment2D. The appearance is described by the concatenation of the PathSegments beginning at the startNode of the GeometrySegment2D.

2.8.11 Class Transformation2D

A *Transformation2D* is used to position an object in 2D space (e.g. on a drawing).

It consists of a reference to a *CartesianPoint2D* as **origin** and a 2×2 matrix **a**. The matrix is given by the attributes **[a11...a22]**, with the numbers representing the index of the matrix element a_{ij} where *i* is the row and *j* the column index of the matrix.

To place an object with a *Transformation2D* its origin is translated in space to the *CartesianPoint2D* **origin** and every point $P = (x,y)$ of the object is transformed to P' by applying the matrix multiplication: $P' = a \cdot P$

With a suitable definition of the transformation matrix, a wide variety of geometric operations can be mapped. Which of these operations are permissible depends on the respective application and cannot be defined in general terms. For example, scaling and mirroring are probably not permissible operations when creating drawings in true scale.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
a11	Double	1	NA
a12	Double	1	NA
a21	Double	1	NA
a22	Double	1	NA

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
origin	CartesianPoint2D	1	Y	N	0..*	N	References the CartesianPoint2D that is the origin of the Transformation2D.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
orientation	0..1	Y	ComponentNodeViewItem	1	Specifies the position / placement of the ConnectionNodeViewItem within the layout.
orientation	0..1	Y	NetworkNodeViewItem	1	Specifies the position / placement of the NetworkNodeViewItem within the layout.
orientation	0..1	Y	OccurrenceOrUsageViewItem2D	1	Specifies the orientation of the view item.
orientation	0..1	Y	ConnectionNodeViewItem	1	Specifies the position / placement of the ConnectionNodeViewItem within the layout.

2.9 Module geo_3d

2.9.1 Class BuildingBlockPositioning3D

Defines the position of a BuildingBlock3D in the HarnessGeometry.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique (optional) identification of the BuildingBlockPositioning3D. The identification is guaranteed to be unique within the HarnessGeometrySpecification3D.

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
positioning	Transformation3D	0..1	Y	N	0..1	Y	Specifies the positioning of the building block in the harness geometry.
referenced3DBuildingBlock	BuildingBlockSpecification3D	1	Y	N	0..*	N	References the building block that is positioned.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
buildingBlockPositionings	0..*	Y	HarnessGeometrySpecification3D	1	Specifies the BuildingBlockPositioning3Ds that are forming the HarnessGeometrySpecification3D.

2.9.2 Class BuildingBlockSpecification3D

Specification for the description of a three-dimensional building block. A building block is a reusable section of a geometry.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
baseUnit	Unit	1	Y	N	0..*	N	NA
cartesianPoint	CartesianPoint3D	0..*	Y	N	0..1	Y	Specifies the CartesianPoint3Ds that are used in the BuildingBlockSpecification3D.
geometryNode	GeometryNode3D	0..*	Y	N	1	Y	Specifies the GeometryNode3Ds defined by the BuildingBlockSpecification3D.
geometrySegment	GeometrySegment3D	0..*	Y	N	1	Y	Specifies the GeometrySegment3Ds defined by the BuildingBlockSpecification3D.
placedElementViewItem3D	OccurrenceOrUsageViewItem3D	0..*	Y	N	1	Y	Specifies the view items for OccurrenceOrUsages in a BuildingBlockSpecification3D.
topologyZone	TopologyZone	0..1	Y	N	0..*	N	References the Zone that is building block represents. This shall be a TopologyZone with the type "DmuZone".

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
referenced3DBuildingBlock	1	N	BuildingBlockPositioning3D	0..*	References the building block that is positioned.

2.9.3 Class CartesianPoint3D

A CartesianPoint3D is a point that is defined by its coordinates in a rectangular three-dimensional Cartesian coordinate system.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
x	Double	1	Specifies the value of x-coordinate of the Cartesian point. The unit of this value is the baseUnit of containing BuildingBlockSpecification3D.
y	Double	1	Specifies the value of y-coordinate of the Cartesian point. The unit of this value is the baseUnit of containing BuildingBlockSpecification3D.
z	Double	1	Specifies the value of z-coordinate of the Cartesian point. The unit of this value is the baseUnit of containing BuildingBlockSpecification3D.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cartesianPoint	1	N	LocalPosition	0..*	NA
cartesianPoint	0..*	Y	BuildingBlockSpecification3D	0..1	Specifies the CartesianPoint3Ds that are used in the BuildingBlockSpecification3D.
cartesianPoint	1	N	GeometryNode3D	0..*	References the CartesianPoint3D where the GeometryNode3D is located.
cartesianPoint	0..*	Y	LocalGeometrySpecification	0..1	All <i>CartesianPoint3Ds</i> that are used in this <i>LocalGeometrySpecification</i> . All <i>CartesianPoint3Ds</i> are defined in relation to the coordinate system of the component.
cartesianPoint3D	1	N	NURBSControlPoint	0..*	The <i>CartesianPoint3D</i> that defines the position of this <i>NURBSControlPoint</i> .
insertionPosition	0..1	Y	CavityPositionDetail	1	The position of the center of the cavity in the entrance surface of the connector during the insertion process.
origin	1	N	Transformation3D	0..*	Specifies the coordinates of the translation.

2.9.4 Class CartesianVector3D

A Cartesian vector in three-dimensional space.

General Information

Base Classifier	CartesianVector
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
x	Double	1	Specifies the x-coordinate in 3D space.

y	Double	1	Specifies the y-coordinate in 3D space.
z	Double	1	Specifies the z-coordinate in 3D space.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
insertionVector	0..1	Y	CavityPositionDetail	NA	A unit vector that points in the direction in which a terminal is inserted from the <i>InsertionPosition</i> into the connector.

2.9.5 Class Curve3D

The *Curve3D* is an abstract representation of a curve, which defines the three-dimensional appearance of the centerline of a segment. The concrete type of the curve (e.g. *NURBSCurve*) defines the mathematical function that applies to curve and stores the required parameter set for this function in the VEC.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
curve	0..*	Y	GeometrySegment3D	NA	Specifies an ordered list of curves which describe the centerline of the segment. If a segment is described by more than one curve, the centerlines of the individual curves are aligned in the order of this association.

2.9.6 Class GeometryNode3D

A *GeometryNode3D* is the geometric representation of a *TopologyNode* in 3D-space.

General Information

Base Classifier	GeometryNode
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cartesianPoint	CartesianPoint3D	1	Y	N	0..*	N	References the CartesianPoint3D where the GeometryNode3D is located.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment

endNode	1	N	GeometrySegment3D	0..*	References the GeometryNode3D where the GeometrySegment3D ends.
geometryNode	0..*	Y	BuildingBlockSpecification3D	1	Specifies the GeometryNode3Ds defined by the BuildingBlockSpecification3D.
startNode	1	N	GeometrySegment3D	0..*	References the GeometryNode3D where the GeometrySegment3D starts.

2.9.7 Class GeometrySegment3D

A GeometrySegment3D is the geometric representation of a TopologySegment in 3D-space.

General Information

Base Classifier	GeometrySegment
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
startVector	CartesianVector3D	1	Specifies the start vector of the geometry segment. The start vector is a tangent to the segment at the start position.
endVector	CartesianVector3D	1	Specifies the end vector of the geometry segment. The end vector is a tangent to the segment at the end position.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
curve	Curve3D	0..*	Y	Y	NA	Y	Specifies an ordered list of curves which describe the centerline of the segment. If a segment is described by more than one curve, the centerlines of the individual curves are aligned in the order of this association.
endNode	GeometryNode3D	1	Y	N	0..*	N	References the GeometryNode3D where the GeometrySegment3D ends.
startNode	GeometryNode3D	1	Y	N	0..*	N	References the GeometryNode3D where the GeometrySegment3D starts.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
geometrySegment	0..*	Y	BuildingBlockSpecification3D	1	Specifies the GeometrySegment3Ds defined by the BuildingBlockSpecification3D.

2.9.8 Class HarnessGeometrySpecification3D

The HarnessGeometrySpecification3D specifies a three-dimensional model of a harness. A harness model is composed of one or more BuildingBlockSpecification3D which are placed in the model.

General Information

Base Classifier	Specification
Applied Stereotype	NA

Is Abstract	false
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Attributes

Name	Type	Mult	Comment
type	GeometryType	1	Specifies the type of the harness geometry.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
buildingBlockPositionings	BuildingBlockPositioning3D	0..*	Y	N	1	Y	Specifies the BuildingBlockPositioning3Ds that are forming the HarnessGeometrySpecification3D.

2.9.9 Class NURBSControlPoint

Represents a control point of a *NURBSCurve*. It consists of a referenced *CartesianPoint3D* for the position and a *weight*.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
weight	Double	1	The weight of the NURBSControlPoint.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cartesianPoint3D	CartesianPoint3D	1	Y	N	0..*	N	The <i>CartesianPoint3D</i> that defines the position of this <i>NURBSControlPoint</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
controlPoint	0..*	Y	NURBSCurve	NA	Defines the control points of the NURBS curve (for details see the class description of <i>NURBSCurve</i>)

2.9.10 Class NURBSCurve

The *NURBSCurve* represents the parameter set of a NURBS (Non-Uniform rational B-Spline) in the VEC. For a complete definition of NURBS see for example (https://en.wikipedia.org/wiki/Non-uniform_rational_B-spline).

Basically, a NURBS curve is defined by:

- a **degree**
- a list of **control points**: with at least *degree* + 1 points.
- a **weight** for each control point.
- a **knots** vector: a list of numbers, with (*degree* + #*controlpoints* + 1) elements. Every number must be equal or greater than its predecessor, and the same value must not be repeated more than *degree* times. It seems that modern NURBS algorithms just require (*degree* + #*controlpoints* - 1) control points.

Commonly used default assignments for the parameters are:

- **weight = 1** for all control points: In this case the curve is called "non-rational".
- **knot** vector: equidistant and increasing values in the knot vector (e.g. 1,2,3,4,5,6,7) means the curve is "uniform" which exists in two variants.
 - **clamped** (or pinned): If the knot vector starts and ends with *degree* times the same value (e.g. degree = 3, knots = [0,0,0,1,2,3,4,5,5,5]), then it is *clamped*. This has the effect, that the first and last control point coincide with the start and end point of the curve.
 - **unclamped** (or unpinned): If there are no repeated values in the knot vector (e.g. 1,2,3,4,5,6,7) it is unclamped.

The VEC *NURBSCurve* corresponds to removed *BSplineCurve* (VEC Version <= 1.1.3 and KBL). However, the *BSplineCurve* did not define weight and knot vector, so default assignments were assumed. Existing implementations are using "uniform non-rational b splines", unfortunately some implementations use "uniform clamped" and some "unclamped".

Since a NURBS cannot be rendered correctly without the knowledge of all parameters and to avoid further misconceptions the VEC allows the definition of all parameters of a NURBS. Furthermore, it requires the specification of all parameters, even if some known default assignment (e.g. non-rational) is used.

General Information

Base Classifier	Curve3D
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
degree	Integer	1	Defines the degree of the NURBS (for details see the class description of <i>NURBSCurve</i>).
knots	Double	0..*	Defines the knot-vector of the NURBS (for details see the class description of <i>NURBSCurve</i>).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
controlPoint	NURBSControlPoint	0..*	Y	Y	NA	Y	Defines the control points of the NURBS curve (for details see the class description of <i>NURBSCurve</i>)

2.9.11 Class OccurrenceOrUsageViewItem3D

An *OccurrenceOrUsageViewItem3D* specifies the existence and representation of an *OccurrenceOrUsage* in a 3D-model.

The definition of the existence is necessary because a 150% model of a harness might contain different geometric variants and not all of them must be represented in the same *BuildingBlockSpecification3D*.

There are two different cases for the representation of *OccurrenceOrUsages* in a 3D model. There are components that can be represented explicitly by a 3D model of the component (e.g. connectors, cable ducts, fixings) and there are components that are represented implicitly by a generic visualization and their placement on the topology (e.g. tapes, tubes). If an *OccurrenceOrUsage* has a 3D model and it shall be placed explicitly in 3D space, the *OccurrenceOrUsageViewItem3D* defines a *Transformation3D* in the *orientation* role. If no *orientation* is defined the *OccurrenceOrUsage* is represented implicitly.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>OccurrenceOrUsageViewItem</i> . The identification is guaranteed to be unique within the <i>BuildingBlockSpecification3D</i> .
aliasId	AliasIdentification	0..*	Specifies additional identifiers for the <i>OccurrenceOrUsageViewItem3D</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
occurrenceOrUsage	OccurrenceOrUsage	0..*	Y	N	0..*	N	Specifies the <i>OccurrenceOrUsages</i> which are represented by the view item. Important: To use one <i>OccurrenceOrUsageViewItem</i> for multiple <i>OccurrenceOrUsages</i> is only valid, if the referenced items are true alternatives to each other. That means, they must have an identical placement, the geometrical models used for each item must be substitutable and the item must be mutually exclusive to each other.
orientation	Transformation3D	0..1	Y	N	0..1	Y	Specifies the orientation of the view item.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
placedElementViewItem3D	0..*	Y	BuildingBlockSpecification3D	1	Specifies the view items for <i>OccurrenceOrUsages</i> in a <i>BuildingBlockSpecification3D</i> .

2.9.12 Class Transformation3D

A *Transformation3D* is used to position an object in 3D space (e.g. on a DMU model).

It consists of a reference to a *CartesianPoint3D* as **origin** and a 3×3 matrix **a**. The matrix is given by the attributes **[a11...a33]**, with the numbers representing the index of the matrix element a_{ij} where **i** is the row and **j** the column index of the matrix.

To place an object with a *Transformation3D* its origin is translated in space to the *CartesianPoint3D* **origin** and every point $P = (x, y, z)$ of the object is transformed to P' by applying the matrix multiplication: $P' = a \cdot P$.

With a suitable definition of the transformation matrix, a wide variety of geometric operations can be mapped. Which of these operations are permissible depends on the respective application and cannot be defined in general terms. For

example, scaling and mirroring are probably not permissible operations in a normal DMU process with harmonized 3D models.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
a11	Double	1	NA
a12	Double	1	NA
a13	Double	1	NA
a21	Double	1	NA
a22	Double	1	NA
a23	Double	1	NA
a31	Double	1	NA
a32	Double	1	NA
a33	Double	1	NA

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
origin	CartesianPoint3D	1	Y	N	0..*	N	Specifies the coordinates of the translation.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
boundingBoxPositioning	0..1	Y	LocalGeometrySpecification	0..1	The transformation that defines the positioning of the bounding box in coordinate system of the component.
orientation	0..1	Y	OccurrenceOrUsageViewItem3D	0..1	Specifies the orientation of the view item.
positioning	0..1	Y	BuildingBlockPositioning3D	0..1	Specifies the positioning of the building block in the harness geometry.

2.9.13 Enumeration GeometryType

Enumeration for the definition of the GeometryType.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Dmu	The geometry is a DMU model of the vehicle.
Formboard	The geometry is a model of a formboard.

2.10 Module instancing**2.10.1 Class OccurrenceOrUsage**

An OccurrenceOrUsage is an abstract appearance of a part in the harness. This can either be a concrete part (with a part number or something similar) or the description (specification / requirements) of a part that should be used at that position. In the first case it would be a PartOccurrence in the second case a PartUsage.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>OccurrenceOrUsage</i> . The identification is guaranteed to be unique within the context. For all VEC-documents an <i>OccurrenceOrUsage</i> -instance can be trusted to be the same if the context-instance is the same and the identification of the <i>OccurrenceOrUsage</i> is the same.
aliasId	AliasIdentification	0..*	Room to specify additional identifiers for the <i>OccurrenceOrUsage</i> .
abbreviation	LocalizedString	0..*	Specifies an abbreviation of the <i>OccurrenceOrUsage</i> . Normally this a human readable short name.
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the <i>OccurrenceOrUsage</i> .
quantity	NumericalValue	0..1	Specifies the actual amount of material consumed by this <i>OccurrenceOrUsage</i> . Allows modelling in arbitrary units (e.g. grams, centimetres, piece). If omitted the default is 1 piece. Use whenever the <i>OccurrenceOrUsage</i> does not consume exactly one discrete item (e.g. 5 gram of glue). There are specialized roles (e.g. <i>WireRole</i> or <i>WireProtectionRole</i>) that define their own, more specific quantity information (e.g. <i>WireLength</i>). Those definitions have precedence over this <i>quantity</i> attribute.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
installationInstruction	Instruction	0..*	Y	N	0..1	Y	Room to specify InstallationInstruction(s) for the OccurrenceOrUsage.
realizedUsageNode	UsageNode	0..1	Y	N	0..*	N	References the <i>UsageNode</i> that is realized by this <i>OccurrenceOrUsage</i> .
referenceElement	OccurrenceOrUsage	0..*	Y	N	0..*	N	References the <i>OccurrenceOrUsage</i> for which this <i>OccurrenceOrUsage</i> is an accessory / supplementary component. If an <i>OccurrenceOrUsage</i> defines more than one

							<i>referenceElement</i> , the semantic is that it is related to the combination all "reference elements" and not to each individually. In other words, in a variable product definition the <i>OccurrenceOrUsage</i> only exists if all <i>referenceElements</i> exist.
role	Role	0..*	Y	N	0..1	Y	Specifies the different roles of the <i>OccurrenceOrUsage</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
completionComponents	1..*	N	ModuleList	0..*	References the components that are used as completion, if any of the Modules in the ModuleList appears in a configuration.
inBillOfMaterial	0..*	N	PartStructureSpecification	0..*	References the PartOccurrences that are building the bill of material of a composite part.
occurrenceOrUsage	0..*	N	OccurrenceOrUsageViewItem3D	0..*	Specifies the <i>OccurrenceOrUsages</i> which are represented by the view item. Important: To use one <i>OccurrenceOrUsageViewItem</i> for multiple <i>OccurrenceOrUsages</i> is only valid, if the referenced items are true alternatives to each other. That means, they must have an identical placement, the geometrical models used for each item must be substitutable and the item must be mutually exclusive to each other.
occurrenceOrUsage	0..*	N	OccurrenceOrUsageViewItem2D	0..*	Specifies the <i>OccurrenceOrUsages</i> which are represented by the view item. Important: To use one <i>OccurrenceOrUsageViewItem</i> for multiple <i>OccurrenceOrUsages</i> is only valid, if the referenced items are true alternatives to each other. That means, they must have an identical placement, the geometrical models used for each item must be substitutable and the item must be mutually exclusive to each other.
referenceElement	0..*	N	OccurrenceOrUsage	0..*	References the <i>OccurrenceOrUsage</i> for which this <i>OccurrenceOrUsage</i> is an accessory / supplementary component. If an <i>OccurrenceOrUsage</i> defines more than one <i>referenceElement</i> , the semantic is that it is related to the combination all "reference elements" and not to each individually. In other words, in a variable product definition the <i>OccurrenceOrUsage</i> only exists if all <i>referenceElements</i> exist.
subComponent	0..*	N	PartWithSubComponentsRole	0..*	References the subcomponents that belong to this instance of a PartWithSubComponents.
usedSupplementaryParts	0..*	N	PlacementPointReference	0..*	NA
usedSupplementaryParts	0..*	N	SlotReference	0..*	This is the relationship of the supplementary parts actually used in this <i>SlotReference</i> . This is the actual selection of the potential supplementary parts defined in the corresponding <i>Slot.supplementaryParts</i> association.
usedSupplementaryParts	0..*	N	CableLeadThroughOutletReference	NA	This is the relationship of the supplementary parts actually used in this <i>CableLeadThroughOutletReference</i> . This is the actual selection of the potential supplementary parts defined in the corresponding <i>Slot.supplementaryParts</i> association.

2.10.2 Class PartWithSubComponentsRole

A PartWithSubComponentsRole defines the instance specific properties and relationships of a part with subcomponents. A PartWithSubComponents is a composite part like an Assembly, a Module, Harness.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
partStructureSpecification	PartStructureSpecification	1	Y	N	0..*	N	References the <i>PartStructureSpecification</i> that is instantiated by this <i>PartWithSubComponentsRole</i> .
subComponent	OccurrenceOrUsage	0..*	Y	N	0..*	N	References the subcomponents that belong to this instance of a PartWithSubComponents.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
moduleInFamily	1..*	N	ModuleFamily	0..*	References the Modules that belong to the ModuleFamily.
moduleInList	1..*	N	ModuleList	0..*	References the Modules that belong to the ModuleList. If any of the referenced Modules participates in a configuration the completion components participate, too.

2.10.3 Class Role

A Role is the corresponding mechanism for OccurrenceOrUsages to the PartOrUsageRelatedSpecifications for PartVersions or PartUsages. The PartOrUsageRelatedSpecifications are describing a certain aspect of the master data of a part. A Role describes the corresponding properties and relationships for instances of a part (e.g. the usage specific properties of a wire occurrence like the length or the contacting).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the Role. The identification is guaranteed to be unique within the OccurrenceOrUsage.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment

role	0..*	Y	OccurrenceOrUsage	0..1	Specifies the different roles of the OccurrenceOrUsage.
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2.10.4 Class SpecificRole

A *SpecificRole* is the possibility to define instance specific properties with custom properties (see *ExtendableElement*). This is necessary, if the part is described by custom properties of a *PartOrUsageRelatedSpecification*.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
specificRoleType	String	1	Specifies the type for role.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
specification	PartOrUsageRelatedSpecification	1	Y	N	0..*	N	References the <i>PartOrUsageRelatedSpecification</i> that is instantiated by this <i>SpecificRole</i> .

2.11 Module instancing_electrical_parts

2.11.1 Class AbstractSlotReference

An *AbstractSlotReference* represents the usage of an AbstractSlot in the context of PartUsage or PartOccurrence.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the SlotReference. The identification is guaranteed to be unique within the ConnectorHousingRole. The cavity & slot number is defined by the associated cavity and slot.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

referencedSlot	AbstractSlot	1	Y	N	0..*	N	Points to the slot referenced by the slot reference.
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Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
firstSlot	1	N	SlotCoupling	0..*	NA
secondSlot	1	N	SlotCoupling	0..*	NA
slotReference	0..*	Y	ConnectorHousing Role	1	Specifies the SlotReferences used in the ConnectorHousingRole.

2.11.2 Class AntennaRole

An *AntennaRole* defines the instance specific properties and relationships of an antenna.

General Information

Base Classifier	EEComponentRole
Applied Stereotype	NA
Is Abstract	false

2.11.3 Class BatteryRole

An *BatteryRole* defines the instance specific properties and relationships of a battery.

General Information

Base Classifier	EEComponentRole
Applied Stereotype	NA
Is Abstract	false

2.11.4 Class BoltTerminalRole

A *BoltTerminalRole* defines the instance specific properties and relationships of a bolt terminal.

General Information

Base Classifier	TerminalRole
Applied Stereotype	NA
Is Abstract	false

2.11.5 Class BridgeTerminalRole

A *BridgeTerminalRole* defines the instance specific properties and relationships of a bridge terminal (see *BridgeTerminalSpecification*).

A bridge terminal is a part that behaves like terminal but has no *WireReceptions*. It is used to create short circuit between different pins in a connector. In its use, it can realize a schematic connection on its own and independently of other components.

General Information

Base Classifier	TerminalRole
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connection	Connection	0..*	Y	N	0..*	N	References the <i>Connection</i> that is realized by this <i>BridgeTerminalRole</i> .

2.11.6 Class CableSealEntryReference

Provides an identifier for the *CableSealEntryReference*. The *Identification* shall be unique within a *CableSealRole*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	NA
positionOnWire	Double	0..1	Specifies the position of this <i>CableSealEntryReference</i> on its associated <i>WireElementReference</i> . This value is analogous to <i>WireEnd.PositionOnWire</i> (see that documentation for details) and must be between 0 and 1. Note that it is not a fraction of the wire's total length used to compute an exact physical location. Its primary purpose is to indicate which side of the wire the seal entry is associated with. By definition, all <i>WireEnd</i> instances sharing the same <i>PositionOnWire</i> lie on the same side of the wire - and the same rule applies to <i>CableSealEntryReference.PositionOnWire</i> . Accordingly, properties such as <i>sealInsertionDepth</i> are always measured relative to the <i>WireEnd</i> with the matching <i>PositionOnWire</i> . In cases where the seal's position is determined by other topology elements (for example, the placement of a grommet) rather than a specific <i>WireEnd</i> , the <i>PositionOnWire</i> attribute may be omitted.
sealInsertionDepth	NumericalValue	0..1	Defines how far the cable seal is pushed onto the wire, measured from the tip of the corresponding <i>WireEnd</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
associatedWireElementRef	WireElementReference	0..*	Y	N	0..*	N	NA
cableSealEntry	CableSealEntry	1	Y	N	NA	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment

cableSealEntryReference	1..*	Y	CableSealRole	NA	NA
pluggedEntry	0..1	N	SealingPinReference	NA	NA

2.11.7 Class CableSealPlugRole

A *CableSealPlugRole* defines the instance specific properties and relationships of a cable seal plug.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cableSealPlugSpecification	CableSealPlugSpecification	1	Y	N	NA	N	NA
sealingPinReference	SealingPinReference	1..*	Y	N	NA	Y	NA

2.11.8 Class CableSealRole

A *CableSealRole* defines the instance specific properties and relationships of a cable seal.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cableSealEntryReference	CableSealEntryReference	1..*	Y	N	NA	Y	NA
cableSealSpecification	CableSealSpecification	1	Y	N	NA	N	NA

2.11.9 Class CapacitorRole

An *CapacitorRole* defines the instance specific properties and relationships of a capacitor.

General Information

Base Classifier	EEComponentRole
Applied Stereotype	NA

Is Abstract	false
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2.11.10 Class CavityAccessoryRole

A *CavityAccessoryRole* defines the instance specific properties and relationships of a cavity accessory.

General Information

Base Classifier	CavityPartRole
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavityAccessorySpecification	CavityAccessorySpecification	1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavityAccessory	0..*	N	CavityMounting	0..*	NA

2.11.11 Class CavityPartRole

NA

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	true

2.11.12 Class CavityPlugRole

A *CavityPlugRole* defines the instance specific properties and relationships of a cavity plug.

General Information

Base Classifier	CavityPartRole
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

cavityPlugSpecification	CavityPlugSpecification	1	Y	N	0..*	N	References the <i>CavityPlugSpecification</i> that is instantiated by this <i>CavityPlugRole</i> .
pluggedCavityRef	CavityReference	0..*	Y	N	0..*	N	Defines which cavity / cavities in a connector instance is sealed by the plug.

Incoming Relations

This End			Other End		General	
Role	Mult.	Agg.	Type	Mult.	Comment	
replacedPlug	0..*	N	CavityMounting	0..*	References the cavity plugs that are obsolete if the cavity mounting is realized.	
usedPlugs	0..*	N	CableLeadThroughReference	0..*	References the plugs that are used with this CableLeadThroughReference. This association might be a 150% selection.	

2.11.13 Class CavityReference

A CavityReference represents the usage of a Cavity in the context of PartUsage or PartOccurrence.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the CavityReference. The identification is guaranteed to be unique within the ConnectorHousingRole. The cavity & slot number is defined by the associated cavity and slot.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentPort	ComponentPort	0..*	Y	N	0..*	N	References the <i>ComponentPort</i> that is implemented by this <i>CavityReference</i> .
integratedTerminalRole	TerminalRole	0..1	Y	N	0..1	Y	Contains the terminal role if the cavity has an integrated terminal (e.g. an IDC).
referencedCavity	Cavity	1	Y	N	0..*	N	Points to the cavity referenced by the cavity reference.

Incoming Relations

This End			Other End		General	
Role	Mult.	Agg.	Type	Mult.	Comment	
cavityReference	0..*	Y	SlotReference	1	Specifies the CavityReferences used in the SlotReference.	
equippedCavityRef	1	N	CavityMountingDetail	0..*	References the cavity that is used for the detailed description of the cavity mounting.	
equippedCavityRef	1..*	N	CavityMounting	0..*	References the cavities that are used for the cavity mounting.	
firstCavity	1	N	CavityCoupling	0..*	NA	

pluggedCavityRef	0..*	N	CavityPlugRole	0..*	Defines which cavity / cavities in a connector instance is sealed by the plug.
secondCavity	1	N	CavityCoupling	0..*	NA

2.11.14 Class CavitySealRole

A *CavitySealRole* defines the instance specific properties and relationships of a cavity seal.

General Information

Base Classifier	CavityPartRole
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavitySealSpecification	CavitySealSpecification	1	Y	N	0..*	N	References the <i>CavitySealSpecification</i> that is instantiated by this <i>CavitySealRole</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
mountedCavitySeal	0..1	N	WireMounting	0..*	References the cavity seal that is used for the wire mounting.
usedSeals	0..*	N	CableLeadThroughReference	0..*	

2.11.15 Class CoatingRole

A *CoatingRole* defines the instance specific properties of a coating for a *WireEnd*.

General Information

Base Classifier	WireEndAccessoryRole
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
length	NumericalValue	0..1	The length over which the coating is applied at the <i>WireEnd</i> , measured from the wire tip.
weight	NumericalValue	0..1	The weight of material that shall be used for the coating of <i>WireEnd</i> .
volume	NumericalValue	0..1	The volume of material that shall be used for the coating of <i>WireEnd</i> .

2.11.16 Class ConnectorHousingCapRole

A *ConnectorHousingCapRole* defines the instance-specific properties of a cap for a connector (backshells).

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connectorHousingCapSpecification	ConnectorHousingCapSpecification	1	Y	N	0..*	N	References the <i>ConnectorHousingCapSpecification</i> that is instanced by this <i>ConnectorHousingCapRole</i> .

2.11.17 Class ConnectorHousingCoverRole

A *ConnectorHousingCoverRole* defines the instance-specific properties of a cover for a connector.

The association to the *ConnectorHousing* is established with the *OccurrenceOrUsage.ReferenceElement* association.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
transportOnly	Boolean	0..1	Specifies if the cover is only used during transportation of the harness. Conversely, <i>transportOnly=false</i> means that the cover might remain in the vehicle, depending on the specific circumstances (e.g. the variant configuration of the vehicle).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connectorHousingCoverSpecification	ConnectorHousingCoverSpecification	1	Y	N	0..*	N	NA

2.11.18 Class ConnectorHousingRole

A *ConnectorHousingRole* defines the instance specific properties and relationships of a connector housing.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
sealState	SealState	0..1	Specifies if this instance of a connector housing should be sealed (waterproof).
cpaState	ConnectorPositionAssuranceState	0..1	The state of the connector position assurance (CPA), if applicable. This is the state of the connector in the integration level described by the model to which the <i>ConnectorHousingRole</i> belongs. For example, if the <i>ConnectorHousingRole</i> is part of a harness specification from the OEM to the Tier1, it is the state the OEM expects for the delivered product, not the final state in the vehicle.
lockingState	ConnectorLockingState	0..1	The state of the primary locking mechanism of the connector, if applicable. This is the state of the connector in the integration level described by the model to which the <i>ConnectorHousingRole</i> belongs. For example, if the <i>ConnectorHousingRole</i> is part of a harness specification from the OEM to the Tier1, it is the state the OEM expects for the delivered product, not the final state in the vehicle.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentConnector	ComponentConnector	0..*	Y	N	0..*	N	References the ComponentConnector that is realized by the referenced ConnectorHousing (OccurrenceOrUsage with ConnectorHousingRole). This can especially be relevant for inliners. KBLFRM-341.
connectorHousingSpecification	ConnectorHousingSpecification	1	Y	N	0..*	N	References the <i>ConnectorHousingSpecification</i> that is instanced by this <i>ConnectorHousingRole</i> .
slotReference	AbstractSlotReference	0..*	Y	N	1	Y	Specifies the SlotReferences used in the ConnectorHousingRole.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
connectorHousingRole	0..1	Y	HousingComponentReference	0..1	NA
firstConnector	0..1	N	CouplingPoint	0..*	NA
secondConnector	0..1	N	CouplingPoint	0..*	NA
usedInserts	0..*	N	ModularSlotReference	0..*	References the inserts that are used in this <i>ModularSlotReference</i> . More than one insert is valid in the case variant dependent equipment of the slot.

2.11.19 Class DiodeRole

An *DiodeRole* defines the instance specific properties and relationships of a diode.

General Information

Base Classifier	EEComponentRole
Applied Stereotype	NA
Is Abstract	false

2.11.20 Class EECOMPONENTROLE

An EECOMPONENTROLE defines the instance specific properties and relationships of an EE-component.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentNode	ComponentNode	0..*	Y	N	0..*	N	References the ComponentNode that is realized by the referenced EECOMPONENT (OccurrenceOrUsage with EECOMPONENTROLE). KBLFRM-341
EEComponentSpecification	EEComponentSpecification	1	Y	N	0..*	N	References the <i>EEComponentSpecification</i> that is instantiated by this <i>EEComponentRole</i> .
extensionSlotRef	ExtensionSlotReference	0..*	Y	N	1	Y	Specifies the ExtensionSlotReferences used in the EECOMPONENTROLE.
housingComponentRef	HousingComponentReference	0..*	Y	N	1	Y	Specifies the HousingComponentReferences used in the EECOMPONENTROLE. (KBLFRM-401)

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
usedInserts	0..*	N	ExtensionSlotReference	0..*	Defines the inserts used for extension slot in a defined instance. These can be more than one EECOMPONENTROLE, because of variance in a 150% specification.

2.11.21 Class EXTENSIONSLOTREFERENCE

An EXTENSIONSLOTREFERENCE represents the usage of an ExtensionSlot in the context of a PartUsage or PartOccurrence.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	NA

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
extensionSlot	ExtensionSlot	1	Y	N	NA	N	NA
usedInserts	EEComponentRole	0..*	Y	N	0..*	N	Defines the inserts used for extension slot in a defined instance. These can be more than one EEComponentRole, because of variance in a 150% specification.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
extensionSlotRef	0..*	Y	EEComponentRole	1	Specifies the ExtensionSlotReferences used in the EEComponentRole.

2.11.22 Class FuseRole

An *FuseRole* defines the instance specific properties and relationships of a fuse.

General Information

Base Classifier	EEComponentRole
Applied Stereotype	NA
Is Abstract	false

2.11.23 Class HoleTerminalRole

An *HoleTerminalRole* defines the instance specific properties and relationships of a hole terminal.

General Information

Base Classifier	TerminalRole
Applied Stereotype	NA
Is Abstract	false

2.11.24 Class HousingComponentReference

A *HousingComponentReference* represents the usage of a *HousingComponent* in the context of a *PartUsage* or *PartOccurrence*. (KBLFRM-401)

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the <i>HousingComponentReference</i> . The identification is guaranteed to be unique within the <i>EEComponentRole</i> .

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentConnector	ComponentConnector	0..*	Y	N	0..*	N	References the ComponentConnector that is realized by the referenced HousingComponentReference.
connectorHousingRole	ConnectorHousingRole	0..1	Y	N	0..1	Y	NA
housingComponent	HousingComponent	1	Y	N	0..*	N	Points to the HousingComponent referenced by the HousingComponent reference. (KBLFRM-401)
pinComponentRef	PinComponentReference	0..*	Y	N	1	Y	Specifies the PinComponentReferences used in the HousingComponentReference. (KBLFRM-401)

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
housingComponentRef	0..*	Y	EEComponentRole	1	Specifies the HousingComponentReferences used in the EEComponentRole. (KBLFRM-401)

2.11.25 Class LabelingRole

If a component is labeled in its concrete usage, label information is defined with the *LabelingRole*.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
labelType	LabelType	0..1	Specifies the type of a label printed, stamped or integrated on or in component (e.g. alpha numerical, bar code).
labelValue	String	0..1	Specifies the value of a label printed on the component.
labelingTechnology	LabelingTechnology	0..1	Specifies the used labeling technology.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
labelingSpecification	LabelingSpecification	1	Y	N	0..*	N	NA

2.11.26 Class ModularSlotReference

A *ModularSlotReference* represents the usage of a *ModularSlot* in the context of *PartUsage* or *PartOccurrence*.

General Information

Base Classifier	AbstractSlotReference
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
usedInserts	ConnectorHousing Role	0..*	Y	N	0..*	N	References the inserts that are used in this <i>ModularSlotReference</i> . More than one insert is valid in the case variant dependent equipment of the slot.

2.11.27 Class MultiCavityPlugRole

Deprecated since 2.2.0:

Reason: see [MultiCavityPlugSpecification](#)

A *MultiCavityPlugRole* defines the instance specific properties and relationships of a multi cavity plug.

General Information

Base Classifier	CavityPlugRole
Applied Stereotype	Deprecated
Is Abstract	false

2.11.28 Class MultiCavitySealRole

Deprecated since 2.2.0:

Reason: see [MultiCavitySealSpecification](#)

A *MultiCavitySealRole* defines the instance specific properties and relationships of a multi cavity seal.

General Information

Base Classifier	CavitySealRole
Applied Stereotype	Deprecated
Is Abstract	false

2.11.29 Class MultiFuseRole

NA

General Information

Base Classifier	EEComponentRole
Applied Stereotype	NA
Is Abstract	false

2.11.30 Class OpenWireEndTerminalRole

An *OpenWireEndTerminalRole* defines the instance specific properties and relationships of an open wire end.

General Information

Base Classifier	TerminalRole
Applied Stereotype	NA
Is Abstract	false

2.11.31 Class PinComponentReference

A PinComponentReference represents the usage of a PinComponent in the context of a PartUsage or PartOccurrence. (KBLFRM-401)

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the PinComponentReference. The identification is guaranteed to be unique within the HousingComponentReference.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
pinComponent	PinComponent	1	Y	N	0..*	N	Points to the PinComponent referenced by the PinComponentReference.
terminalRole	TerminalRole	0..1	Y	N	0..1	Y	References the TerminalRole of PinComponentReference. This is required to specify a Mating for EECComponents with other EECComponents or a Harness. (KBLFRM-401)

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
pinComponentRef	0..*	Y	HousingComponentReference	1	Specifies the PinComponentReferences used in the HousingComponentReference. (KBLFRM-401)
pinComponentReference	1	N	PinWireMappingPoint	NA	NA

2.11.32 Class PluggableTerminalRole

A PluggableTerminalRole defines the instance specific properties and relationships of a pluggable terminal.

General Information

Base Classifier	TerminalRole
Applied Stereotype	NA

Is Abstract	false
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2.11.33 Class PotentialDistributorRole

An *PotentialDistributorRole* defines the instance specific properties and relationships of a potential distributor.

General Information

Base Classifier	EEComponentRole
Applied Stereotype	NA
Is Abstract	false

2.11.34 Class RelayRole

An *RelayRole* defines the instance specific properties and relationships of a relay.

General Information

Base Classifier	EEComponentRole
Applied Stereotype	NA
Is Abstract	false

2.11.35 Class RingTerminalRole

A *RingTerminalRole* defines the instance specific properties and relationships of a ring terminal.

General Information

Base Classifier	TerminalRole
Applied Stereotype	NA
Is Abstract	false

2.11.36 Class SealingPinReference

A *SealingPinReference* is an instance of a *CableSealEntry*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Provides an identifier for the <i>SealingPinReference</i> . The <i>Identification</i> shall be unique within a <i>CableSealPlugRole</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

pluggedEntry	CableSealEntryReference	0..1	Y	N	NA	N	NA
sealingPin	SealingPin	1	Y	N	NA	N	NA

Incoming Relations

This End			Other End			General	
Role	Mult.	Agg.	Type		Mult.	Comment	
sealingPinReference	1..*	Y	CableSealPlugRole		NA	NA	

2.11.37 Class SlotReference

A *SlotReference* represents the usage of a *Slot* in the context of *PartUsage* or *PartOccurrence*.

General Information

Base Classifier	AbstractSlotReference
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
tpaState	TerminalPositionAssuranceState	0..1	The state of the terminal position assurance (TPA, syn. secondary locking), if applicable. This is the state of the slot in the integration level described by the model to which the <i>SlotReference</i> belongs. For example, if the <i>SlotReference</i> is part of a harness specification from the OEM to the Tier1, it is the state the OEM expects for the delivered product, not the final state in the vehicle.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavityReference	CavityReference	0..*	Y	N	1	Y	Specifies the CavityReferences used in the SlotReference.
usedSupplementaryParts	OccurrenceOrUsage	0..*	Y	N	0..*	N	This is the relationship of the supplementary parts actually used in this <i>SlotReference</i> . This is the actual selection of the potential supplementary parts defined in the corresponding <i>Slot.supplementaryParts</i> association.

2.11.38 Class SpliceTerminalRole

A *SpliceTerminalRole* defines the instance specific properties and relationships of a splice terminal.

General Information

Base Classifier	TerminalRole
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
spliceType	SpliceType	0..1	Specifies the type of splice (inline, end).
insulationState	InsulationState	0..1	Specifies the insulation state of the splice, in other words is electrically insulated or not.

2.11.39 Class TerminalProtectionRole

A *TerminalProtectionRole* defines the instance specific properties and relationships of a *TerminalProtection*.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
terminalProtection Specification	TerminalProtectionSpecification	1	Y	N	0..*	N	NA

2.11.40 Class TerminalReceptionReference

A *TerminalReceptionReference* is an instance of a *TerminalReception*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique <i>identification</i> of the <i>TerminalReceptionReference</i> . The <i>identification</i> is guaranteed to be unique within the <i>TerminalRole</i> (this might be for example a reception number).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
terminalReception	TerminalReception	1	Y	N	0..*	N	References the <i>TerminalReception</i> that is instanced by this <i>TerminalReceptionReference</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
firstTerminalReception	1	N	MatingDetail	0..*	References the first terminal reception that is mated.
secondTerminalReception	1	N	MatingDetail	0..*	References the second terminal reception that is mated.
terminalReceptionReference	1	N	CavityMountingDetail	0..*	References the TerminalReception that is used for the detailed description of the cavity mounting.
terminalReceptionReference	0..*	Y	TerminalRole	1	Specifies the <i>TerminalReceptionReferences</i> of this <i>TerminalRole</i> .

2.11.41 Class TerminalRole

A TerminalRole defines the instance specific properties and relationships of a terminal.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
sealState	SealState	0..1	Specifies if this instance of a terminal should be sealed (waterproof). This applies for example to SpliceTerminals.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentPort	ComponentPort	0..*	Y	N	0..*	N	References the ComponentPort that is realized by the referenced Terminal (OccurrenceOrUsage with TerminalRole). KBLFRM-341
terminalReceptionReference	TerminalReceptionReference	0..*	Y	N	1	Y	Specifies the <i>TerminalReceptionReferences</i> of this <i>TerminalRole</i> .
terminalSpecification	TerminalSpecification	1	Y	N	0..*	N	References the <i>TerminalSpecification</i> that is instantiated by this <i>TerminalRole</i> .
wireReceptionReference	WireReceptionReference	0..*	Y	N	1	Y	Specifies the <i>WireReceptionReferences</i> of this <i>TerminalRole</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
firstTerminalRole	1	N	MatingPoint	0..*	References the first terminal that is mated.
integratedTerminalRole	0..1	Y	CavityReference	0..1	Contains the terminal role if the cavity has an integrated terminal (e.g. an IDC).
mountedTerminal	0..1	N	ContactPoint	0..*	References the terminal that is used for contacting defined by the ContactPoint.

secondTerminalRole	1	N	MatingPoint	0..*	References the second terminal that is mated.
terminalRole	0..1	Y	PinComponentReference	0..1	References the TerminalRole of PinComponentReference. This is required to specify a Mating for EECComponents with other EECComponents or a Harness. (KBLFRM-401)

2.11.42 Class WireElementReference

A WireElementReference represents the usage of a WireElement in the context of a PartUsage or PartOccurrence. For contacting purposes, a WireElementReference has WireEnds. KBLFRM-384

General Information

Base Classifier	RoutableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the WireElementReference. The identification is guaranteed to be unique within the WireRole.
unconnected	Boolean	0..1	If this attribute is <i>true</i> , the <i>WireElementReference</i> is not connected (unused) on purpose. This can be the case for example if a multi core is used, but not all cores are necessary in a specific situation. However, for all <i>WireElements</i> defined in the <i>WireSpecification</i> a corresponding <i>WireElementReference</i> shall exist. This attribute can be used to mark these unused cores explicitly.
labelValue	String	0..1	Specifies the value of label, if this <i>WireElementReference</i> is labeled.
labelType	LabelType	0..1	NA
labelPosition	LabelPosition	0..1	Specifies the position of label, if this <i>WireElementReference</i> is labeled.
labelingTechnology	LabelingTechnology	0..1	Specifies the used <i>LabelingTechnology</i> , if this <i>WireElementReference</i> is labeled.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connection	Connection	0..*	Y	N	0..*	N	References the Connection that is realized by the referenced WireElement (WireElementReference). KBLFRM-341
connectionGroup	ConnectionGroup	0..*	Y	N	NA	N	References the <i>ConnectionGroup</i> that is realized by this <i>WireElementReference</i> . This applies normally to <i>WireElementReference</i> that have <i>subWireElements</i> .
referencedWireElement	WireElement	1	Y	N	0..*	N	References the WireElement that is represented by the WireElementReference.
signal	Signal	0..1	Y	N	0..*	N	References the signal that is transmitted by the WireElementReference.

wireEnd	WireEnd	0..*	Y	N	1	Y	Specifies the ends of the WireElementReference for contacting purposes.
wireLength	WireLength	0..*	Y	N	1	Y	Specifies the different length of a wire.

Incoming Relations

This End			Other End		General	
Role	Mult.	Agg.	Type	Mult.	Comment	
associatedWireElementRef	0..*	N	CableSealEntryReference	0..*	NA	
relatedWireElementReference	0..*	N	WireGrouping	0..*	References the concrete wire elements (<i>WireElementReference</i>) that are grouped by the WireGrouping.	
wireElementReference	0..*	Y	WireRole	1	Specifies the WireElementReferences used in the WireRole. For multi core wires more than one WireElementReference is needed.	

2.11.43 Class WireEnd

A WireEnd is the end of a wire. This class mainly needed for the definition of a contacting. As a wire can be contacted on more than two ends (e.g. IDC) the WireEnd has a position on the wire.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the WireEnd. The identification is guaranteed to be unique within the WireElementReference.
positionOnWire	Double	1	Specifies the order of the <i>WireEnds</i> on the <i>WireElementReference</i> . This must be a value between 0 and 1. The <i>PositionOnWire</i> is defined as a floating-point number, to allow the ordering of an arbitrary number of <i>WireEnds</i> while having constant known values for the first and last <i>WireEnd</i> ("0.0" & "1.0"). The <i>PositionOnWire</i> is <u>not</u> defined as a relative factor with respect to the <i>WireLength</i> to determine an exact position of the <i>WireEnd</i> on the wire. This would not be defined unambiguously, since a <i>WireElementReference</i> can define multiple lengths of different <i>WireLengthTypes</i> . If such an interpretation can be made, it is tool or process specific definition. In case of <i>WireRole</i> with multiple <i>WireElementReferences</i> the ordering of the <i>WireEnds</i> shall be consistent for all <i>WireElementReferences</i> . That means that all <i>WireEnds</i> with <i>PositionOnWire</i> = 0.0 are on the same side of the <i>WireRole</i> .
cutBackLength	NumericalValue	0..1	Defines the length, by which this <i>WireEnd</i> is cut back relative to the cut length of the outer most <i>WireElementReference</i> (see <i>strippingLength</i> for an example).
strippingLength	NumericalValue	0..1	Defines the length by which an insulation on this stripped relative to this <i>WireEnd</i> , defined by the <i>cutBackLength</i> . E.g. a <i>cutBackLength</i> of 20mm and a <i>strippingLength</i> of 10mm means, that this <i>WireElementReference</i> is cutback by 20mm relative to length of whole wire, on the side of the defining <i>WireEnd</i> . Form the remaining <i>WireElementReference</i> are then 10mm insulation removed.
insulationPullbackLength	NumericalValue	0..1	Defines a partial stripping process in which the insulation is pulled back by the <i>insulationPullbackLength</i> after being cut at the <i>strippingLength</i> and is left on the <i>WireEnd</i> e.g. for logistic reasons.

			The <i>insulationPullbackLength</i> must be shorter than the <i>strippingLength</i> ; otherwise, it would not qualify as partial stripping.
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Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connectionEnd	ConnectionEnd	0..*	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
associatedWireEnds	0..*	N	WireTupleTermination	0..*	NA
referencedWireEnd	1..*	N	WireMounting	0..*	References the wire ends that are used for the wire mounting. The minimum cardinality is one, because a wire mounting without wire end makes no sense. The maximum cardinality is * in order to support multi crimps.
referencedWireEnd	1..*	N	WireMountingDetail	0..*	References the WireEnds that are mounted to referenced WireReception. A cardinality of more than one is allowed in order support parallel connectors, where multiple wire ends are placed on one side of the connector (one wire reception) and the other wire ends are placed on the other side of the connector (the other wire reception).
wireEnd	0..*	Y	WireElementReference	1	Specifies the ends of the WireElementReference for contacting purposes.

2.11.44 Class WireEndAccessoryRole

A *WireEndAccessoryRole* defines the instance specific properties and relationships of a *WireEndAccessory*.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
wireEndAccessorySpecification	WireEndAccessorySpecification	1	Y	N	NA	N	References the <i>WireEndAccessorySpecification</i> that is instantiated by this <i>WireEndAccessoryRole</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment

wireEndAccessory	0..*	N	WireMounting	0..*	NA
------------------	------	---	--------------	------	----

2.11.45 Class WireGrouping

A *WireGrouping* is the definition of a multi core wire in its usage. The elements of a *WireGrouping* are well defined wires (e.g. with a part number). The grouping itself is only created in its concrete usage. The most common use case is the individual definition of unshielded twisted pair wires without creating the full combinatorics of every possible core / insulation / twist combination in a part master data system (and by this creating part numbers for all of them). However, there are other use cases as well.

A *WireGrouping* groups the *relatedWireElementReferences* on an equal level contained *WireGroupings* are on a lower level. So, in order to create something like a shielded twisted pair, one *WireGrouping* "A" that references two *WireElementReferences* is required to represent the twisted pair and another *WireGrouping* "B" that contains *WireGrouping* "A" and references the "shield wire element".

The referenced *WireGroupSpecification* defines the handling of the *WireGrouping* during its assembly (e.g. twist).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>WireGrouping</i> . The identification is guaranteed to be unique within the <i>WireGroupingSpecification</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connectionGroup	ConnectionGroup	0..*	Y	N	NA	N	References the <i>ConnectionGroup</i> that is realized by this <i>WireGrouping</i> .
containedWireGroupings	WireGrouping	0..*	Y	N	0..1	Y	References the <i>WireGroupings</i> that are contained in this <i>WireGrouping</i> .
relatedWireElementReference	WireElementReference	0..*	Y	N	0..*	N	References the concrete wire elements (<i>WireElementReference</i>) that are grouped by the <i>WireGrouping</i> .
wireGroupSpecification	WireGroupSpecification	0..1	Y	N	0..*	N	References the <i>WireGroupSpecification</i> that applies to the <i>WireGrouping</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
containedWireGroupings	0..*	Y	WireGrouping	0..1	References the <i>WireGroupings</i> that are contained in this <i>WireGrouping</i> .
wireGrouping	1..*	Y	WireGroupingSpecification	0..1	Specifies the <i>WireGroupings</i> described by the <i>WireGroupingSpecification</i> .

2.11.46 Class WireGroupingSpecification

Specification for the description of *WireGroupings*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
wireGrouping	WireGrouping	1..*	Y	N	0..1	Y	Specifies the WireGroupings described by the WireGroupingSpecification.

2.11.47 Class WireLength

Defines the length of a WireElementReference. A WireElementReference can have multiple lengths of different types but must not have more than one length of the same type.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
lengthType	WireLengthType	1	The type of the length. Possible values are defined in an open enumeration (see WireLengthType).
lengthValue	NumericalValue	1	The length value for the type.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
wireLength	0..*	Y	WireElementReference	1	Specifies the different length of a wire.

2.11.48 Class WireReceptionReference

A *WireReceptionReference* is an instance of a *WireReception*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
------	------	------	---------

identification	String	0..1	Specifies a unique <i>identification</i> of the <i>WireReceptionReference</i> . The <i>identification</i> is guaranteed to be unique within the <i>TerminalRole</i> (this might be for example a reception number).
----------------	--------	------	---

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
wireReception	WireReception	1	Y	N	0..*	N	References the <i>WireReception</i> that is instantiated by this <i>WireReceptionReference</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
contactedWireReception	1	N	WireMountingDetail	0..*	References the <i>WireReception</i> that is used for the <i>WireMounting</i> .
wireReceptionReference	0..*	Y	TerminalRole	1	Specifies the <i>WireReceptionReferences</i> of this <i>TerminalRole</i> .

2.11.49 Class WireRole

A *WireRole* defines the instance specific properties and relationships of a wire.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
wireElementReference	WireElementReference	0..*	Y	N	1	Y	Specifies the <i>WireElementReferences</i> used in the <i>WireRole</i> . For multi core wires more than one <i>WireElementReference</i> is needed.
wireSpecification	WireSpecification	1	Y	N	0..*	N	References the <i>WireSpecification</i> that is instantiated by this <i>WireRole</i> .

2.11.50 Class WireTupleTermination

A *WireTupleTermination* is to define the handling of wire ends in a tuple (group). This is mostly relevant for twisted pairs / triple / tuple to define the handling of untwisting.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
fixation	Boolean	0..1	Defines whether a fixation to prevent untwisting must be applied to associated <i>WireEnds</i> of this <i>WireTupleTermination</i> .
fixationOffset	NumericalValue	0..1	Specifies the offset of the fixation.
permittedUntwist	NumericalValue	0..1	Specifies the allowed length of untwist for the <i>WireElementReferences</i> on the sides of associated <i>WireEnds</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
associatedWireEnds	WireEnd	0..*	Y	N	0..*	N	NA
wireTupleSpecification	WireTupleSpecification	0..1	Y	N	NA	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
wireTupleTermination	0..*	Y	WireTupleTerminationSpecification	1	NA

2.11.51 Class WireTupleTerminationSpecification

A *WireTupleTerminationSpecification* is the container specification for *WireTupleTerminations*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
wireTupleTermination	WireTupleTermination	0..*	Y	N	1	Y	NA

2.11.52 Enumeration ConnectorLockingState

Defines the valid states for the connector locking.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Open	NA
Closed	NA

2.11.53 Enumeration ConnectorPositionAssuranceState

Defines the valid states for the connector position assurance (CPA).

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
NotInstalled	NA
PreInstalled	NA
Engaged	NA

2.11.54 Enumeration InsulationState

Enumeration for the definition of the insulation state of the splice.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Insulated	The splice is electrically insulated.
Uninsulated	The splice is not electrically insulated.

2.11.55 Enumeration LabelPosition

Open Enumeration of standard label positions on *WireElementReferences*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
End	NA
Continuous	NA

2.11.56 Enumeration SealState

Enumeration for the definition of the sealing state of the splice.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Unsealed	The splice is not waterproof.
Sealed	The splice is waterproof.

2.11.57 Enumeration SpliceType

Enumeration for the definition of type of splice.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
InlineSplice	An <i>InlineSplice</i> is a splice where wires from both sides are connected.
EndSplice	An <i>EndSplice</i> is a splice where wires only from one side are connected.

2.11.58 Enumeration TerminalPositionAssuranceState

Defines the valid states for the terminal position assurance (TPA, syn. secondary locking).

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
NotInstalled	NA
PreInstalled	NA
Engaged	NA

2.11.59 Enumeration WireLengthType

Specifies possible values for the *lengthType* of *WireLength*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
DMU	The length of the wire is calculated from the sum of the lengths of the neutral axes of the corresponding segments in the DMU model.
Drawing	The length is rounded length from the DMU model (shown on the drawing), without any additions
Contract	The agreed length for any negotiations and calculations.

Production	The cutting length for the used in production environments.
Functional	A <i>Functional</i> length is a length specification which must be precisely adhered to in production, and which results from the design, for example due to signal/wave characteristics. This is in contrast to the <i>DMU / Drawing Length</i> which only specifies the length of the routed neutral phase.

2.12 Module instancing_non_electrical_parts

2.12.1 Class BoltMountedFixingRole

An *BoltMountedFixingRole* defines the instance specific properties and relationships of a bolt mounted fixing.

General Information

Base Classifier	FixingRole
Applied Stereotype	NA
Is Abstract	false

2.12.2 Class CableDuctRole

A *CableDuctRole* defines the instance specific properties and relationships of a cable duct.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cableDuctSpecification	CableDuctSpecification	1	Y	N	0..*	N	References the <i>CableDuctSpecification</i> that is instanced by this <i>CableDuctRole</i> .

2.12.3 Class CableLeadThroughOutletReference

A *CableLeadThroughOutletReference* is the instance of a *CableLeadThroughOutlet*. It can define a set of cable seals or plugs as supplementary parts. Plugs are used if no wire is present, seals are used together with a wire.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>CableLeadThroughReference</i> . The identification is guaranteed to be unique within the <i>GrommetRole</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cableLeadThroughOutlet	CableLeadThroughOutlet	1	Y	N	NA	N	NA
usedSupplementaryParts	OccurrenceOrUsage	0..*	Y	N	NA	N	This is the relationship of the supplementary parts actually used in this <i>CableLeadThroughOutletReference</i> . This is the actual selection of the potential supplementary parts defined in the corresponding <i>Slot.supplementaryParts</i> association.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cableLeadThroughOutletReference	0..*	Y	CableLeadThroughReference	NA	NA

2.12.4 Class CableLeadThroughReference

A *CableLeadThroughReference* is the instance of a *CableLeadThrough*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the <i>CableLeadThroughReference</i> . The identification is guaranteed to be unique within the <i>GrommetRole</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cableLeadThrough	CableLeadThrough	0..1	Y	N	NA	N	NA
cableLeadThroughOutletReference	CableLeadThroughOutletReference	0..*	Y	N	NA	Y	NA
usedPlugs	CavityPlugRole	0..*	Y	N	0..*	N	<u>Deprecated since 2.2.0:</u> <u>Reason: Use a cable seal plug as supplementary part instead.</u> References the plugs that are used with this <i>CableLeadThroughReference</i> . This association might be a 150% selection.

usedSeals	CavitySealRole	0..*	Y	N	0..*	N	<i>Deprecated since 2.2.0: Reason: Use a cable seal as supplementary part instead.</i>
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Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cableLeadThrough Reference	0..*	Y	GrommetRole	1	NA

2.12.5 Class CableTieRole

A *CableTieRole* defines the instance specific properties and relationships of a cable tie.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cableTieSpecification	CableTieSpecification	1	Y	N	0..*	N	NA

2.12.6 Class CorrugatedPipeRole

Specific *WireProtectionRole* for instances of *CorrugatedPipeSpecification*.

General Information

Base Classifier	TubeRole
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
cutOnWave	Boolean	0..1	Defines if the corrugated pipe is cut on top of the corrugation. This applies only to corrugated pipes specified as bulk material. For parts with a predefined length the cutting is defined in the <i>CorrugatedPipeSpecification</i> .

2.12.7 Class EdgeMountedFixingRole

An *EdgeMountedFixingRole* defines the instance specific properties and relationships of an edge mounted fixing.

General Information

Base Classifier	FixingRole
Applied Stereotype	NA

Is Abstract	false
-------------	-------

2.12.8 Class FerriteRole

A *FerriteRole* defines the instance specific properties and relationships of a ferrite.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
numberOfWindings	Integer	0..1	Defines the of windings that the wires are wound around the ferrite.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
ferriteSpecification	FerriteSpecification	1	Y	N	NA	N	NA

2.12.9 Class FittingRole

Specific *WireProtectionRole* for instances of *FittingSpecification*.

General Information

Base Classifier	WireProtectionRole
Applied Stereotype	NA
Is Abstract	false

2.12.10 Class FixingRole

A *FixingRole* defines the instance specific properties and relationships of a fixing.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
fixingSpecification	FixingSpecification	1	Y	N	0..*	N	References the <i>FixingSpecification</i> that is instanced by this <i>FixingRole</i> .

2.12.11 Class GrommetRole

A *GrommetRole* defines the instance specific properties and relationships of a grommet.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cableLeadThroughReference	CableLeadThroughReference	0..*	Y	N	1	Y	NA
grommetSpecification	GrommetSpecification	1	Y	N	0..*	N	References the <i>GrommetSpecification</i> that is instantiated by this <i>GrommetRole</i> .

2.12.12 Class HoleMountedFixingRole

An *HoleMountedFixingRole* defines the instance specific properties and relationships of a hole mounted fixing.

General Information

Base Classifier	FixingRole
Applied Stereotype	NA
Is Abstract	false

2.12.13 Class ShrinkableTubeRole

Specific *WireProtectionRole* for instances of *ShrinkableTubeSpecification*.

General Information

Base Classifier	TubeRole
Applied Stereotype	NA
Is Abstract	false

2.12.14 Class StripeRole

Specific *WireProtectionRole* for instances of *StripeSpecification*.

General Information

Base Classifier	WireProtectionRole
Applied Stereotype	NA
Is Abstract	false

2.12.15 Class TapeRole

Specific *WireProtectionRole* for instances of *TapeSpecification*. This *Role* is used for all kinds of taping and the *WindingType* defines the actual type of execution. Not all attributes in the *TapeRole* are relevant for all *WindingTypes*. Generating processes and systems are responsible to fill only attributes relevant for the selected *WindingType*.

General Information

Base Classifier	WireProtectionRole
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
tapeOverlap	NumericalValue	0..1	Specifies the amount of overlap two rounds of taping around a segment have, as an absolute value. A negative value means, that there is a gap between two rounds.
tapeOverlapRelative	Double	0..1	Specifies the amount of overlap two rounds of taping around a segment have, as a relative value. A negative value means, that there is a gap between two rounds. Values are defined as a factor, not as a percentage. Values can be between 1.0 and negative "infinity". Examples: An overlap of 1.0 defines that the second round is placed exactly on top of the first one. An overlap of 0.5 specifies that one half of the second round is on top of the first round (50% overlapping), a value of 0 specifies, that there is no overlap, but also no gap. A value of -2.0 specifies that there is a gap twice the width of the tape between two rounds.
tapingDirection	TapingDirection	0..1	Specifies the direction of the taping.
gradient	ValueWithUnit	0..1	Specifies the gradient of the taping.
windingType	WindingType	0..1	Defines the type of the tape's winding (see <i>WindingType</i>).
windingFirmness	WindingFirmness	0..1	Defines the firmness of the tape's winding (see <i>WindingFirmness</i>).
numberOfTurns	Integer	0..1	Specifies the number of turns, the tape is wrapped around a segment. This is mainly a relevant definition for <i>Spot</i> and <i>Cross</i> tapes.

2.12.16 Class TubeRole

Specific *WireProtectionRole* for instances of *TubeSpecification*.

General Information

Base Classifier	WireProtectionRole
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
slitStyle	TubeSlitStyle	0..1	Specifies the style of the slitting of the tube. If a <i>slitStyle</i> is defined, it implies that <i>isSlit=true</i> . This attribute is defined as an <i>OpenEnumeration</i> .
isSlit	Boolean	0..1	Specifies whether the tube is slit or not. The style of the slitting can be defined with the <i>slitStyle</i> . If a <i>slitStyle</i> is defined, it implies that <i>isSlit=true</i> .

2.12.17 Class WireProtectionRole

A WireProtectionRole defines the instance specific properties and relationships of a wire protection. This is a general-purpose role for instances of all types of *WireProtectionSpecifications* that do not have specific instance attributes. For *TapeSpecifications* the more specific *TapeRole* shall be used.

General Information

Base Classifier	Role
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
protectionLength	NumericalValue	0..1	The <i>ProtectionLength</i> specifies the length of the area covered by the protection. As a consequence, the <i>ProtectionLength</i> may differ from the real length of the material (e.g. for tapes).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
materialLength	ProtectionMaterialLength	0..*	Y	N	1	Y	NA
wireProtectionSpecification	WireProtectionSpecification	1	Y	N	0..*	N	References the <i>WireProtectionSpecification</i> that is instanced by this <i>WireProtectionRole</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
relatedWireProtections	1..*	N	WireProtectionGroup	0..1	References the members in the <i>WireProtectionGroup</i> .

2.12.18 Enumeration TapingDirection

Defines the direction in relation to the start & end-Location of the corresponding placement. If no TapingDirection is defined it is arbitrary.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
FromStart	NA
FromEnd	NA

2.12.19 Enumeration WindingFirmness

Defines the firmness with which a tape is applied to a segment.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Limp	The taping is applied limp (loose taping).
Tight	The taping is applied tight.

2.12.20 Enumeration WindingType

Defines the type of taping.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
OnSpace	<i>OnSpace</i> describes a taping with gaps between the rounds (German: Luecke)
Spare	<i>Spare</i> describes a taping with larger gaps between the rounds (German: Spar)
Overlap	<i>Overlap</i> describes a taping where each round overlaps the preceding round.
DoubleOverlap	<i>DoubleOverlap</i> describes a taping where each round overlaps the preceding round and the taping is done twice (forward and backwards).
Spiral	<i>Spiral</i> defines a taping with the least overlapping (German: Spiral).
Longitudinal	<i>Longitudinal</i> defines a taping with where the tape is folded around a segment.
Spot	<i>Spot</i> defines a taping where all rounds are placed at the same location (SpotTape).
Cross	<i>Cross</i> describes a special taping method at intersections where the rounds around the segment form a cross (German: Kreuzwickel)

2.13 Module instructions**2.13.1 Class DocumentBasedInstruction**

A DocumentBasedInstruction is an Instruction to a SheetOrChapter in a DocumentVersion or to a complete DocumentVersion.

General Information

Base Classifier	Instruction
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

referencedDocument	DocumentVersion	1	Y	N	0..*	N	References the DocumentVersion that is used as an Instruction.
referencedSheetOrChapter	SheetOrChapter	0..1	Y	N	0..*	N	References the SheetOrChapter that is used as an Instruction.

2.13.2 Class FileBasedInstruction

A FileBasedInstruction is an Instruction that references a file packaged (VEC-Package) together with a VEC-file. Such a file can be for example an image.

General Information

Base Classifier	Instruction
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
fileName	String	1	The name of the file as it appears in the VEC-Package, including the folder structure (fully qualified name) that contains this <i>FileBasedInstruction</i> . The usage of this attribute is only valid, if the original document is distributed along with the VEC-file in a VEC-Package. It must not point to any file location which is not part of the VEC-Package (e.g. a file on a central server file share). The <i>fileName</i> is relative to the VEC-Package root. It MUST NOT contain a drive or device letter, or a leading slash. All slashes MUST be forward slashes '/' (UNIX-style).
lastModified	Date	0..1	The last modified timestamp of the file.
dataFormat	String	0..1	The dataFormat specifies the format of the FileReference. The dataFormat has to be a valid "Internet Media Type" (see https://www.iana.org/assignments/media-types/media-types.xhtml).

2.13.3 Class Instruction

Abstract super class to specify different types of instructions. Possible instructions are text, file or document based.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
installationInstruction	0..*	Y	OccurrenceOrUsage	0..1	Room to specify InstallationInstruction(s) for the OccurrenceOrUsage.

2.13.4 Class TextBasedInstruction

A TextBasedInstruction is used to specify human readable instructions in text.

General Information

Base Classifier	Instruction
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
installationInstruction	LocalizedString	1..*	Specifies the instruction in a human readable form. Different languages are possible.

2.14 Module local_geometry

2.14.1 Class LocalGeometrySpecification

A *LocalGeometrySpecification* is responsible to create a relationship between the 3D model of a component (e.g. a connector, a cable duct or a fixing) and entities of the VEC. The 'Local' in the name refers to the fact that all definitions within this specification are 'local' to the 3D model of a specific component (a component in a library, not in a specific usage).

Specifically, it defines a transformation to transform the *BoundingBox* of a component into to coordinate system of the component and it defines the positions of *Placement*- and *MeasurementPoints* in this coordinate system.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
baseUnit	Unit	1	Y	N	NA	N	The <i>Unit</i> in which all coordinates (e.g. cartesian points) are defined. Shall be a unit of length (e.g. millimetre).
boundingBoxPositioning	Transformation3D	0..1	Y	N	0..1	Y	The transformation that defines the positioning of the bounding box in coordinate system of the component.
cartesianPoint	CartesianPoint3D	0..*	Y	N	0..1	Y	All <i>CartesianPoint3Ds</i> that are used in this <i>LocalGeometrySpecification</i> . All <i>CartesianPoint3Ds</i> are defined in relation to the coordinate system of the component.
positions	LocalPosition	0..*	Y	N	1	Y	All position defined by this <i>LocalGeometrySpecification</i> .

2.14.2 Class LocalPosition

A *LocalPosition* defines the position of a relevant point of component within the coordinate system of the component.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Outgoing Relations

Other End	This	General
-----------	------	---------

Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cartesianPoint	CartesianPoint3D	1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
positions	0..*	Y	LocalGeometrySpecification	1	All position defined by this <i>LocalGeometrySpecification</i> .

2.14.3 Class MeasurePointPosition

Defines the position of a MeasurementPoint within the coordinate system of the component. MeasurementPoint onto which dimension can be defined.

General Information

Base Classifier	LocalPosition
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
measurementPoint	MeasurementPoint	1	Y	N	0..*	N	NA

2.14.4 Class PlacementPointPosition

Defines the position of a PlacementPoint within the coordinate system of the component. PlacementPoints are points where a component is attached to a *TopologySegment*. Therefor a *PlacementPointPosition* can define a tangent vector (in the coordinate system of the component) for a segment that is connected to the *PlacementPoint*.

General Information

Base Classifier	LocalPosition
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
tangent	CartesianVector3D	0..1	A vector in the direction of the tangent on the <i>TopologySegment</i> that is attached to the <i>PlacementPoint</i> represented by this instance. The vector is defined in the coordinate system of the 3D model of the component.

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
placementPoint	PlacementPoint	1	Y	N	0..*	N	NA

2.15 Module mapping

2.15.1 Class CavityMapping

Defines the mapping of two cavities contained Slot A & B of the containing SlotMapping-object.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identificationA	String	1	The identification of the Cavity on side A
identificationB	String	1	The identification of the Cavity on side B

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cavityMapping	0..*	Y	SlotMapping	1	NA

2.15.2 Class Mapping

The Mapping defines the concrete mapping two parts aliased as A & B. For performance reasons the roles PartSideA and PartSideB are abbreviated to A & B.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identificationA	String	0..1	Identification of the <i>ConnectorHousingSpecification</i> on side A.
identificationB	String	0..1	Identification of the <i>ConnectorHousingSpecification</i> on side A.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

A	PartVersion	1	Y	N	0..*	N	NA
B	PartVersion	1	Y	N	0..*	N	NA
slotMapping	SlotMapping	0..*	Y	N	1	Y	NA

Incoming Relations

This End			Other End		General		
Role	Mult.	Agg.	Type	Mult.	Comment		
mapping	0..*	Y	MappingSpecification	1	NA		

2.15.3 Class MappingSpecification

Specification for the description of a *CavityMappings*. A *CavityMapping* allows the mapping between the *Cavities* of two independent *ConnectorHousingSpecifications*. This is for example needed to determine the mating for an inliner or for the mating between an EE-Component and a ConnectorHousing. The *CavityMapping* is a part master data.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
mapping	Mapping	0..*	Y	N	1	Y	NA

2.15.4 Class SlotMapping

Defines the mapping of two slots contained PartVersion A & B of the containing Mapping-object.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identificationA	String	1	The identification of the Slot on side A
identificationB	String	1	The identification of the Slot on side B
layoutIdentificationA	String	0..1	The identification of a ModularSlotLayout, if the Slot on side A refers to a <i>ModularSlot</i> .
layoutIdentificationB	String	0..1	The identification of a ModularSlotLayout, if the Slot on side B refers to a <i>ModularSlot</i> .

Outgoing Relations

Other End	This	General
-----------	------	---------

Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cavityMapping	CavityMapping	0..*	Y	N	1	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
slotMapping	0..*	Y	Mapping	1	NA

2.16 Module modules

2.16.1 Class ModuleFamily

A ModuleFamily is a mechanism to group mutually exclusive modules. This could be for example something like "audio equipment", "diesel engine".

In other words, a module family groups different variants of the same basic feature. In a real car configuration only one member of the family can participate. For the example module family "audio equipment" the members may be named: "Basic Audio Equipment", "Standard Audio Equipment", "Premium / High End Audio Equipment".

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the ModuleFamily. The identification is guaranteed to be unique within the ModuleFamilySpecification. For all VEC-documents a ModuleFamily-instance can be trusted to be the same if the ModuleFamilySpecification-instance is the same and the identification of the ModuleFamily is the same.
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the ModuleFamily.
mandatory	Boolean	0..1	If ' <i>mandatory=true</i> ' applies, then exactly one <i>PartWithSubComponentsRole</i> from the <i>ModuleFamily</i> the must be contained in a valid vehicle configuration, provided that the <i>ModuleFamily</i> itself is valid for the vehicle configuration. This is implied by the fact that the <i>ModuleFamily</i> itself is a <i>ConfigurableElement</i> . If this attribute is false or not defined, then at most one <i>PartWithSubComponentsRole</i> can be selected from the <i>ModuleFamily</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
moduleInFamily	PartWithSubComponentsRole	1..*	Y	N	0..*	N	References the Modules that belong to the ModuleFamily.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
moduleFamily	1..*	Y	ModuleFamilySpecification	1	Specifies the ModuleFamilies defined in the ModuleFamilySpecification.

2.16.2 Class ModuleFamilySpecification

Specification for the description of module families (see ModuleFamily).

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
moduleFamily	ModuleFamily	1..*	Y	N	1	Y	Specifies the ModuleFamilies defined in the ModuleFamilySpecification.

2.16.3 Class ModuleList

A ModuleList is a mechanism to control additional / completion PartOccurrences. This means for a car configuration, if at least one of the modules in the list participates in the configuration, the "completionComponent" participates, too.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the ModuleList. The identification is guaranteed to be unique within the ModuleListSpecification. For all VEC-documents a ModuleList-instance can be trusted to be the same if the ModuleListSpecification-instance is the same and the identification of the ModuleList is the same.
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the ModuleList.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

completionComponents	OccurrenceOrUsage	1..*	Y	N	0..*	N	References the components that are used as completion, if any of the Modules in the ModuleList appears in a configuration.
moduleInList	PartWithSubComponentsRole	1..*	Y	N	0..*	N	References the Modules that belong to the ModuleList. If any of the referenced Modules participates in a configuration the completion components participate, too.

Incoming Relations

This End			Other End		General	
Role	Mult.	Agg.	Type	Mult.	Comment	
moduleListConfiguration	1..*	Y	ModuleListSpecification	1	Specifies the ModuleLists defined in the ModuleListSpecification.	

2.16.4 Class ModuleListSpecification

Specification for the description of module lists (see ModuleList).

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
moduleListConfiguration	ModuleList	1..*	Y	N	1	Y	Specifies the ModuleLists defined in the ModuleListSpecification.

2.17 Module net**2.17.1 Class Net**

A Net is an undirected link between *NetworkPorts*. It defines that the *NetworkPorts* are related to each other with the *Net*.

A *Net* is normally an instance of a *NetType*. E.g. if "CAN-BUS" is defined as a *NetType* typical *Nets* would be "BODY-CAN", "AUDIO-CAN".

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the Net. The identification is guaranteed to be unique within the NetSpecification.
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the Net.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
netType	NetType	0..1	Y	N	NA	N	NA
networkPort	NetworkPort	1..*	Y	N	1	N	References the NetworkPorts that are connected by the Net.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
net	2..*	N	NetGroup	0..1	References the Nets that are grouped by the NetGroup.
net	0..*	Y	NetSpecification	1	Specifies the Nets defined in the NetSpecification.
net	0..1	N	Connection	0..*	References the Net that is realized by the Connection.

2.17.2 Class NetGroup

Defines a logical grouping of specific *Nets*. For example, it can be used to identify all *Nets* of specific CAN domain, a function, a requirement level (e.g. Safety & Security).

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the NetGroup. The identification is guaranteed to be unique within the NetSpecification.
netGroupType	String	0..1	Specifies the type of the group.
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the NetGroup.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
net	Net	2..*	Y	N	0..1	N	References the Nets that are grouped by the NetGroup.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment

netGroup	0..*	Y	NetSpecification	1	Specifies the NetGroup defined in the NetSpecification.
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2.17.3 Class NetSpecification

Specification for the description of electrological nets.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
net	Net	0..*	Y	N	1	Y	Specifies the Nets defined in the NetSpecification.
netGroup	NetGroup	0..*	Y	N	1	Y	Specifies the NetGroup defined in the NetSpecification.
netType	NetType	0..*	Y	N	1	Y	NA
networkNode	NetworkNode	0..*	Y	N	1	Y	Specifies the NetworkNodes defined in the NetSpecification.

2.17.4 Class NetType

A *NetType* defines the different types of Nets used in the *NetSpecification*. The level of detail of the *NetTypes* for the can be process dependent.

A *NetType* could be just used to differentiate between conventional (analogue) communication and bus communication (digital), it can also already define the different types of digital communication (e.g. CAN, MOST, Ethernet).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the NetType. The identification is guaranteed to be unique within the NetSpecification.
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the <i>NetType</i> .
signalType	SignalType	0..1	NA
signalSubType	SignalSubType	0..1	NA
signalInformationType	SignalInformationType	0..1	NA
signalTransmissionMediumType	SignalTransmissionMediumType	0..1	Specifies the type of the transmission medium for signals of this type.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
netType	0..*	Y	NetSpecification	1	NA
netType	0..1	N	Signal	NA	NA
netType	0..1	N	NetworkPort	0..*	NA
netType	0..1	N	Net	NA	NA

2.17.5 Class NetViewSpecification

The *NetViewSpecification* supports the exchange of a basic layout for a *NetSpecification*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cartesianPoint	CartesianPoint2D	0..*	Y	N	0..1	Y	Specifies all the <i>CartesianPoint2Ds</i> that are used within the <i>NetViewSpecification</i> .
placedViewItems	NetworkNodeViewItem	0..*	Y	N	1	Y	Specifies all <i>NetworkNodeViewItems</i> that are displayed within a <i>NetViewSpecification</i> .

2.17.6 Class NetworkNode

A *NetworkNode* is a representative for an actor in the electric system, e.g. an actuator, a sensor, an ECU.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>NetworkNode</i> . The identification is guaranteed to be unique within the <i>NetSpecification</i> .
abbreviation	LocalizedString	0..*	Room for a short name of the <i>NetworkNode</i> .
networkNodeType	NetworkNodeType	0..1	Specifies the type of a <i>NetworkNode</i> . Common values are agreed as an <i>OpenEnumeration</i> .
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the <i>NetworkNode</i> .
subType	NetworkNodeSubType	0..1	Specifies the sub type of a <i>NetworkNode</i> . The sub type allows a differentiation within a specific type. E.g. an actuator can be differentiated into lamps, speakers, motors.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
port	NetworkPort	0..*	Y	N	1	Y	Specifies the NetworkPorts of a NetworkNode.
realizedUsageNode	UsageNode	0..1	Y	N	0..*	N	References the <i>UsageNode</i> that is realized by this <i>NetworkNode</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
networkNode	1	N	NetworkNodeViewItem	0..*	References the <i>NetworkNode</i> that is represented by this <i>NetworkNodeViewItem</i> .
networkNode	0..*	Y	NetSpecification	1	Specifies the NetworkNodes defined in the NetSpecification.
networkNode	0..1	N	ComponentNode	0..*	References the NetworkNode that is realized by the ComponentNode.

2.17.7 Class NetworkNodeViewItem

A *NetworkNodeViewItem* represents a *NetworkNode* within a *NetViewSpecification* and defines its position within the layout.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>NetworkNodeViewItem</i> . The identification is guaranteed to be unique within the <i>NetViewSpecification</i> .
gridSquare	String	0..1	Specifies the grid square in which the <i>NodeViewItem</i> is placed (e.g. E/40).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
displayedPort	NetworkPortViewItem	0..*	Y	Y	1	Y	Specifies all <i>NetworkPortViewItems</i> that are displayed on this <i>NetworkNodeViewItem</i> . The order of this association defines the clockwise arrangement of the ports on the node. As each <i>NetworkPortViewItem</i> can also define the side on which it is placed, side definitions take precedence over order. Caution: This association is <u>ordered</u> !

networkNode	NetworkNode	1	Y	N	0..*	N	References the <i>NetworkNode</i> that is represented by this <i>NetworkNodeViewItem</i> .
orientation	Transformation2D	0..1	Y	N	1	Y	Specifies the position / placement of the <i>NetworkNodeViewItem</i> within the layout.

Incoming Relations

This End			Other End			General	
Role	Mult.	Agg.	Type		Mult.		Comment
placedViewItems	0..*	Y	NetViewSpecification		1		Specifies all <i>NetworkNodeViewItems</i> that are displayed within a <i>NetViewSpecification</i> .

2.17.8 Class NetworkPort

NetworkPort is the source or the receiver of a of a Net.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the NetworkPort. The identification is guaranteed to be unique within the NetSpecification.
aliasId	AliasIdentification	0..*	Specifies additional identifiers for the element.
signalDirection	SignalDirection	0..1	Specifies the direction of the signal on this NetworkPort.
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the NetworkPort.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
netType	NetType	0..1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End			General	
Role	Mult.	Agg.	Type		Mult.		Comment
networkPort	1	N	NetworkPortViewItem		0..*		References the <i>NetworkPort</i> that is represented by this <i>NetworkPortViewItem</i> .
networkPort	1..*	N	Net		1		References the NetworkPorts that are connected by the Net.
networkPort	0..1	N	ComponentPort		0..*		References the NetworkPort that is realized by the ComponentPort.
port	0..*	Y	NetworkNode		1		Specifies the NetworkPorts of a NetworkNode.

2.17.9 Class NetworkPortViewItem

A *NetworkPortViewItem* represents a *NetworkPort* within a *NetworkNodeViewItem* and defines its position on the node.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
side	NetworkPortViewItemSide	0..1	Specifies the side of the <i>NetworkNodeViewItem</i> on which the port should be displayed.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
networkPort	NetworkPort	1	Y	N	0..*	N	References the <i>NetworkPort</i> that is represented by this <i>NetworkPortViewItem</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
displayedPort	0..*	Y	NetworkNodeViewItem	1	Specifies all <i>NetworkPortViewItems</i> that are displayed on this <i>NetworkNodeViewItem</i> . The order of this association defines the clockwise arrangement of the ports on the node. As each <i>NetworkPortViewItem</i> can also define the side on which it is placed, side definitions take precedence over order. Caution: This association is <u>ordered</u> !

2.17.10 Enumeration NetworkNodeSubType

Defines agreed values for *NetworkNodeSubTypes*. Not all combinations of *NetworkNodeSubTypes* and *NetworkNodeTypes* are semantically correct (e.g. Lamp, Microphone, Speaker, Motor are all Actuators).

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Lamp	NA
Relay	NA
Fuse	NA
Microphone	NA

Speaker	NA
Motor	NA

2.17.11 Enumeration NetworkNodeType

Defines the common agreed values for the Types of a NetworkNode.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
ECU	Electronic Control Unit. An electronic control unit (ECU) is any embedded system in automotive electronics that controls one or more of the electrical systems or subsystems in a vehicle.
Sensor	A sensor is a device that detects changes to the environment and sends them as information to other devices (e.g. an ECU).
Actuator	An actuator is a device that is responsible for controlling or moving a mechanism.
CouplingDevice	A CouplingDevice is the (virtual) device that separates / connects two or more wiring harnesses. "Virtual" because it can be interpreted as a device / interface definition between the harnesses, where one harness behaves like an E/E component from the point of view of the other harness.
EnergyStorage	A device that stores energy in some kind of way (e.g. a battery).
Generator	A device that can generate energy.
PowerDistribution	A device that distributes power to other devices (e.g. a fuse box).
Switch	A "Switch" is a device that can change its internal connectivity in reaction to some external action (e.g. connect / disconnect some pins). A switch has no "active" logic (in contrast to an ECU).
OpenEnd	Defines that this NetworkNode is the end point for some unconnected nets that require wires and routings in the resulting harness (e.g. an antenna). A NetworkNode of this type is used whenever nets shall not be connected (on one side).
Ground	Defines that this NetworkNode is a grounding point.

2.17.12 Enumeration NetworkPortViewItemSide

NA

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Top	NA
Bottom	NA
Left	NA
Right	NA

2.17.13 Enumeration SignalDirection

Enumeration for the definition of SignalDirections.

General Information

Applied Stereotype	ClosedEnumeration
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Enumeration Literals

Name	Comment
In	NA
Out	NA
InOut	NA

2.18 Module non_electrical_parts

2.18.1 Class BoltMountedFixingSpecification

Specification for fixings that are mounted onto a bolt. This means the fixing itself has got a hole, which is mounted onto a bolt of the surrounding.

General Information

Base Classifier	FixingSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
boltType	BoltMountedFixingBoltType	0..1	Specifies the type of the bolt on which the fixing can be mounted.
boltShape	BoltMountedFixingBoltShape	0..1	Specifies the shape of the bolt on which the fixing can be mounted.
boltDiameter	NumericalValue	0..1	Specifies the diameter of the bolt on which the fixing can be mounted.
boltHeight	NumericalValue	0..1	Specifies the height of the bolt on which the fixing can be mounted.

2.18.2 Class CableDuctOutlet

Specifies one outlet of the cable duct.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies the identification of the Outlet. This must be unique within a CableDuctSpecification.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
placementPoint	PlacementPoint	0..1	Y	N	0..*	N	Specifies the <i>PlacementPoint</i> that represents this <i>CableDuctOutlet</i> in a <i>PlaceableElementSpecification</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
outlet	0..*	Y	CableDuctSpecification	1	NA

2.18.3 Class CableDuctSpecification

Specification for cable ducts. A cable duct can have one or more outlets.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
outlet	CableDuctOutlet	0..*	Y	N	1	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cableDuctSpecification	1	N	CableDuctRole	0..*	References the <i>CableDuctSpecification</i> that is instantiated by this <i>CableDuctRole</i> .

2.18.4 Class CableLeadThrough

A *CableLeadThrough* specifies a hole in the grommet through which wires can pass through the grommet. There can be different technical realizations of a lead through, e.g. it can be realized with a shrinking material or an additional seal. The properties of a *CableLeadThrough* are defined in the referenced *CableLeadThroughSpecification*.

For a precise placement in topology, a *CableLeadThrough* specifies outlets.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies the identification of the CableLeadThrough. This must be unique within a GrommetSpecification.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cableLeadThroughSpecification	CableLeadThroughSpecification	0..1	Y	N	NA	N	References the <i>CableLeadThroughSpecification</i> that defines the technical properties of this <i>CableLeadThrough</i> .
outlet	CableLeadThroughOutlet	0..*	Y	N	1	Y	Specifies the outlets of a <i>CableLeadThrough</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cableLeadThrough	0..1	N	CableLeadThroughReference	NA	NA
cableLeadThrough	0..*	Y	GrommetSpecification	1	Specifies the CableLeadThroughs of the Grommet.

2.18.5 Class CableLeadThroughOutlet

Specifies one outlet of a *CableLeadThrough*. For most *CableLeadThroughs*, there will be two.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies the identification of the <i>CableLeadThroughOutlet</i> . This must be unique within a <i>CableLeadThrough</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
placementPoint	PlacementPoint	0..1	Y	N	0..*	N	Specifies the <i>PlacementPoint</i> that represents this <i>CableLeadThroughOutlet</i> in a <i>PlaceableElementSpecification</i> .
supplementaryParts	PartRelation	0..*	Y	N	NA	N	References the <i>PartRelations</i> that specify supplementary parts for this <i>CableLeadThroughOutlet</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cableLeadThroughOutlet	1	N	CableLeadThroughOutletReference	NA	NA
outlet	0..*	Y	CableLeadThrough	1	Specifies the outlets of a <i>CableLeadThrough</i> .

2.18.6 Class CableLeadThroughSpecification

A *CableLeadThrough* specifies a hole in the grommet through which wires can pass through the grommet. The *CableLeadThroughSpecification* defines the technical properties of a *CableLeadThrough* (or a set of similar ones).

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	CableLeadThroughType	0..1	Defines the type of a cable lead through. Standardized values are defined in an <i>OpenEnumeration</i> .
geometry	CableLeadThroughGeometry	0..1	Defines the geometry of a cable lead through in the sealing area. Standardized values are defined in an <i>OpenEnumeration</i> .
sealingDimension	Size	0..1	Specifies the dimension of the cable lead through in the sealing area.
minSegmentOutsideDiameter	NumericalValue	0..1	Specifies the minimum diameter a segment can have to fit through the cable lead through. This definition is necessary, since segments that are too small might cause movements and unacceptable torsion forces or they are not sealable.
maxSegmentOutsideDiameter	NumericalValue	0..1	Specifies the maximum diameter a segment can have to fit into the cable lead through.
sealable	Boolean	0..1	Specifies if the cable lead through is sealable.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cableLeadThroughSpecification	0..1	N	CableLeadThrough	NA	References the <i>CableLeadThroughSpecification</i> that defines the technical properties of this <i>CableLeadThrough</i> .

2.18.7 Class CableTieSpecification

Specification for the definition of cable ties.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
------	------	------	---------

length	NumericalValue	0..1	NA
width	NumericalValue	0..1	NA
thickness	NumericalValue	0..1	NA
tensioningStrength	NumericalValue	0..1	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
cableTieSpecification	1	N	CableTieRole	0..*	NA

2.18.8 Class CorrugatedPipeSpecification

Specification for the definition of corrugated pipes.

General Information

Base Classifier	TubeSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
corrugationHeight	NumericalValue	0..1	Specifies the height of a corrugation of the pipe.
corrugationWidth	NumericalValue	0..1	Specifies the width of a corrugation of the pipe.
corrugationGradient	NumericalValue	0..1	Specifies the gradient of a corrugation of the pipe.
cutOnWave	Boolean	0..1	Defines if the corrugated pipe is cut on top of the corrugation. This applies only to corrugated pipes that have a defined length. For bulk material in the cutting is defined in the <i>CorrugatedPipeRole</i> .

2.18.9 Class EdgeMountedFixingSpecification

Specification for fixings that are mounted onto an edge.

General Information

Base Classifier	FixingSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
edgeThickness	ValueRange	0..1	Defines a range of valid thicknesses, onto which the fixing can be mounted.

2.18.10 Class FerriteSpecification

Specification for the definition of ferrites.

General Information

Base Classifier	PartOrUsageRelatedSpecification
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Applied Stereotype	NA
Is Abstract	false

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
ferriteSpecification	1	N	FerriteRole	NA	NA

2.18.11 Class FittingOutlet

Specifies one outlet of the fitting.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies the identification of the Outlet. This must be unique within a FittingSpecification.
innerDiameter	NumericalValue	0..1	Specifies the inner diameter of the outlet.
outerDiameter	NumericalValue	0..1	Specifies the outer diameter of the outlet.
nominalSize	String	0..1	Defines the nominal size of a tube. The nominal size is a name for the size of the tube that is somehow related to the inner diameter of the tube. However, it is not the inner diameter (e.g. "10.5").

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
placementPoint	PlacementPoint	0..1	Y	N	0..*	N	Specifies the <i>PlacementPoint</i> that represents this <i>FittingOutlet</i> in a <i>PlaceableElementSpecification</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
outlet	0..*	Y	FittingSpecification	1	NA

2.18.12 Class FittingSpecification

Specification for the definition of fittings. A fitting is a part that is used to connect pipes or tube. A fitting has a specific form (e.g. Y,T)

General Information

Base Classifier	WireProtectionSpecification
Applied Stereotype	NA

Is Abstract	false
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Attributes

Name	Type	Mult	Comment
form	FittingForm	0..1	Specifies the form of the fitting (e.g. Y, T).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
outlet	FittingOutlet	0..*	Y	N	1	Y	NA

2.18.13 Class FixingSpecification

Specification for the definition of fixings. A fixing is used to fix the harness in a certain position (e.g. at the car body, a seat, an ECU etc.). The FixingSpecification describes how the fixing is attached to the "non-harness" element. The attachment to harness is described by a PlaceableElementSpecification.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
offset	NumericalValue	0..1	Offset between the origin of a fixing (which is defined by its installed position on the car body), and the placement point on the harness. See Diagram "Fixing Coordinates and Dimensions" for details.
nominalSize	NumericalValue	0..1	NA
maximumWidthA	NumericalValue	0..1	The maximum width of the fixing, measured from the placement plane (XZ-plane) in the direction of the Y-axis (direction of the harness, passing through the fixing). See Diagram "Fixing Coordinates and Dimensions" for details.
maximumWidthB	NumericalValue	0..1	The maximum width of the fixing, measured from the placement plane (XZ-plane) in the opposite direction of the Y-axis (direction of the harness, passing through the fixing). See Diagram "Fixing Coordinates and Dimensions" for details.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
fixingSpecification	1	N	FixingRole	0..*	References the <i>FixingSpecification</i> that is instanced by this <i>FixingRole</i> .

2.18.14 Class GrommetSpecification

Specification for the definition of grommets.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
hardness	NumericalValue	0..1	Specifies the hardness of the grommet.
holeDiameter	NumericalValue	0..1	Specifies the valid diameter of a hole into which the grommet fits.
plateThickness	ValueRange	0..1	Specifies valid the plate thickness at the hole into which the grommet fits.
grommetType	GrommetType	0..1	Specifies the type of the grommet.
mountingType	GrommetMountingType	0..1	Defines how the grommet is mounted onto the harness (not the vehicle body).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cableLeadThrough	CableLeadThrough	0..*	Y	N	1	Y	Specifies the CableLeadThroughs of the Grommet.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
grommetSpecification	1	N	GrommetRole	0..*	References the <i>GrommetSpecification</i> that is instantiated by this <i>GrommetRole</i> .

2.18.15 Class HoleMountedFixingSpecification

Specification for fixings that are mounted into a hole. This means the fixing has got a bolt which is mounted into a hole of the surrounding.

General Information

Base Classifier	FixingSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
holeDiameter	NumericalValue	0..1	Specifies the diameter of the hole in which the fixing is mounted.
plateThickness	ValueRange	0..1	Specifies the thickness of the plate in which the hole is positioned.
holeType	HoleMountedFixingHoleType	0..1	Specifies the type of how the fixing is fixated in the hole. In other words, it defines what features the hole has to provide to allow a fixation of the fixing.

holeShape	HoleMountedFixingHoleShape	0..1	Specifies the shape of the hole in horizontal direction (the plain on which the fixing is fixated).
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2.18.16 Class ProtectionMaterialLength

Defines the material length of a *WireProtectionRole*. A *WireProtectionRole* can have multiple lengths of different types but must not have more than one length of the same type.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
lengthValue	NumericalValue	1	NA
valueDetermination	ValueDetermination	1	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
materialLength	0..*	Y	WireProtectionRole	1	NA

2.18.17 Class ShrinkableTubeSpecification

Specification of tubes that are shrinkable.

General Information

Base Classifier	TubeSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
shrinkingFactor	Double	0..1	Defines the factor of shrinking for the tube.
maximumLongitudinalShrinkage	NumericalValue	0..1	Defines the shrinkage in longitudinal direction.
resin	Material	0..*	Defines the material of the resin usable for this shrinkable tube.
waterAbsorbtion	NumericalValue	0..1	Defines the water absorption of the shrinkable tube specification.

2.18.18 Class StripeSpecification

Specifies a stripe which has fixed length and width. A stripe is a textile, foam or similar piece with fixed length & width that is wrapped around the harness.

General Information

Base Classifier	WireProtectionSpecification
Applied Stereotype	NA

Is Abstract	false
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Attributes

Name	Type	Mult	Comment
length	NumericalValue	0..1	Specifies the length of the stripe.
segmentDiameter	ValueRange	0..1	Specifies the valid segment diameter range for which the stripe can be used.
width	NumericalValue	0..1	Specifies the width of the stripe.
thickness	NumericalValue	0..1	Specifies the thickness of the stripe (adhesive + backing).

2.18.19 Class TapeSpecification

Specification for the description of tapes.

General Information

Base Classifier	WireProtectionSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
backing	Material	0..*	Specifies the material of carrier of the tape, on which the adhesive material is applied.
adhesive	Material	0..*	Specifies the adhesive material of the tape.
width	NumericalValue	0..1	Specifies the width of the tape.
thickness	NumericalValue	0..1	Specifies the thickness of the tape (adhesive + backing).
coilCoreDiameter	NumericalValue	0..1	Specifies the inner diameter of the coil on which the tape is delivered.

2.18.20 Class TubeSpecification

Specifies tubes.

General Information

Base Classifier	WireProtectionSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
bendRadius	NumericalValue	0..1	Specifies the bend radius of the tube.
innerDiameter	NumericalValue	0..1	Defines the inner diameter of a tube. In the case of a shrinkable tube, it is the diameter of the tube in the unshrunk state.
wallThickness	NumericalValue	0..1	Specifies the thickness of the wall of the tube.
isSlit	Boolean	0..1	Specifies whether the tube is slit or not. The style of the slitting can be defined with the <i>slitStyle</i> . If a <i>slitStyle</i> is defined, it implies that <i>isSlit=true</i> .

slitStyle	TubeSlitStyle	0..1	Specifies the style of the slitting of the tube. If a <i>slitStyle</i> is defined, it implies that <i>isSlit=true</i> . This attribute is defined as an <i>OpenEnumeration</i> .
nominalSize	String	0..1	Defines the nominal size of a tube. The nominal size is a name for the size of the tube that is somehow related to the inner diameter of the tube. However, it is not the inner diameter (e.g. "10.5").
secondaryNominalSize	String	0..1	Defines the secondary nominal size of a tube. The nominal size is a name for the size of the tube that is somehow related to the inner diameter of the tube. However, it is not the inner diameter (e.g. "10.5"). The secondary nominal size shall only be used for two-parted tubes (see TubeSlitStyle = TwoParts). The secondary nominal size defines the size of the outer (larger) tube of a two-parted tube.
shape	TubeShape	0..1	Specifies the shape of the cross section of the tube. This attribute is defined as an <i>OpenEnumeration</i>
outerDiameter	NumericalValue	0..1	Specifies the outer diameter of the tube. The outer diameter of a tube shall only be used for circular tubes (shape = Circular). For other shapes, height and width shall be used.
height	NumericalValue	0..1	Specifies the height of the tube. If the height is defined, a width shall be defined, too. The height and width of a tube shall only be used for tubes that are not circular (shape != Circular). For circular shapes, the outside diameter shall be used.
width	NumericalValue	0..1	Specifies the width of the tube. If the width is defined, a height shall be defined, too. The height and width of a tube shall only be used for tubes that are not circular (shape != Circular). For circular shapes, the outside diameter shall be used.
length	NumericalValue	0..1	Specifies the length of the tube if it is a predefined value.

2.18.21 Class WireProtectionGroup

A *WireProtectionGroup* is the definition of a set of *WireProtectionRoles* to which the following rules apply:

- The *WireProtectionGroup* defines a processing instruction, not a precise product specification.
- All transitions between adjacent elements in group shall be implemented continuous, without interruption, wherever technically possible.
- The elements of a *WireProtectionGroup* can be subject to variance depending on the scenario in a specific 100% configuration. Meaning, some elements of the group might only be applicable in some scenarios, e.g. a cross-taping might only occur, if at least three segments with wires in them exist in a configuration.
- The existence of the elements of a group in a specific 100% configuration is not controlled by the *WireProtectionGroup*, but by the individual variance control applied to the participating *Roles* and the corresponding *OccurrenceOrUsages*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique <i>identification</i> of the <i>WireProtectionGroup</i> . The <i>identification</i> is guaranteed to be unique within the <i>WireProtectionGroupSpecification</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
relatedWireProtecti ons	WireProtectionRol e	1..*	Y	N	0..1	N	References the members in the <i>WireProtectionGroup</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
wireProtectionGrou p	0..*	Y	WireProtectionGro upSpecification	0..1	NA

2.18.22 Class WireProtectionGroupSpecification

Specification for the description of *WireProtectionGroups*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
wireProtectionGrou p	WireProtectionGro up	0..*	Y	N	0..1	Y	NA

2.18.23 Class WireProtectionSpecification

Specification for the description of wire protections.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
soundDampingClass	SoundDampingClas s	0..*	Specifies the class of sound damping. According to the VDA this is a value between A & E. KBLFRM-311

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment

wireProtectionSpecification	1	N	WireProtectionRole	0..*	References the <i>WireProtectionSpecification</i> that is instantiated by this <i>WireProtectionRole</i> .
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2.18.24 Enumeration BoltMountedFixingBoltShape

OpenEnumeration for the shapes of the bolts on which the fixing can be mounted.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Circular	NA
Square	NA
Oval	NA

2.18.25 Enumeration BoltMountedFixingBoltType

OpenEnumeration for the types of the bolts on which the fixing can be mounted.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
MetricThread	A bolt with a metric scale thread.
ImperialThread	A bolt with an imperial scale thread.
Plain	A plain bolt without any specific features.
Regular	A bolt with some kind of corrugation for fixing to lock on.

2.18.26 Enumeration CableLeadThroughGeometry

Defines valid values for the geometry of a *CableLeadThrough* in the sealing area.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Square	NA
Circular	NA
Oval	NA

2.18.27 Enumeration CableLeadThroughType

Defines valid types for CableLeadThroughs.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
ShrinkingTube	
ShrinkingTubeWithSealingGlue	NA
SealingGlue	NA
Overmolded	NA
Foamed	NA
WithSealingComponent	Literal used when the <i>CableLeadThrough</i> is sealed with an additional sealing component (a "cavity seal").

2.18.28 Enumeration FittingForm

OpenEnumeration to define different fitting forms.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Tee	NA
Cross	NA
Straight	Straight 1:1 coupling e.g. used for the coupling of different pipe sections.
Cap	Cap for the closure of tubes and pipes.
EndPiece	For covering pipe ends when no screw fitting is used. Allows cables to be inserted level to avoid damage.
Plug	Plug for the closure of tubes and pipes.
Reducer	Straight 1:1 coupling used for changes of the nominal size.
Elbow	Angled 1:1 coupling.
Wye	Symmetric division (2 ways).
Lateral	A "wye" with one straight and one lateral outlet.
M	A "wye" with more than 2 outlets (one inlet).
A	A pipe / tube connector (1:1) with an additional outlet for wires (without a pipe / tube).

2.18.29 Enumeration HoleMountedFixingHoleShape

Open Enumeration of standard hole shapes (see *HoleMountedFixingSpecification.holeShape*).

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Circular	NA
Square	NA
Oval	NA
Slotted	NA

2.18.30 Enumeration HoleMountedFixingHoleType

Open Enumeration of standard hole types (see HoleMountedFixingSpecification.holeType).

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
MetricThread	The hole provides a metric thread for fixation.
ImperialThread	The hole provides an imperial thread for fixation.
Plain	The hole itself is plain and provides no specific features.
Through	The hole is in some kind metal sheet or similar. The fixing is stuck through the hole and expands in some way on the other side to reach a fixation.

2.18.31 Enumeration TubeShape

Defines valid shapes of the cross section of a tube.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Circular	The tube has circular cross section.
NonCircular	The tube has a cross section that is not circular.

2.18.32 Enumeration TubeSlitStyle

Defines valid types / styles of a slitted tube.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Slit	The tube has just a simple slit.
SelfClosing	The slit of the tube is self-closing.
Closable	The slit can be closed manually.

Overlapping	The edges of the slit are overlapping.
TwoParts	Two-parted tubes consist of inner tube and an outer tube (normally defined as one <i>PartNumber</i>). Both tubes are slit and are combined into each other during assembly, thus creating one closed tube.

2.19 Module part_structure

2.19.1 Class CompositionSpecification

The CompositionSpecification is used to define a set of occurrences required to describe unambiguously the design of a composite part. This does not have to be necessarily the same occurrences which are building the bill of material. Example: A company might want to regard an antenna cable as one part out of a bill of material perspective. However, at the same time it may be useful for the company to be able to describe the contacting of the antenna cable within the VEC. (see also PartStructureSpecification)

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
component	PartOccurrence	0..*	Y	N	1	Y	Specifies the PartOccurrences defined in the CompositionSpecification.

2.19.2 Class PartOccurrence

A PartOccurrence is an instance of a component with a specified part number (PartVersion).

General Information

Base Classifier	OccurrenceOrUsage
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
isSecondaryAlternative	Boolean	0..1	If a PartUsage is realized by more than one PartOccurrence it is possible to specify which one is the preferred. (see KBLFRM-264)

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

alternativeOccurrence	PartOccurrence	0..*	Y	N	0..*	N	References the PartOccurrences that are an alternative for this PartOccurrence.
instanciatedOccurrence	PartOccurrence	0..*	Y	N	0..*	N	References the PartOccurrence which is instantiated by the PartOccurrence. This reference is for example needed in the case of usage of assemblies.
part	PartVersion	0..1	Y	N	0..*	N	References the PartVersion that is instantiated by this PartOccurrence.
realizedPartUsage	PartUsage	0..*	Y	N	0..*	N	References the PartUsages that are realized by the PartOccurrence.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
alternativeOccurrence	0..*	N	PartOccurrence	0..*	References the PartOccurrences that are an alternative for this PartOccurrence.
component	0..*	Y	CompositionSpecification	1	Specifies the PartOccurrences defined in the CompositionSpecification.
instanciatedOccurrence	0..*	N	PartOccurrence	0..*	References the PartOccurrence which is instantiated by the PartOccurrence. This reference is for example needed in the case of usage of assemblies.

2.19.3 Class PartStructureSpecification

Specification for the description of a part structure. This specification defines the PartOccurrences that are in the bill of material of the described PartOrUsage. This is necessary, because it is possible, that for a definite description a Part more PartOccurrences are needed than the ones that are in the bill of material (see CompositionSpecification).

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
content	PartStructureContent Type	0..1	Specifies the type of content of the bill of material (e.g. module, harness complete set)
definedAsEmpty	Boolean	0..1	If <i>true</i> the <i>PartStructureSpecification</i> defines a part that has no <i>OccurrenceOrUsage</i> in its <i>inBillOfMaterial</i> on purpose. Such parts are sometimes required to satisfy certain variant management methods. This attribute serves as flag to clearly separate such parts from the ones whose bill of material is not yet defined.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
inBillOfMaterial	OccurrenceOrUsage	0..*	Y	N	0..*	N	References the PartOccurrences that are building the bill of material of a composite part.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
partStructureSpecification	1	N	PartWithSubComponentsRole	0..*	References the <i>PartStructureSpecification</i> that is instantiated by this <i>PartWithSubComponentsRole</i> .

2.19.4 Enumeration PartStructureContentType

OpenEnumeration that defines standard content types for the *PartStructureSpecification*.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Assembly	Definition of part / component without variance, which has a bill of material (100%).
ConfigurableAssembly	Definition of part / component that has variance, which has a bill of material (150%).
Module	A module / option in a customer specific or configurable harness (KSK / Stufenleitungssatz). The content of a module is normally free of variance but does not represent a functional subset on its own. (10%)
Variant	A concrete variant of a harness (100%).
Harness	A definition of harness, containing variance (150%).

2.20 Module part_usage**2.20.1 Class PartUsage**

PartUsages shall be used for the specification of the elements on an electrical system wiring plan and for the specification of the elements on a pure geometry description. PartUsages shall more than ever be used in cases where it is necessary to describe a certain instance of a part or part group, possibly together with certain technical properties, but where it is at the same time yet not possible to define a concrete part number.

General Information

Base Classifier	OccurrenceOrUsage
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
primaryPartUsageType	PrimaryPartType	1	The primary type of the PartUsage defines the type of the described part (e.g. ConnectorHousing, Fixing, etc.) Since the VEC supports dual use parts (e.g. Fixing & WireProtection) there is no direct connection between the primaryPartUsageType and the allowed specifications for the description of a PartUsage.

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
instanciatedUsage	PartUsage	0..*	Y	N	0..*	N	NA
partOrUsageRelatedSpecification	PartOrUsageRelatedSpecification	0..*	Y	N	0..*	N	References the PartOrUsageRelatedSpecification(s) that describe the PartOrUsageRelatedSpecification. KBLFRM-399
partSubstitution	PartSubstitutionSpecification	0..1	Y	N	NA	N	<u>Deprecated since 2.2.0:</u> <i>Reason:</i> Replaced by the common concept that the <i>PartSubstitutionSpecification</i> is a <i>PartOrUsageRelatedSpecification</i> . NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
instanciatedUsage	0..*	N	PartUsage	0..*	NA
partUsage	0..*	Y	PartUsageSpecification	1	Specifies the PartUsages defined by the PartUsageSpecification.
realizedPartUsage	0..*	N	PartOccurrence	0..*	References the PartUsages that are realized by the PartOccurrence.

2.20.2 Class PartUsageSpecification

Specification for the definition of PartUsages.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
partUsage	PartUsage	0..*	Y	N	1	Y	Specifies the PartUsages defined by the PartUsageSpecification.

2.21 Module pdm**2.21.1 Class Approval**

Defines the approval of an ItemVersion. This consists of the StatusOfApproval, and the Permissions issued for the approval.

General Information

Base Classifier	ExtendableElement
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Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the approval. The identification is guaranteed to be unique over all VEC-documents. Normally this would reference to a company specific approval number or something similar. KBLFRM-349
description	AbstractLocalizedString	0..*	Room for additional information about the item.
companyInScope	String	0..*	Room to specify for which companies the Approval is valid. It might be e.g. that an approved Item may only be delivered by some specific company.
status	StatusOfApproval	1	The approval level concerning approval status. Predefined are the values: NotYetApproved, Approved and Withdrawn. The status refers to the status of the <i>Approval</i> , not the status of the <i>ItemVersion</i> . So, e.g. withdrawn means the approval (with its corresponding level) has been withdrawn, not the <i>ItemVersion</i> itself.
levelOfApproval	LevelOfApproval	0..1	Relates to activities that are allowed with the ItemVersion after release for example building pilot tools or production tools. The levelOfApproval applies to the <i>ItemVersion</i> itself without further detailing or additional context. So, for example "Free" means, that from a component's perspective the corresponding <i>PartVersion</i> has satisfied all qualification procedures and can be used within its specified limits without restriction. The levels <i>Planned</i> , <i>Free</i> , <i>Invalid</i> refer to a single approval level. The levels <i>Develop</i> and <i>Restricted</i> refer to a category of approval levels in the lifecycle of an <i>ItemVersion</i> that all belong to the same phase but are highly company specific. For example, <i>develop</i> approvals might be a "a start of construction approval" or a "building of prototypes or tools allowed approval". "Restricted" approvals might be "only for special purpose vehicles", "spare part only" or "residual parts may be used up". In these cases, the <i>additionalLevelInformation</i> can be used to provide further information (e.g. a company specific approval level).
additionalLevelInformation	String	0..1	Additional potentially company specific information about the level of approval (e.g. further detailing of a "Restricted" approval.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
permission	Permission	0..*	Y	N	1	Y	Specifies the permission issued with the approval.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
approval	0..*	Y	ItemVersion	1	Specifies the approval information of the ItemVersion.

2.21.2 Class ChangeDescription

A ChangeDescription describes the implemented issues that are reason for the aggregating ItemVersion to be either an initial or successor version. A ChangeDescription can optionally define the person who has approved the change.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies an identifier for the ChangeDescription. This can be the ID of a change order or an ID which indicates model upgrading. (see KBLFRM-249)
label	String	0..1	Specifies the label of the change on a drawing for example. If more than one change is issued with one ItemVersion (e.g. change 0001, 0002), in some cases the different changes are labelled on the drawing (e.g. A, B, C).
description	AbstractLocalizedString	0..*	Specifies additional, human readable, information about the ChangeDescription.
approver	Person	0..1	Specifies the person who has approved the change.
changeDate	Date	0..1	Specifies the date when the change was performed.
responsibleDesigner	Person	0..1	Specifies the design engineer who is/was responsible to perform the change.
relatedChangeRequest	String	0..*	Specifies the identification of a corresponding change request. (see KBLFRM-249)

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
changeDescription	0..*	Y	ItemVersion	0..1	Specifies the change history of the ItemVersion.
changeDescription	0..*	Y	SheetOrChapter	0..1	Specifies the change history of the SheetOrChapter.

2.21.3 Class Contract

A Contract-instance describes the relationship between an ItemVersion-instance and a Company-instance additionally defining the role the company takes in reference to the ItemVersion.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	constant
Is Abstract	false

Attributes

Name	Type	Mult	Comment
companyName	String	1	Specifies the company which acts in the specified Role in the Contract Relationship.

contractRole	ContractRole	1	The role the company takes in reference to the associated ItemVersion. Predefined are the values: OEM, Supplier and Manufacturer.
--------------	--------------	---	---

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
contract	0..*	Y	VecContent	1	Specifies the contracts used in the VEC-file.
contract	0..*	N	ItemVersion	0..*	References the contracts that apply to an ItemVersion.

2.21.4 Class CopyrightInformation

A CopyrightInformation-instance specifies copyright information for one or more Items.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	constant
Is Abstract	false

Attributes

Name	Type	Mult	Comment
copyrightNote	LocalizedString	1..*	An informal text which specifies copyright information.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
copyrightInformation	0..1	N	ItemVersion	0..*	References the <i>CopyrightInformation</i> that is in effect for this <i>ItemVersion</i> . If no <i>CopyrightInformation</i> is referenced by the <i>ItemVersion</i> , the <i>CopyrightInformation</i> that is referenced by the <i>VecContent</i> (if defined) shall be considered as in effect for this <i>ItemVersion</i> .
copyrightInformation	0..*	Y	VecContent	1	Specifies the CopyrightInformation used in the VEC-file.
standardCopyrightInformation	0..1	N	VecContent	0..*	References the <i>CopyrightInformation</i> that is in effect for the complete content of this <i>VecContent</i> . It is applied to all <i>ItemVersions</i> that do not references their own individual <i>CopyrightInformation</i> .

2.21.5 Class Creation

A Creation-instance provides additional information to the owning ItemVersion stating personal information on the creator and the creation date.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
------	------	------	---------

creationDate	Date	1	Specifies the date when the associated ItemVersion was created. (see KBLFRM-241)
creator	Person	0..1	Specifies the person who has created the Item.
responsibleDesigner	Person	0..1	Specifies the person, which is the responsible designer for the ItemVersion at the point of creation.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
creation	0..1	Y	ItemVersion	1	Specifies the information about the creation of the ItemVersion.

2.21.6 Class ItemEquivalence

Defines two or more ItemVersions to be equivalent out of the view of a certain company. The ItemEquivalence class will most likely be used by a company to express which PartNumber a certain PartVersion has in the context of other companies (same applies to *DocumentVersions*). However, for every other company separate ItemEquivalences are needed as the statement of equivalence can be very subjective.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult.	Comment
companyName	String	1	Specifies the company which states the ItemEquivalence.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
item	ItemVersion	2..*	Y	N	0..*	N	References all ItemVersion that are considered to be equivalent by the ItemEquivalence. A single <i>ItemEquivalence</i> shall only reference <i>ItemVersions</i> of the same class (either <i>DocumentVersions</i> or <i>PartVersions</i>).

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
ItemEquivalence	0..*	Y	DocumentVersion	1	Specifies ItemEquivalences defined by the DocumentVersion.

2.21.7 Class ItemHistoryEntry

An ItemHistoryEntry defines the direct relationship between ItemVersions in the terms of predecessor and successor. There are two possible types of relationships between ItemVersions, derivation and sequence. Derivation means for example for parts, that the successor version is a new part developed on the base of the predecessor version. Sequence means the successor version is an improvement of the predecessor version. By the combination of more than one ItemHistoryEntry a linear sequence of ItemVersions can be represented.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	constant
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	HistoryEntryType	1	Specifies the type of relationship between the ItemVersions.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
predecessorVersion	ItemVersion	1	Y	N	0..*	N	References the ItemVersion that is the predecessor in the ItemHistoryEntry.
successorVersion	ItemVersion	1	Y	N	0..*	N	References the ItemVersion that is the successor in the ItemHistoryEntry.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
itemHistoryEntry	0..*	Y	VecContent	1	Specifies the ItemVersionHistoryEntries for ItemVersions contained in the VEC-file.

2.21.8 Class Permission

Describes an act of acceptance together with information about the responsible person, department and company who directly provoked the approval level and status as described in the referenced Approval-instance. (see KBLFRM-229)

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
permission	TypeOfPermission	1	Specifies the type of permission. Predefined values are: Seen, Checked, Released.
permissionDate	Date	0..1	Specifies the date when the permission was stated.
permitter	Person	0..1	Specifies the person who was involved in the approval process giving a certain Permission.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
permission	0..*	Y	Approval	1	Specifies the permission issued with the approval.

2.21.9 Class Person

Specifies all relevant data of a person.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
companyName	String	0..1	Specifies the name of the company the person belongs to.
department	String	0..1	Specifies the department the person belongs to.
firstName	String	0..1	Specifies the person's first name.
lastName	String	1	Specifies the person's last name.
phoneNumber	String	0..*	Specifies the person's phone number.
emailAddress	String	0..*	Specifies the person's email address.
aliasId	AliasIdentification	0..*	Specifies identifiers for the <i>Person</i> in different contexts.

2.21.10 Class Project

Specifies a certain vehicle project. Instances of this class are assumed to be constant. Thus, this is located directly under VEC-root element.

A vehicle project can be some abstract or concrete node in the product structure, addressed by the car classification levels.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	constant
Is Abstract	false

Attributes

Name	Type	Mult	Comment
carClassificationLevel2	String	0..1	Provides a classification according to "CC8 Recommended Practices Specification and Configuration, Product Structures". Car classification is the identification of a set of similar cars to be offered to the market. Level 2 stands for "Technical information / platform" and reflects the level of a product class in a BoM system which represents a main technical product base (e.g. project, platform, engineering series etc.). In some cases, this level carries a complete BoM ("Maximum BoM") for a project, platform, engineering series etc. This level is in some cases called technical documentation.
carClassificationLevel3	String	0..1	Provides a classification according to "CC8 Recommended Practices Specification and Configuration, Product Structures". Car classification is the identification of a set of similar cars to be offered to the market. Level 3 stands for "Configuration information / product family" where all variant control mechanisms are attached.
carClassificationLevel4	String	0..1	Provides a classification according to "CC8 Recommended Practices Specification and Configuration, Product Structures". Car classification is the identification of a set of similar cars to be offered to the market. Level 4 stands for "Furthest pre-configured abstract product class" and represents the furthest configured class of a product, which is not yet a real product. E.g. this could

			be a complete vehicle, engine, gearbox etc. which has not been evaluated against customer special choices or an abstract vehicle, engine, gearbox etc. which could become a real one after the associated BoM is evaluated. The purpose of this level of a product class instance is in any case to reflect that level of product class of a BoM system which leads to the individual BoM for a single product.
identification	String	0..1	Specifies the development order number (car or engine project)
description	AbstractLocalizedString	0..*	Room for additional information about the item.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
project	0..*	N	UsageConstraint	0..*	References the <i>Projects</i> to which the <i>UsageConstraint</i> applies. This means the described PartVersion is allowed / denied in the referenced UsageConstraint.
project	0..*	N	ApplicationConstraint	NA	Defines the projects for which the application constraint applies.
project	0..*	Y	VecContent	1	Specifies the Projects used in the VEC-file.
project	0..1	N	PartVersion	0..*	References the project that develops the PartVersion.
usedInProject	0..*	N	UsageNode	0..*	Specifies the <i>Projects</i> in which the <i>UsageNode</i> can be used.

2.21.11 Enumeration ContractRole

Enumeration for the definition of roles a contractor has in a contract.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Oem	NA
Supplier	NA
Manufacturer	NA

2.21.12 Enumeration HistoryEntryType

Enumeration for the definition of the type of relationship represented by an ItemHistoryEntry. (see KBLFRM-271)

General Information

Applied Stereotype	ClosedEnumeration
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Enumeration Literals

Name	Comment
Derivation	NA
Sequence	NA

2.21.13 Enumeration LevelOfApproval

The LevelOfApproval standardizes the approval levels most engineering processes have in common. See *Approval.levelOfApproval* for more details.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Planned	The <i>ItemVersion</i> has been created.
Develop	An approval during the development process. There might process specific different approvals out of this category. See <i>Approval.additionalLevelInformation</i> for more information about the details.
Free	The <i>ItemVersion</i> can be used without restriction.
Restricted	The Usage of the <i>ItemVersion</i> is restricted in some way. See <i>additionalLevelInformation</i> for more details.
Invalid	The <i>ItemVersion</i> shall not be used anymore.

2.21.14 Enumeration StatusOfApproval

Enumeration for the definition of the state of an approval.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
NotYetApproved	NA
Approved	NA
Withdrawn	NA

2.21.15 Enumeration TypeOfPermission

Enumeration for the definition of the type of a permission.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Seen	NA
Checked	NA
Released	NA

2.22 Module physical_information

2.22.1 Class BoundingBox

The bounding box is defined for the processed state of the component, as it appears in the finished product. Therefore, it is valid to use the bounding box as simplified geometry of the component for viewing or simple DMU operations. For correct 3D positioning, the bounding might require a transformation in the coordinate system of the component (see *LocalGeometrySpecification.BoundingBoxPositioning*).

The bounding box defines the smallest cuboid (box) that can contain a described part completely. If a component has multiple variants, the bounding box is the smallest cuboid that can contain every variant, while maintaining the spatial orientation of the bounding box and component variants.

It is valid to use the *BoundingBox* to describe the dimensions of a component, even if not all dimensions are known (e.g. only length and width). However, it must be possible to transform such a partial bounding box into a complete bounding box by adding the missing dimensions.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
x	NumericalValue	0..1	Defines the extent of the bounding box in the direction of x (length).
y	NumericalValue	0..1	Defines the extent of the bounding box in the direction of y (width).
z	NumericalValue	0..1	Defines the extent of the bounding box in the direction of z (height).

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
boundingBox	0..1	Y	GeneralTechnicalPartSpecification	NA	Defines the bounding box of the part.

2.22.2 Class Color

Specifies a color value. A color is always defined by a key value. What color is meant by this key value is defined by a standard reference system (e.g. RAL).

For example, if a RAL color should be expressed in the terms of the VEC the *referenceSystem* would be "RAL", the *key* would be the RAL number defined by the standard (e.g. "1003" for signal yellow).

Color-attributes normally have the multiplicity [0..*]. This means that such an attribute can have a value for different referenceSystems (e.g. RAL, RGB,...). It must not have multiple values for the same ReferenceSystem.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
key	String	1	The key of the color in the corresponding color reference system.
referenceSystem	ColorReferenceSystem	1	The identification of the color reference system, which is defining possible values and the semantic of color keys. (see KBLFRM-

			315). For common color reference systems, the literals defined in the open enumeration <i>ColorReferenceSystem</i> shall be used.
description	LocalizedString	0..*	On optional human readable description of the color (e.g. the name).

2.22.3 Class CompositeUnit

Defines a unit as a composition of other units. The composition is done by multiplying the different other units. By this way combined units like kg/m can be formed.

General Information

Base Classifier	Unit
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
factors	Unit	1..*	Y	N	NA	N	References the <i>Units</i> that are used as factors to create the <i>CompositeUnit</i> .

2.22.4 Class CustomUnit

A CustomUnit can be used to define "Free Text"-Units. Custom units must not be used for units that can be expressed by any of the other subclasses of Unit. Custom units are only allowed if a unit is needed that cannot be handled by any of the other classes.

General Information

Base Classifier	Unit
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	A unique identification of the custom unit.

2.22.5 Class DataPoint

A *DataPoint* represents one set of values in a *MultiDimensionalValue*. It contains multiple *DataPointValues*, one for each *QuantityKind* referenced as *dimension*. Each *DataPoint* thereby defines a complete point in the value space.

Each *DataPoint* shall define a *DataPointValue* for **all** *dimensions* of its *MultiDimensionalValue*.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
dataPointValue	DataPointValue	1..*	Y	N	NA	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
dataPoint	1..*	Y	MultiDimensionalValue	1	NA

2.22.6 Class DataPointValue

A *DataPointValue* assigns a concrete numeric value to one *dimension* within a *DataPoint*. Each *DataPointValue* must only reference a *QuantityKind* that is defined as *Dimension* of its owning *MultiDimensionalValue*.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
value	Double	1	NA

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
dimension	QuantityKind	1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
dataPointValue	1..*	Y	DataPoint	NA	NA

2.22.7 Class IECUnit

The *IECUnit* class can define quantities in the terms of the IEC-Unit-System by specifying the corresponding IEC prefix (optional) and an IEC unit name.

General Information

Base Classifier	Unit
Applied Stereotype	NA

Is Abstract	false
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Attributes

Name	Type	Mult	Comment
iecUnitName	IECUnitName	1	Specifies the name of the IEC unit.
iecPrefix	IECPrefix	0..1	Specifies the prefix of the IEC unit.

2.22.8 Class ImperialUnit

The ImperialUnit class can define quantities in the terms of the Imperial-Unit-System by specifying the corresponding Imperial unit name.

General Information

Base Classifier	Unit
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
imperialUnitName	ImperialUnitName	1	Specifies the name of the imperial unit.

2.22.9 Class MassInformation

Allows the definition of mass information. Attributes of the type MassInformation normally have the multiplicity [0..*]. This means that such an attribute can have mass values for different *determinationTypes* and *valueSources*. It must not have multiple values for the same *determinationType* and *valueSource*.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
determinationType	ValueDetermination	0..1	Specifies the determination type of the mass information.
value	NumericalValue	1	Specifies the mass as numerical value.
valueSource	MassInformationSource	0..1	The <i>valueSource</i> defines in an OpenEnumeration the source from which the <i>MassInformation</i> has been retrieved. This information, in combination with the <i>ValueDetermination</i> gives a hint about the reliability of the <i>MassInformation</i>

2.22.10 Class Material

Allows the definition of material information. *Material*-attributes normally have the multiplicity [0..*]. This means that such an attribute can have material values for different *referenceSystems*. It must not have multiple values for the same *referenceSystems*.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive

Is Abstract	false
--------------------	-------

Attributes

Name	Type	Mult	Comment
key	String	1	The key of the material in the corresponding material reference system.
referenceSystem	String	1	The identification of the material reference system, which is defining possible values and the semantic of material keys.
description	LocalizedString	0..*	On optional human readable description of the material (e.g. the name).

2.22.11 Class MaterialComposition

Parts and components are often composed from a material-mix. The *MaterialComposition* allows the definition of the various *Materials* in a part and their mass.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
material	Material	0..*	Specifies the material of this composition entry with the syntax and nomenclature of an arbitrary reference system.
massInformation	MassInformation	0..*	Specifies the mass of this corresponding <i>Material</i> in a part per Unit of quantity (e.g. g/piece or g/meter...) of the part.

2.22.12 Class MultiDimensionalValue

The *MultiDimensionalValue* represents a set of values defined across multiple *Dimensions*. It is used to describe properties that vary along several axes, such as characteristic curves. Each *DataPoint* assigned to a *MultiDimensionalValue* must define one *DataPointValue* per *Dimension*.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
dataPoint	DataPoint	1..*	Y	N	1	Y	NA
dimension	QuantityKind	1..*	Y	N	0..*	N	NA

2.22.13 Class NumericalValue

A quantity expressed with a numerical value and a unit.

General Information

Base Classifier	ValueWithUnit
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
valueComponent	Double	1	Specifies the value of the numerical value.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
tolerance	Tolerance	0..1	Y	N	0..1	Y	Specifies the tolerance for the dimension.

2.22.14 Class OtherUnit

The OtherUnit class can be used to define a unit, which is necessary in the context of data exchange but not contained in the standard systems (e.g. Piece).

General Information

Base Classifier	Unit
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
otherUnitName	OtherUnitName	1	Specifies the name of the unit.

2.22.15 Class QuantityKind

A *QuantityKind* is any observable property that can be measured and quantified numerically. Familiar examples include physical properties such as length, mass, time, force, energy, power, electric charge, etc. A quantity (not represented as class in the VEC) would always be associated with the context of measurement (i.e. the thing measured, the measured value, the accuracy of measurement, etc.) whereas the underlying *QuantityKind* is independent of any particular measurement.

A *QuantityKind* provides an identifier and an optional human-readable name and is associated with a *Unit* that specifies how magnitudes of this *QuantityKind* are expressed within this VEC.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	constant
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>QuantityKind</i> . The identification is guaranteed to be unique within a VEC.

			Furthermore, the same <i>QuantityKind</i> shall have the same <i>Identification</i> for different VECs in the same a data exchange context. For example, if using the identification <code>trippingTime</code> for the time- <i>Quantity</i> of a <i>FuseCharacteristic</i> all appearances of this <i>QuantityKind</i> shall have the same <i>Identification</i> , when created by the same publisher.
name	LocalizedString	0..*	A human readable name of the <i>QuantityKind</i> for example used as axis label in a visual diagram.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
unit	Unit	1	Y	N	0..*	N	The <i>Unit</i> in which the magnitudes of this <i>QuantityKind</i> are defined.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
dimension	1..*	N	MultiDimensionalValue	0..*	NA
dimension	1	N	DataPointValue	0..*	NA
quantityKind	0..*	Y	VecContent	1	Specifies the <i>QuantityKind</i> used in the VEC-file.
time	1	N	FuseCharacteristic	0..1	NA
trippingCurrent	1	N	FuseCharacteristic	0..1	NA

2.22.16 Class RobustnessProperties

Allows the definition of robustness properties. Robustness of a component is specified as a level of robustness against a specific influence (e.g. oil, water, UV-light). The influence is specified by the *class* and the level is specified by the *classKey*. Valid robustness classes and keys are specified by the reference system. Attributes of the type *RobustnessProperties* normally have the multiplicity [0..*]. This means that such an attribute can have *RobustnessProperties* entries for different *classReferenceSystems* and *classes*.

Constraints for the multiplicity of entries for the same *class* and *classReferenceSystem* are defined by the reference system. In most cases an element can have only one single value for a *class* and *classReferenceSystem*. This applies for example to cumulative ratings where a "higher" rating includes all "lower" ratings. However, there are reference systems where this is not true, for example in the "ISO 20653 - Liquid Ingress Protection" a component rated as IPX7 is not automatically rated IPX5 and vice versa. In these cases, multiple values (*classKey*) for a single *class* and *classReferenceSystem* are allowed.

Note: Most reference systems just define one class or at least some of the possible classes, but not all (e.g. the ISO 20653 defines "Solid Particle Protection" and "Liquid Ingress Protection", whereas the ISO 6722 defines "Ambient Temperature" among others).

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
class	RobustnessClass	1	Specifies the identifier of a robustness class defined by the robustness class reference system. Robustness classes are for

			example: oil, petrol, UV, water. Specific known and used classes are defined in an open enumeration.
classKey	String	0..1	Specifies a key for the robustness level defined in the specified robustness class (e.g. A, B, C). The classKey is the key as specified in the reference system and no additional syntax or prefixes, e.g. for ISO 20653 (Ingress Protection) it is "9K" and not "IPX9K".
classReferenceSystem	RobustnessClassReferenceSystem	1	The identification of the robustness class reference system, which is defining possible values and the semantic of robustness classes and robustness class keys. Specific known and used reference systems are defined in an open enumeration.
hasRobustness	Boolean	1	Specifies if the described element has a robustness in the specified robustness class. (see KBLFRM-260)
description	LocalizedString	0..*	On optional human readable description of the robustness (e.g. the name).

2.22.17 Class SIUnit

The SIUnit class can define quantities in the terms of the SI-Unit-System by specifying the corresponding SI prefix (optional) and a SI unit name. The usage of SI units must be the preferred way of expressing units, since these units can be easily translated into other SI units.

General Information

Base Classifier	Unit
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
siUnitName	SiUnitName	1	Specifies the name of SI unit (e.g. metre, second,...)
siPrefix	SiPrefix	0..1	Specifies the prefix of the SI unit (e.g. milli, centi, micro,...)

2.22.18 Class Size

Defines the size of an element by width & height. Per definition is width >= height.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
width	NumericalValue	1	The width of the element.
height	NumericalValue	1	The height of the element.

2.22.19 Class SoundDampingClass

Allows the definition of a sound damping class. The sound damping class of a component is specified as a level of sound damping. The level is specified by the *classKey*. Valid keys are specified by the *referenceSystem*. Attributes of the type *SoundDampingClass* normally have the multiplicity [0..*]. This means that such an attribute can have *SoundDampingClass* entries for different *referenceSystems*. It must not have multiple entries for the same *referenceSystem*.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
classKey	String	1	Specifies a key for the level defined in the sound damping class (e.g. A, B, C).
referenceSystem	String	1	The identification of the sound damping class reference system, which is defining possible values and the semantic of sound damping keys.

2.22.20 Class TemperatureInformation

Defines valid temperature ranges for a general technical part. A part can have different valid temperature ranges in different scenarios e.g. operating temperature, storage temperature, processing temperature, environment temperature. A class can own multiple TemperatureInformations but must not have more than one TemperatureInformations of the same temperatureType.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
temperatureRange	ValueRange	0..1	Specifies the allowed temperature range for this type of temperature.
temperatureType	TemperatureType	0..1	The type of a TemperatureInformation.

2.22.21 Class Tolerance

A *Tolerance* defines the permissible limit or limits of variation for defined values (e.g. a *NumericalValue*). The tolerated value is defined by the context in which the *Tolerance* is contained / used.

The values of the limits of the *Tolerance*, *LowerBoundary* and *UpperBoundary*, shall be interpreted as "modifiers" to the actual value. To obtain an absolute range of valid values, the values of boundaries shall be added to the actual value, regardless of the *Upper* or *Lower* prefix. For example, to define a value of 100mm with a tolerated variation between 95mm and 105mm, the definition would be *Value* = 100 mm, *LowerBoundary*=-5, *UpperBoundary*=+5. The *Unit* of the *Tolerance* boundaries is always the same as in the defining context.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
lowerBoundary	Double	1	Specifies the lower boundary for the tolerance.

upperBoundary	Double	1	Specifies the upper boundary for the tolerance.
---------------	--------	---	---

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
tolerance	0..1	Y	NumericalValue	0..1	Specifies the tolerance for the dimension.
tolerance	0..1	Y	Dimension	0..1	NA
toleranceIndication	1	Y	DefaultDimension	0..1	NA

2.22.22 Class Unit

A precisely specified quantity in terms of which the magnitudes of other quantities of the same kind can be stated. The different systems to define units are represented by the subclasses of this class (e.g. SIUnit, ImperialUnit).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	constant
Is Abstract	true

Attributes

Name	Type	Mult	Comment
exponent	Integer	0..1	Defines the exponent with which this unit instance should be used. In order to define square meters for example, the SIUnit "metre" with an exponent of 2 will be used. If no exponent is defined it is equivalent to the value 1.
unEceCode	String	0..1	The UNECE provides a list of " Codes for Units of Measurement used in the International Trade ". This attribute shall be used for traceability of <i>Units</i> defined in the VEC into the UNECE system only. It provides <u>additional</u> meta information about the unit but shall not be used as the only definition of a <i>Unit</i> . Specifying a UNECE code does not exempt the creating system from generating a correct VEC mapping of the corresponding unit. In case of doubt, the VEC mapping has priority.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
baseUnit	1	N	LocalGeometrySpecification	NA	The <i>Unit</i> in which all coordinates (e.g. cartesian points) are defined. Shall be a unit of length (e.g. millimetre).
baseUnit	1	N	BuildingBlockSpecification3D	0..*	NA
baseUnit	1	N	BuildingBlockSpecification2D	0..*	NA
baseUnit	1	N	CavityPositionDetail	0..*	The <i>Unit</i> in which all coordinates (e.g. cartesian points) of this <i>CavityPositionDetail</i> are defined. Shall be a unit of length (e.g. millimetre).
factors	1..*	N	CompositeUnit	NA	References the <i>Units</i> that are used as factors to create the <i>CompositeUnit</i> .
unit	0..*	Y	VecContent	1	Specifies the Units used in the VEC-file.
unit	1	N	QuantityKind	0..*	The <i>Unit</i> in which the magnitudes of this <i>QuantityKind</i> are defined.
unitComponent	1	N	ValueWithUnit	0..*	References the unit of the value.
unitComponent	1	N	Dimension	0..*	NA

2.22.23 Class USUnit

The USUnit class can define quantities in the terms of the US-Unit-System by specifying the corresponding US unit name. The US Unit System is quite similar to the imperial unit system; however, some units are defined slightly different.

General Information

Base Classifier	Unit
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
usUnitName	USUnitName	1	Specifies the name of the <i>USUnit</i> .

2.22.24 Class ValueRange

A pair of numerical values representing a value range.

General Information

Base Classifier	ValueWithUnit
Applied Stereotype	StructuredPrimitive
Is Abstract	false

Attributes

Name	Type	Mult	Comment
minimum	Double	1	Lower bound of the value range.
maximum	Double	1	Upper bound of the value range.

2.22.25 Class ValueWithUnit

Abstract class either for a single numerical measure or a range of numerical measures with upper, lower, or upper and lower bounds.

General Information

Base Classifier	NA
Applied Stereotype	StructuredPrimitive
Is Abstract	true

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
unitComponent	Unit	1	Y	N	0..*	N	References the unit of the value.

2.22.26 Enumeration ColorReferenceSystem

Defines the literals that shall be used for specific color reference systems.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
IEC 60757	The literal shall be used for the IEC 60757 "Electrotechnical engineering; code for designation of colours"
RAL	NA
RGB	NA

2.22.27 Enumeration IECPrefix

Enumeration for the definition of IEC unit prefixes.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Yobi	NA
Zebi	NA
Exbi	NA
Pebi	NA
Tebi	NA
Gibi	NA
Mebi	NA
Kibi	NA

2.22.28 Enumeration IECUnitName

Enumeration for the definition of IEC unit names.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Bit	NA
Byte	NA

2.22.29 Enumeration ImperialUnitName

Enumeration for the definition of imperial unit names.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Thou	NA
Inch	NA
Foot	NA
Yard	NA
Chain	NA
Furlong	NA
Mile	NA
League	NA
FluidOunce	NA
Gill	NA
Pint	NA
Quart	NA
Gallon	NA
Grain	NA
Drachm	NA
Ounce	NA
Pound	NA
Stone	NA
Quarter	NA
HundredWeight	NA
Ton	NA
Perch	NA
Rood	NA
Acre	NA

2.22.30 Enumeration MassInformationSource

Defines possible sources for *MassInformations*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Prototype	A <i>MassInformation</i> with this <i>valueSource</i> has been determined based on a preproduction part.
Series	A <i>MassInformation</i> with this <i>valueSource</i> has been determined based on a serial production part.
IMDS	A <i>MassInformation</i> with this <i>valueSource</i> has been retrieved from the <i>IMDS</i> (see http://www.mdssystem.com).

2.22.31 Enumeration OtherUnitName

Enumeration for the definition of other unit names.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Pi	NA
Piece	NA
ArcDegree	ArcDegree is a unit of angle ($1^\circ = \pi/180$ rad). This unit belongs to the special group of units called "non-SI units that are accepted for use with the SI".
ArcMinute	ArcMinute is a unit of angle (see ArcDegree).
ArcSecond	ArcSecond is a unit of angle (see ArcDegree).

2.22.32 Enumeration RobustnessClass

Defines the literals that shall be used for specific robustness classes.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
AmbientTemperature	NA
SolidParticleProtection	NA
LiquidIngressProtection	NA
ImpactProtection	NA
AbrasionResistance	NA

2.22.33 Enumeration RobustnessClassReferenceSystem

Defines the literals that shall be used for specific robustness class reference systems.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
ISO 20653	The literal that shall be used for the ISO 20653 "Road vehicles - Degrees of protection (IP code) - Protection of electrical equipment against foreign objects, water and access".
ISO 6722	The literal that shall be used for the ISO 6722 "Road vehicles -- 60 V and 600 V single-core cables".

	<i>Note: ISO 6722-1 & 6722-2 define the same temperature classes, except that 6722-1 defines two additional (G & H). Therefore 6722-1 & 6722-2 are not differentiated here.</i>
LV 112	The literal that shall be used for the LV 112.
LV 214	OEM Connector Test Specification
LV 215	OEM Specification for High Voltage contacting.
LV 312	Protective systems for wiring harnesses in motor vehicles Hoses

2.22.34 Enumeration SiPrefix

Enumeration for the definition of SI unit prefixes.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Yotta	NA
Zetta	NA
Exa	NA
Peta	NA
Tera	NA
Giga	NA
Mega	NA
Kilo	NA
Hecto	NA
Deca	NA
Deci	NA
Centi	NA
Milli	NA
Micro	NA
Nano	NA
Pico	NA
Femto	NA
Atto	NA
Zepto	NA
Yocto	NA

2.22.35 Enumeration SiUnitName

Enumeration for the definition of SI unit names. In difference to the SI-System gram is used as literal for the measurement of mass, instead of kilogram. Since a unit is formed with prefix and unit name gram would have to be defined as milli kilogram otherwise.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Metre	NA
Gram	NA
Second	NA
Ampere	NA
Kelvin	NA
Mole	NA
Candela	NA
Radian	NA
Steradian	NA
Hertz	NA
Newton	NA
Pascal	NA
Joule	NA
Watt	NA
Coulomb	NA
Volt	NA
Farad	NA
Ohm	NA
Siemens	NA
Weber	NA
Tesla	NA
Henry	NA
DegreeCelsius	NA
Lumen	NA
Lux	NA
Becquerel	NA
Gray	NA
Sievert	NA
Katal	NA

2.22.36 Enumeration TemperatureType

Defines the type (scenario) for which a TemperatureInformation applies.

General Information

Applied Stereotype	OpenEnumeration
---------------------------	---------------------------------

Enumeration Literals

Name	Comment
AmbientTemperature	NA
OperatingTemperature	NA
StorageTemperature	NA
ProcessingTemperature	NA

2.22.37 Enumeration USUnitName

Enumeration for the definition of US unit names.

General Information

Applied Stereotype	ClosedEnumeration
---------------------------	-----------------------------------

Enumeration Literals

Name	Comment
Thou	NA
Inch	NA
Foot	NA
Yard	NA
Chain	NA
Furlong	NA
Mile	NA
League	NA
FluidOunce	NA
Gill	NA
Pint	NA
Quart	NA
Gallon	NA
Grain	NA
Drachm	NA
Ounce	NA
Pound	NA
Stone	NA
Quarter	NA
HundredWeight	NA
Ton	NA
Perch	NA

Rood	NA
Acre	NA
AWG	American wire gauge (AWG), also known as the Brown & Sharpe wire gauge, is a standardized wire gauge system used in the United States and Canada.

2.22.38 Enumeration ValueDetermination

Enumeration for the definition of a value determination. (see KBLFRM-316)

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Calculated	The value is determined by a calculation algorithm.
Measured	The value is determined by an exact measurement.
Estimated	The value is estimated by a person.

2.23 Module pin_wire_mapping

2.23.1 Class PinWireMappingPoint

The *PinWireMappingPoint* creates a single variance free mapping between a *ContactPoint* and a *PinComponentReference* within a *PinWireMappingSpecification* (more details there).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
contactPoint	ContactPoint	1	Y	N	NA	N	NA
pinComponentReference	PinComponentReference	1	Y	N	NA	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
pinWireMappingPoint	0..*	Y	PinWireMappingSpecification	NA	NA

2.23.2 Class PinWireMappingSpecification

A *PinWireMappingSpecification* can be used create **variance free** mappings between a wire (represented by the contact point) and the pin of an E/E component. This is a possibility to create a shortcut in the model between a wire and its connected E/E-component (e.g. a fuse) that might be only indirectly connected to a wire (e.g. via a fuse and relay carrier). This is a relevant information for e.g. the validation of fusing. See the Pin Wire Mapping Diagram for more details.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
pinWireMappingPoint	PinWireMappingPoint	0..*	Y	N	NA	Y	NA

2.24 Module placeable_element

2.24.1 Class MeasurementPoint

Defines a reference point on a component that can be used to apply a Dimension. This is normally a significant point of the component e.g. an edge, a hole, a bolt or something similar.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the MeasurementPoint. The identification is guaranteed to be unique within a component.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
measurementPoint	1	N	MeasurementPointPosition	0..*	NA
measurementPoint	0..*	Y	PlaceableElementSpecification	1	Specifies the <i>MeasurementPoints</i> of a <i>PlaceableElement</i> .
measurementPoint	1	N	MeasurementPointReference	0..*	References the <i>MeasurementPoint</i> that is instanced by this <i>MeasurementPointReference</i> .

2.24.2 Class PlaceableElementSpecification

Specification for the general aspects of a component that are enabling the component to be placed in a topology. All components that should have the ability to be placed on a certain position in the topology must have a *PlaceableElementSpecification*.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
validPlacementTypes	PlacementType	1..2	Defines the <i>PlacementTypes</i> that are valid for this element.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
measurementPoint	MeasurementPoint	0..*	Y	N	1	Y	Specifies the <i>MeasurementPoints</i> of a <i>PlaceableElement</i> .
placementPoint	PlacementPoint	0..*	Y	N	1	Y	Specifies the <i>PlacementPoints</i> of a <i>PlaceableElementSpecification</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
placeableElementSpecification	1	N	PlaceableElementRole	0..*	References the <i>PlaceableElementSpecification</i> that is instanced by this <i>PlaceableElementRole</i> .

2.24.3 Class PlacementPoint

Defines a point on the component which can be placed explicitly in the topology (e.g. opening in a grommet, the entry point of a connector housing). If a *PlacementPoint* requires further specification for a certain component type (e.g. the entry point of a connector housing), this is done by an element specific for the component type, which related to the placement point.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>PlacementPoint</i> . The identification is guaranteed to be unique within a component.
segmentDiameter	ValueRange	0..1	Specifies the valid segment diameter range for which this <i>PlacementPoint</i> is suitable.

Outgoing Relations

Other End	This	General
-----------	------	---------

Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
supplementaryParts	PartRelation	0..*	Y	N	NA	N	References the <i>PartRelations</i> that specify supplementary parts for this <i>PlacementPoint</i> , e.g. cable ties for attaching the component on the harness.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
placementPoint	0..1	N	FittingOutlet	0..*	Specifies the <i>PlacementPoint</i> that represents this <i>FittingOutlet</i> in a <i>PlaceableElementSpecification</i> .
placementPoint	0..1	N	SegmentConnectionPoint	0..*	Specifies the <i>PlacementPoint</i> that represents this <i>SegmentConnectionPoint</i> in a <i>PlaceableElementSpecification</i> .
placementPoint	1	N	PlacementPointReference	0..*	References the <i>PlacementPoint</i> that is instanced by this <i>PlacementPointReference</i> .
placementPoint	0..1	N	WireReception	0..*	Specifies the <i>PlacementPoint</i> that represents this <i>WireReception</i> in a <i>PlaceableElementSpecification</i> .
placementPoint	0..1	N	CableDuctOutlet	0..*	Specifies the <i>PlacementPoint</i> that represents this <i>CableDuctOutlet</i> in a <i>PlaceableElementSpecification</i> .
placementPoint	0..1	N	CableLeadThroughOutlet	0..*	Specifies the <i>PlacementPoint</i> that represents this <i>CableLeadThroughOutlet</i> in a <i>PlaceableElementSpecification</i> .
placementPoint	1	N	PlacementPointPosition	0..*	NA
placementPoint	0..1	N	SegmentConnectionPointHC	0..*	Specifies the <i>PlacementPoint</i> that represents this <i>SegmentConnectionPoint</i> in a <i>PlaceableElementSpecification</i> .
placementPoint	0..*	Y	PlaceableElementSpecification	1	Specifies the <i>PlacementPoints</i> of a <i>PlaceableElementSpecification</i> .

2.24.4 Enumeration PlacementType

Defines the type for which a *PlaceableElement* can be used.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
OnWay	<i>OnWay</i> refers to an <i>OnWayPlacement</i> .
OnPoint	<i>OnPoint</i> refers to an <i>OnPointPlacement</i> .

2.25 Module placement

2.25.1 Class DefaultDimension

A *DefaultDimension* defines a tolerance value that shall be applied to a part, if no explicit tolerance value has been defined.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
dimensionValueRange	ValueRange	1	dimensionValueRange defines the magnitude of measurements for which the tolerance applies (e.g. length from 500mm - 1500mm).
dimensionType	DefaultDimensionType	1	The <i>dimensionType</i> defines to which measurements this <i>DefaultDimension</i> applies.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
toleranceIndication	Tolerance	1	Y	N	0..1	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
defaultDimension	0..*	Y	DefaultDimensionSpecification	NA	NA

2.25.2 Class DefaultDimensionSpecification

A *DefaultDimensionSpecification* defines tolerances that shall be applied to a part, if no explicit tolerance value has been defined.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
defaultDimension	DefaultDimension	0..*	Y	N	NA	Y	NA

2.25.3 Class Dimension

A Dimension defines a measurement for the distance between two Locations. An acceptable tolerance can be specified additionally. If the Locations are not located on adjacent topology-elements it is possible to specify a Path for the dimension along which the measurement must be taken.

The value for the Dimension is not specified as NumericalValue (which can define a Tolerance as well). This is because the "valueComponent" of the NumericalValue is mandatory. For Dimensions it shall be optional since there are scenarios where the dimension only specifies a Tolerance for a distance defined by the topology (segment length + placement information).

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the Dimension. The identification is guaranteed to be unique within the containing PlacementSpecification.
valueComponent	Double	0..1	Defines the value of the dimension. This value can be null, if it shall be calculated and only a tolerance shall be defined.
valueCalculated	Boolean	0..1	Defines if the value of the <i>Dimension</i> was calculated (e.g. the sum of segment lengths in the topology) or if it was defined manually.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
definedLocations	Location	0..2	Y	N	1	Y	NA
dimensionAnchor	DimensionAnchor	1	Y	N	0..*	N	References the location that is used as dimension point for the dimensioning (e.g. the entry point of a bundle into a connector housing). See KBLFRM-329 and KBLFRM-391.
path	Path	0..1	Y	N	0..1	Y	Specifies a path in the topology along which the dimension is defined.
referenceAnchor	DimensionAnchor	1	Y	N	0..*	N	References the location that is used as reference point for the dimensioning (e.g. the location of a fixing as this relates to a fixpoint of the body in white). See KBLFRM-329 and KBLFRM-391.
tolerance	Tolerance	0..1	Y	N	0..1	Y	NA
unitComponent	Unit	1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
dimension	0..*	Y	PlacementSpecification	1	Specifies the Dimensions defined by the PlacementSpecification.

2.25.4 Class DimensionAnchor

A *DimensionAnchor* represents an abstract anchor onto which a *Dimension* can be specified.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	true

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
dimensionAnchor	1	N	Dimension	0..*	References the location that is used as dimension point for the dimensioning (e.g. the entry point of a bundle into a connector housing). See KBLFRM-329 and KBLFRM-391.
referenceAnchor	1	N	Dimension	0..*	References the location that is used as reference point for the dimensioning (e.g. the location of a fixing as this relates to a fixpoint of the body in white). See KBLFRM-329 and KBLFRM-391.

2.25.5 Class MeasurementPointReference

A *MeasurementPointReference* is the instance of a *MeasurementPoint* in the context of an *OccurrenceOrUsage*.

General Information

Base Classifier	DimensionAnchor
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>MeasurementPointReference</i> . The identification is guaranteed to be unique within the containing <i>PlaceableElementRole</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
measurementPoint	MeasurementPoint	1	Y	N	0..*	N	References the <i>MeasurementPoint</i> that is instanced by this <i>MeasurementPointReference</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
measurementPointReference	0..*	Y	PlaceableElementRole	1	NA

2.25.6 Class OnPointPlacement

An OnPointPlacement is a placement of an OccurrenceOrUsage that places it onto discrete points, in most cases one point. In some cases, it is necessary to place a component (with more than one reference point) onto multiple points (e.g. a cable duct).

General Information

Base Classifier	Placement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
location	Location	1..*	Y	N	0..1	Y	References the Locations where Placement places the reference points of the placed element.

2.25.7 Class OnWayPlacement

An OnWayPlacement places an OccurrenceOrUsage onto an area of the Topology (e.g. a tape or a tube). The area is defined by a startLocation and an endLocation. If startLocation and endLocation are not located on the same TopologySegment it is possible to specify a Path of TopologySegments over which the OnWayPlacement goes.

The names start- and endLocation are used to distinguish between the two ends. It does **not** indicate a direction as property of the product (e.g. for tapes).

General Information

Base Classifier	Placement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
endLocation	Location	1	Y	N	0..1	Y	References the Location where OnWayPlacement ends.
path	Path	0..1	Y	N	0..1	Y	Specifies the topology path defining the way the OnWayPlacement takes in the topology.
startLocation	Location	1	Y	N	0..1	Y	References the Location where OnWayPlacement starts.

2.25.8 Class PlaceableElementRole

A PlaceableElementRole defines the instance specific properties and relationships of a PlaceableElement.

General Information

Base Classifier	Role
-----------------	----------------------

Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
measurementPointReference	MeasurementPointReference	0..*	Y	N	1	Y	NA
placeableElementSpecification	PlaceableElementSpecification	1	Y	N	0..*	N	References the <i>PlaceableElementSpecification</i> that is instantiated by this <i>PlaceableElementRole</i> .
placementPointReference	PlacementPointReference	0..*	Y	N	1	Y	

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
placedElement	1..*	N	Placement	0..*	References the PlaceableElementRoles placed by the Placement.

2.25.9 Class Placement

A placement defines the placement of a PlaceableElementRole onto a Topology. For the definition of the place on the Topology, Locations are used. A Placement can either be a placement on discrete points (OnPointPlacement) or on an area of the topology (OnWayPlacement).

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the Location. The identification is guaranteed to be unique within the PlacementSpecification.
type	String	0..1	Room to specify additional type information of the placement.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
isOnTopOf	Placement	0..*	Y	N	0..*	N	Specifies constraints on ordering of Placements. All referenced Placements must be below (nearer to the center of the Segment) this Placement. (see KBLFRM-171)

placedElement	PlaceableElement Role	1..*	Y	N	0..*	N	References the PlaceableElementRoles placed by the Placement.
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Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
isOnTopOf	0..*	N	Placement	0..*	Specifies constraints on ordering of Placements. All referenced Placements must be below (nearer to the center of the Segment) this Placement. (see KBLFRM-171)
placement	0..*	Y	PlacementSpecification	1	Specifies the Placements defined by the PlacementSpecification.

2.25.10 Class PlacementPointReference

A *PlacementPointReference* is the instance of a *PlacementPoint* in the context of an *OccurrenceOrUsage*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the PlacementPointReference. The identification is guaranteed to be unique within the containing PlaceableElementRole.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
placementPoint	PlacementPoint	1	Y	N	0..*	N	References the <i>PlacementPoint</i> that is instanced by this <i>PlacementPointReference</i> .
usedSupplementaryParts	OccurrenceOrUsage	0..*	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
placedPlacementPoints	0..*	N	Location	0..*	References the <i>PlacementPointReference</i> that is placed by this location.
placementPointReference	0..*	Y	PlaceableElement Role	1	

2.25.11 Class PlacementSpecification

Specification for the description of placements. *Placements* are used to connect *OccurrenceOrUsages* with a topology.

General Information

Base Classifier	Specification
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Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
dimension	Dimension	0..*	Y	N	1	Y	Specifies the Dimensions defined by the PlacementSpecification.
placement	Placement	0..*	Y	N	1	Y	Specifies the Placements defined by the PlacementSpecification.

2.25.12 Enumeration DefaultDimensionType

The *dimensionType* defines to which measurements this *DefaultDimension* applies.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
MeasurementChains	Applies to all measurements without specific tolerance definitions or a more specific DefaultDimensionType. Sums of individual dimensions must also not exceed these tolerances.
SegmentLengths	Applies to length values for segments.
FixingDistances	Applies to the measurements between individual fixings.
WireLengths	Applies to length values for <i>WireElementReferences</i> .

2.26 Module requirements_conformance**2.26.1 Class RequirementsConformanceSpecification**

A *RequirementsConformanceSpecification* can be used to express the conformance of a *PartVersion* (or a group of *PartVersions*). The *PartVersions* to which this *RequirementsConformanceSpecification* applies are expressed through the *describedPart* reference.

General Information

Base Classifier	PartOrUsageRelatedSpecification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

conformanceStatement	RequirementsConformanceStatement	0..*	Y	N	1	Y	NA
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2.26.2 Class RequirementsConformanceStatement

A *RequirementsConformanceStatement* states that the *PartVersions* referenced by the parent *RequirementsConformanceSpecification* satisfy or do not satisfy the requirements defined in the associated *DocumentVersion* (via the *requirementsSpecification* association).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
satisfies	Boolean	1	Defines if the <i>describedParts</i> satisfy (satisfy = true) or explicitly fail (satisfy = false) to conform with the <i>requirementsSpecification</i> .
description	LocalizedString	0..*	A free text description / additional information / comment for the <i>RequirementsConformanceStatement</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
documentVersion	DocumentVersion	1	Y	N	NA	N	References the <i>DocumentVersion</i> that contains the requirements to which a conformance statement shall be expressed.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
conformanceStatement	0..*	Y	RequirementsConformanceSpecification	1	NA

2.27 Module reuseage

2.27.1 Class Reusage

A *Reusage* defines a link for of information reuse between model elements that are reused in a different context. The source of information (*template*) and the destination (*usage*) shall have the same type (VEC model class). For more information see Chapter "Information Reuse".

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
template	ExtendableElement	1	Y	N	0..*	N	The <i>template</i> references a model element that is used as "source" of information.
usage	ExtendableElement	1	Y	N	0..*	N	The <i>usage</i> references a model element that is the reuse of the <i>template</i> . The types of elements referenced by <i>usage</i> and <i>template</i> shall be the same.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
reusage	0..*	Y	ReusageSpecification	1	The <i>Reusages</i> that define the actual mapping between information source and

2.27.2 Class ReusageSpecification

A *ReusageSpecification* enables the traceability of information reuse or derivation in different contexts. A mapping between the source (*template*) and the destination (*usage*) model elements is created with the contained *Reusage*-Elements. For more information see Chapter "Information Reuse".

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
reusage	Reusage	0..*	Y	N	1	Y	The <i>Reusages</i> that define the actual mapping between information source and

2.28 Module routing**2.28.1 Class Routing**

A Routing is the assignment of a RoutableElement (Connection or WireElementReference) to a path in the topology.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
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identification	String	0..1	Specifies a unique identification of the Routing. The identification is guaranteed to be unique within the RoutingSpecification.
specialRoutedComment	LocalizedString	0..1	Allows the specification of an explanation why this routing has been routed in a special way.
specialRouted	Boolean	0..1	Specifies that routing has been created in a special way. This means it has not been calculated in the standard way, because for some reason the result of the standard calculation has been inconvenient.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
mandatorySegment	TopologySegment	0..*	Y	N	0..*	N	Specifies some constraints for the routing. If the path of the routing is recalculated the referenced segments must be visited.
path	Path	1	Y	N	0..1	Y	Specifies a Path on the topology that is used for the routing.
routedElement	RoutableElement	1	Y	N	0..*	N	Specifies the Element that is routed.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
routing	0..*	Y	RoutingSpecification	1	Specifies the Routings described by the RoutingSpecification.

2.28.2 Class RoutingSpecification

Specification for the description of Routings.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
routing	Routing	0..*	Y	N	1	Y	Specifies the Routings described by the RoutingSpecification.

2.29 Module schematic

2.29.1 Class ComponentConnector

A *ComponentConnector* is a grouping of *ComponentPorts* and represents a logical abstraction of a connector of an *EEDComponent*. When the *ComponentNode* is realized by an *EEDComponentRole* the *ComponentConnector* will result in a *HousingComponent*.

It is permitted that a *ComponentConnector* is a 150% definition of connector.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the ComponentConnector. The identification is guaranteed to be unique within the ComponentNode.
description	AbstractLocalizedString	0..*	NA

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentPort	ComponentPort	0..*	Y	N	1	Y	Specifies the ComponentPorts of the ComponentConnector.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
componentConnector	0..1	N	HousingComponent	0..*	The <i>ComponentConnector</i> that represents this <i>HousingComponent</i> in the internal connectivity of the <i>EEComponentSpecification</i> .
componentConnector	0..*	N	ConnectorHousingRole	0..*	References the ComponentConnector that is realized by the referenced ConnectorHousing (OccurrenceOrUsage with ConnectorHousingRole). This can especially be relevant for inliners. KBLFRM-341.
componentConnector	0..*	Y	ComponentNode	1	Specifies the ComponentConnectors of a ComponentNode.
componentConnector	0..*	N	HousingComponentReference	0..*	References the ComponentConnector that is realized by the referenced HousingComponentReference.

2.29.2 Class ComponentNode

A ComponentNode is a node where an electrological component is located. It is a representative for an element in the electric system, e.g. an actuator, a sensor, an ECU. In this way it is quite similar to a NetworkNode and may even reference the corresponding NetworkNode in this case. However, a ComponentNode is more likely to be used as a representative of an inliner or a splice. Moreover, a ComponentNode can define childNodes in order to describe its internal structure.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the ComponentNode. The identification is guaranteed to be unique within the ConnectionSpecification.
abbreviation	LocalizedString	0..*	Room for a short name of the ComponentNode.
componentNodeType	ComponentNodeType	0..1	Specifies the type of the ComponentNode.
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the ComponentNode.
subType	ComponentNodeSubType	0..1	Specifies the sub type of a ComponentNode. The sub type allows a differentiation within a specific type. E.g. an actuator can be differentiated into lamps, speakers, motors.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
childNode	ComponentNode	0..*	Y	N	0..1	Y	Specifies the ComponentNodes that are a child of this ComponentNode.
componentConnector	ComponentConnector	0..*	Y	N	1	Y	Specifies the ComponentConnectors of a ComponentNode.
networkNode	NetworkNode	0..1	Y	N	0..*	N	References the NetworkNode that is realized by the ComponentNode.
realizedUsageNode	UsageNode	0..1	Y	N	0..*	N	References the <i>UsageNode</i> that is realized by this <i>ComponentNode</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
childNode	0..*	Y	ComponentNode	0..1	Specifies the ComponentNodes that are a child of this ComponentNode.
componentNode	1	N	ComponentNodeViewItem	0..*	References the <i>ComponentNode</i> that is represented by this <i>ComponentNodeViewItem</i> .
componentNode	1	N	ComponentNodeViewItem	0..*	NA
componentNode	0..*	Y	ConnectionSpecification	0..1	Specifies the ComponentNodes defined by the ConnectionSpecification.
componentNode	0..1	N	EEDComponentSpecification	0..*	The <i>ComponentNode</i> that defines the internal connectivity of this <i>EEDComponentSpecification</i> .
componentNode	0..*	N	EEDComponentRole	0..*	References the ComponentNode that is realized by the referenced EEDComponent (OccurrenceOrUsage with EEDComponentRole). KBLFRM-341

2.29.3 Class ComponentNodeViewItem

A *ComponentNodeViewItem* represents a *ComponentNode* within a *ConnectionViewSpecification* and defines its position within the layout.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>ComponentNodeViewItem</i> . The identification is guaranteed to be unique within the <i>ConnectionViewSpecification</i> .
gridSquare	String	0..1	Specifies the grid square in which the <i>ComponentNodeViewItem</i> is placed (e.g. E/40).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentNode	ComponentNode	1	Y	N	0..*	N	NA
displayedPort	ComponentPortViewItem	0..*	Y	Y	1	Y	Specifies all <i>ComponentPortViewItems</i> that are displayed on this <i>ComponentNodeViewItem</i> . The order of this association defines the clockwise arrangement of the ports on the node. As each <i>ComponentPortViewItem</i> can also define the side on which it is placed, side definitions take precedence over order. Caution: This association is <u>ordered</u> !
orientation	Transformation2D	0..1	Y	N	1	Y	Specifies the position / placement of the <i>ConnectionNodeViewItem</i> within the layout.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
placedNodeViewItems	0..*	Y	ConnectionViewSpecification	1	Specifies all <i>ComponentNodeViewItems</i> that are displayed within a <i>ConnectionViewSpecification</i> .

2.29.4 Class ComponentPort

Defines a port of ComponentNode. A ComponentPort is usually the realization of a NetworkPort. Electrological connections are defined between two or more ComponentPorts.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the ComponentPort. The identification is guaranteed to be unique within the ComponentConnector.
aliasId	AliasIdentification	0..*	Specifies additional identifiers for the element.
signalDirection	SignalDirection	0..1	Specifies the direction of the signal on this ComponentPort.
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the ComponentPort.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
networkPort	NetworkPort	0..1	Y	N	0..*	N	References the NetworkPort that is realized by the ComponentPort.
signal	Signal	0..1	Y	N	0..*	N	References the <i>Signal</i> that is associated with the <i>ComponentPort</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
componentPort	1	N	ComponentPortViewItem	0..*	References the <i>ComponentPort</i> that is represented by this <i>ComponentPortViewItem</i> .
componentPort	0..*	Y	ComponentConnector	1	Specifies the ComponentPorts of the ComponentConnector.
componentPort	0..1	N	PinComponent	0..*	The <i>ComponentPort</i> that represents this <i>PinComponent</i> in the internal connectivity of the <i>EEComponentSpecification</i> .
componentPort	0..*	N	CavityReference	0..*	References the <i>ComponentPort</i> that is implemented by this <i>CavityReference</i> .
componentPort	0..*	N	TerminalRole	0..*	References the ComponentPort that is realized by the referenced Terminal (OccurrenceOrUsage with TerminalRole). KBLFRM-341
connectedComponentPort	1	N	ConnectionEnd	0..*	References the ComponentPort that is connected by the ConnectionEnd.

2.29.5 Class ComponentPortViewItem

A *ComponentPortViewItem* represents a *ComponentPort* within a *ComponentNodeViewItem* and defines its position on the node.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
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side	ComponentPortViewItemSide	0..1	Specifies the side of the <i>ComponentNodeViewItem</i> on which the port should be displayed.
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Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentPort	ComponentPort	1	Y	N	0..*	N	References the <i>ComponentPort</i> that is represented by this <i>ComponentPortViewItem</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
displayedPort	0..*	Y	ConnectionNodeViewItem	1	Specifies all <i>ComponentPortViewItems</i> that are displayed on this <i>ComponentNodeViewItem</i> . The order of this association defines the clockwise arrangement of the ports on the node. As each <i>ComponentPortViewItem</i> can also define the side on which it is placed, side definitions take precedence over order. Caution: This association is <u>ordered</u> !
displayedPort	0..*	Y	ComponentNodeViewItem	1	Specifies all <i>ComponentPortViewItems</i> that are displayed on this <i>ComponentNodeViewItem</i> . The order of this association defines the clockwise arrangement of the ports on the node. As each <i>ComponentPortViewItem</i> can also define the side on which it is placed, side definitions take precedence over order. Caution: This association is <u>ordered</u> !

2.29.6 Class Connection

A Connection is an electrological connection between two or more ComponentPorts.

General Information

Base Classifier	RoutableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the Connection. The identification is guaranteed to be unique within the ConnectionSpecification.
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the Connection.

installationInstruction	Instruction	0..*	Specifies installation instruction for the connection.
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Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connectionEnd	ConnectionEnd	2..*	Y	N	1	Y	Specifies the ConnectionEnds of the Connection.
net	Net	0..1	Y	N	0..*	N	References the Net that is realized by the Connection.
signal	Signal	0..1	Y	N	0..*	N	References the signal that is transmitted by the connection.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
connection	0..*	N	MatingDetail	0..*	References the <i>Connection</i> that is realized by this <i>MatingPointDetail</i> . For example, when a connection is realized by directly plugging or screwing two E/E components together. The definition at level of the <i>MatingDetail</i> might be required if the <i>TerminalRole</i> of the <i>MatingPoint</i> carries multiple different potentials (e.g. Coax).
connection	0..*	Y	ConnectionSpecification	1	Specifies the Connection defined by the ConnectionSpecification.
connection	0..*	N	WireElementReference	0..*	References the Connection that is realized by the referenced WireElement (WireElementReference). KBLFRM-341
connection	0..*	N	BridgeTerminalRole	0..*	References the <i>Connection</i> that is realized by this <i>BridgeTerminalRole</i> .
connection	0..*	N	MatingPoint	0..*	References the <i>Connection</i> that is realized by this <i>MatingPoint</i> . For example, when a connection is realized by directly plugging or screwing two E/E components together.
connection	2..*	N	ConnectionGroup	0..*	References the Connections that are grouped by the ConnectionGroup.
switchedLogicalConnections	1..*	N	SwitchingState	NA	Specifies the <i>Connections</i> in the internal connectivity of the <i>EEDComponentSpecification</i> , that are switched by this <i>SwitchingState</i> .

2.29.7 Class ConnectionEnd

A connection end is the end of a Connection at a ComponentPort.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
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identification	String	1	Specifies a unique identification of the ConnectionEnd. The identification is guaranteed to be unique within the ConnectionSpecification.
isExternalEnd	Boolean	1	Specifies if the ConnectionEnd is connected to the internal or the external side of the ComponentPort.
gender	ConnectionEndGender	0..1	Specifies if the ConnectionEnd is male or female. This may be e.g. important in case of an inliner.
installationInstruction	Instruction	0..*	Specifies installation instruction for the <i>ConnectionEnd</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connectedComponentPort	ComponentPort	1	Y	N	0..*	N	References the ComponentPort that is connected by the ConnectionEnd.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
connectionEnd	2..*	Y	Connection	1	Specifies the ConnectionEnds of the Connection.
connectionEnd	0..*	N	WireEnd	0..*	NA

2.29.8 Class ConnectionGroup

A ConnectionGroup references two or more Connections expressing that the physical realization of the referenced Connection shall be somehow grouped e.g. twisted. For complex structures a ConnectionGroup can specify subgroups.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the ConnectionGroup. The identification is guaranteed to be unique within the ConnectionSpecification.
connectionGroupType	ConnectionGroupType	1	Specifies the type of the connectionGroup, valid literals are defined in the open enumeration <i>ConnectionGroupType</i> .
installationInstruction	Instruction	0..*	Specifies additional InstallationInstructions for the ConnectionGroup.
description	AbstractLocalizedString	0..*	Room for additional, human readable information about the ConnectionGroup.

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
connection	Connection	2..*	Y	N	0..*	N	References the Connections that are grouped by the ConnectionGroup.
subGroup	ConnectionGroup	0..*	Y	N	0..1	Y	Specifies the ConnectionGroups that are a subgroup of this ConnectionGroup.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
connectionGroup	0..*	N	WireElementReference	NA	References the <i>ConnectionGroup</i> that is realized by this <i>WireElementReference</i> . This applies normally to <i>WireElementReference</i> that have <i>subWireElements</i> .
connectionGroup	0..*	Y	ConnectionSpecification	0..1	Specifies the ConnectionGroup defined by the ConnectionSpecification.
connectionGroup	0..*	N	WireGrouping	NA	References the <i>ConnectionGroup</i> that is realized by this <i>WireGrouping</i> .
subGroup	0..*	Y	ConnectionGroup	0..1	Specifies the ConnectionGroups that are a subgroup of this ConnectionGroup.

2.29.9 Class ConnectionNodeViewItem*Deprecated since 2.0.2:**Reason: Class was accidentally named incorrectly in 2.0.0. Use ComponentNodeViewItem instead.*

A *ComponentNodeViewItem* represents a *ComponentNode* within a *ConnectionViewSpecification* and defines its position within the layout.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	Deprecated
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>ComponentNodeViewItem</i> . The identification is guaranteed to be unique within the <i>ConnectionViewSpecification</i> .
gridSquare	String	0..1	Specifies the grid square in which the <i>ComponentNodeViewItem</i> is placed (e.g. E/40).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentNode	ComponentNode	1	Y	N	0..*	N	References the <i>ComponentNode</i> that is represented by this <i>ComponentNodeViewItem</i> .

displayedPort	ComponentPortViewItem	0..*	Y	Y	1	Y	Specifies all <i>ComponentPortViewItems</i> that are displayed on this <i>ComponentNodeViewItem</i> . The order of this association defines the clockwise arrangement of the ports on the node. As each <i>ComponentPortViewItem</i> can also define the side on which it is placed, side definitions take precedence over order. Caution: This association is <u>ordered</u> !
orientation	Transformation2D	0..1	Y	N	1	Y	Specifies the position / placement of the <i>ConnectionNodeViewItem</i> within the layout.

Incoming Relations

This End			Other End		General	
Role	Mult.	Agg.	Type	Mult.	Comment	
placedViewItems	0..*	Y	ConnectionViewSpecification	1	Specifies all <i>ComponentNodeViewItems</i> that are displayed within a <i>ConnectionViewSpecification</i> .	

2.29.10 Class ConnectionSpecification

A *ConnectionSpecification* is used to define electrological connectivity.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
componentNode	ComponentNode	0..*	Y	N	0..1	Y	Specifies the <i>ComponentNodes</i> defined by the <i>ConnectionSpecification</i> .
connection	Connection	0..*	Y	N	1	Y	Specifies the <i>Connection</i> defined by the <i>ConnectionSpecification</i> .
connectionGroup	ConnectionGroup	0..*	Y	N	0..1	Y	Specifies the <i>ConnectionGroup</i> defined by the <i>ConnectionSpecification</i> .

2.29.11 Class ConnectionViewSpecification

The *ConnectionViewSpecification* supports the exchange of a basic layout for a *ConnectionSpecification*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
cartesianPoint	CartesianPoint2D	0..*	Y	N	0..1	Y	Specifies all the <i>CartesianPoint2Ds</i> that are used within the <i>ConnectionViewSpecification</i> .
placedNodeViewItems	ComponentNodeViewItem	0..*	Y	N	1	Y	Specifies all <i>ComponentNodeViewItems</i> that are displayed within a <i>ConnectionViewSpecification</i> .
placedViewItems	ConnectionNodeViewItem	0..*	Y	N	1	Y	<u>Deprecated since 2.0.2:</u> <u>Reason: Target class was accidentally named incorrectly in 2.0.0. Use <i>ComponentNodeViewItem</i> instead.</u> Specifies all <i>ComponentNodeViewItems</i> that are displayed within a <i>ConnectionViewSpecification</i> .

2.29.12 Enumeration ComponentNodeSubType

Defines agreed values for *ComponentNodeSubTypes*. Not all combinations of *ComponentNodeSubTypes* and *ComponentNodeTypes* are semantically correct (e.g. Lamp, Microphone, Speaker, Motor are all Actuators).

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Lamp	NA
Relay	NA
Fuse	NA
Microphone	NA
Speaker	NA
Motor	NA

2.29.13 Enumeration ComponentNodeType

Defines the common agreed values for the types of a *ComponentNode*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
ECU	Electronic Control Unit. An electronic control unit (ECU) is any embedded system in automotive electronics that controls one or more of the electrical systems or subsystems in a vehicle.
Sensor	A sensor is a device that detects changes to the environment and sends them as information to other devices (e.g. an ECU).
Actuator	An actuator is a device that is responsible for controlling or moving a mechanism.
CouplingDevice	A CouplingDevice is the (virtual) device that separates / connects two or more wiring harnesses. "Virtual" because it can be interpreted as a device / interface definition between the

	harnesses, where one harness behaves like an E/E component from the point of view of the other harness.
EnergyStorage	A device that stores energy in some kind of way (e.g. a battery).
Generator	A device that can generate energy.
PowerDistribution	A device that distributes power to other devices (e.g. a fuse box).
Switch	A "Switch" is a device that can change its internal connectivity in reaction to some external action (e.g. connect / disconnect some pins). A switch has no "active" logic (in contrast to an ECU).
Lamp	<u>Deprecated since 1.2.0:</u> <u>Reason: Use ComponentNodeType Actuator instead and ComponentSubType Lamp.</u>
Relay	<u>Deprecated since 1.2.0:</u> <u>Reason: Use instead ComponentSubType Relay.</u>
Fuse	<u>Deprecated since 1.2.0:</u> <u>Reason: Use instead ComponentSubType Fuse.</u>
Ground	Defines that this ComponentNode is a grounding point.
OpenEnd	Defines that this ComponentNode is the end point for some unconnected connections that require wires and routings in the resulting harness (e.g. an antenna). A component node of this type is used whenever connections shall not be connected (on one side), regardless of if it is an individual wire or a core of a multi core wire.
OpenLink	Defines that this ComponentNode is an OpenLink. In the description of partial systems, it can be necessary to reference a ComponentNode that is not defined in the scope of the partial system (e.g. vehicle infrastructure like power, ground, bus systems). OpenLinks must be resolved and replaced by a determined ComponentNode when a partial system is integrated into a vehicle system.
PotentialDistributor	A passive device that distributes one or more potentials. Splits a network into different Connections with the same Signal.

2.29.14 Enumeration ComponentPortViewItemSide

NA

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
Top	NA
Bottom	NA
Left	NA
Right	NA

2.29.15 Enumeration ConnectionEndGender

Defines the gender of a connection end.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Male	NA
Female	NA
Unspecified	NA

2.29.16 Enumeration ConnectionGroupType

Defines the type of physical realization that is expressed by a *ConnectionGroup*.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Twisted	NA
Shielded	NA
Coaxial	NA

2.30 Module signal

2.30.1 Class Signal

Specifies a signal.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the Signal. The identification is guaranteed to be unique within the SignalSpecification.
signalName	String	0..1	Name of the Signal, which is not guaranteed to be unique and is normally function oriented.
clampName	String	0..1	Specifies the name of the clamp e.g. KL15, KL30, KL31.
aliasId	AliasIdentification	0..*	Specifies additional identifiers for the <i>Signal</i> .
description	AbstractLocalizedString	0..*	A human readable description of the signal.
signalInformationType	SignalInformationType	0..1	Specifies the information type of the signal.

signalTransmissionMediumType	SignalTransmissionMediumType	0..1	Specifies the type of the transmission medium for the signal.
signalForm	SignalForm	0..1	Specifies the form of the signal.
signalCurve	SignalCurve	0..1	Specifies the curve of the signal.
signalType	SignalType	0..1	NA
signalSubType	SignalSubType	0..1	NA
currentType	CurrentType	0..1	NA
nominalVoltage	NominalVoltage	0..1	NA
dataRate	NumericalValue	0..1	Defines the data rate of the signal. This applies only to signals with <i>signalType</i> = 'information' and <i>signalInformationType</i> = 'digital'. For the numerical value, an appropriate IECUnit combination shall be used (e.g. GBit / Second).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
netType	NetType	0..1	Y	N	NA	N	NA
recommendedConductorSpecification	ConductorSpecification	0..1	Y	N	NA	N	Defines a recommended Specification for the cores that implement this signal.
recommendedInsulationSpecification	InsulationSpecification	0..1	Y	N	NA	N	Defines a recommended Specification for the insulation (e.g. the color) that implements this signal.
wireTupleRequirements	WireTupleSpecification	0..1	Y	N	NA	N	Defines a requirement for wire tuples.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
signal	0..1	N	Connection	0..*	References the signal that is transmitted by the connection.
signal	0..1	N	ComponentPort	0..*	References the <i>Signal</i> that is associated with the <i>ComponentPort</i> .
signal	0..1	N	WireElementReference	0..*	References the signal that is transmitted by the <i>WireElementReference</i> .
signal	0..1	N	PinComponentBehavior	0..*	Specifies the <i>Signal</i> associated with the pin in this behavior.
signal	0..*	Y	SignalSpecification	1	Specifies the signals.

2.30.2 Class SignalSpecification

Specification for the definition of a list of valid signals.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
signal	Signal	0..*	Y	N	1	Y	Specifies the signals.

2.30.3 Enumeration CurrentType

NA

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
ACOnePhase	NA
ACTwoPhase	NA
ACThreePhase	NA
DC	NA

2.30.4 Enumeration NominalVoltage*OpenEnumeration* defines the nominal voltage levels currently known and used in vehicles.**General Information**

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
12V	NA
24V	NA
48V	NA
HV	HV defines all voltages that are dangerous to health or life, regardless of alternating (AC) or direct current (DC). This literal should be used if the concrete class is not known, not defined or not specified.
HV1	HV Class 1 (200V DC)
HV2	HV Class 2 (300V DC)
HV3	HV Class 3 (600V DC)
HV4	HV Class 4 (900V DC)
HV5	HV Class 5 (1200V DC)

2.30.5 Enumeration SignalCurve

NA

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
ContinuousSignal	NA
DiscreteSignal	NA

2.30.6 Enumeration SignalForm

NA

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
SinusWave	NA
SquareWave	NA

2.30.7 Enumeration SignalInformationType

NA

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Analog	NA
Digital	NA

2.30.8 Enumeration SignalSubType

NA

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
CAN	Controller Area Network
LIN	Local Interconnect Network
FlexRay	NA
MOST	NA
Ethernet	NA

BroadR-Reach	NA
RTPGE	NA
APIX	Automotive Pixel Link
APIX2	NA
APIX3	NA
FBAS	NA
USB	Universal Serial Bus Version 1.X: The "USB" literal represents all USB 1.X Versions for USB 2.0 or USB 3.X the corresponding literals shall be used.
USB2	Universal Serial Bus Version 2.0
USB3.X	Universal Serial Bus Version 3.X
LVDS	low-voltage differential signalling
RGB	NA
BTLE	Bluetooth Low Energy
NFC	Near Field Communication
IEEE802.11	also: Wi-Fi, Wireless LAN
SignalGround	NA
PowerGround	NA

2.30.9 Enumeration SignalTransmissionMediumType

NA

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Electrical	NA
Optical	NA
Hydraulic	NA
Pneumatic	NA
Acoustic	NA
Inductive	NA
RadioTransmission	Signal transmission via electromagnetic waves (e.g. Wi-Fi, 4G, 5G)

2.30.10 Enumeration SignalType

NA

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Information	NA
Energy	NA
Ground	NA

2.31 Module signal_graph

2.31.1 Class SignalGraph

A *SignalGraph* is a directed graph of elements in the vehicle network that participate in the transmission of a signal / function / power / ground etc. A *SignalGraph* is expected to be at least a unilaterally connected graph, meaning it contains a directed path from *u* to *v* or a directed path from *v* to *u* for every pair *u* and *v* of *SignalGraphNode*s in the *SignalGraph*.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique <i>identification</i> of the <i>SignalGraph</i> . The <i>identification</i> is guaranteed to be unique within the <i>SignalGraphSpecification</i> .
aliasId	AliasIdentification	0..*	Defines additional identifiers for the <i>SignalGraph</i> .
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the <i>SignalGraph</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
edge	SignalGraphEdge	0..*	Y	N	1	Y	The <i>SignalGraphEdges</i> in the <i>SignalGraph</i> .
node	SignalGraphNode	0..*	Y	N	1	Y	The <i>SignalGraphNodes</i> in the <i>SignalGraph</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
from	1	N	SignalGraphRelation	NA	NA
signalGraph	0..*	Y	SignalGraphSpecification	1	NA
to	1	N	SignalGraphRelation	NA	NA

2.31.2 Class SignalGraphEdge

A *SignalGraphEdge* connects two *SignalGraphNodes* directed from "fromNode" to "toNode" and represents the direction of signal flow.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique <i>identification</i> of the <i>SignalGraphEdge</i> . The <i>identification</i> is guaranteed to be unique within the <i>SignalGraph</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
fromNode	SignalGraphNode	1	Y	N	0..*	N	NA
toNode	SignalGraphNode	1	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
edge	0..*	Y	SignalGraph	1	The <i>SignalGraphEdges</i> in the <i>SignalGraph</i> .

2.31.3 Class SignalGraphNode

A *SignalGraphNode* represent some physical element that participates in the signal transmission related to a *SignalGraph*. If the *representedElement* fails or is not present in a network, some paths in the *SignalGraph* maybe disrupted.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique <i>identification</i> of the <i>SignalGraphNode</i> . The <i>identification</i> is guaranteed to be unique within the <i>SignalGraph</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
representedElement	ConfigurableElement	1	Y	N	NA	N	References the <i>ConfigurableElement</i> that is represented by a <i>SignalGraphNode</i> . This is expected to be some physical element of the vehicle network, which participates in signal transmission. In many cases this will be an

							<i>OccurrenceOrUsage</i> , however in some cases sub elements of the <i>OccurrenceOrUsage</i> might be relevant, e.g. a <i>PinComponentReference</i> or a <i>WireElementReference</i> .
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Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
fromNode	1	N	SignalGraphEdge	0..*	NA
node	0..*	Y	SignalGraph	1	The <i>SignalGraphNode</i> s in the <i>SignalGraph</i> .
toNode	1	N	SignalGraphEdge	0..*	NA

2.31.4 Class SignalGraphRelation

A *SignalGraphRelation* represents a directed association between two *SignalGraphs*. The direction of the relationship is defined as "from" --> "to" and the semantic is defined by the *type*.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique <i>identification</i> of the <i>SignalGraphRelation</i> . The <i>identification</i> is guaranteed to be unique within the <i>SignalGraphSpecification</i> .
aliasId	AliasIdentification	0..*	Defines additional identifiers for the <i>SignalGraphRelation</i> .
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the <i>SignalGraphRelation</i> .
type	SignalGraphRelation Type	1	Defines the semantic of the relationship between the "from"- and "to"- <i>SignalGraph</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
from	SignalGraph	1	Y	N	NA	N	NA
to	SignalGraph	1	Y	N	NA	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
signalGraphRelation	0..*	Y	SignalGraphSpecification	1	NA

2.31.5 Class SignalGraphSpecification

Specification to define a list of *SignalGraphs* in a vehicle network.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
signalGraph	SignalGraph	0..*	Y	N	1	Y	NA
signalGraphRelation	SignalGraphRelation	0..*	Y	N	1	Y	NA

2.31.6 Enumeration SignalGraphRelationType

This *OpenEnumeration* defines a set of predefined semantics for the *SignalGraphRelation* between two *SignalGraph* ("from" & "to").

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
RedundancyFor	<i>RedundancyFor</i> specifies, that "from"- <i>SignalGraph</i> of a <i>SignalGraphRelation</i> is a redundant implementation for the "to"- <i>SignalGraph</i> .

2.32 Module terminal_pairing**2.32.1 Class TerminalPairing**

A *TerminalPairing* is a standard reference setup of exactly two terminals and a defined length of a core contacted to both terminals. The *TerminalPairing* defines physical properties that apply to this combination (e.g. *contactResistance*). Other values might be independent from a specific conductor (e.g. *matingForce*). Therefore, the *referencedCoreSpecification* is optional.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
contactResistance	NumericalValue	0..1	Specifies the resistance of the terminal pairing.
matingForce	NumericalValue	0..1	Specifies the joining force of the two terminals.

unmatingForce	NumericalValue	0..1	Specifies the force required to unmate the two terminals.
matingCycles	Integer	0..1	The maximum number of mating cycles, this terminal combination is certified for.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
firstTerminal	PartVersion	1	Y	N	0..*	N	References the first terminal of the TerminalPairing.
referencedCoreSpecification	ConductorSpecification	0..1	Y	N	0..*	N	References the CoreSpecification that is used on both sides of the ContactSystem.
secondTerminal	PartVersion	1	Y	N	0..*	N	References the second terminal of the TerminalPairing (first and second does not imply any specific order).

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
terminalPairing	0..*	Y	TerminalPairingSpecification	1	Specifies the TerminalPairings described by this TerminalPairingSpecification.

2.32.2 Class TerminalPairingSpecification

Specification for the definition of TerminalPairings.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
terminalPairing	TerminalPairing	0..*	Y	N	1	Y	Specifies the TerminalPairings described by this TerminalPairingSpecification.

2.33 Module topology**2.33.1 Class CartesianVector**

Abstract super class for vectors, either 2D or 3D.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA

Is Abstract	true
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2.33.2 Class GeometryNode

A GeometryNode is the geometric representation of a TopologyNode. A TopologyNode may be represented by more than one GeometryNodes, but only within different *BuildingBlockSpecification2D/3D*. That means, in a single BuildingBlockSpecification a *TopologyNode* shall only be represent once (or not).

A GeometryNode is either a GeometryNode2D or a GeometryNode3D.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the GeometryNode. The identification is guaranteed to be unique within the corresponding BuildingBlockSpecification. For all VEC-documents a GeometryNode-instance can be trusted to be the same if the BuildingBlockSpecification-instance is the same (see BuildingBlockSpecification) and the identification of the GeometryNode is the same.
aliasId	AliasIdentification	0..*	Specifies additional identifiers for the GeometryNode.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
referenceNode	TopologyNode	0..1	Y	N	0..*	N	References the TopologyNode that is represented by the GeometryNode.

2.33.3 Class GeometrySegment

A GeometrySegment is the geometric representation of a TopologySegment. A TopologySegment may be represented by more than one GeometrySegments, but only within different *BuildingBlockSpecification2D/3D*. That means, in a single BuildingBlockSpecification a *TopologySegment* shall only be represent once (or not).

Furthermore, the definition of the *GeometrySegment* shall be consistent to the definition in the *TopologySpecification*. That means, that a *GeometrySegment* shall have those *GeometryNodes* as start- & endNode that represent the *TopologyNodes* referenced from the corresponding *TopologySegment*.

A GeometrySegment is either a GeometrySegment2D or a GeometrySegment3D.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the GeometrySegment. The identification is guaranteed to be unique within the corresponding

			BuildingBlockSpecification. For all VEC-documents a GeometrySegment-instance can be trusted to be the same if the BuildingBlockSpecification-instance is the same (see BuildingBlockSpecification) and the identification of the GeometrySegment is the same.
aliasId	AliasIdentification	0..*	Specifies additional identifiers for the GeometrySegment.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
referenceSegment	TopologySegment	0..1	Y	N	0..*	N	References the TopologySegment that is represented by the GeometrySegment.

2.33.4 Class Location

A Location specifies a distinct position on a topology. Locations can be used for the placement of components or for the definition of Dimensions.

General Information

Base Classifier	DimensionAnchor
Applied Stereotype	NA
Is Abstract	true

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the Location. The identification is guaranteed to be unique within the PlacementSpecification.
matchingId	String	0..1	Specifies an identification for matching the location with a reference point of component (e.g. a cable channel).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
placedPlacementPoints	PlacementPointReference	0..*	Y	N	0..*	N	References the <i>PlacementPointReference</i> that is placed by this location.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
definedLocations	0..2	Y	Dimension	1	NA
endLocation	1	Y	OnWayPlacement	0..1	References the Location where OnWayPlacement ends.
firstLocation	1	Y	ZoneCoverage	0..1	NA
location	1..*	Y	OnPointPlacement	0..1	References the Locations where Placement places the reference points of the placed element.

mappedPosition	1	Y	NodeMapping	0..1	NA
secondLocation	1	Y	ZoneCoverage	0..1	NA
startLocation	1	Y	OnWayPlacement	0..1	References the Location where OnWayPlacement starts.

2.33.5 Class NodeLocation

Specifies a TopologyNode as a Location.

General Information

Base Classifier	Location
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
referencedNode	TopologyNode	1	Y	N	0..*	N	References the TopologyNode on which the Location is located.

2.33.6 Class NodeMapping

Defines the relationship of an inner node to its outer topology. The relationship to the outer topology is expressed with a *Location*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
innerNode	TopologyNode	1	Y	N	NA	N	NA
mappedPosition	Location	1	Y	N	0..1	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
nodeMapping	0..*	Y	TopologyMapping Specification	1	NA

2.33.7 Class Path

Describes a path in the topology. A *Path* is a continuous way through a topology without interruptions, defined by an ordered list of *TopologySegments*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
segment	TopologySegment	0..*	N	Y	0..*	N	Specifies an ordered list of TopologySegments the routing goes through.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
mappedPosition	1	Y	SegmentMapping	0..1	NA
path	0..1	Y	Dimension	0..1	Specifies a path in the topology along which the dimension is defined.
path	0..1	Y	OnWayPlacement	0..1	Specifies the topology path defining the way the OnWayPlacement takes in the topology.
path	1	Y	Routing	0..1	Specifies a Path on the topology that is used for the routing.
restrictedPath	1	Y	TopologyBendingRestriction	0..1	The path that defines the restricted way in the topology.

2.33.8 Class SegmentCrossSectionArea

Specifies the cross-section area of a segment. For the data exchange the cross-section area is used and not the diameter, because the diameter is only a valid measure for circular segments. For circular segments, the diameter and the cross-section area can be translated into each other without the loss of information. Attributes of the type SegmentCrossSectionArea normally have the multiplicity [0..*]. This means that such an attribute can have SegmentCrossSectionArea entries for different crossSectionAreaTypes and valueDeterminations. It must not have multiple entries with the same crossSectionAreaType and valueDetermination.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
area	NumericalValue	1	Specifies the value of the cross-section area of the segment.
valueDetermination	ValueDetermination	1	Specifies the method for determination of the value.
crossSectionAreaType	SegmentCrossSectionAreaType	0..1	Specifies the type of the cross-section area of the segment. Different types are for example needed to differentiate between the

			designed space of a segment and the required space (e.g. calculated from the wires going through the segment). Attribute is defined as an <i>OpenEnumeration</i> .
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Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
crossSectionAreaInformation	0..*	Y	TopologySegment	1	Specifies the different SegmentCrossSectionAreas of the TopologySegment.

2.33.9 Class SegmentLength

Specifies the length of a *TopologySegment*. The length of a *TopologySegment* is defined as the length of the centerline of the segment.

Attributes of the type SegmentLength normally have the multiplicity [0..*]. This means that such an attribute can have SegmentLength-entries for different classifications. It must not have multiple entries with the same classification.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
length	NumericalValue	1	Specifies the length of the TopologySegment.
classification	LengthClassification	1	Specifies the classification of the segment length.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
lengthInformation	0..*	Y	TopologySegment	1	Specifies the different SegmentLengths of the TopologySegment.

2.33.10 Class SegmentLocation

Specifies a point on a TopologySegment as a Location.

General Information

Base Classifier	Location
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
offset	NumericalValue	1	Specifies the offset / distance of the Location to the anchor of the location.
anchor	AnchorType	1	Specifies if the location on the <i>TopologySegment</i> is defined as on offset relative to the startNode of the TopologySegment or the endNode.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
referencedSegment	TopologySegment	1	Y	N	0..*	N	References the <i>TopologySegment</i> on which the Location is located.

2.33.11 Class SegmentMapping

Defines the relationship of an inner segment to its outer topology. The relationship to the outer topology is expressed with a *Path*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
innerSegment	TopologySegment	1	Y	N	NA	N	NA
mappedPosition	Path	1	Y	N	0..1	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
segmentMapping	0..*	Y	TopologyMapping Specification	NA	NA

2.33.12 Class TopologyBendingRestriction

As the name implies, a *TopologyBendingRestriction* defines restrictions on the bendability a path in the *Topology*. There are multiple reasons, why such restrictions exist, for example:

- There is a bending restriction for a wire in the segment.
- The number of wires and the segment diameter is such, that excessive bending causes intolerable torsion forces on wires in the segment.
- Other technical reasons

Since these restrictions can be determined using a variety of methods the VEC provides a concept to store this information for later use. This makes the information available in the downstream processes, without detailed knowledge of the determination procedure (e.g. during form board design, packaging or installation).

The restriction applies to a path of segments as this can cover different case:

- a single segment
- two adjacent segments at a node

- the complete path of a wire

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
minBendRadiusDynamic	NumericalValue	0..1	Specifies the minimum bend radius for the restricted path, if it is used in a dynamic environment, where it is bended repeatedly (e.g. in the grommet of the back door).
minBendRadiusStatic	NumericalValue	0..1	Specifies the minimum bend radius for the restricted path, if it is used in a static environment, where it is bended once during installation. After that it remains unchanged in its bended position during usage.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
restrictedPath	Path	1	Y	N	0..1	Y	The path that defines the restricted way in the topology.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
topologyBendingRestriction	0..*	Y	TopologyBendingRestrictionSpecification	1	NA

2.33.13 Class TopologyBendingRestrictionSpecification

A *TopologyBendingRestrictionSpecification* can be used to define *TopologyBendingRestrictions*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
topologyBendingRestriction	TopologyBendingRestriction	0..*	Y	N	1	Y	NA

2.33.14 Class TopologyGroupSpecification

A TopologyGroupSpecification defines a new Topology based on the grouped Topologies. (see KBLFRM-240)

General Information

Base Classifier	TopologySpecification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
topologySpecification	TopologySpecification	0..*	Y	N	0..*	N	NA

2.33.15 Class TopologyMappingSpecification

A *TopologyMappingSpecification* allows the definition of hierarchical topologies. It relates an outer topology with an enclosed inner topology.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
innerTopology	TopologySpecification	1	Y	N	NA	N	NA
nodeMapping	NodeMapping	0..*	Y	N	1	Y	NA
outerTopology	TopologySpecification	1	Y	N	NA	N	NA
segmentMapping	SegmentMapping	0..*	Y	N	NA	Y	NA

2.33.16 Class TopologyNode

A TopologyNode is a point in the Topology where TopologySegments are starting and ending.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
aliasId	AliasIdentification	0..*	Specifies additional identifiers for the TopologyNode. Example: TopologyNode Ids may vary from one CAD system export to another. Therefore, the CAD system Id is improper for identification attribute. The identification shall have a value which is unique within the Topology. AliasId may be used for the CAD system Id.
identification	String	1	Specifies a unique identification of the TopologyNode. The identification is guaranteed to be unique within the TopologySpecification. For all VEC-documents a TopologyNode-instance can be trusted to be the same if the TopologySpecification-instance is the same (see TopologySpecification) and the identification of the TopologyNode is the same.
matchingPointId	String	0..1	Specifies an identifier that can be used for matching <i>TopologyNodes</i> that belong to different <i>TopologySpecifications</i> and are actually representing the same node. Example: There are two <i>TopologySpecifications</i> , each specifying the topology of a certain zone of the car. If the zones are adjacent, it is possible that there are <i>TopologyNodes</i> in both zones where the two topologies are connected / have contact. These "connection-nodes" would then define the same <i>matchingPointId</i> .
processingInstruction	LocalizedString	0..*	Specifies processing instructions for the TopologyNode.
nodeType	NodeType	0..1	Specifies the type of the TopologyNode. A Node can either be an EndNode, a Junction or an Inliner.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
instantiatedNode	TopologyNode	0..1	Y	N	0..*	N	<u>Deprecated since 2.2:</u> <u>Reason: Replaced by general concept, see "ReusageSpecification"</u> If this <i>TopologyNode</i> is an instance of another <i>TopologyNode</i> that is defined elsewhere (e.g. the topology of an assembly), then the instantiated may be referenced here.
realizedUsageNode	UsageNode	0..1	Y	N	0..*	N	References the <i>UsageNode</i> that is realized by this <i>TopologyNode</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
endNode	1	N	TopologySegment	0..*	References the TopologyNode where the TopologySegment ends.
innerNode	1	N	NodeMapping	NA	NA
instantiatedNode	0..1	N	TopologyNode	0..*	If this <i>TopologyNode</i> is an instance of another <i>TopologyNode</i> that is defined elsewhere (e.g. the topology of an assembly), then the instantiated may be referenced here.
referencedNode	1	N	NodeLocation	0..*	References the TopologyNode on which the Location is located.

referenceNode	0..1	N	GeometryNode	0..*	References the TopologyNode that is represented by the GeometryNode.
startNode	1	N	TopologySegment	0..*	References the TopologyNode where the TopologySegment starts.
topologyNode	0..*	Y	TopologySpecification	1	Specifies the TopologyNodes defined by the TopologySpecification.

2.33.17 Class TopologySegment

A *TopologySegment* is a distance in the Topology where no intermediate electrical contacts appear. If a Topology contains routed wires, then the wire at the beginning of a TopologySegment must be the same as in the ending.

TopologySegments are a logical construct to describe the physical representation of a wiring harness topology. Therefore, a *TopologySegment* is only valid if it has a physical manifestation. *TopologySegments* with a length of 0 or less do not have a physical manifestation and are therefore not valid.

Additionally, the usage of *TopologySegments* with a length of 0 create problems in the overall process. For example, the synchronization of 3D / 2D systems becomes harder or even impossible and the handling of wire protections on those *TopologySegment* is also unclear.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the TopologySegment. The identification is guaranteed to be unique within the TopologySpecification. For all VEC-documents a TopologySegment-instance can be trusted to be the same if the TopologySpecification-instance is the same (see TopologySpecification) and the identification of the TopologySegment is the same.
form	SegmentForm	0..1	Specifies the form of the TopologySegment. A TopologySegment can either be circular or noncircular.
aliasId	AliasIdentification	0..*	Specifies additional identifiers for the TopologySegment. Example: TopologySegment Ids may vary from one CAD system export to another. Therefore, the CAD system Id is improper for identification attribute. The identification shall have a value which is unique within the Topology. AliasId may be used for the CAD system Id.
processingInstruction	LocalizedString	0..*	Specifies processing instructions for the TopologySegment.

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
crossSectionAreaInformation	SegmentCrossSectionArea	0..*	Y	N	1	Y	Specifies the different SegmentCrossSectionAreas of the TopologySegment.
endNode	TopologyNode	1	Y	N	0..*	N	References the TopologyNode where the TopologySegment ends.

instantiatedSegment	TopologySegment	0..1	Y	N	0..*	N	<i>Deprecated since 2.2:</i> <i>Reason: Replaced by general concept, see "ReusageSpecification"</i> If this <i>TopologySegment</i> is an instance of another <i>TopologySegment</i> that is defined elsewhere (e.g. the topology of an assembly), then the instantiated may be referenced here.
lengthInformation	SegmentLength	0..*	Y	N	1	Y	Specifies the different SegmentLengths of the TopologySegment.
startNode	TopologyNode	1	Y	N	0..*	N	References the TopologyNode where the TopologySegment starts.

Incoming Relations

This End			Other End		General	
Role	Mult.	Agg.	Type	Mult.	Comment	
assignedSegment	1	N	ZoneAssignment	0..*	The <i>TopologySegment</i> that is assigned to <i>TopologyZone</i> with this <i>ZoneAssignment</i> .	
innerSegment	1	N	SegmentMapping	NA	NA	
instantiatedSegment	0..1	N	TopologySegment	0..*	If this <i>TopologySegment</i> is an instance of another <i>TopologySegment</i> that is defined elsewhere (e.g. the topology of an assembly), then the instantiated may be referenced here.	
mandatorySegment	0..*	N	Routing	0..*	Specifies some constraints for the routing. If the path of the routing is recalculated the referenced segments must be visited.	
referencedSegment	1	N	SegmentLocation	0..*	References the <i>TopologySegment</i> on which the Location is located.	
referenceSegment	0..1	N	GeometrySegment	0..*	References the TopologySegment that is represented by the GeometrySegment.	
segment	0..*	N	Path	0..*	Specifies an ordered list of TopologySegments the routing goes through.	
topologySegment	0..*	Y	TopologySpecification	1	Specifies the TopologySegments defined by the TopologySpecification.	

2.33.18 Class TopologySpecification

Specification for the definition of a topology. A topology consists of TopologyNodes, TopologySegments.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
topologyNode	TopologyNode	0..*	Y	N	1	Y	Specifies the TopologyNodes defined by the TopologySpecification.
topologySegment	TopologySegment	0..*	Y	N	1	Y	Specifies the TopologySegments defined by the TopologySpecification.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
innerTopology	1	N	TopologyMapping Specification	NA	NA
outerTopology	1	N	TopologyMapping Specification	NA	NA
topologySpecification	0..*	N	TopologyGroupSpecification	0..*	NA

2.33.19 Class TopologyZone

A *TopologyZone* divides a topology in different sections / design spaces. A *TopologyZone* may be subdivided in further *TopologyZone*. There can be multiple reasons for creating an orthogonal sectioning on a *Topology*. Therefore, *TopologyZones* can be overlapping.

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the Zone (normally the name). The identification is guaranteed to be unique within the TopologySpecification. For all VEC-documents a Zone-instance can be trusted to be the same if the TopologySpecification-instance is the same (see TopologySpecification) and the identification of the Zone is the same.
type	TopologyZoneType	0..1	The type of the TopologyZone. Valid values are defined in an OpenEnumeration.
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the zone.
ambientTemperature	TemperatureInformation	0..1	Defines the ambient temperature that can occur in this zone. This can result in specific requirements for the used components.
requiredRobustnessProperties	RobustnessProperties	0..*	Defines the robustness properties that are required in this zone. This can result in specific requirements for the used components (e.g. the ability for sealing).

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
assignment	ZoneAssignment	0..*	Y	N	NA	Y	The assignments of specific topology elements to this zone.
subZone	TopologyZone	0..*	Y	N	0..1	Y	Specifies the sub <i>TopologyZones</i> that are part the <i>TopologyZone</i> . All <i>ZoneAssignments</i> defined for subZones are automatically inherited by the parent zone.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
subZone	0..*	Y	TopologyZone	0..1	Specifies the sub <i>TopologyZones</i> that are part the <i>TopologyZone</i> . All <i>ZoneAssignments</i> defined for subZones are automatically inherited by the parent zone.
topologyZone	0..1	N	BuildingBlockSpecification3D	0..*	References the Zone that is building block represents. This shall be a TopologyZone with the type "DmuZone".
zone	0..*	Y	TopologyZoneSpecification	0..1	Specifies the <i>TopologyZones</i> that are part of the <i>TopologyZoneSpecification</i> .

2.33.20 Class TopologyZoneSpecification

Specification for the definition of TopologyZones.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
zone	TopologyZone	0..*	Y	N	0..1	Y	Specifies the <i>TopologyZones</i> that are part of the <i>TopologyZoneSpecification</i> .

2.33.21 Class ZoneAssignment

A *ZoneAssignment* defines that a specific TopologySegment is affected by the *TopologyZone*. If a coverage is defined, only the parts of the Segment that have a coverage are affected. If no coverage is defined, the whole segment is affected.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
assignedSegment	TopologySegment	1	Y	N	0..*	N	The <i>TopologySegment</i> that is assigned to <i>TopologyZone</i> with this <i>ZoneAssignment</i> .
coverage	ZoneCoverage	0..*	Y	N	1	Y	Contains a set of <i>ZoneCoverages</i> that define the areas of a <i>TopologySegment</i> that is affected by the TopologyZone. If no coverage is defined, the complete segment is affected.

							Multiple coverages can be necessary if the <i>TopologySegment</i> zigzags in and out of the <i>TopologyZone</i> .
--	--	--	--	--	--	--	---

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
assignment	0..*	Y	TopologyZone	NA	The assignments of specific topology elements to this zone.

2.33.22 Class ZoneCoverage

A *ZoneCoverage* defines an area on a *TopologySegment* that lies within a *TopologyZone*. The area is defined with two *Locations*. *Locations* are the same mechanism that is used to define placements for components. The area that is in the *TopologyZone* is the area between the two locations. There is no semantic in the direction of the definition, so the assignment of first & second *Location* is completely arbitrary.

However, there are some restrictions for the definition of the locations. All locations have to be in relation to the *TopologySegment* that is referenced by the containing *ZoneAssignment*. This means the *Locations* have to be either a *SegmentLocation* on the respective *TopologySegment* or a *NodeLocation* on the start or end node of this particular *TopologySegment*. A *ZoneCoverage* from start to end node of a *TopologySegment* is equivalent to the complete omission of *ZoneCoverages* for a particular *ZoneAssignment*.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
firstLocation	Location	1	Y	N	0..1	Y	NA
secondLocation	Location	1	Y	N	0..1	Y	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
coverage	0..*	Y	ZoneAssignment	1	Contains a set of <i>ZoneCoverages</i> that define the areas of a <i>TopologySegment</i> that is affected by the <i>TopologyZone</i> . If no coverage is defined, the complete segment is affected. Multiple coverages can be necessary if the <i>TopologySegment</i> zigzags in and out of the <i>TopologyZone</i> .

2.33.23 Enumeration AnchorType

Enumeration for the definition of *AnchorType* of the *SegmentLocation*.

General Information

Applied Stereotype	ClosedEnumeration
--------------------	-----------------------------------

Enumeration Literals

Name	Comment
FromStartNode	The offset of the location is measured from the startNode of the TopologySegment.
FromEndNode	The offset of the location is measured from the endNode of the TopologySegment.

2.33.24 Enumeration LengthClassification

Enumeration for the definition of a LengthClassification.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Designed	A designed length means that the value is derived automatically in a CAD Tool (e.g. from a 3D Geometry).
Adapted	An adapted length means that the value is not the exact value taken from the CAD tool but is adapted in some way. Adapted values are supposed for further use in the process, especially as basis for the product specification. Adapted values are normally created for example by rounding the designed values.

2.33.25 Enumeration NodeType

Enumeration for the definition of the type of a TopologyNode.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
EndNode	Electrical components are normally placed on an EndNode.
Junction	A Junction is a TopologyNode where no electrical component is placed.
Inliner	An Inliner is a TopologyNode where one section of the electrical system is connected to another section.

2.33.26 Enumeration SegmentCrossSectionAreaType

Defines valid values the type of the cross-section area of a *TopologySegment*, since a *TopologySegment* can have different cross section areas with different meanings in the process.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
Reserved	<i>Reserved</i> is the type for cross section areas that define a reserved space in the DMU for the <i>TopologySegment</i> .
Real	<i>Real</i> is the type for cross section areas that can be observed for <i>TopologySegments</i> with variants of a Harness that are producible.

2.33.27 Enumeration SegmentForm

Enumeration for the definition of the SegmentForm.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Circular	NA
NonCircular	NA

2.33.28 Enumeration TopologyZoneType

There can be various reasons to define TopologyZones. These can be differentiated with these type literals.

General Information

Applied Stereotype	OpenEnumeration
--------------------	---------------------------------

Enumeration Literals

Name	Comment
DmuZone	NA
EnvironmentZone	NA

2.34 Module usage_constraint

2.34.1 Class UsageConstraint

Specifies a constraint of the possible usages for PartVersion or PartUsages. UsageConstraints are of part master data information. UsageConstraints specify general restrictions for the application of a PartVersion or PartUsage (e.g. a certain period of time in which the part is allowed to be used). (see KBLFRM-229)

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	UsageConstraintType	1	Defines if the usage constraint is positive (allowance) or negative (denial).
fromDate	Date	0..1	Specifies the lower bound of the time period to which the usage constraint applies.
toDate	Date	0..1	Specifies the upper bound of the time period to which the usage constraint applies.
fromSerialNumber	String	0..1	Specifies the lower bound of a serial number range to which the usage constraint applies.
toSerialNumber	String	0..1	Specifies the upper bound of a serial number range to which the usage constraint applies.

projectPhase	String	0..*	Specifies the project phases to which the usage constraint applies.
fromEffectivityControlKey	String	0..1	In the production logistics of complex products (like vehicles), the control of operation and cancellation dates of changes, features, components or the like is usually not carried out via concrete dates but via "date keys". These keys are then mapped in planning systems with concrete time specifications for certain plants or production steps. In the VEC these keys are called <i>EffectivityControlKey</i> . With the <i>FromEffectivityControlKey</i> and <i>ToEffectivityControlKey</i> a validity interval can be defined. If one of the two values is defined and the other is omitted, the meaning shall be "unbounded".
toEffectivityControlKey	String	0..1	See <i>FromEffectivityControlKey</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
project	Project	0..*	Y	N	0..*	N	References the <i>Projects</i> to which the <i>UsageConstraint</i> applies. This means the described <i>PartVersion</i> is allowed / denied in the referenced <i>UsageConstraint</i> .
usageNode	UsageNode	0..*	Y	N	0..*	N	References the <i>UsageNode</i> to which the <i>UsageConstraint</i> applies. This means the described <i>PartVersion</i> is allowed / denied in the referenced <i>UsageNode</i> .

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
partUsageConstraint	1..*	Y	UsageConstraintSpecification	1	Specifies the <i>UsageConstraints</i> that apply to the <i>PartVersion</i> or <i>PartUsages</i> described by the <i>UsageConstraintSpecification</i> . The ordering of this association is relevant. The elements shall be arranged in the order of ascending priority. That means, elements further back in the collection have a higher priority.

2.34.2 Class UsageConstraintSpecification

Specification for the definition of usage constraints. The associated *UsageConstraints* are restricting the possible usages of the associated *PartVersions* and *PartUsages*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

constrainedParts	PartVersion	0..*	Y	N	0..*	N	References the <i>PartVersions</i> to which this <i>UsageConstraintSpecification</i> applies.
partUsageConstraint	UsageConstraint	1..*	Y	Y	1	Y	Specifies the <i>UsageConstraints</i> that apply to the <i>PartVersion</i> or <i>PartUsages</i> described by the <i>UsageConstraintSpecification</i> . The ordering of this association is relevant. The elements shall be arranged in the order of ascending priority. That means, elements further back in the collection have a higher priority.

2.34.3 Enumeration UsageConstraintType

Enumeration for the definition of the type of a *UsageConstraint*. Valid values are *Allow* and *Deny*.

General Information

Applied Stereotype	ClosedEnumeration
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Enumeration Literals

Name	Comment
Allow	NA
Deny	NA

2.35 Module usage_node

2.35.1 Class UsageNode

A *UsageNode* represents a position in an abstract vehicle. For example, the "Head Light Left". *UsageNodes* belong to the master data, and they are defined on some companywide level. They can be used to enforce consistent naming over different projects and different development streams (e.g. between Geometry and Electrolologic).

A *UsageNode* can be realized by different elements in the VEC (e.g. *NetworkNode*, *OccurrenceOrUsage*, *TopologyNode*, *ComponentNode*).

General Information

Base Classifier	ConfigurableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
abbreviation	LocalizedString	0..*	Specifies an abbreviation of the <i>UsageNode</i> . Normally this a human readable short name.
identification	String	1	Specifies a unique <i>identification</i> of the <i>UsageNode</i> . The <i>identification</i> is guaranteed to be unique within the context. For all VEC-documents a <i>UsageNode-instance</i> can be trusted to be the same if the context-instance is the same and the <i>identification</i> of the <i>UsageNode</i> is the same.
description	AbstractLocalizedString	0..*	Specifies additional, human readable information about the <i>UsageNode</i> .
usageNodeType	UsageNodeType	0..1	Defines the type of the <i>UsageNode</i> . The type determines how the <i>UsageNode</i> is handled in the latter processes.

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
subUsageNodes	UsageNode	0..*	Y	N	0..1	Y	NA
usedInProject	Project	0..*	Y	N	0..*	N	Specifies the <i>Projects</i> in which the <i>UsageNode</i> can be used.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
realizedUsageNode	0..1	N	OccurrenceOrUsage	0..*	References the <i>UsageNode</i> that is realized by this <i>OccurrenceOrUsage</i> .
realizedUsageNode	0..1	N	TopologyNode	0..*	References the <i>UsageNode</i> that is realized by this <i>TopologyNode</i> .
realizedUsageNode	0..1	N	NetworkNode	0..*	References the <i>UsageNode</i> that is realized by this <i>NetworkNode</i> .
realizedUsageNode	0..1	N	ComponentNode	0..*	References the <i>UsageNode</i> that is realized by this <i>ComponentNode</i> .
subUsageNodes	0..*	Y	UsageNode	0..1	NA
usageNode	0..*	N	UsageConstraint	0..*	References the <i>UsageNode</i> to which the <i>UsageConstraint</i> applies. This means the described <i>PartVersion</i> is allowed / denied in the referenced <i>UsageNode</i> .
usageNodes	0..*	Y	UsageNodeSpecification	0..1	Specifies the <i>UsageNodes</i> defined by this <i>UsageNodeSpecification</i> .

2.35.2 Class UsageNodeSpecification

A *UsageNodeSpecification* is a container for the specification of *UsageNodes*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
usageNodes	UsageNode	0..*	Y	N	0..1	Y	Specifies the <i>UsageNodes</i> defined by this <i>UsageNodeSpecification</i> .

2.35.3 Enumeration UsageNodeType

Enumeration for the different types of *UsageNodes*.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
ECU	NA
Sensor	NA
Actuator	NA
CouplingDevice	NA
EnergyStorage	NA
Generator	NA
PowerDistribution	NA
Switch	NA
Lamp	NA
Relay	NA
Fuse	NA
Ground	Defines that this ComponentNode is a grounding point.
Splice	NA
Protection	NA
Fixing	NA
Grommet	NA
CableDuct	NA

2.36 Module variants

2.36.1 Class ApplicationConstraint

An *ApplicationConstraint* defines a condition with which it is possible to express the inclusion or exclusion of ConfigurableElements in different variants of a product. The *ApplicationConstraint* is focused to express validity rules based on time periods or elements of the product hierarchy in a concise way (attributes and relationships). It is complementary to the *VariantConfiguration* which expresses validity rules based on control strings.

An *ApplicationConstraint* can reference another *ApplicationConstraint* as *baseInclusion*. In this case, an *ApplicationConstraint* can only be satisfied if its *baseInclusion* is also satisfied.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
type	ApplicationConstraintType	1	Defines if the application constraint is positive (allowance) or negative (denial). If the <i>ApplicationConstraint</i> has a <i>baseInclusion</i> that <i>baseInclusion</i> shall define the same type.
fromDate	Date	0..1	Specifies the lower bound of the time period to which the application constraint applies.

toDate	Date	0..1	Specifies the upper bound of the time period to which the application constraint applies.
fromSerialNumber	String	0..1	Specifies the lower bound of a serial number range to which the application constraint applies.
toSerialNumber	String	0..1	Specifies the upper bound of a serial number range to which the application constraint applies.
projectPhase	String	0..*	Specifies the project phases to which the application constraint applies.
fromEffectivityControlKey	String	0..1	In the production logistics of complex products (like vehicles), the control of operation and cancellation dates of changes, features, components or the like is usually not carried out via concrete dates but via "date keys". These keys are then mapped in planning systems with concrete time specifications for certain plants or production steps. In the VEC these keys are called <i>EffectivityControlKey</i> . With the <i>FromEffectivityControlKey</i> and <i>ToEffectivityControlKey</i> a validity interval can be defined. If one of the two values is defined and the other is omitted, the meaning shall be "unbounded".
toEffectivityControlKey	String	0..1	See <i>FromEffectivityControlKey</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
baseInclusion	ApplicationConstraint	0..1	Y	N	0..*	N	An <i>ApplicationConstraint</i> can only be satisfied if its <i>baseInclusion</i> is satisfied as well.
project	Project	0..*	Y	N	NA	N	Defines the projects for which the application constraint applies.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
applicationConstraint	0..*	N	ConfigurableElement	NA	References the application constraints that apply to the ConfigurableElement.
applicationConstraint	0..*	N	ConfigurationConstraint	NA	References the application constraint that applies to the ConfigurationConstraint.
applicationConstraint	1..*	Y	ApplicationConstraintSpecification	1	Specifies the ApplicationConstraints.
baseInclusion	0..1	N	ApplicationConstraint	0..*	An <i>ApplicationConstraint</i> can only be satisfied if its <i>baseInclusion</i> is satisfied as well.

2.36.2 Class ApplicationConstraintSpecification

Specification for the definition of application constraints. The associated ApplicationConstraints are restricting the possible usages of the associated *ConfigurableElements*.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
applicationConstraint	ApplicationConstraint	1..*	Y	N	1	Y	Specifies the ApplicationConstraints.

2.36.3 Class VariantCode

VariantCodes are defining the literals on which VariantConfiguration are stated. Possible VariantCodes might be different for steering types, optional equipment, engine-power class.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the variant code. This is also the literal value for the VariantCode, which shall be used in the logisticControlExpressions of VariantConfigurations.
aliasId	AliasIdentification	0..*	Room to specify additional identifiers for the VariantCode.
abbreviation	LocalizedString	0..1	Room for a human readable short name, title etc. of the VariantConfiguration.
codeType	VariantCodeType	0..1	Allows the classification of a VariantCodes.
description	AbstractLocalizedString	0..*	On optional human readable description of the variant code.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
variantCode	0..*	Y	VariantCodeSpecification	1	Specifies the individual VariantCodes defined in the VariantCodeSpecification.
variantCode	0..*	N	VariantGroup	0..*	References the VariantCodes that are member of the VariantGroup.

2.36.4 Class VariantCodeSpecification

Specification for the definition of variant codes.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End			This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
variantCode	VariantCode	0..*	Y	N	1	Y	Specifies the individual VariantCodes defined in the VariantCodeSpecification.

2.36.5 Class VariantConfiguration

A variant configuration defines a condition with which it is possible to express the inclusion or exclusion of ConfigurableElements in different variants of a product.

A *VariantConfiguration* can reference another *VariantConfiguration* as *baseInclusion*. In this case, a *VariantConfiguration* can only be satisfied if its *baseInclusion* is also satisfied.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	0..1	Specifies a unique identification of the variant configuration. The identification is guaranteed to be unique within the specification and does not change over the time.
description	AbstractLocalizedString	0..*	On optional human readable description of the variant configuration.
logisticControlString	String	0..1	Specifies a logisticControlString which can be used if the variant management is not done by Boolean logic.
logisticControlExpression	String	0..1	Specifies a logisticControlExpression expressed as Boolean term.
configurationType	VariantConfigurationType	0..1	Allows the classification of a VariantConfiguration. (see KBLFRM-250, KBLFRM-314, KBLFRM-290)

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
baseInclusion	VariantConfiguration	0..1	Y	N	NA	N	A <i>VariantConfiguration</i> can only be satisfied if its <i>baseInclusion</i> is satisfied as well.

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
baseInclusion	0..1	N	VariantConfiguration	NA	A <i>VariantConfiguration</i> can only be satisfied if its <i>baseInclusion</i> is satisfied as well.
configInfo	0..1	N	ConfigurableElement	0..*	References the configuration information that applies to the ConfigurableElement.
configInfo	0..1	N	ConfigurationConstraint	NA	References the configuration information that applies to the ConfigurationConstraint.

variantConfiguration	0..*	Y	VariantConfigurationSpecification	1	Specifies the individual VariantConfigurations defined in the VariantConfigurationSpecification.
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2.36.6 Class VariantConfigurationSpecification

Specification for the definition of variant configurations.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
variantConfiguration	VariantConfiguration	0..*	Y	N	1	Y	Specifies the individual VariantConfigurations defined in the VariantConfigurationSpecification.

2.36.7 Class VariantGroup

With a VariantGroup it is possible to group VariantCodes. The semantic of this grouping should be defined with the groupType (e.g. composition, choice, etc.).

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the variant code.
aliasId	AliasIdentification	0..*	Room to specify additional identifiers for the VariantGroup.
abbreviation	LocalizedString	0..1	Room for a human readable short name, title etc. of the VariantGroup.
description	AbstractLocalizedString	0..*	On optional human readable description of the variant group.
groupType	VariantGroupType	0..1	Allows the classification of a VariantGroups into different types. For example: - composition (e.g. winter package) - choice (e.g. right hand / left hand driving). Agreed literals for this attribute are defined in the OpenEnumeration <i>VariantGroupType</i> .

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment

variantCode	VariantCode	0..*	Y	N	0..*	N	References the VariantCodes that are member of the VariantGroup.
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Incoming Relations

This End			Other End			General
Role	Mult.	Agg.	Type	Mult.	Comment	
containedGroups	0..*	N	VariantStructureNode	0..*	NA	
variantGroup	0..*	Y	VariantGroupSpecification	1	Specifies the individual VariantGroups defined in the VariantGroupSpecification.	

2.36.8 Class VariantGroupSpecification

Specification for the definition of variant groups.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
variantGroup	VariantGroup	0..*	Y	N	1	Y	Specifies the individual VariantGroups defined in the VariantGroupSpecification.

2.36.9 Class VariantStructureNode

VariantStructureNodes can be used to define a hierarchical structure on *VariantGroups*. Every *VariantStructureNodes* can reference *VariantGroups* and *VariantStructureNodes* as children.

General Information

Base Classifier	ExtendableElement
Applied Stereotype	NA
Is Abstract	false

Attributes

Name	Type	Mult	Comment
identification	String	1	Specifies a unique identification of the <i>VariantStructureNode</i> .
aliasId	AliasIdentification	0..*	Room to specify additional identifiers for the <i>VariantStructureNode</i> .
abbreviation	LocalizedString	0..1	Room for a human readable short name, title etc. of the <i>VariantStructureNode</i> .
description	AbstractLocalizedString	0..*	On optional human readable description of the <i>VariantStructureNode</i> .

Outgoing Relations

Other End	This	General
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Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
childNodes	VariantStructureNode	0..*	Y	N	NA	Y	NA
containedGroups	VariantGroup	0..*	Y	N	0..*	N	NA

Incoming Relations

This End			Other End		General
Role	Mult.	Agg.	Type	Mult.	Comment
childNodes	0..*	Y	VariantStructureNode	NA	NA
rootNode	1	Y	VariantStructureSpecification	0..1	

2.36.10 Class VariantStructureSpecification

Specification to define any hierarchical structure on variant groups (e.g. by the means of a functional organization). The hierarchy starts with a single root node.

General Information

Base Classifier	Specification
Applied Stereotype	NA
Is Abstract	false

Outgoing Relations

Other End					This	General	
Role	Type	Multiplicity	Unique	Ordered	Multiplicity	Aggregation	Comment
rootNode	VariantStructureNode	1	Y	N	0..1	Y	

2.36.11 Enumeration ApplicationConstraintType

Enumeration for the definition of the type of an ApplicationConstraints. Valid values are *Allow* and *Deny*.

General Information

Applied Stereotype	ClosedEnumeration
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Enumeration Literals

Name	Comment
Allow	NA
Deny	NA

2.36.12 Enumeration VariantCodeType

OpenEnumeration for the available types of the variant codes.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Serial	
Prototype	

2.36.13 Enumeration VariantConfigurationType

Defines an OpenEnumeration for VariantConfigurationTypes. The configuration type defines in which "variant code universe" the logistic control expression is defined.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Logistic	The <i>VariantConfiguration</i> is a concrete logistic key (e.g. within a vehicle project) that used for example in order / delivery systems.
Technical	The <i>VariantConfiguration</i> is a technical (feature or function oriented) key normally used in a building block set. In a concrete vehicle project those keys are mapped to specific logistic keys within the project.

2.36.14 Enumeration VariantGroupType

Defines valid values for the type of VariantGroups.

General Information

Applied Stereotype	OpenEnumeration
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Enumeration Literals

Name	Comment
Family	A group with the type "Family" defines a set of variant codes that are mutually exclusive, and a valid configuration <u>must</u> contain one variant code from the group.
OptionalFamily	A group with the type "OptionalFamily" defines a set of variant codes that are mutually exclusive, and a valid configuration can, but does not have to contain one variant code from the group.
LogicalGroup	Defines a logical grouping on variant codes. However, no semantic constraint is implied.

2.37 Module VEC

2.37.1 Primitive Boolean

Primitive type for attributes with Boolean values. During schema generation this is translated to *xs:boolean*.

General Information

Applied Stereotype	NA
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2.37.2 Primitive Date

Primitive type for attributes with date and time values. During schema generation this is translated to *xs:dateTime*.

General Information

Applied Stereotype	NA
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2.37.3 Primitive Double

Primitive type for attributes with double values. During schema generation this is translated to *xs:double*.

General Information

Applied Stereotype	NA
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2.37.4 Primitive Integer

Primitive type for attributes with integer values. During schema generation this is translated to *xs:integer*.

General Information

Applied Stereotype	NA
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2.37.5 Primitive String

Primitive type for attributes with string values. During schema generation this is translated to *xs:string*.

General Information

Applied Stereotype	NA
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